

Cloudera on premises / CDP Private Cloud (PvC)

Installation & Setup

:: Deployment Guide ::

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CLOU_DERA

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Audience

The intended audience for this document includes, but is not limited to, sales engineers, field consultants, professional services, IT managers, IT engineers, partners, and customers who are interested in learning about and deploying the Cloudera Data Platform Private Cloud (Cloudera on premises) on the bare metal/ virtual machines for digital transformation through cloud-native modern data analytics and AI/ML.

Purpose of this Document

This document describes the architecture, installation, configuration, and validated use cases for the on premise platforms using Cloudera Data Platform Private Cloud Base on bare metal based servers or virtual machines. A reference architecture is provided to configure the Cloudera Platform on VMs.

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Author History

Name	Version	Date
Kuldeep Sahu	1.0	23-May-2024
Yash Gulati	2.0	23-Jan-2026

Cloudera on premises Installation and Configuration on on-premises Infrastructure:

This document provides all the required information for setup and install Cloudera on premises.

Important URLs:

Install Cloudera on premises Base and Data Service Clusters:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-installation.html>

<https://docs.cloudera.com/cdp-private-cloud-data-services/latest/installation-ecs/topics/cdppvc-installation-ecs-steps.html>

Uninstall and cleanup Cloudera on premises Base, Data Service Clusters and PostgreSQL DB:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-uninstallation.html>

<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/installation-ecs/topics/cdppvc-installation-ecs-uninstall-pvc.html>

<https://kb.objectrocket.com/postgresql/how-to-completely-uninstall-postgresql-757>

Internal documentation: Prerequisites list by Dennis Lee and PvC AWS Setup by Puneet Joshi

<https://dennislee22.github.io/docs/cdppvc>

<https://docs.google.com/document/d/1OSKBChSTbc8NhuQ8YXRN-YxFnaVBj47Lz4cWro-zTVs/edit>

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Introduction:

Cloudera on premises is an integrated analytics and data management platform deployed in private data centers. Cloudera Data Platform is a single platform that has two form factors CDP Public and Cloudera on premises.

It consists of Cloudera on premises Base and Cloudera on premises Data Services and offers broad data analytics and artificial intelligence functionality along with secure user access and data governance features.

Cloudera on premises (PvC) data services components run on containerized clusters and thus require a container orchestration engine to manage all the workloads.

There will be two major components in Cloudera on premises Installation:

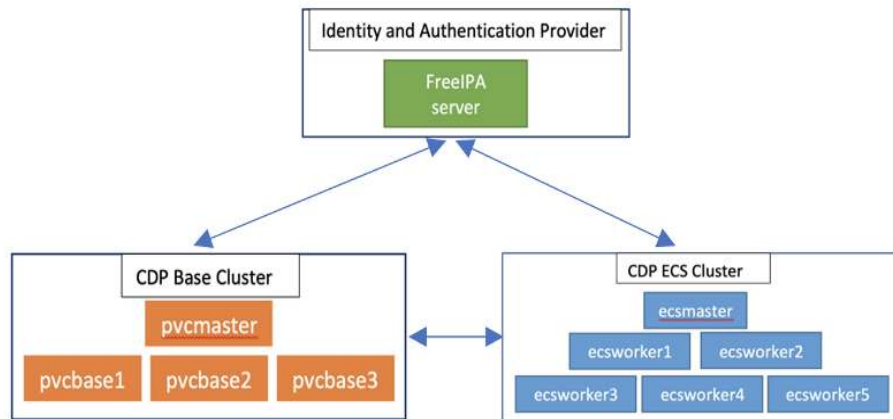
- Cloudera on premises Base Cluster
- Cloudera on premises Data Services Cluster

Cloudera on premises DS offers installation with two orchestration engines.

- RedHat Openshift Container Platform (OCP)
- **Embedded Container Service (Cloudera managed-ECS)**

In this document, we focus on **Cloudera on premises Data Service Cluster** setup with **ECS**.

Cloudera on premises setup consists of the following three parts.



- **FreeIPA / AD server:-** It provides the Identity and Authentication to the cluster. It includes Kerberos as the authentication provider and LDAP as directory service provider. All the cluster nodes (both Base and ECS) act as FreeIPA agents. (FreeIPA server includes Private DNS Server, MIT Kerberos KDC, Directory Server, Chrony, Dogtag certificate system, SSSD)
- **CDP Base Cluster:-** It consists of all the prerequisite services that form the basis for CDP Data Lake for Data Services.
 - Atlas
 - Solr
 - HBase
 - HDFS
 - Hive (Metastore Server)
 - Hive-on-Tez (HiveServer2)
 - Hue (Required for CDW data service)
 - Iceberg Replication
 - Impala(Used as Client)
 - Kafka
 - Ozone (Required for CDE data service)
 - Phoenix
 - Ranger
 - Spark on YARN (Spark 2)
 - Spark 3
 - Tez
 - YARN
 - Yarn Queue Manager (Optional)
 - ZooKeeper

Missing Components which we aren't considering as part of this setup guide but can be installed additionally, if needed:

- Flink
- Knox
- Kudu
- Livy

- Nifi
- Nifi Registry
- QueueManager
- S3 Connector
- Zeppelin

- **CDP ECS Data Services Cluster:-** This is the Rancher (RKE) based Kubernetes cluster that forms the basis for all the containerized workloads of CDP Data Services. It consists of ECS master, ECS agents, and Docker servers.

- **Service Dependencies:**

Service	Dependencies
Atlas	• HDFS • HBase • Kafka (Kafka broker role only) • Solr
DataViz	-
HBase	• HDFS • ZooKeeper
HDFS	• Hive • Spark • Yarn
Hive	• HDFS • YARN • Tez
Hive-on-Tez	• HDFS • YARN • Hive • Tez
Hue	• HDFS • Hive
Impala	• HDFS • Hive
Kafka	ZooKeeper

Service	Dependencies
Livy	• Spark • Impala
Nifi	• Knox • Ranger
Nifi Registry	• Nifi • Knox • Ranger
Oozie	• YARN
Ozone	-
Ranger	• HDFS • Solr • Atlas
Solr	• HDFS • ZooKeeper
Spark on YARN	• YARN
Streams Messaging Manager	• Kafka
Streams Replication Manager	• Kafka
Tez	• YARN
YARN	• HDFS • ZooKeeper
Zeppelin	• HDFS • Spark-on-YARN • YARN
ZooKeeper	-

Let’s have a look at the prerequisites before proceeding with the actual setup.

Solution Summary

This RA document details the process of installing Cloudera on premises on bare metal or VM based servers and configuration details of fully tested and validated workloads in the cluster.

Prerequisites:

Entitlements

Your License key must have the PvC DS entitlement. A current key without the entitlement will block access to ECS bits. Please raise a ticket or reach out to the Cloudera POC to get the necessary entitlements.

Virtual Machines

Administrator access to virtual machines.

Infrastructure Setup: Hardware and Software Requirements

Below table summarizes the machines used for this POC. This is a minimum requirement, One can increase the number of machines to achieve High Availability and Fault Tolerance. If this cluster is not meant to perform any benchmarking or performance test, one can proceed ahead with this infrastructure.

Note: *The cluster configurations used in this document are designed and decided considering the installation/configuration and management of all 3 data services' i.e. CAI, CDW, CDE. with minimalistic workloads on a single ECS data services cluster for the PoC purpose. The hardware specs should be redetermined and recalculated for the clusters to set up for a different purpose from above mentioned.*

Note: *Screenshots shown and versions used in this document are just for the reference purpose and may differ from the version to version used.*

Table 1.

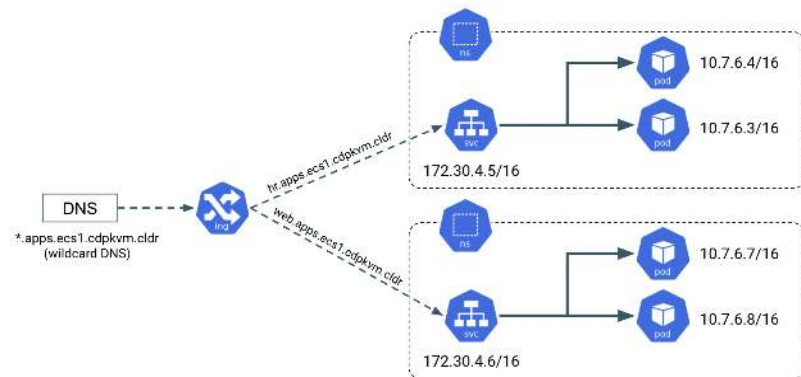
Count	CDP Role
1	FreeIPA or AD Server (Will be used for FreeIPA, Kerberos, Private DNS, LDAP, NTP, KDC, and will be used as an ansible controller node for automation purpose)
1	Cloudera-Manager Server (with external PostgreSQL Database server, will be used for downloading bits as well)
4	CDP Base Cluster (1 Master and 3 Worker Nodes)
3-11	CDP ECS Data Services Cluster (1 Master and 3-10 Worker Nodes)

DNS Server (In case we are not going with FreeIPA)

An external DNS server must contain the forward and reverse zones of the company domain name. The external DNS server must be able to resolve the hostname of all Cloudera on premises hosts and the 3rd party components (includes Kerberos, LDAP server, external database, NFS server) and perform reverse DNS lookup.

Wildcard DNS entry must be configured; e.g. ***.apps.cldrsetup.local**. This helps to reduce Day-2 operational tasks to set separate DNS entries for each newly provisioned external-facing application/service.

The external DNS server is expected to be ready prior to installing the Cloudera on premises solution and its installation procedure is not covered in this document.



Kerberos + LDAP Server/AD + Certificate (Required only, in case we are not going with FreeIPA)

An external Kerberos server and the Kerberos key distribution center (KDC) (with a realm established) must be available to provide authentication to CDP services, users and hosts.

An external secured LDAP-compliant identity/directory server (LDAPS) is required to enable the Cloudera on premises solution to look up for the user accounts and groups in the directory. This is expected to be ready prior to installing the Cloudera on premises solution and its installation procedure is not covered in this document.

Auto-TLS should be enabled using certificates created and managed by a Cloudera Manager certificate authority (CA), or certificates signed by a trusted public CA or your own internal CA. Prepare the certificate of your choice.

The total number of CA certificates must not exceed 10. Otherwise, pods will be evicted during initialization due to limited memory (1Gi) to process the configmap file.

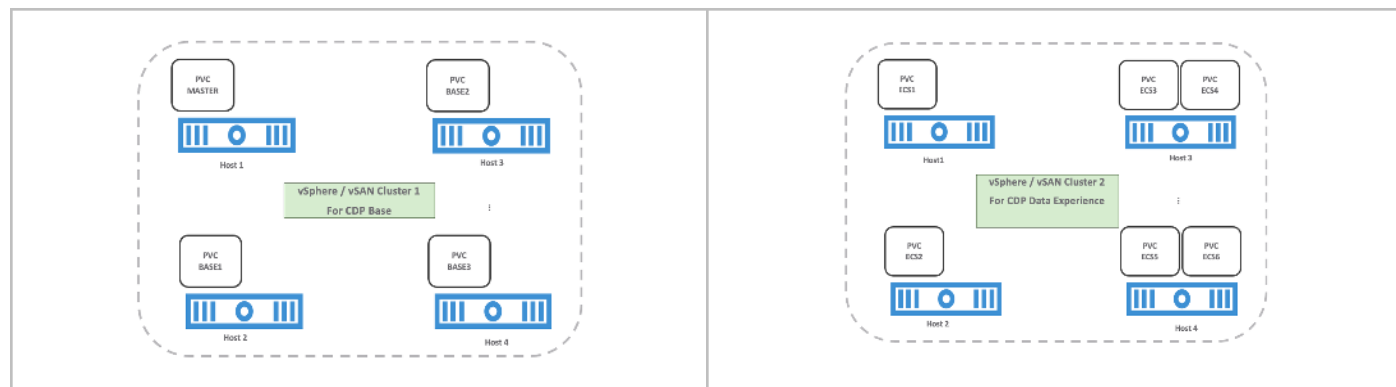
Note: For the Nutanix setup, this document will use the existing **Active Directory (AD) server** available within the Nutanix cluster for configuring Cloudera Base and Cloudera Data Services.

External NFS (Preferable but optional; needed for CAI use case)

CAI requires an external NFS server to store the project files and directories. NFS version 4.1 must be supported.

The external NFS storage is expected to be ready prior to installing the Cloudera on premises solution. External NFS storage installation is not covered in this document.

This document covers the Cloudera on premises setup and testing of the Data Services.



Hardware Requirements

**Hardware specs e.g. CPU, memory, disk, etc. should be analyzed and re-determined as per the setup requirement e.g. POC, demo, HA, DR etc. Current setup is for POC/Demo purpose only.

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-requirements-supported-versions.html>

FreeIPA/Kerberos & Private DNS Server: (In case we are not going with FreeIPA, External Kerberos/KDC is Req'd.)

Role	HostName	CPU	RAM	Disk	Partitions (recommended but not mandatory)
ipaserver/AD +ansible-controller	ipaserver	16	32GB	OS disk (250GB)	root partition
cldr-mngr, postgres db, bits	cldr-mngr	32	64GB	800GB	root partition, /var=400GB /opt=400GB
Bastion server	jump	8	16GB	200GB	root partition

Note: Each of the nodes in the below configurations require a dedicated minimum allocation of 450GB each for /docker and /var, important to consider if dedicated mounts(disks) are used.

PvC Base Cluster we will be installing a 4-node cluster on VMs:

** Here BaseMaster Node will also host Gateway and Utility hosts' services as per public documentation at

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-runtime-cluster-hosts-role-assignments.html>

** The Role assignment strategy for Control Plane Services' (e.g. HDFS, YARN, Spark, etc.) is discussed in the later steps of PvC Base Cluster Setup.

Role	HostName	CPU	RAM	Disk	Partitions
BASE CLUSTER					
Base-Master	pvcbase-master	32	64GB	root partition (600GB)	/hdfs /opt /var /yarn
Base-Worker	pvcbase-worker1	32	64GB	root partition (600GB)	/hdfs /opt /var /yarn
Base-Worker	pvcbase-worker2	32	64GB	root partition (600GB)	/hdfs /opt /var /yarn
Base-Worker	pvcbase-worker3	32	64GB	root partition (600GB)	/hdfs /opt /var /yarn

PvC Data Service (ECS) Cluster with CDW: we will be installing a 4-node cluster on VMs:

** Specs upgraded for concurrent tests and higher data volume tests and assumes only CDW services will be deployed

** Assuming Specs for 1 CDW Data Catalog, 1 CDV (DataViz) Small Instance, 1 Hive LLAP and 1 Impala Virtual Warehouse each with 1 coordinator and 2 executors.

Role	HostName	CPU	RAM	Disk	Partitions
ECS DS CLUSTER	CDW				
ECS-Master	pvcecs-master	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker1	32	256GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker2	32	256GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker3	32	256GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var

PvC Data Service (ECS) Cluster with CAI: we will be installing a 3-node cluster on VMs:

** Specs upgraded for concurrent tests and higher data volume tests and assumes only CAI services will be deployed

** Assuming Specs for 1 CAI Workbench with 10 small and 2 Average sized CAI Concurrent Sessions.



Role	HostName	CPU	RAM	Disk	Partitions
ECS DS CLUSTER	CAI				
ECS-Master	pvcecs-master	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var
ECS-Worker	pvcecs-worker1	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var
ECS-Worker	pvcecs-worker2	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var

PvC Data Service (ECS) Cluster with CDE: we will be installing a 6-node cluster on VMs:

**** Specs upgraded for concurrent tests and higher data volume tests and assumes only CDE services will be deployed.**

**** Assuming Specs for 1 CDE Virtual service along with 1 Virtual Cluster with 5 small and 2 Average sized CDE Concurrent Jobs.**

Role	HostName	CPU	RAM	Disk	Partitions
ECS DS CLUSTER	CDE				
ECS-Master	pvcecs-master	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var
ECS-Worker	pvcecs-worker1	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var
ECS-Worker	pvcecs-worker2	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var
ECS-Worker	pvcecs-worker3	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var
ECS-Worker	pvcecs-worker4	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var
ECS-Worker	pvcecs-worker5	32	128GB	root partition + 1.3 TB	/docker, /lhdata, /var

PvC Data Service(ECS) Cluster with CDW+CDE+CAI: we will be installing a 11-node cluster on VMs:

**** Specs upgraded for concurrent tests and higher data volume tests and assumes all 3 services will be deployed (CDW, CDE, CAI)**

**** Assuming Specs for 1 CDW Data Catalog, 1 CDV (DataViz Small) Instance, 1 Hive LLAP and 1 Impala Virtual Warehouse each with 1 coordinator and 2 executors.**

**** Assuming Specs for 1 CAI Workbench with 10 small and 2 Average sized CAI Concurrent Sessions.**

**** Assuming Specs for 1 CDE Virtual service along with 1 Virtual Cluster with 5 small and 2 Average sized CDE Concurrent Jobs.**

Role	HostName	CPU	RAM	Disk	Partitions
ECS DS CLUSTER	CAI+CDW+CDE				
ECS-Master	pvcecs-master	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker1	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker2	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker3	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker4	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker5	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker6	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker7	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker8	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker9	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var
ECS-Worker	pvcecs-worker10	32	128GB	root partition + 1.3 TB	/cdwdata, /docker, /lhdata, /var

Reference Architecture

Data Lake (Cloudera on premises Base) Reference Architecture

- Cloudera Data Platform: Cloudera Base on premises **7.3.2 CHF1**
- Cloudera Data Platform: Cloudera Data Services on premises **1.5.5 SP3 CHF3**

This RA document explains the architecture and deployment procedures for Cloudera Data Platform Cloudera on premises on cluster using on premise Infrastructure for Big Data and Analytics. The solution provides the details to configure Cloudera on premises on the bare metal RHEL9 based infrastructure.

Software Requirements

Note: This document is written for the below specified versions and commands to be executed are specific for those versions. If you are planning to use some different versions, commands may need to be updated separately.

[Table 1](#) lists the software components and the versions required for a single cluster of the Servers running in on premise, as tested, and validated in this document.

Below table summarizes the list of softwares/packages and their use for setting up Cloudera on premises cluster.

Table 2. Software Distributions and Firmware Versions

Software Component	Version or Release	Host to be Installed
OS: Red Hat Enterprise Linux Server (RHEL)	9.7 (Verify with SupportMatrix first)	All Servers
OpenJDK	17.0.14.0.7-1 >=	All Servers
Python3	3.9.22 >=	All Servers
PostgreSQL DB	14 >=	Cldr-Mngr
Psycopg2-binary	2.9.11 >=	All Servers
Postgres-JDBC-Connector	42.7.7 >=	All Servers
Cloudera Manager	7.13.2.100 (7.13.2.100-81323056)	Cldr-Mngr
Cloudera on premises Base (RunTime)	7.3.2.100 (7.3.2-1.cdh7.3.2.p100.81274879)	PvC Base Cluster Nodes
Cloudera on premises Data Services	1.5.5 SP3 CHF3 (1.5.5-h3300-b26)	PvC Data Service Cluster Nodes
Hadoop (Includes YARN and HDFS)	3.4.2.7.3.2.100-296	PvC Base Cluster Nodes
Spark3	3.5.4.7.3.2.100-296	PvC Base Cluster Nodes
Ozone	2.0.0.7.3.2.100-296	PvC Base Cluster Nodes
FreeIPA Server (Only if you are going with FreeIPA)	Latest: 4.13.0	IPA server node
FreeIPA Client (Only if you are going with FreeIPA)	Latest: 4.13.0	All nodes except ipaserver
NFS Utility Package	Latest: 2.8.3	PvC Data Service Cluster Nodes
TLS	AutoTLS (Self-signed)	ipa server node
Kerberos + LDAP(IdP) + DNS	FreeIPA	ipa server node
Data Lake Storage	HDFS (All) Ozone Iceberg v2 (with HDFS & Ozone)	
ECS DB Configuration	Embedded	ECS Only
Vault	Embedded	ECS Only
Docker Registry Type	Embedded/ Cloudera Default	ECS Only
NFS for CAI	Embedded i.e Internal	ECS Only

Note: Please check the Cloudera on premises requirements and supported versions for information about hardware, operating system, and database requirements, as well as product compatibility matrices, here: <https://supportmatrix.cloudera.com/> and here:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/index.html>

Note: For Cloudera on premises Base and Experiences versions and supported features, go to: https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/private-release-notes/topics/rt-runtime-component-versions_732.html

Note: For Cloudera on premises Base requirements and supported version, go to: <https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-requirements-supported-versions.html>

Note: Dedicated *NVMe/SSD drives* are recommended to store *Ozone metadata*, *Ozone mgmt* configuration for the admin/mgmt. nodes and worker/data nodes and *CDW data service storage* for virtual warehouses for local attached Storage Tiering Cache.

Summary

The below table contains the names assigned to the VM instances and to some other required components. Going forward in this document will refer to them by name.

Note: The domain name, and the hostnames mentioned here are just for reference. You may choose to have the hostnames as per your requirements.

Table 3.

NodeName	Details
pvcbase-master	Cloudera on premises Base Master
pvcbase-worker1 to pvcbase-worker3	CDP Base Cluster Worker Nodes
ipaserver (OR Existing LDAP/AD + DNS + Kerberos + KDC)	FreeIPA / Existing AD Server
cldr-mngr	Cloudera-Manager and PostgreSQL DB Server
pvcecs-master	ECS Master Node
pvcecs-worker1 to pvcecs-worker10	ECS Worker Nodes
CLDRSETUP.LOCAL (Replace with your ORG DOMAIN)	Dummy Domain For POC Purpose

Once you have familiarized yourself with all the information mentioned above, you can start with the preliminary work for CDP Base Cluster setup.

Creating Virtual Machines (VMs) on Nutanix Hardware using Prism Central

Note: Follow these steps only if you have existing Nutanix Hardware and want to deploy Cloudera on premises on Nutanix (Skip this section, if using any other bare-metal or Hypervisor platform)

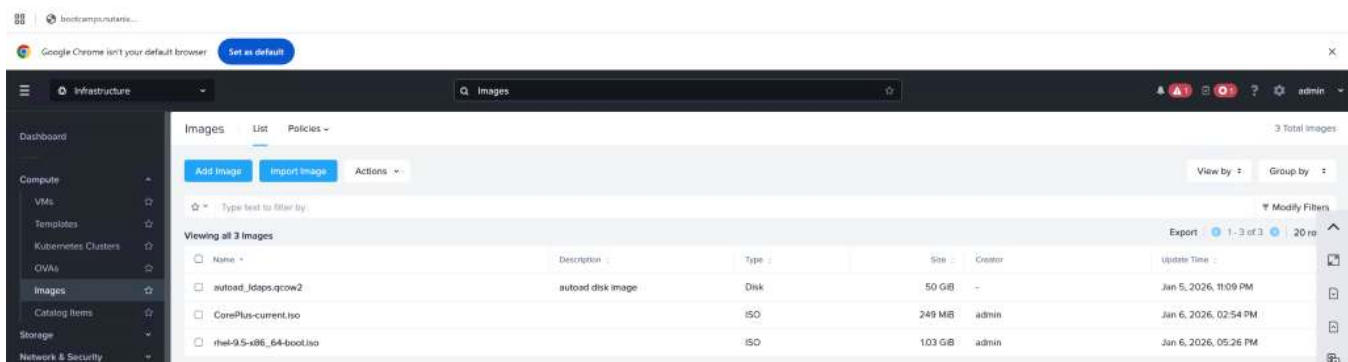
This document outlines the step-by-step procedure to create virtual machines (VMs) on Nutanix hardware/clusters using Prism Central. The process includes image creation, VM configuration, disk attachment, networking, and powering on the VM.

Prerequisites

- Access to Prism Central
- Required VM image (ISO or disk image)
- Appropriate permissions to create images and VMs
- Available storage container and subnet

Step 1: Create an Image

1. Log in to the Prism Central Dashboard.
2. From the left navigation panel, go to Compute → Images.
3. Click Add Image.



4. You will see the following options to create an image:
 - Upload an image file
 - Provide an ISO image URL
 - Select an existing VM disk
5. Choose the appropriate option and click **Next**.
6. Complete the image creation process.



1

Select Image

2

Select Location

Image Source

☒ Image File☐ URL☐ VM Disk

+ Add File

Cancel

Next

Step 2: Create a Virtual Machine

1. From the Prism Central Dashboard, navigate to Compute → VMs.
2. Click Create VM.
3. Enter the following details:
 - VM Name
 - Number of VMs
 - CPUs
 - Cores per CPU
 - Memory
4. Click Next.

Create VM

- 1 Configuration 2 Resources 3 Management 4 Review

General

Name

pvcbase-worker

Description (Optional)

Project

_internal

Cluster

RNO-POC005

Number of VMs

1

VM Properties

CPUs

16

vCPUs

Cores Per CPU

2

Cores

Memory

128

GiB

Cancel

Next

Step 3: Attach Disks

3.1 Attach Boot Disk (CD-ROM)

1. Click Attach Disk.
2. Select Type as CD-ROM.
3. Under Operation, select Clone from Image.
4. Choose the image created in Step 1.
5. Set Bus Type to SATA.
6. Click Save.



Create VM

Configuration

Resources

Management

Review

Any storage policy applied later, will manage the storage properties for all VM disks. Data placement will remain unaffected. [Learn More](#)

Disks

Attach Disk

Network

+ Attach to Subnet

More Network Options

Boot Configuration

UEFI BIOS Mode

UEFI BIOS Mode supports enhanced Shield VM security settings.

Legacy BIOS Mode

Back

Cancel

Next

Attach Disk

Type

CD-ROM

Operation

Clone from Image

Image

rhel-9.5-x86_64-boot.iso

Bus Type

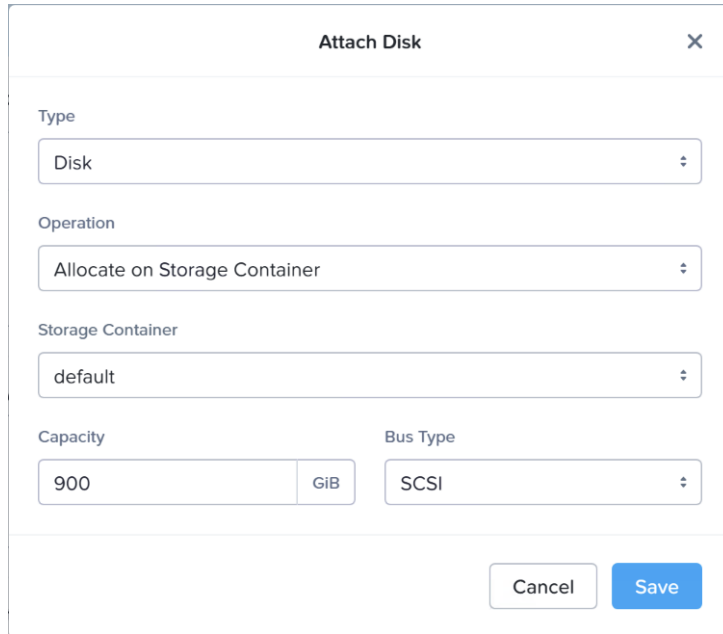
SATA

Cancel

Save

3.2 Attach Storage Disk

1. Click Attach Disk again.
2. Select Type as Disk.
3. Under Operation, select Allocate on Storage Container.
4. Choose the storage container (for example, **default**).
5. Enter the disk capacity (for example, 900 GB).
6. Set Bus Type to SCSI.
7. Click Save.

The image shows a dialog box titled "Attach Disk" with a close button (X) in the top right corner. The dialog contains several configuration fields: "Type" is a dropdown menu with "Disk" selected; "Operation" is a dropdown menu with "Allocate on Storage Container" selected; "Storage Container" is a dropdown menu with "default" selected; "Capacity" is a text input field with "900" entered, followed by a unit selector with "GiB" selected; and "Bus Type" is a dropdown menu with "SCSI" selected. At the bottom right of the dialog are two buttons: "Cancel" and "Save".

Attach Disk

Type

Disk

Operation

Allocate on Storage Container

Storage Container

default

Capacity

900

GiB

Bus Type

SCSI

Cancel Save

Step 4: Configure Networking

1. Click Attach to Subnet.
2. Select the desired subnet (for example, **aux1**).
3. Choose the IP assignment type:
 - DHCP (IP assigned dynamically)
 - Static IP (manually enter the IP address)
4. Keep the remaining settings as default.
5. Click Save.

Attach to Subnet

Subnet Attachment

Subnet

aux-1

VLAN ID	IPAM	Virtual Switch
35	Managed	br0

Network Connection State

Connected

Private IP Assignment

Network Address / Prefix	Free IPs (in Subnet)	Free IPs (in Pool)
10.8.16.128/25	108	105

Assignment Type

Assign with DHCP

IP Address

NIC Configuration

Attachment Type

Access

Cancel

Save

Step 5: Boot Configuration

1. Set Boot Configuration to UEFI BIOS Mode.
2. Keep all other settings as default.
3. Click Next.



Create VM

✓ Configuration

2 Resources

3 Management

4 Review

Any storage policy applied later, will manage the storage properties for all VM disks. Data placement will remain unaffected. [Learn More](#)

Disks

Attach Disk

#	Type	Source	Size	Bus Type	Actions
1	CD-ROM	rhel-9.5-x86_64-boot.iso Image	1.04 GiB	SATA	
2	Disk	default	900 GiB	SCSI	

Network

+ Attach to Subnet More Network Options

Normal Nic

Type	Subnet	VLAN ID / VPC	Private IP	Public IP	
Normal	aux-1	35	Auto-Assign	None	

Boot Configuration

UEFI BIOS Mode

UEFI BIOS Mode supports enhanced Shield VM security settings.

Legacy BIOS Mode

Shield VM Security Settings

☐ Secure Boot

☐ Windows® Defender Credential Guard

☐ Attach vTPM

< Back

Cancel Next

< Back

Cancel Next

Step 6: Management Settings

1. Under the Management tab, keep all settings unchanged.
2. Click Next.

Step 7: Review and Create VM

1. Review all VM configurations and settings.
2. Click Create VM to provision the VM.



Create VM

✓ Configuration

✓ Resources

✓ Management

4 Review

Configuration

Edit

VM Name

pvcbase-worker

Description (Optional)

-

Project

_internal

Cluster

RNO-POC005

Number of VMs

1

Instance Properties

8 vCPUs, 2 Cores, 16 GB

Memory Overcommit

Disabled

Advanced processor compatibility

-

Resources

Edit

Disks

#	Type	Source	Size	Bus Type
1	CD-ROM	rhel-9.5-x86_64-boot.iso	100 MB	SCSI

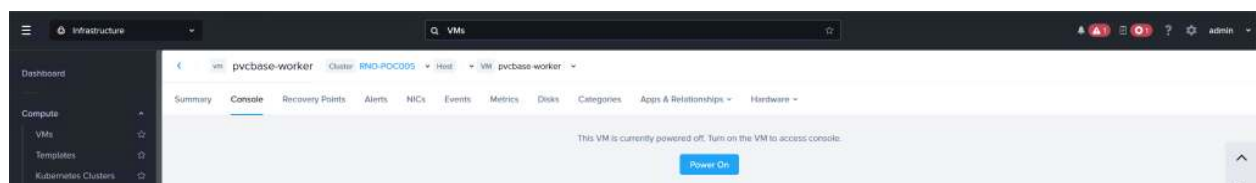
< Back

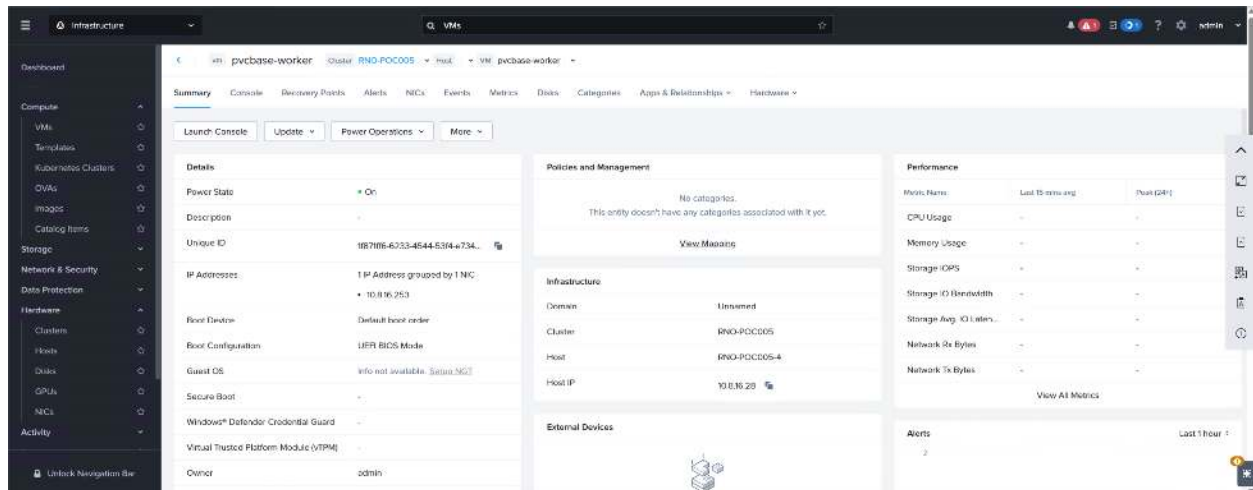
Cancel

Create VM

Step 8: Power On and Access the VM

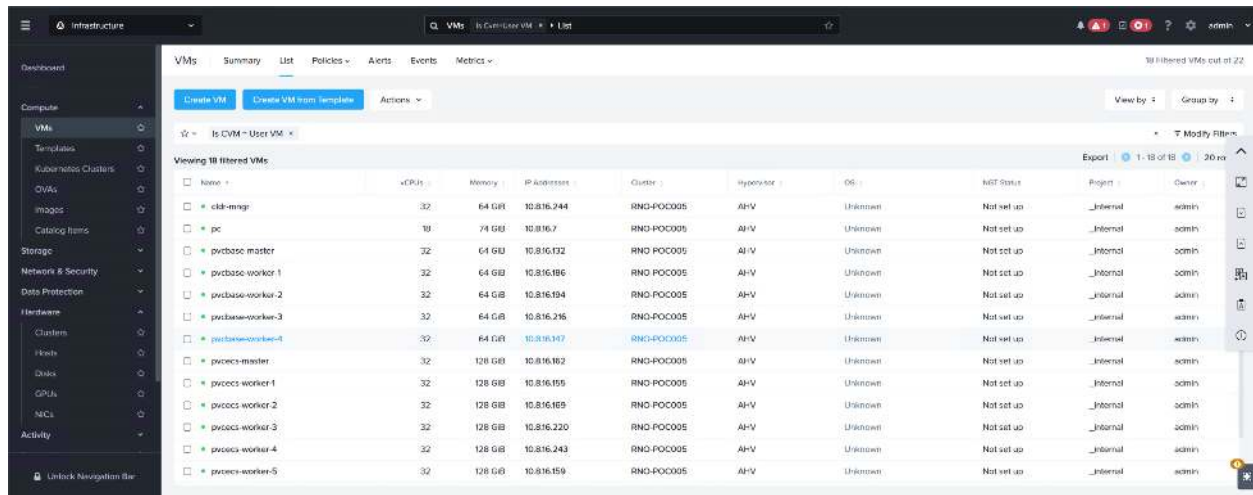
1. Once the VM is created, click on the VM name.
2. Go to the Console tab.
3. Click Power On to start the VM.
4. Access the VM either:
 - From the Console tab, or
 - From the Summary tab by clicking Launch Console.





Additional Notes

- Follow the same steps to create all required VMs for PVC Base and PVC ECS master and worker nodes.
- Ensure sufficient CPU, memory, and storage are available before VM creation.



Post OS Installation

Preliminary Work

Before getting into the actual installation of Cloudera on premises Base & Data Services clusters, we need to prepare our machines and perform some steps to meet the prerequisites.

Choose one of the nodes of the cluster or a separate node as the Ansible Admin/Controller Node for management. In this document, we configured the ipaserver for this purpose.

Procedure 1. Configure individual servers' static hostnames and prepare /etc/hosts file

Step 1. Setup the hostname *for each individual node* by logging in using the IP addresses provided by the infrastructure team, so we can refer to them with names instead of IP addresses for simplicity and ease of identification. While you are configuring the hostname, also follow **Step 2** while logging in to each host. **Replace your ORG DOMAIN**

```
[root@ipaserver ~]# sudo hostnamectl set-hostname --static ipaserver.cldrsetup.local
[root@cldr-mngr ~]# sudo hostnamectl set-hostname --static cldr-mngr.cldrsetup.local

[root@pvcbase-master ~]# sudo hostnamectl set-hostname --static pvcbase-master.cldrsetup.local
[root@pvcbase-worker1 ~]# sudo hostnamectl set-hostname --static pvcbase-worker1.cldrsetup.local
[root@pvcbase-worker2 ~]# sudo hostnamectl set-hostname --static pvcbase-worker2.cldrsetup.local
[root@pvcbase-worker3 ~]# sudo hostnamectl set-hostname --static pvcbase-worker3.cldrsetup.local

[root@pvcecs-master ~]# sudo hostnamectl set-hostname --static pvcecs-master.cldrsetup.local
[root@pvcecs-worker1 ~]# sudo hostnamectl set-hostname --static pvcecs-worker1.cldrsetup.local
[root@pvcecs-worker2 ~]# sudo hostnamectl set-hostname --static pvcecs-worker2.cldrsetup.local
[root@pvcecs-worker3 ~]# sudo hostnamectl set-hostname --static pvcecs-worker3.cldrsetup.local
[root@pvcecs-worker4 ~]# sudo hostnamectl set-hostname --static pvcecs-worker4.cldrsetup.local
[root@pvcecs-worker5 ~]# sudo hostnamectl set-hostname --static pvcecs-worker5.cldrsetup.local
[root@pvcecs-worker6 ~]# sudo hostnamectl set-hostname --static pvcecs-worker6.cldrsetup.local
[root@pvcecs-worker7 ~]# sudo hostnamectl set-hostname --static pvcecs-worker7.cldrsetup.local
[root@pvcecs-worker8 ~]# sudo hostnamectl set-hostname --static pvcecs-worker8.cldrsetup.local
[root@pvcecs-worker9 ~]# sudo hostnamectl set-hostname --static pvcecs-worker9.cldrsetup.local
[root@pvcecs-worker10 ~]# sudo hostnamectl set-hostname --static pvcecs-worker10.cldrsetup.local
```

Step 2. While you set the hostnames by logging in to each individual hosts, make sure to run the dnf update and install python3 dependencies as well, since these are fresh nodes:

**** Python3** can be installed manually on bare minimum (ipaserver/ansible admin) and can be later installed using ansible on the rest of the nodes. (Only, If you don't want it to install on each individual node)

Note: By default, running **dnf update** on RHEL may upgrade the system to the latest minor release (e.g., 9.8).

E.g., If you want to stay on RHEL 9.7 and receive only security and bug-fix updates, not a minor version upgrade.

Solution 1 - Lock to Your Current Minor Version (RHEL 9.7)

Run the following:

```
sudo subscription-manager release --set=9.7
```

Then verify:

```
sudo subscription-manager release --show
```

Expected output:

Release: 9.7

This locks your system repositories to RHEL 9.5 content - updates will stay within 9.5 and not move to 9.6.

```
[root@ipaserver ~]# sudo dnf -y update
[root@ipaserver ~]# sudo dnf -y install wget telnet net-tools bind-utils iproute traceroute nc

##### Verify If Python3 and Pip3 are already installed.
[root@ipaserver ~]# python3 --version
[root@ipaserver ~]# pip3 --version

##### Install python3.9 - ON ALL NODES (if not Present already):
##### Check if Python could be installed using dnf command directly:
[root@ipaserver ~]# sudo dnf -y install python39 python3-pip
##### If Python could not be installed using dnf command directly:
[root@ipaserver ~]# sudo dnf -y groupinstall "Development Tools"
[root@ipaserver ~]# sudo dnf -y install epel-release openssl-devel bzip2-devel libffi-devel xz-devel
[root@ipaserver ~]# VERSION=3.9.22
[root@ipaserver ~]# wget https://www.python.org/ftp/python/$VERSION/Python-$VERSION.tgz
[root@ipaserver ~]# tar xvf Python-$VERSION.tgz
[root@ipaserver ~]# cd Python-$VERSION/
[root@ipaserver ~]# ./configure --enable-optimizations
[root@ipaserver ~]# sudo make altinstall
[root@ipaserver ~]# python3 --version
[root@ipaserver ~]# cd -
[root@ipaserver ~]# rm -rvf Python-$VERSION/ Python-$VERSION.tgz

##### Install pip3 - ON ALL NODES (if not Present already):
[root@ipaserver ~]# dnf install -y python3-pip
[root@ipaserver ~]# pip3 install --upgrade pip
[root@ipaserver ~]# pip3 --version
[root@ipaserver ~]# pip3 install psycpg2-binary

##### Verify Python and Pip Versions
[root@ipaserver ~]# python3 -V
Python 3.9.22
[root@ipaserver ~]# pip3 --version
```

Step 3. Log into the ipaserver Node using IP provided previously by the infrastructure team.

```
[root@ipaserver ~]# ssh 172.31.24.240
```

Step 4. Setup /etc/hosts on the ipaserver node; this is a pre-configuration to setup Private DNS as shown in the next section. In large scale production grade deployment, DNS server setup is highly recommended.

Populate the host file with IP addresses and corresponding hostnames on the ipaserver node by taking the private IP of machine and add an entry in /etc/hosts file as follows: **(All of below mentioned IPs are private IP addresses)**

(We will later copy the same hosts file to all other nodes with the help of ansible)

```
[root@ipaserver ~]# sudo vi /etc/hosts
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost localhost.localdomain localhost6 localhost6.localdomain6

# Free-IPA Server
172.31.24.240 ipaserver.cldrsetup.local ipaserver

# Cloudera Manager Server
172.31.1.38 cldr-mngr.cldrsetup.local cldr-mngr
172.31.1.38 postgresdb.cldrsetup.local postgresdb

# PvC Base Cluster Nodes
172.31.1.34 pvcbase-master.cldrsetup.local pvcbase-master
172.31.1.35 pvcbase-worker1.cldrsetup.local pvcbase-worker1
172.31.1.36 pvcbase-worker2.cldrsetup.local pvcbase-worker2
172.31.1.37 pvcbase-worker3.cldrsetup.local pvcbase-worker3

# PvC Data Services ECS Cluster Nodes
172.31.30.239 pvcecs-master.cldrsetup.local pvcecs-master
```



```
172.31.22.43 pvcecs-worker1.cldrsetup.local pvcecs-worker1
172.31.30.249 pvcecs-worker2.cldrsetup.local pvcecs-worker2
172.31.26.24 pvcecs-worker3.cldrsetup.local pvcecs-worker3
172.31.24.198 pvcecs-worker4.cldrsetup.local pvcecs-worker4
172.31.24.53 pvcecs-worker5.cldrsetup.local pvcecs-worker5
172.31.22.44 pvcecs-worker6.cldrsetup.local pvcecs-worker6
172.31.30.250 pvcecs-worker7.cldrsetup.local pvcecs-worker7
172.31.26.25 pvcecs-worker8.cldrsetup.local pvcecs-worker8
172.31.24.199 pvcecs-worker9.cldrsetup.local pvcecs-worker9
172.31.24.54 pvcecs-worker10.cldrsetup.local pvcecs-worker10
```

Step 5. Perform the basic validation of OS version and hostname/IP configurations:

Ensure that the OS version is RHEL 9.x.

To verify the version, run the below command. It should return RedHat Linux version 9.x.

```
[root@ipaserver ~]# cat /etc/*rel* | grep -E 'NAME|VERSION'
NAME="Red Hat Enterprise Linux"
VERSION="9.5 (Plow)"
VERSION_ID="9.5"
PRETTY_NAME="Red Hat Enterprise Linux 9.5 (Plow)"
CPE_NAME="cpe:/o:redhat:enterprise_linux:9::baseos"
REDHAT_BUGZILLA_PRODUCT_VERSION=9.5
REDHAT_SUPPORT_PRODUCT_VERSION="9.5"
```

Verify Hostname and IP addresses

```
[root@ipaserver ~]# hostname -f
ipaserver.cldrsetup.local
```

```
[root@ipaserver ~]# hostname -i
172.31.24.240
```

```
[root@ipaserver ~]# cat /etc/hostname
ipaserver.cldrsetup.local
```

```
[root@ipaserver ~]# ip addr show eth0 | grep -e inet
inet 10.0.2.2/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
inet6 fe80::c3:16ff:fe00:9/64 scope link noprefixroute
```

please ignore if the below command fails due to instance having single network interface attached

```
[root@ipaserver ~]# ip addr show eth1 | grep -e inet
inet 172.31.24.240/24 brd 172.31.1.255 scope global dynamic noprefixroute eth1
inet6 fe80::65b3:25c6:8b2a:b4ae/64 scope link noprefixroute
```

```
[root@ipaserver ~]# ip addr show | grep $(hostname -i)
inet 172.31.24.240/24 brd 172.31.1.255 scope global dynamic noprefixroute eth1
```

```
[root@ipaserver ~]# host -v -t A $(hostname) | grep -A2 ANSWER
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0
```

```
;; QUESTION SECTION:
```

```
--
```

```
;; ANSWER SECTION:
```

```
ipaserver.cldrsetup.local. 1200 IN A 172.31.24.240
```

```
[root@ipaserver ~]#
```

```
[root@ipaserver ~]# uname -a
```

```
Linux ipaserver.cldrsetup.local 5.14.0-427.26.1.el9_4.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Jul 5 11:34:54 EDT 2024 x86_64 x86_64 x86_64 GNU/Linux
```

Procedure 2. Setup ipaserver (which includes Private DNS Server, MIT Kerberos KDC, Directory Server, Chrony, Dogtag certificate system, SSSD)

Note: You can skip this procedure if you are going with existing **AD server** instead of FreeIPA

Install and Setup of IPA services

In this step, a Private DNS server and other services like KDC, Directory Service will be configured on the ipaserver. Also, please note that the hostnames used in this installation can be modified as per your requirements.

Follow the on screen instructions and provide the inputs for the parameters as per the table below.

Parameter	Value
Server host name [ipaserver.cldrsetup.local]:	ipaserver.cldrsetup.local
Please confirm the domain name [cldrsetup.local]:	cldrsetup.local
Please provide a realm name [CLDRSETUP.LOCAL]:	CLDRSETUP.LOCAL
Directory Manager password:	<Password For Directory Manager> (cloudera123)
Password (confirm):	<Confirm Password> (cloudera123)
IPA admin password:	<Password For IPA Admin> (cloudera123)
Password (confirm):	<Confirm Password> (cloudera123)
Do you want to configure DNS forwarders? [yes]:	<ENTER>
Do you want to search for missing reverse zones?[yes]:	no
NetBIOS domain name [CLDRSETUP]:	CLDRSETUP
Do you want to configure chrony with NTP server or pool address? [no]:	yes
Enter NTP source server addresses separated by comma, or press Enter to skip:	<ENTER>
Enter a NTP source pool address, or press Enter to skip:	<ENTER>
Continue to configure the system with these values?[no]:	yes

Please keep the same password for both Directory manager and IPA admin so that there is no confusion in future while using the same. Also, note down the password separately.

Step 1. Login to IPAServer node and Install ipa-server packages:

```
# Install ipa server dependencies packages through dnf using the below command.  
[root@ipaserver ~]# sudo dnf install -y ipa-server bind bind-dyndb-ldap ipa-server-dns firewallld
```

```
# If required, use below command to set the java version  
[root@ipaserver ~]# update-alternatives --config java
```

```
# Configure ipa-server and DNS by using command: ipa-server-install --setup-dns
```

```
[root@ipaserver ~]# ipa-server-install --setup-dns
```

```
The log file for this installation can be found in /var/log/ipaserver-install.log
```

```
=====
```

```
This program will set up the IPA Server.  
Version 4.11.0
```

```
This includes:
```

- * Configure a stand-alone CA (dogtag) for certificate management
- * Configure the NTP client (chronyd)
- * Create and configure an instance of Directory Server
- * Create and configure a Kerberos Key Distribution Center (KDC)
- * Configure Apache (httpd)
- * Configure DNS (bind)
- * Configure SID generation
- * Configure the KDC to enable PKINIT

```
To accept the default shown in brackets, press the Enter key.
```

Enter the fully qualified domain name of the computer on which you're setting up server software. Using the form <hostname>.<domainname>
Example: ipaserver.cldrsetup.local

Server host name [ipaserver.cldrsetup.local]: **<ENTER>**

Warning: skipping DNS resolution of host ipaserver.cldrsetup.local
The domain name has been determined based on the host name.

Please confirm the domain name [cldrsetup.local]: **<ENTER>**

The kerberos protocol requires a Realm name to be defined.
This is typically the domain name converted to uppercase.

Please provide a realm name [CLDRSETUP.LOCAL]: **<ENTER>**
Certain directory server operations require an administrative user.
This user is referred to as the Directory Manager and has full access to the Directory for system management tasks and will be added to the instance of directory server created for IPA.
The password must be at least 8 characters long.

Directory Manager password: **<cloudera123>**
Password (confirm): **<cloudera123>**

The IPA server requires an administrative user, named 'admin'.
This user is a regular system account used for IPA server administration.

IPA admin password: **<cloudera123>**
Password (confirm): **<cloudera123>**

Checking DNS domain cldrsetup.local., please wait ...
Do you want to configure DNS forwarders? [yes]: **no**
No DNS forwarders configured
Do you want to search for missing reverse zones? [yes]: **no**
Trust is configured but no NetBIOS domain name found, setting it now.
Enter the NetBIOS name for the IPA domain.
Only up to 15 uppercase ASCII letters, digits and dashes are allowed.
Example: EXAMPLE.

NetBIOS domain name [CLDRSETUP]: **<ENTER>**

Do you want to configure chrony with NTP server or pool address? [no]: **yes**
Enter NTP source server addresses separated by comma, or press Enter to skip: **<ENTER>**
Enter a NTP source pool address, or press Enter to skip: **<ENTER>**

The IPA Master Server will be configured with:
Hostname: ipaserver.cldrsetup.local
IP address(es): 172.31.24.240
Domain name: cldrsetup.local
Realm name: CLDRSETUP.LOCAL

The CA will be configured with:
Subject DN: CN=Certificate Authority,O=CLDRSETUP.LOCAL
Subject base: O=CLDRSETUP.LOCAL
Chaining: self-signed

BIND DNS server will be configured to serve IPA domain with:
Forwarders: No forwarders
Forward policy: only
Reverse zone(s): No reverse zone

Continue to configure the system with these values? [no]: **yes**

The following operations may take some minutes to complete.
Please wait until the prompt is returned.

```
Disabled p11-kit-proxy
Synchronizing time
No SRV records of NTP servers were found and no NTP server or pool address was provided.
Using default chrony configuration.
Attempting to sync time with chronyc.
Time synchronization was successful.
Configuring directory server (dirsrv). Estimated time: 30 seconds
[1/43]: creating directory server instance
Validate installation settings ...
Create file system structures ...
Perform SELinux labeling ...
Create database backend: dc=cldrsetup,dc=local ...
Perform post-installation tasks ...
[2/43]: tune ldbm plugin
[3/43]: adding default schema

-----
-----
-----
-----

[6/8]: restarting Directory Server to take MS PAC and LDAP plugins changes into account
[7/8]: adding fallback group
[8/8]: adding SIDs to existing users and groups
This step may take a considerable amount of time, please wait..
Done.
Configuring client side components
This program will set up an IPA client.
Version 4.11.0
```

```
Using the existing certificate '/etc/ipa/ca.crt'.
Client hostname: ipaserver.cldrsetup.local
Realm: CLDRSETUP.LOCAL
DNS Domain: cldrsetup.local
IPA Server: ipaserver.cldrsetup.local
BaseDN: dc=cldrsetup,dc=local
```

```
Configured /etc/sss/sss.conf
Systemwide CA database updated.
Adding SSH public key from /etc/ssh/ssh_host_rsa_key.pub
Adding SSH public key from /etc/ssh/ssh_host_ecdsa_key.pub
Adding SSH public key from /etc/ssh/ssh_host_ed25519_key.pub
SSSD enabled
Configured /etc/openldap/ldap.conf
Configured /etc/ssh/ssh_config
Configured /etc/ssh/sshd_config.d/04-ipa.conf
Configuring cldrsetup.local as NIS domain.
Client configuration complete.
The ipa-client-install command was successful
```

```
=====
Setup complete
```

Next steps:

1. You must make sure these network ports are open:

TCP Ports:

- * 80, 443: HTTP/HTTPS
- * 389, 636: LDAP/LDAPS
- * 88, 464: kerberos
- * 53: bind

UDP Ports:

- * 88, 464: kerberos
- * 53: bind
- * 123: ntp

2. You can now obtain a kerberos ticket using the command: 'kinit admin'
This ticket will allow you to use the IPA tools (e.g., ipa user-add)
and the web user interface.

Be sure to back up the CA certificates stored in /root/cacert.p12
These files are required to create replicas. The password for these
files is the Directory Manager password
The ipa-server-install command was successful
[root@ipaserver ~]#

Disable the firewall on ipaserver to be able to connect from rest of hosts

```
[root@ipaserver ~]# systemctl stop firewalld
[root@ipaserver ~]# systemctl disable firewalld
Removed "/etc/systemd/system/multi-user.target.wants/firewalld.service".
Removed "/etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service".
[root@ipaserver ~]#
```

If Fail, do: If the installation fails, then run the below command to uninstall and retry with the above command for installation.

```
[root@ipaserver ~]# ipa-server-install --uninstall
[root@ipaserver ~]# ipa-server-install --setup-dns (again)
```

The setup will take 10-15 Minutes. If everything goes fine then you should get an output similar to the below screenshot.

```
The ipa-client-install command was successful

=====
Setup complete

Next steps:
  1. You must make sure these network ports are open:
      TCP Ports:
        * 80, 443: HTTP/HTTPS
        * 389, 636: LDAP/LDAPS
        * 88, 464: kerberos
        * 53: bind
      UDP Ports:
        * 88, 464: kerberos
        * 53: bind
        * 123: ntp

  2. You can now obtain a kerberos ticket using the command: 'kinit admin'
     This ticket will allow you to use the IPA tools (e.g., ipa user-add)
     and the web user interface.

Be sure to back up the CA certificates stored in /root/cacert.p12
These files are required to create replicas. The password for these
files is the Directory Manager password
```

Step 2. Verify KDC setup: kerberos ticket is working fine by generating a ticket for the admin user.

Run the kinit admin command to authenticate as admin and enter the directory password provided during ipa server installation. The command should generate the ticket and should be listed by executing klist -e.

```
[root@ipaserver ~]# kinit admin
Password for admin@CLDRSETUP.LOCAL: <cloudera123>
```

```
[root@ipaserver ~]# klist -e
Ticket cache: KCM:0
Default principal: admin@CLDRSETUP.LOCAL
```

```
Valid starting Expires Service principal
05/13/2024 04:07:15 05/14/2024 03:30:48 krbtgt/CLDRSETUP.LOCAL@CLDRSETUP.LOCAL
Etype (skey, tkt): aes256-cts-hmac-sha384-192, aes256-cts-hmac-sha384-192
```

try kinit admin@CLDRSETUP.LOCAL

(if fails anytime, run below commands)

```
[root@ipaserver ~]# ipactl stop && ipactl start && ipactl status
```

Verify the status of ipa services installed

```
[root@ipaserver ~]# ipactl status
Directory Service: RUNNING
krb5kdc Service: RUNNING
kadmin Service: RUNNING
named Service: RUNNING
httpd Service: RUNNING
ipa-custodia Service: RUNNING
```

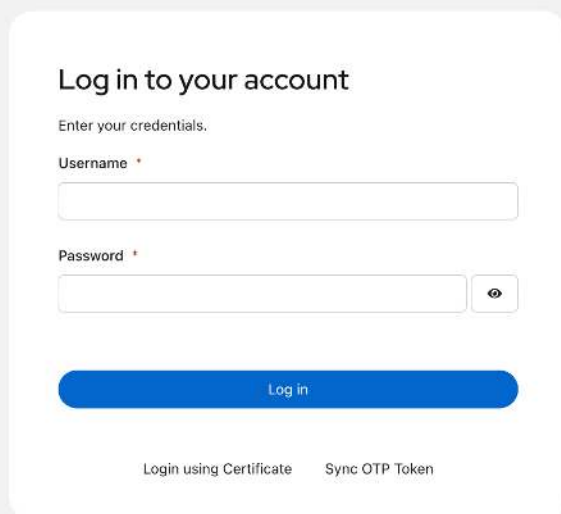
```
pki-tomcatd Service: RUNNING
ipa-otpd Service: RUNNING
ipa-dnssyncd Service: RUNNING
ipa: INFO: The ipactl command was successful
```

This command should return the below output:

```
[root@ipaserver centos]# kinit admin
Password for admin@CDPPVCDS.COM:
[root@ipaserver centos]# klist -e
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: admin@CDPPVCDS.COM

Valid starting    Expires          Service principal
04/05/23 10:49:29 04/06/23 10:49:26 krbtgt/CDPPVCDS.COM@CDPPVCDS.COM
    Etype (skey, tkt): aes256-cts-hmac-sha1-96, aes256-cts-hmac-sha1-96
[root@ipaserver centos]#
```

You can also access the FreeIPA WebUI at <https://ipaserver.cldrsetup.local> using the same FreeIPA CLI credentials created in earlier steps during installation.

The image shows a web browser window displaying the FreeIPA WebUI login page. The page has a light gray background. On the left, there is a white rounded rectangle containing the login form. The form has a title "Log in to your account" and a subtitle "Enter your credentials.". Below this, there are two input fields: "Username *" and "Password *". The "Password *" field has a small eye icon to its right. Below the input fields is a blue button labeled "Log in". At the bottom of the white rectangle, there are two links: "Login using Certificate" and "Sync OTP Token". On the right side of the page, there is a section titled "RED HAT* IDENTITY MANAGEMENT". Below this title, there are three bullet points: "· To log in with username and password, enter them in the corresponding fields, then click 'Log in'."; "· To log in with Kerberos, please make sure you have valid tickets (obtainable via kinit) and configured the browser correctly, then click 'Log in'."; and "· To log in with certificate, please make sure you have valid personal certificate.". At the bottom of this section, there are two links: "Browser Kerberos setup" and "FreeIPA website".

Procedure 3. Set Up Password-less Login

To manage all the nodes in a cluster from the admin/controller node, password-less login needs to be set up. It assists in automating common tasks with Ansible, and shell-scripts without having to use passwords.

Enable the passwordless login across all the nodes when Red Hat Linux is installed across all the nodes in the cluster.

Step 1. Log into the ipaserver Node.

```
[root@ipaserver ~]# ssh 172.31.24.240
```

Step 2. Run the ssh-keygen command to create both public and private SSH key-pair on the ansible-controller node.

```
[root@ipaserver ~]# ssh-keygen -N '' -f ~/.ssh/id_rsa
[root@ipaserver ~]# ls -l /root/.ssh
[root@ipaserver ~]# chmod 600 /root/.ssh/id_rsa
```

```
> ssh-keygen -t rsa -f /root/.ssh/id_rsa_new
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
```

Step 3. Run the following command from the ansible-controller/ipaserver node to copy the public key id_rsa.pub to all the nodes of the cluster. ssh-copy-id appends the keys to the remote-hosts .ssh/authorized_keys.

(NA in AWS EC2, need to be done manually, as right now password based authentication for non-root users is disabled)

```
[root@ipaserver ~]# for i in {1}; do echo "copying ipaserver.cldrsetup.local"; ssh-copy-id -i ~/.ssh/id_rsa.pub root@ipaserver.cldrsetup.local; done;
[root@ipaserver ~]# for i in {1}; do echo "copying cldr-mngr.cldrsetup.local"; ssh-copy-id -i ~/.ssh/id_rsa.pub root@cldr-mngr.cldrsetup.local; done;
[root@ipaserver ~]# for i in {1}; do echo "copying pvcbase-master.cldrsetup.local"; ssh-copy-id -i ~/.ssh/id_rsa.pub root@pvcbase-master.cldrsetup.local; done;
[root@ipaserver ~]# for i in {1..3}; do echo "copying pvcbase-worker$i.cldrsetup.local"; ssh-copy-id -i ~/.ssh/id_rsa.pub root@pvcbase-worker$i.cldrsetup.local; done;
[root@ipaserver ~]# for i in {1}; do echo "copying pvcecs-master.cldrsetup.local"; ssh-copy-id -i ~/.ssh/id_rsa.pub root@pvcecs-master.cldrsetup.local; done;
[root@ipaserver ~]# for i in {1..10}; do echo "copying pvcecs-worker$i.cldrsetup.local"; ssh-copy-id -i ~/.ssh/id_rsa.pub root@pvcecs-worker$i.cldrsetup.local; done;
```

Alternate way is to add pub key to authorized_keys file manually on ipaserver node and copy the entire .ssh directory to all other NODES; otherwise login into each hosts and manually update authorized_keys:

```
[root@ipaserver ~]# cat /root/.ssh/id_rsa.pub >> /root/.ssh/authorized_keys
```

copy the entire .ssh directory to all NODES

```
[root@ipaserver ~]# scp -r /root/.ssh root@cldr-mngr.cldrsetup.local:/root/.
```

(provide root user password when prompted)

Download the id_rsa and id_rsa.pub to your local machine by either using scp or sftp (as it will be required later)

Step 4. Enter yes for *Are you sure you want to continue connecting (yes/no)?*

Step 5. Enter the password of the remote host.

Procedure 4. Set up Ansible (We will be using Bastion as ansible controller/admin node)

Step 1. Login to Bastion node and Install ansible-core

```
[root@jump ~]# dnf install -y ansible-core
[root@jump ~]# ansible --version
ansible [core 2.14.14]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /bin/ansible
  python version = 3.9.22 (main, Jan 24 2024, 00:00:00) [GCC 11.4.1 20231218 (Red Hat 11.4.1-3)] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
[root@jump ~]# echo "export ANSIBLE_HOST_KEY_CHECKING=False" >> ~/.bashrc && source ~/.bashrc
```

Step 2. Prepare the host inventory file for Ansible as shown below. Various host groups have been created based on any specific installation requirements of certain hosts.

```
[root@jump ~]# vi /etc/ansible/hosts
```

```
[admin]
ipaserver.cldrsetup.local
```

```

[ipaserver]
ipaserver.cldrsetup.local

[cldr-mngr]
cldr-mngr.cldrsetup.local

[namenodes]
pvcbase-master.cldrsetup.local

[datanodes]
pvcbase-worker1.cldrsetup.local
pvcbase-worker2.cldrsetup.local
pvcbase-worker3.cldrsetup.local

[ecsmasternodes]
pvcecs-master.cldrsetup.local

[ecsnodes]
pvcecs-worker1.cldrsetup.local
pvcecs-worker2.cldrsetup.local
pvcecs-worker3.cldrsetup.local
pvcecs-worker4.cldrsetup.local
pvcecs-worker5.cldrsetup.local
pvcecs-worker6.cldrsetup.local
pvcecs-worker7.cldrsetup.local
pvcecs-worker8.cldrsetup.local
pvcecs-worker9.cldrsetup.local
pvcecs-worker10.cldrsetup.local

[nodes]
pvcbase-master.cldrsetup.local
pvcbase-worker1.cldrsetup.local
pvcbase-worker2.cldrsetup.local
pvcbase-worker3.cldrsetup.local
pvcecs-master.cldrsetup.local
pvcecs-worker1.cldrsetup.local
pvcecs-worker2.cldrsetup.local
pvcecs-worker3.cldrsetup.local
pvcecs-worker4.cldrsetup.local
pvcecs-worker5.cldrsetup.local
pvcecs-worker6.cldrsetup.local
pvcecs-worker7.cldrsetup.local
pvcecs-worker8.cldrsetup.local
pvcecs-worker9.cldrsetup.local
pvcecs-worker10.cldrsetup.local

```

Step 3. Verify the host group by running the following commands.

```

[root@jump ~]# ansible datanodes -m ping
pvcbase-worker2.cldrsetup.local | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
pvcbase-worker3.cldrsetup.local | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
pvcbase-worker1.cldrsetup.local | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },

```



```
"changed": false,
"ping": "pong"
}
```

Step 4. Copy `/etc/hosts` file to each node part of the cloudera deployment to resolve fqdn across the cluster

```
[root@jump ~]# ansible all -m copy -a "src=/etc/hosts dest=/etc/hosts"
```

Procedure 5. Set up Network configuration files and DNS Zones/Records

Step 1. We will update the network configuration file `/etc/resolv.conf` on the IPA / AD server to use the Name Server created in previous steps and will later copy this file to the rest of nodes using ansible (after installing freeipa-client, as it overrides `resolv.conf` and may lead to rework) to make them able to resolve FQDNs across the cluster:

(Open the file `/etc/resolv.conf` in edit mode and add the following. Make sure the new entry is added above any other nameserver entry. The contents of the file must look similar to the below.)

Note: Make sure that the `/etc/resolv.conf` file on the ECS hosts *contains a maximum of 2 active search domains*.

<https://docs.cloudera.com/data-warehouse/1.5.5/release-notes/topics/dw-private-cloud-known-issues-ecs-environments.html>

```
[root@jump ~]# cp /etc/resolv.conf /etc/resolv.conf.orig
[root@jump ~]# cat /etc/resolv.conf
search ap-southeast-1.compute.internal cldrsetup.local
nameserver 172.31.24.240 # PrivateIP of FreeIPA /AD Server must be first nameserver entry after search
nameserver 172.31.0.2
nameserver 127.0.0.1
[root@jump ~]#
```

Step 2. We will update the network configuration file `/etc/sysconfig/network` on the IPA server to use the Name Server created in previous steps and will later copy this file to the rest of nodes to make them able to resolve FQDNs across the cluster:

(The changes in `/etc/resolv.conf` above are temporary and would get overwritten if the machine is rebooted. In order to keep the nameserver entry persistent, open the file `/etc/sysconfig/network` in edit mode and add below entries.)

```
[root@jump ~]# cp /etc/sysconfig/network /etc/sysconfig/network.orig
[root@jump ~]# cat /etc/sysconfig/network
NETWORKING=yes
NISDOMAIN=cldrsetup.local # our DNS DOMAIN
DNS1=172.31.24.240 # PRIVATE_IP_OF_IPASERVER / AD Server
NOZEROCONF=yes
[root@jump ~]#
```

```
NETWORKING=yes
NISDOMAIN=cdppvcds.com
DNS1=172.31.40.119
NOZEROCONF=yes
```

Step 3. Copy `/etc/resolv.conf` file to each node to make them able to resolve FQDNs across the cluster:

```
[root@ipaserver ~]# ansible all -m copy -a "src=/etc/resolv.conf dest=/etc/resolv.conf" --become
```

Step 4. Copy `/etc/sysconfig/network` file to each node to make them able to resolve FQDNs across the cluster: (`/etc/resolv.conf` changes may vanished after the reboot, so to persist those changes, we need the below configuration)

```
[root@ipaserver ~]# ansible all -m copy -a "src=/etc/sysconfig/network dest=/etc/sysconfig/network" --become
```

Step 5. Update Network config to make sure DNS entries persist, even after reboot:

```
# Extract DNS entries
[root@jump ~]# grep '^nameserver' /etc/resolv.conf | awk '{print "DNS" NR "=" $2}' > /tmp/dns_entries.txt

# Update ifcfg-eth0 with DNS entries
[root@jump ~]# while IFS= read -r line; do
  ansible all -m lineinfile -a "path=/etc/sysconfig/network-scripts/ifcfg-eth0 line='${line}' state=present backup=true" --become
done < /tmp/dns_entries.txt
```

```
# Clean up
[root@jump ~]# rm -vf /tmp/dns_entries.txt
```

Step 6. Setup Reverse DNS Zone on ipaserver / AD server, –from-ip is VPC-CIDR In this step we will be setting up a reverse DNS zone on the FreeIPA server for reverse lookup:

Note: In case of **AD server**, this would be already created/present (No need to perform the below steps, jump to step 8)

Take the CIDR block of the network in which the instances are created and create a reverse DNS zone by executing the below command on the IPA Server machine.

ipa dnszone-add --name-from-ip=<YOUR_VPC_CIDR>

If your VPC has a CIDR 172.16.0.0/16, then the command looks as below.

```
[root@ipaserver ~]# ipa dnszone-add --name-from-ip=172.31.0.0/16
```

Zone name [31.172.in-addr.arpa.]:

Zone name: 31.172.in-addr.arpa.

Active zone: True

Authoritative nameserver: ipaserver.cldrsetup.local.

Administrator e-mail address: hostmaster

SOA serial: 1715598489

SOA refresh: 3600

SOA retry: 900

SOA expire: 1209600

SOA minimum: 3600

BIND update policy: grant CLDRSETUP.LOCAL krb5-subdomain 31.172.in-addr.arpa. PTR;

Dynamic update: False

Allow query: any;

Allow transfer: none;

Once you execute the above command, accept the default value by hitting the enter key. It will create a reverse DNS zone by name 16.172.in-addr.arpa. (with a trailing dot)

```
[root@ipaserver centos]# ipa dnszone-add --name-from-ip=172.31.0.0/16
Zone name [31.172.in-addr.arpa.]:
Zone name: 31.172.in-addr.arpa.
Active zone: TRUE
Authoritative nameserver: ipaserver.cdppvcds.com.
Administrator e-mail address: hostmaster
SOA serial: 1680093921
SOA refresh: 3600
SOA retry: 900
SOA expire: 1209600
SOA minimum: 3600
BIND update policy: grant CDPPVCDS.COM krb5-subdomain 31.172.in-addr.arpa. PTR;
Dynamic update: FALSE
Allow query: any;
Allow transfer: none;
```

Step 7. Disable krb5 ccache config and verify:

OPEN /etc/krb5.conf on IPASERVER and comment ccache conf: (this step is not needed on any cluster node, as CDP will manage the krb5.conf in further steps config)

After the setup is complete, we need to make a kerberos config change which gets enabled automatically post the ipa server setup.

Open the file /etc/krb5.conf in edit mode and comment out the line related to ccache_name as shown below.

```
[root@ipaserver ~]# vi /etc/krb5.conf
```

Comment the below ccache config

```
#default_ccache_name = KEYRING:persistent:%{uid}
```

After any changes of /etc/krb5.conf anytime, do run the below commands to restart all the IPA services.

```
[root@ipaserver ~]# ipactl restart
```

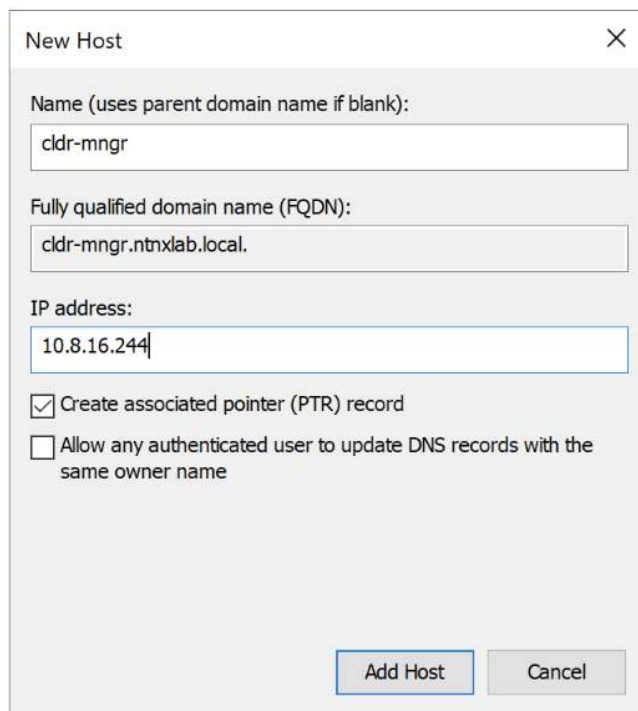
```
[root@ipaserver centos]# cat /etc/krb5.conf
includedir /etc/krb5.conf.d/
includedir /var/lib/sss/pubconf/krb5.include.d/

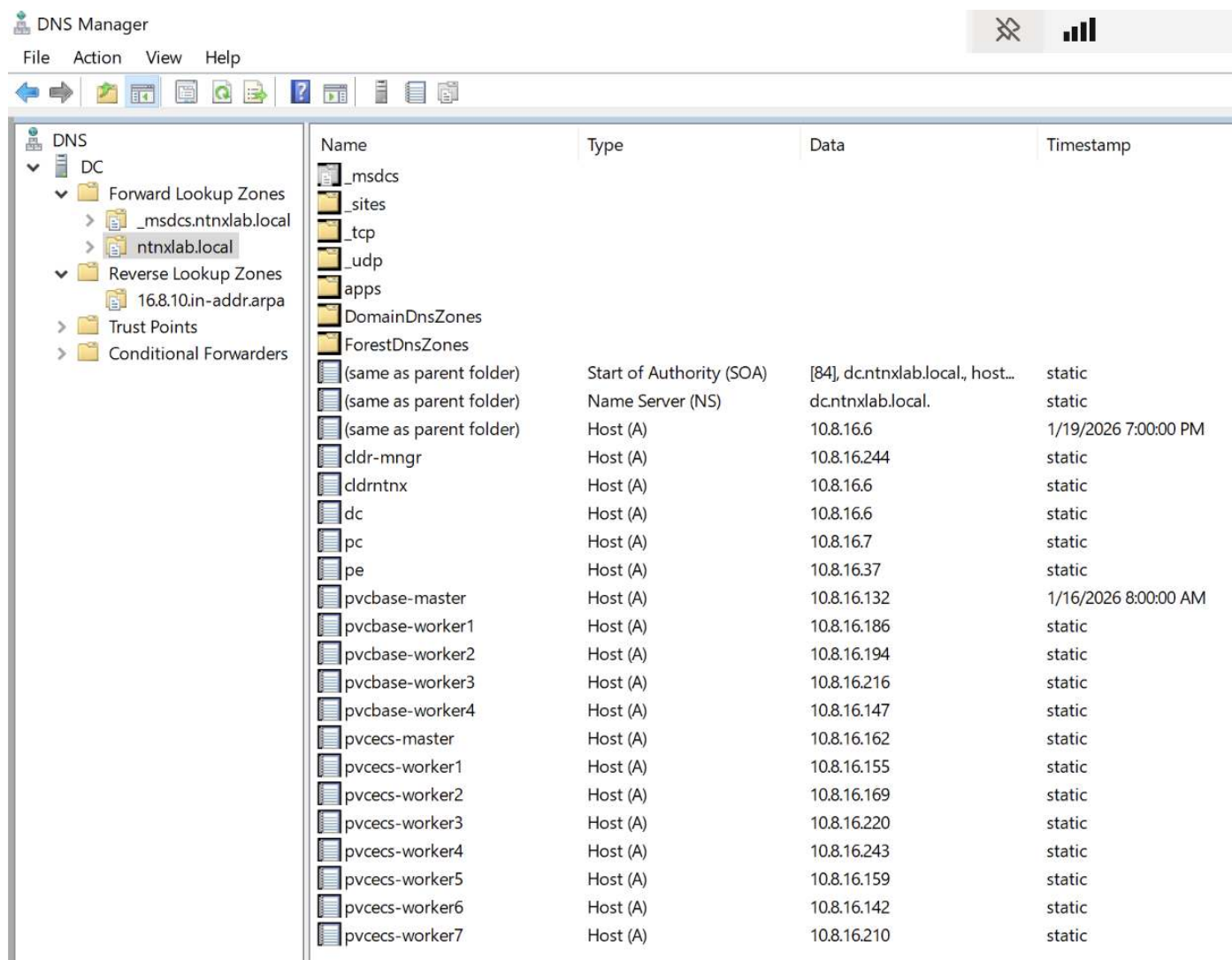
[logging]
default = FILE:/var/log/krb5libs.log
kdc = FILE:/var/log/krb5kdc.log
admin_server = FILE:/var/log/kadmind.log

[libdefaults]
default_realm = CDPPVCDS.COM
dns_lookup_realm = false
dns_lookup_kdc = true
rdns = false
ticket_lifetime = 24h
forwardable = true
udp_preference_limit = 0
# default_ccache_name = KEYRING:persistent:%{uid}
```

Step 8. Prepare the commands for adding dnsrecord and configuring reverse lookup:

Note: If you are using an **AD server** instead of FreeIPA, you must manually create the required PTR records in both the forward lookup zones and the reverse lookup zones on the AD server. If you check the option “Create associated pointer (PTR) record” while creating the records in forward lookup zones, records will automatically be created in reverse lookup zones. (You need to do this for all the nodes)

A screenshot of a 'New Host' dialog box. The dialog has a title bar with 'New Host' and a close button. It contains three text input fields: 'Name (uses parent domain name if blank):' with 'cldr-mngr', 'Fully qualified domain name (FQDN):' with 'cldr-mngr.ntnslab.local.', and 'IP address:' with '10.8.16.244'. Below these fields are two checkboxes: 'Create associated pointer (PTR) record' which is checked, and 'Allow any authenticated user to update DNS records with the same owner name' which is unchecked. At the bottom right are two buttons: 'Add Host' and 'Cancel'.



(Skip below steps if using **AD server**)

ADD The entry of all individual machines (separate IP separate command) to reverse DNS zone:

We need to create a record for each machine in the reverse DNS zone, created previously.

Use the below command as reference and make changes as per your configuration/machine's private IP and Hostname.

Add the entry of this e.g. IPA server machine to the reverse DNS zone.

We need to add the IPV4 address in reverse order. The first two octets are already added in the reverse zone above. Now we need to create a record for this machine inside that zone by using the last two octets.

In the command below you need to add the record by providing the last two octets of your machine's private IPV4 in reverse order. Include the trailing dot after the machine name FQDN in the above command.

Generate the command as shown below and run the same for all the FreeIPA agents, that includes all the nodes of Base and ECS cluster.

```
ipa dnsrecord-add <2nd>.<1st>.in-addr.arpa. <4th>.<3rd> --ptr-rec <server FQDN>.
```

Example:

```
ipa dnsrecord-add 16.172.in-addr.arpa. 226.31 --ptr-rec ipaserver.cldrsetup.local.
```

Following the same, The record for the machine should be created in the Reverse DNS zone.

```
[root@ipaserver centos]# ipa dnsrecord-add 31.172.in-addr.arpa. 119.40 --ptr-rec ipaserver.cdppvcds.com.  
Record name: 119.40  
PTR record: ipaserver.cdppvcds.com.
```

```
[root@cdpbase centos]# ipa dnsrecord-add 31.172.in-addr.arpa. 234.0 --ptr-rec cdpbase.cdppvcds.com.  
Record name: 234.0  
PTR record: cdpbase.cdppvcds.com.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 240.24 --ptr-rec ipaserver.cldrsetup.local.  
Record name: 240.24  
PTR record: ipaserver.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 139.27 --ptr-rec cldr-mngr.cldrsetup.local.  
Record name: 139.27  
PTR record: cldr-mngr.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 104.21 --ptr-rec pvcbase-master.cldrsetup.local.  
Record name: 104.21  
PTR record: pvcbase-master.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 185.16 --ptr-rec pvcbase-worker1.cldrsetup.local.  
Record name: 185.16  
PTR record: pvcbase-worker1.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 0.23 --ptr-rec pvcbase-worker2.cldrsetup.local.  
Record name: 0.23  
PTR record: pvcbase-worker2.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 240.18 --ptr-rec pvcbase-worker3.cldrsetup.local.  
Record name: 240.18  
PTR record: pvcbase-worker3.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 239.30 --ptr-rec pvcecs-master.cldrsetup.local.  
Record name: 239.30  
PTR record: pvcecs-master.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 43.22 --ptr-rec pvcecs-worker1.cldrsetup.local.  
Record name: 43.22  
PTR record: pvcecs-worker1.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 249.30 --ptr-rec pvcecs-worker2.cldrsetup.local.  
Record name: 249.30  
PTR record: pvcecs-worker2.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 24.26 --ptr-rec pvcecs-worker3.cldrsetup.local.  
Record name: 24.26  
PTR record: pvcecs-worker3.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 198.24 --ptr-rec pvcecs-worker4.cldrsetup.local.  
Record name: 198.24  
PTR record: pvcecs-worker4.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 53.24 --ptr-rec pvcecs-worker5.cldrsetup.local.  
Record name: 53.24  
PTR record: pvcecs-worker5.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 43.22 --ptr-rec pvcecs-worker6.cldrsetup.local.  
Record name: 43.22  
PTR record: pvcecs-worker6.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 249.30 --ptr-rec pvcecs-worker7.cldrsetup.local.  
Record name: 249.30  
PTR record: pvcecs-worker7.cldrsetup.local.
```

```
[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 24.26 --ptr-rec pvcecs-worker8.cldrsetup.local.
```

Record name: 24.26
PTR record: pvcecs-worker8.cldrsetup.local.

[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 198.24 --ptr-rec pvcecs-worker9.cldrsetup.local.

Record name: 198.24
PTR record: pvcecs-worker9.cldrsetup.local.

[root@ipaserver ~]# ipa dnsrecord-add 31.172.in-addr.arpa. 53.24 --ptr-rec pvcecs-worker10.cldrsetup.local.

Record name: 53.24
PTR record: pvcecs-worker10.cldrsetup.local.

Step 9. Verify the DNS records have been added successfully:

[root@ipaserver ~]# ipa dnsrecord-find 31.172.in-addr.arpa.

Record name: @
NS record: ipaserver.cldrsetup.local.

Record name: 240.24
PTR record: ipaserver.cldrsetup.local.

Record name: 139.27
PTR record: cldr-mngr.cldrsetup.local.

Record name: 104.21
PTR record: pvcbase-master.cldrsetup.local.

Record name: 185.16
PTR record: pvcbase-worker1.cldrsetup.local.

Record name: 0.23
PTR record: pvcbase-worker2.cldrsetup.local.

Record name: 240.18
PTR record: pvcbase-worker3.cldrsetup.local.

Record name: 239.30
PTR record: pvcecs-master.cldrsetup.local.

Record name: 43.22
PTR record: pvcecs-worker1.cldrsetup.local.

Record name: 249.30
PTR record: pvcecs-worker2.cldrsetup.local.

Record name: 198.24
PTR record: pvcecs-worker3.cldrsetup.local.

Record name: 53.24
PTR record: pvcecs-worker4.cldrsetup.local.

Record name: 24.26
PTR record: pvcecs-worker5.cldrsetup.local.

Record name: 43.22
PTR record: pvcecs-worker6.cldrsetup.local.

Record name: 0.23
PTR record: pvcecs-worker7.cldrsetup.local.

Record name: 198.24
PTR record: pvcecs-worker8.cldrsetup.local.

Record name: 53.24
PTR record: pvcecs-worker9.cldrsetup.local.

Record name: 24.26
PTR record: pvcecs-worker10.cldrsetup.local.

Number of entries returned 18

[root@ipaserver ~]#

Procedure 6. Configure freeipa-client on all other nodes to get them managed by ipa-server

Option 1: If using FreeIPA

Step 1: Install free-ipa client along with other packages needed on all hosts except ipaserver:

Note: Setup ipaserver client and krb5 libs on each node before copying resolv.conf, as installation of ipa-client will override this. (**UDP port 123 and TCP port 389 need to be enabled for ipa services, ntp and timesync**)

Note: Remove chrony from all hosts using ansible as it creates issues in installing and configuring ipa services successfully.

Note: Please review JAVA requirement in Cloudera on premises Base Requirements and Supported Versions sections: (We installed OpenJDK11 for this solution validation, ipa-client will also require and auto install java 11 on all hosts, if it is not present or any different version is installed e.g. java17)

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-java-requirements.html>

```
[root@jump ~]# ansible all -m shell -a "sudo subscription-manager repos --enable=rhel-9-for-x86_64-baseos-rpms && sudo subscription-manager repos --enable=rhel-9-for-x86_64-appstream-rpms && sudo dnf install -y java-17-openjdk java-17-openjdk-devel python3-pip wget telnet mlocate tar traceroute net-tools bind-utils traceroute nc && java -version && python3 -V && pip3 install --upgrade pip && pip3 -V && pip3 install pycopg2-binary && pip3 list |grep psy"
```

```
[root@jump ~]# ansible all -m shell -a "sudo dnf install -y https://download.postgresql.org/pub/repos/yum/repos/rpms/EL-9-x86_64/pgdg-redhat-repo-latest.noarch.rpm && sudo dnf install -y postgresql14"
```

```
[root@jump ~]# ansible all -m shell -a "sudo subscription-manager repos --enable=rhel-9-for-x86_64-baseos-rpms && sudo subscription-manager repos --enable=rhel-9-for-x86_64-appstream-rpms && sudo dnf install -y freeipa-client openldap-clients krb5-workstation krb5-libs && chronyc tracking && chronyc sources" -I 'all:admin'
```

Step 2: Install and Setup IPA services by configuring the free-ipa client on all machines (except ipaserver) and add all the machines to the DNS server, by running the command **"ipa-client-install"** to set up the IPA client.

Enter the values for these parameters as below. After entering these values, it should return the message as **"The ipa-client-install command was successful"**.

Parameter	Value
Do you want to configure chrony with NTP server or pool address? [no]:	yes
Enter NTP source server addresses separated by comma, or press Enter to skip:	<ENTER>
Enter a NTP source pool address, or press Enter to skip:	<ENTER>
Continue to configure the system with these values? [no]:	yes
User authorized to enroll computers:	admin
Password for admin@<Your_Domain>:	<Password created earlier> (cloudera123)

```
[root@pvcbase-master ~]# ipa-client-install --force-join
```

This program will set up IPA client.
Version 4.11.0

Discovery was successful!

Do you want to configure chrony with NTP server or pool address? [no]: **yes**

Enter NTP source server addresses separated by comma, or press Enter to skip: **<ENTER>**

Enter a NTP source pool address, or press Enter to skip: **<ENTER>**

Client hostname: cldr-mngr.cldrsetup.local

Realm: CLDRSETUP.LOCAL
DNS Domain: cldrsetup.local
IPA Server: ipaserver.cldrsetup.local
BaseDN: dc=cldrsetup,dc=local

Continue to configure the system with these values? [no]: **yes**

Synchronizing time

No SRV records of NTP servers were found and no NTP server or pool address was provided.

Using default chrony configuration.

Attempting to sync time with chronyc.

Time synchronization was successful.

User authorized to enroll computers: **<admin>**

Password for admin@CLDRSETUP.LOCAL: **<cloudera123>**

Successfully retrieved CA cert

Subject: CN=Certificate Authority,O=CLDRSETUP.LOCAL

Issuer: CN=Certificate Authority,O=CLDRSETUP.LOCAL

Valid From: 2024-05-13 10:59:53+00:00

Valid Until: 2044-05-13 10:59:53+00:00

Enrolled in IPA realm CLDRSETUP.LOCAL

Created /etc/ipa/default.conf

Configured /etc/sss/sss.conf

Systemwide CA database updated.

Hostname (pvcbase-master.cldrsetup.local) does not have A/AAAA record.

Adding SSH public key from /etc/ssh/ssh_host_rsa_key.pub

Adding SSH public key from /etc/ssh/ssh_host_ecdsa_key.pub

Adding SSH public key from /etc/ssh/ssh_host_ed25519_key.pub

SSSD enabled

Configured /etc/openldap/ldap.conf

Configured /etc/ssh/ssh_config

Configured /etc/ssh/sshd_config.d/04-ipa.conf

Configuring cldrsetup.local as NIS domain.

Configured /etc/krb5.conf for IPA realm CLDRSETUP.LOCAL

Client configuration complete.

The ipa-client-install command was successful

[root@pvcbase-master ~]#


```

[root@cdpbase centos]# ipa-client-install --force-ntpd
Discovery was successful!
Client hostname: cdpbase.cdppvcds.com
Realm: CDPPVCD.S.COM
DNS Domain: cdppvcds.com
IPA Server: ipaserver.cdppvcds.com
BaseDN: dc=cdppvcds,dc=com

Continue to configure the system with these values? [no]: yes
Synchronizing time with KDC...
Attempting to sync time using ntpd. Will timeout after 15 seconds
Attempting to sync time using ntpd. Will timeout after 15 seconds
Unable to sync time with NTP server, assuming the time is in sync. Please check that 123 UDP port is opened.
User authorized to enroll computers: admin
Password for admin@CDPPVCD.S.COM:
Successfully retrieved CA cert
  Subject: CN=Certificate Authority,0=CDPPVCD.S.COM
  Issuer: CN=Certificate Authority,0=CDPPVCD.S.COM
  Valid From: 2023-03-29 11:23:01
  Valid Until: 2043-03-29 11:23:01

Enrolled in IPA realm CDPPVCD.S.COM
Created /etc/ipa/default.conf
New SSSD config will be created
Configured sudoers in /etc/nsswitch.conf
Configured /etc/sss/sssd.conf
Configured /etc/krb5.conf for IPA realm CDPPVCD.S.COM
trying https://ipaserver.cdppvcds.com/ipa/json
[try 1]: Forwarding 'schema' to json server 'https://ipaserver.cdppvcds.com/ipa/json'
trying https://ipaserver.cdppvcds.com/ipa/session/json
[try 1]: Forwarding 'ping' to json server 'https://ipaserver.cdppvcds.com/ipa/session/json'
[try 1]: Forwarding 'ca_is_enabled' to json server 'https://ipaserver.cdppvcds.com/ipa/session/json'
Systemwide CA database updated.
Hostname (cdpbase.cdppvcds.com) does not have A/AAAA record.
Missing reverse record(s) for address(es): 172.31.0.234.
Adding SSH public key from /etc/ssh/ssh_host_rsa_key.pub
Adding SSH public key from /etc/ssh/ssh_host_ecdsa_key.pub
Adding SSH public key from /etc/ssh/ssh_host_ed25519_key.pub
[try 1]: Forwarding 'host_mod' to json server 'https://ipaserver.cdppvcds.com/ipa/session/json'
SSSD enabled
Configured /etc/openldap/ldap.conf
NTP enabled
Configured /etc/ssh/ssh_config
Configured /etc/ssh/sshd_config
Configuring cdppvcds.com as NIS domain.
Client configuration complete.
The ipa-client-install command was successful

```

Step 3: Verify KDC setup: kerberos ticket generation is working fine by generating a ticket for the admin user from all individual hosts.

Run the kinit admin command to authenticate as admin and enter the directory password provided during ipa server installation. The command should generate the ticket and should be listed by executing klist -e.

```

[root@ipaserver ~]# kinit admin
Password for admin@CLDRSETUP.LOCAL: <cloudera123>

```

```

[root@ipaserver ~]# klist -e
Ticket cache: KCM:0
Default principal: admin@CLDRSETUP.LOCAL

```

```

Valid starting Expires Service principal
05/13/2024 04:07:15 05/14/2024 03:30:48 krbtgt/CLDRSETUP.LOCAL@CLDRSETUP.LOCAL
Etype (skey, tkt): aes256-cts-hmac-sha384-192, aes256-cts-hmac-sha384-192

```

try kinit admin@CLDRSETUP.LOCAL
 ##### (if fails anytime, run below commands

```

[root@ipaserver ~]# ipactl stop && ipactl start && ipactl status

```

Verify the status of ipa services installed

```

[root@ipaserver ~]# ipactl status
Directory Service: RUNNING
krb5kdc Service: RUNNING
kadmin Service: RUNNING
named Service: RUNNING
httpd Service: RUNNING
ipa-custodia Service: RUNNING
ntpd Service: RUNNING
pki-tomcatd Service: RUNNING

```

```
ipa-otpd Service: RUNNING
ipa-dnskeysyncd Service: RUNNING
ipa: INFO: The ipactl command was successful
```

This command should return the below output:

```
[root@ipaserver centos]# kinit admin
Password for admin@CDPPVCDS.COM:
[root@ipaserver centos]# klist -e
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: admin@CDPPVCDS.COM

Valid starting    Expires          Service principal
04/05/23 10:49:29 04/06/23 10:49:26  krbtgt/CDPPVCDS.COM@CDPPVCDS.COM
           Etype (skey, tkt): aes256-cts-hmac-sha1-96, aes256-cts-hmac-sha1-96
[root@ipaserver centos]#
```

Option 2: If using AD Server

- If you are using **AD server** as Kerberos instead of FreeIPA, run the following command on all the client machines and skip the ipa-client-install steps

```
[root@pvcbase-master ~]# realm join --user=administrator ntnxlab.local

# Non-interactive command
[root@pvcbase-master ~]# echo 'password' | realm join --user=administrator ntnxlab.local
```

- Verify KDC setup: kerberos ticket generation is working fine by generating a ticket for the administrator user from all individual hosts.

```
[root@pvcbase-master ~]# kinit administrator@NTNXLAB.LOCAL
Password for administrator@NTNXLAB.LOCAL:
[root@pvcbase-master ~]# klist -e
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: administrator@NTNXLAB.LOCAL

Valid starting    Expires          Service principal
01/22/2026 07:05:46 01/22/2026 17:05:46  krbtgt/NTNXLAB.LOCAL@NTNXLAB.LOCAL
           renew until 01/29/2026 07:05:40, Etype (skey, tkt): aes256-cts-hmac-sha1-96, aes256-cts-hmac-sha1-96
[root@pvcbase-master ~]#
```

Step 4: Verify the network configuration file `/etc/resolv.conf` on the Bastion server to use the Name Server created in previous steps (after installing freeipa-client, as it overrides resolv.conf and may lead to rework) to make them able to resolve FQDNs across the cluster:

(Open the file `/etc/resolv.conf` in edit mode and verify the following. Make sure the new entry is added above any other nameserver entry. The contents of the file must look similar to the below.)

Note: Make sure that the `/etc/resolv.conf` file on the ECS hosts *contains a maximum of 2 active search domains*.

<https://docs.cloudera.com/data-warehouse/1.5.5/release-notes/topics/dw-private-cloud-known-issues-ecs-environments.html>

```
[root@jump ~]# cp /etc/resolv.conf /etc/resolv.conf.orig
[root@jump ~]# cat /etc/resolv.conf
search ap-southeast-1.compute.internal cldrsetup.local
nameserver 172.31.24.240 # PrivateIP of FreeIPA Server must be first nameserver entry after search
nameserver 172.31.0.2 # DNS of AWS i.e. in case of PVC Configured on EC2
nameserver 127.0.0.1
[root@jump ~]#
```

```
; generated by /usr/sbin/dhclient-script
search ap-south-1.compute.internal cdppvcds.com
nameserver 172.31.40.119
```

Step 5: Verify the network configuration file `/etc/sysconfig/network` on the Bastion server to use the Name Server created in previous steps:

(The changes in `/etc/resolv.conf` above are temporary and would get overwritten if the machine is rebooted. In order to keep the nameserver entry persistent, open the file `/etc/sysconfig/network` in edit mode and verify the entries below.)

```
[root@jump ~]# cp /etc/sysconfig/network /etc/sysconfig/network.orig
[root@jump ~]# cat /etc/sysconfig/network
NETWORKING=yes
NISDOMAIN=cldrsetup.local      # our DNS DOMAIN
DNS1=172.31.24.240             # PRIVATE_IP_OF_IPASERVER
NOZEROCONF=yes
[root@jump ~]#
```

```
NETWORKING=yes
NISDOMAIN=cdppvcds.com
DNS1=172.31.40.119
NOZEROCONF=yes
```

Step 6: Copy `/etc/resolv.conf` file to each node again, to make them able to resolve FQDNs across the cluster:

```
[root@jump ~]# ansible all -m copy -a "src=/etc/resolv.conf dest=/etc/resolv.conf"
```

Step 7: Copy `/etc/sysconfig/network` file again, to each node to make them able to resolve FQDNs across the cluster: (`/etc/resolv.conf` changes may vanished after the reboot, so to persist those changes, we need the below configuration)

```
[root@jump ~]# ansible all -m copy -a "src=/etc/sysconfig/network dest=/etc/sysconfig/network"
```

Step 8: Enable permissions for HDFS and for PAM Authentication:

```
[root@jump ~]# ansible all -m shell -a "chmod 1777 /tmp && chmod 444 /etc/shadow"
```

Step 9: Login to pvcbase-master node and verify forward and reverse DNS lookup is working fine from each machine:

Note: In case of **AD server**, you can verify on any client machine like pvcbase-master

```
[root@pvcbase-master ~]# nslookup cldr-mngr.cldrsetup.local
Server:      172.31.24.240
Address:     172.31.24.240#53

Name: cldr-mngr.cldrsetup.local
Address: 172.31.27.139

#(forward lookup)Running the below command should return the IPV4 of the machine in the Answer Section.

# dig <FQDN of the SERVER> A
# dig $(hostname) A | grep -A2 ANSWER
# Ex:- dig ipaserver.cldrsetup.local A

[root@pvcbase-master ~]# dig $(hostname -f) A | grep -A2 ANSWER
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
--
;; ANSWER SECTION:
pvcbase-master.ntnslab.local. 3600 IN A    10.8.16.132

#(reverse lookup) Running the below command should return the hostname of the machine in the Answer Section.
```

```
# dig -x <Private_IP_of_SERVER>
# dig -x $(hostname -i) | grep -A2 ANSWER
# Ex:- dig -x 172.31.40.119

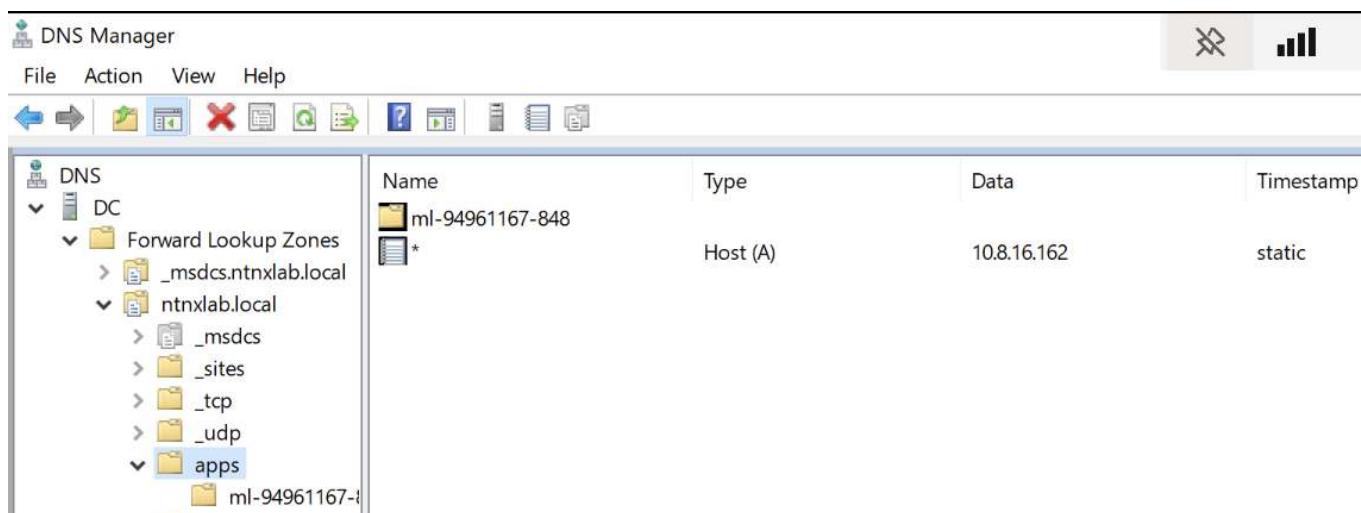
[root@pvcbase-master ~]# dig -x $(hostname -i) | grep -A2 ANSWER
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
--
;; ANSWER SECTION:
132.16.8.10.in-addr.arpa. 3600 IN PTR pvcbase-master.ntnslab.local.

[root@pvcbase-master ~]$
```

Step 10: Login on ipaserver, configure and validate wildcard DNS record is working fine and resolvable, which is required later for the ECS data service cluster (if not set properly, chances of ECS installation getting corrupt):

Note: In case of **AD server**, you need to create the wildcard DNS record on the AD DNS server manually in the forward and reverse lookup zones and then you can verify by running the command **nslookup console-cdp.apps.cldrsetup.local** pvcecs-master node.



```
[root@pvcecs-master ~]# nslookup console-cdp.apps.ntnslab.local
Server:          10.8.16.6
Address:         10.8.16.6#53

Name:   console-cdp.apps.ntnslab.local
Address: 10.8.16.162
```

Skip below steps if you are using **AD server** (Jump to step 11)

```
[root@ipaserver ~]# ipa dnsrecord-add cldrsetup.local *.apps
Please choose a type of DNS resource record to be added
The most common types for this type of zone are: A, AAAA

DNS resource record type: A
A IP Address: 172.31.30.239   #Provide the IP address of ecs-master node
Record name: *.apps
A record: 172.31.30.239

[root@ipaserver ~]# nslookup console-cdp.apps.cldrsetup.local
Server:      172.31.24.240
```

```

Address: 172.31.24.240#53

Name: console-cdp.apps.cldrsetup.local
Address: 172.31.30.239

[root@ipaserver ~]# dig console-cdp.apps.cldrsetup.local A | grep -A2 ANSWER
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
--
;; ANSWER SECTION:
console-cdp.apps.cldrsetup.local. 86400 IN A 172.31.30.239

[root@ipaserver ~]# dig -x 172.31.30.239 | grep -A2 ANSWER
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
--
;; ANSWER SECTION:
239.30.31.172.in-addr.arpa. 86400 IN PTR pvcecs-master.cldrsetup.local.

[root@ipaserver ~]#

```

Step 11: Download and copy postgresql-jdbc driver to all hosts:

```

[root@jump ~]# wget https://jdbc.postgresql.org/download/postgresql-42.7.13.jar
[root@jump ~]# chmod 644 postgresql-42.7.13.jar
[root@jump ~]# ansible all -m copy -a "src=postgresql-42.7.13.jar dest=/usr/share/java/postgresql-connector-java.jar"
[root@jump ~]# ansible all -m shell -a "sudo ls -l /usr/share/java/postgresql-connector-java.jar"
[root@jump ~]#

```

Procedure 7. Disable the Linux Firewall (Ignore the errors if firewall is not installed)

Note: The default Linux firewall settings are too restrictive for any Hadoop deployment. Since the Cloudera on premises deployment will be in its own isolated network in the on premise environment, there is no need for that additional firewall. **(NA in AWS EC2)**

```

##### Either disable the firewall or update the rules: (ON ALL HOSTS)
[root@jump ~]# ansible all -m command -a "firewall-cmd --zone=public --add-port=80/tcp --permanent"
[root@jump ~]# ansible all -m command -a "firewall-cmd --zone=public --add-port=443/tcp --permanent"
[root@jump ~]# ansible all -m command -a "firewall-cmd --reload"
[root@jump ~]# ansible all -m command -a "systemctl disable firewalld && systemctl stop firewalld && systemctl status firewalld | grep -e disabled -e inactive"
[root@jump ~]#

```

Procedure 8. Disable SELinux

Note: SELinux must be disabled during the install procedure and cluster setup. SELinux can be enabled after installation and while the cluster is running.

Step 1: SELinux can be disabled by editing `/etc/selinux/config` (in some systems it would be `/etc/sysconfig/selinux`) To disable SELinux, change SELINUX=enforcing to SELINUX=disabled or SELINUX=permissive. follow these steps:

```

[root@jump ~]# ansible all -m shell -a "sed -i 's/SELINUX=enforcing/SELINUX=disabled/g' /etc/selinux/config"
[root@jump ~]# ansible all -m shell -a "setenforce 0"
[root@jump ~]# ansible all -m shell -a "getenforce"

```

Note: This command may fail if SELinux is already disabled. This requires reboot to take effect.

Note: While the suggested configuration is to disable SELinux as shown below, if for any reason SELinux needs to be enabled on the cluster, run the following command to make sure that the httpd can read the *Yum* profiles.

```

[root@jump ~]# chcon -R -t httpd_sys_content_t /var/www/html/

```

```
# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
SELINUX=disabled
# SELINUXTYPE= can take one of three values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are protected.
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

Procedure 9. Enable Syslog

Syslog must be enabled on each node to preserve logs regarding killed processes or failed jobs. Modern versions such as syslog-ng and rsyslog are possible, making it more difficult to be sure that a syslog daemon is present.

Step 1. Use one of the following commands to confirm that the service is properly configured:

```
[root@jump ~]# ansible all -m command -a "rsyslogd -v"
[root@jump ~]# ansible all -m command -a "service rsyslog status"
```

Procedure 10. Set ulimit

On each node, ulimit -n specifies the number of inodes that can be opened simultaneously. With the default value of 1024, the system appears to be out of disk space and shows no inodes available. This value should be set to 64000 on every node. Higher values are unlikely to result in an appreciable performance gain.

Step 1. For setting the ulimit on RedHat, edit `/etc/security/limits.conf` on admin node and add the following lines:

```
[root@jump ~]# vi /etc/security/limits.conf
* soft nfile 1048576
* hard nfile 1048576
```

Step 2. Copy the `/etc/security/limits.conf` file from admin node (ipaserver) to all the nodes using the following command:

```
[root@jump ~]# ansible all -m copy -a "src=/etc/security/limits.conf dest=/etc/security/limits.conf"
```

Step 3. Make sure that the `/etc/pam.d/su` file contains the following settings:

```
[root@ipaserver ~]# vi /etc/pam.d/su
##PAM-1.0
auth    required    pam_env.so
auth    sufficient   pam_rootok.so
# Uncomment the following line to implicitly trust users in the "wheel" group.
#auth    sufficient   pam_wheel.so trust use_uid
# Uncomment the following line to require a user to be in the "wheel" group.
#auth    required     pam_wheel.so use_uid
auth    include      system-auth
auth    include      postlogin
account sufficient   pam_succeed_if.so uid = 0 use_uid quiet
account include      system-auth
password include      system-auth
session include      system-auth
session include      postlogin
session optional     pam_xauth.so
```

Step 4. Copy the `/etc/pam.d/su` file from admin node (ipaserver) to all the nodes using the following command:

```
[root@jump ~]# ansible all -m copy -a "src=/etc/pam.d/su dest=/etc/pam.d/su"
```

Note: The `ulimit` values are applied on a new shell, running the command on a node on an earlier instance of a shell will show old values.

Procedure 11. Set TCP Retries

Adjusting the `tcp_retries` parameter for the system network enables faster detection of failed nodes. Given the advanced network-ing features of UCS, this is a safe and recommended change (failures observed at the operating system layer are most likely serious rather than transitory).

Note: On each node, setting the number of TCP retries to 5 can help detect unreachable nodes with less latency.

Step 1. Edit the file `/etc/sysctl.conf` on ipaserver node and add the following lines:

```
[root@jump ~]# vi /etc/sysctl.conf
net.ipv4.tcp_retries2=5
fs.inotify.max_user_instances=512
sysctl -p
```

Step 2. Copy the `/etc/sysctl.conf` file from admin node to all the nodes using the following command:

```
[root@jump ~]# ansible all -m copy -a "src=/etc/sysctl.conf dest=/etc/sysctl.conf"
```

Procedure 12. Disable IPv6 Defaults

Step 1. Run the following command:

```
[root@jump ~]# ansible all -m shell -a "echo 'net.ipv6.conf.all.disable_ipv6 = 1' >> /etc/sysctl.conf" -I 'all:!ecsmasternodes:!ecsnodes'
[root@jump ~]# ansible all -m shell -a "echo 'net.ipv6.conf.default.disable_ipv6 = 1' >> /etc/sysctl.conf" -I 'all:!ecsmasternodes:!ecsnodes'
[root@jump ~]# ansible all -m shell -a "echo 'net.ipv6.conf.lo.disable_ipv6 = 0' >> /etc/sysctl.conf" -I 'all:!ecsmasternodes:!ecsnodes'
```

Procedure 13. Disable Swapping

Step 1. Run the following to set VM swappiness to 1, by updating `/etc/sysctl.conf` file on all nodes:

```
[root@jump ~]# ansible all -m shell -a "echo 'vm.swappiness=1' >> /etc/sysctl.conf"
```

Procedure 14. Disable Memory Overcommit

Step 1. Run the following on all nodes. Variable `vm.overcommit_memory=0`

```
[root@jump ~]# ansible all -m shell -a "echo 'vm.overcommit_memory=0' >> /etc/sysctl.conf"
```

Step 2. Load the settings from default `sysctl` file `/etc/sysctl.conf` and verify the content of `sysctl.conf`:

```
[root@jump ~]# ansible all -m shell -a "sysctl -p" ## Reload sysctl.conf
[root@jump ~]# ansible all -m shell -a "cat /etc/sysctl.conf"
net.ipv4.tcp_retries2=5
net.ipv6.conf.all.disable_ipv6 = 1
net.ipv6.conf.default.disable_ipv6 = 1
net.ipv6.conf.lo.disable_ipv6 = 0
vm.swappiness=1
vm.overcommit_memory=0
[root@jump ~]#
```

Procedure 15. Disable Transparent Huge Pages

Disabling Transparent Huge Pages (THP) reduces elevated CPU usage caused by THP.

Step 1. You must run the following commands for every reboot:

```
[root@jump ~]# ansible all -m shell -a "echo never > /sys/kernel/mm/transparent_hugepage/enabled"
[root@jump ~]# ansible all -m shell -a "echo never > /sys/kernel/mm/transparent_hugepage/defrag"
```

Step 2. On the Ansible-controller/ ipaserver node, run the following commands:

```
[root@jump ~]# rm -f /root/thp_disable
[root@jump ~]# echo "echo never > /sys/kernel/mm/transparent_hugepage/enabled" >> /root/thp_disable
[root@jump ~]# echo "echo never > /sys/kernel/mm/transparent_hugepage/defrag" >> /root/thp_disable

##### Disable IPV6
[root@jump ~]# echo "sysctl -w net.ipv6.conf.all.disable_ipv6=1" >> /root/thp_disable
[root@jump ~]# echo "sysctl -w net.ipv6.conf.default.disable_ipv6=1" >> /root/thp_disable
[root@jump ~]# echo "sysctl -w net.ipv6.conf.lo.disable_ipv6=0" >> /root/thp_disable
```

Step 3. Copy file to each node to copy the command to */etc/rc.d/rc.local* so they are executed automatically for every reboot:

```
[root@jump ~]# ansible all -m copy -a "src=/root/thp_disable dest=/root/thp_disable"
```

Append the content of file *thp_disable* to */etc/rc.d/rc.local*:

```
[root@jump ~]# ansible all -m shell -a "cat /root/thp_disable >> /etc/rc.d/rc.local"
```

```
[root@jump ~]# ansible all -m shell -a "chmod +x /etc/rc.d/rc.local"
```

```
[root@jump ~]# ansible all -m shell -a "cat /etc/rc.d/rc.local"
```

```
#!/bin/bash
```

```
# Please note that you must run 'chmod +x /etc/rc.d/rc.local' to ensure
```

```
# that this script will be executed during boot.
```

```
touch /var/lock/subsys/local
```

```
# Disable Transparent Huge Pages
```

```
echo never > /sys/kernel/mm/transparent_hugepage/enabled
```

```
echo never > /sys/kernel/mm/transparent_hugepage/defrag
```

```
# Disable IPV6
```

```
sysctl -w net.ipv6.conf.all.disable_ipv6=1
```

```
sysctl -w net.ipv6.conf.default.disable_ipv6=1
```

```
sysctl -w net.ipv6.conf.lo.disable_ipv6=0
```

```
[root@jump ~]#
```

```
#!/bin/bash
```

```
# THIS FILE IS ADDED FOR COMPATIBILITY PURPOSES
```

```
#
```

```
# It is highly advisable to create own systemd services or udev rules  
# to run scripts during boot instead of using this file.
```

```
#
```

```
# In contrast to previous versions due to parallel execution during boot  
# this script will NOT be run after all other services.
```

```
#
```

```
# Please note that you must run 'chmod +x /etc/rc.d/rc.local' to ensure  
# that this script will be executed during boot.
```

```
touch /var/lock/subsys/local
```

```
echo never > /sys/kernel/mm/transparent_hugepage/enabled
```

```
echo never > /sys/kernel/mm/transparent_hugepage/defrag
```

```
sysctl -w net.ipv6.conf.all.disable_ipv6=1
```

```
sysctl -w net.ipv6.conf.default.disable_ipv6=1
```

```
sysctl -w net.ipv6.conf.lo.disable_ipv6=0
```


Procedure 16. Disable tuned service

For Cloudera cluster with hosts are running RHEL/CentOS 7.x or 8.x or 9.x, disable the "tuned" service by running the following commands:

Step 1. Ensure that the tuned service is started.

```
[root@jump ~]# ansible all -m shell -a "systemctl start tuned"
```

Step 2. Turn the tuned service off.

```
[root@jump ~]# ansible all -m shell -a "tuned-adm off"
```

Step 3. Ensure that there are no active profiles.

```
[root@jump ~]# ansible all -m shell -a "tuned-adm list"
```

The output should contain the following line:

```
# pvcecs-worker4.cldrsetup.local | CHANGED | rc=0 >>
```

Available profiles:

- accelerator-performance - Throughput performance based tuning with disabled higher latency STOP states
- aws - Optimize for aws ec2 instances
- balanced - General non-specialized tuned profile
- desktop - Optimize for the desktop use-case
- hpc-compute - Optimize for HPC compute workloads
- intel-sst - Configure for Intel Speed Select Base Frequency
- latency-performance - Optimize for deterministic performance at the cost of increased power consumption
- network-latency - Optimize for deterministic performance at the cost of increased power consumption, focused on low latency network performance
- network-throughput - Optimize for streaming network throughput, generally only necessary on older CPUs or 40G+ networks
- optimize-serial-console - Optimize for serial console use.
- powersave - Optimize for low power consumption
- throughput-performance - Broadly applicable tuning that provides excellent performance across a variety of common server workloads
- virtual-guest - Optimize for running inside a virtual guest
- virtual-host - Optimize for running KVM guests

No current active profile.

Step 4. Shutdown and disable the tuned service.

```
[root@jump ~]# ansible all -m shell -a "systemctl stop tuned"
```

```
[root@jump ~]# ansible all -m shell -a "systemctl disable tuned"
```

Procedure 17. Turning off TCP checksum offload on the all ecs nodes' (both master and worker) network interface (Only Applicable for VMWare machines)

The default CNI (Canal) that comes with ECS (RKE2 from SUSE) does not support VM. Customers have reported escalations with the default CNI when they use VMWARE (WF for example). Customers may experience network connectivity issues or degraded performance when deploying ECS clusters on VMWARE environments with the default CNI. So, they use a workaround by turning off TCP checksum offload on the interface.

Note: Disabling TCP checksum offload may have implications on network performance or security. Evaluate the impact of this workaround in the specific environment and consider reverting the changes once a permanent solution or alternative workaround is available. Additionally, consult with the VMWARE documentation or support resources for guidance on network configuration and optimization in VMWARE environments.

Here's how you can perform the workaround for customers experiencing issues with the default CNI in ECS (RKE2 from SUSE) when using VMWARE infrastructure, by disabling TCP checksum offload on the network interface used by the affected ECS nodes:

Step 1: Identify the network interface:

```
##### Determine the network interface used by the affected ECS nodes. This can typically be found using the `ifconfig` or `ip addr` command:
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "ifconfig | grep flags | grep -v lo && ip addr | grep mtu | grep -v lo"
```

```
[root@jump ~]$ ifconfig | grep flags | grep -v lo
```

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
```

```
[root@jump ~]$ ip addr | grep mtu | grep -v lo
```

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
```

```
[root@jump ~]$
```

Step 2: Disable TCP Checksum Offload: Use the appropriate commands or configuration settings to disable TCP checksum offload on the identified network interface. For example, using ethtool:

Replace `<interface_name>` with the name of the network interface identified in step 1.

```
[root@ipserver ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo ethtool -K <interface_name> tx off"
```

After disabling TCP checksum offload, verify that the changes have been applied correctly and that the affected ECS nodes no longer experience the reported issues.

Procedure 18. Create partitions on ECS nodes (master and worker) manually, if not present: (Do df -h) **(Skip this)**

Step 1: Create partitions on disk attached. We require three partitions i.e. /docker, /cdwdata and /lhdata:

```
[root@pvcecs-master ~]# lsblk
```

```
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
```

```
xvda 202:0 0 600G 0 disk
```

```
├─xvda1 202:1 0 1M 0 part
```

```
├─xvda2 202:2 0 200M 0 part /boot/efi
```

```
├─xvda3 202:3 0 600M 0 part /boot
```

```
└─xvda4 202:4 0 599.2G 0 part /
```

```
xvdb 202:16 0 600G 0 disk
```

```
xvdc 202:32 0 600G 0 disk
```

```
xvdd 202:48 0 600G 0 disk
```

```
xvde 202:48 0 600G 0 disk
```

```
[root@pvcecs-master ~]# fdisk -l
```

```
Disk /dev/sda: 268.4 GB, 268435456000 bytes, 524288000 sectors
```

```
Units = sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk label type: dos
```

```
Disk identifier: 0x000a04a5
```

```
Device Boot Start End Blocks Id System
```

```
/dev/sda1 * 2048 2099199 1048576 83 Linux
```

```
/dev/sda2 2099200 524287999 261094400 8e Linux LVM
```

```
/dev/xvdb 2099200 524287999 261094400 8e Linux LVM
```

```
/dev/xvdc 2099200 524287999 261094400 8e Linux LVM
```

```
/dev/xvdd 2099200 524287999 261094400 8e Linux LVM
```

```
/dev/xvde 2099200 524287999 261094400 8e Linux LVM
```

```
Disk /dev/sdb: 644.2 GB, 644245094400 bytes, 1258291200 sectors
```

```
Units = sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
[root@pvcecs-master ~]#
```

```
[root@pvcecs-master ~]# sudo parted /dev/xvdb mklabel gpt
```

```
Information: You may need to update /etc/fstab.
```

```
[root@pvcecs-master ~]# sudo parted /dev/xvdc mklabel gpt
```

```
[root@pvcecs-master ~]# sudo parted /dev/xvdd mklabel gpt
```

```
[root@pvcecs-master ~]# sudo parted -a opt /dev/xvdb mkpart primary ext4 0% 100%
```

```
Information: You may need to update /etc/fstab.
```

```
[root@pvcecs-master ~]# sudo parted -a opt /dev/xvdc mkpart primary ext4 0% 100%
```

```
[root@pvcecs-master ~]# sudo parted -a opt /dev/xvdd mkpart primary ext4 0% 100%
```

```

[root@pvcecs-master ~]# sudo mkfs.ext4 /dev/xvdb
mke2fs 1.46.5 (30-Dec-2021)
Found a gpt partition table in /dev/xvdb
Proceed anyway? (y,N) y
Creating filesystem with 157286400 4k blocks and 39321600 inodes
Filesystem UUID: 934fece5-074e-4102-8481-6ed3a3b10931
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000, 7962624, 11239424, 20480000, 23887872, 71663616, 78675968,
    102400000

Allocating group tables: done
Writing inode tables: done
Creating journal (262144 blocks):
done
Writing superblocks and filesystem accounting information: done

[root@pvcecs-master ~]# sudo mkfs.ext4 /dev/xvdc
[root@pvcecs-master ~]# sudo mkfs.ext4 /dev/xvdd

[root@pvcecs-master ~]# sudo mkdir -p /docker /cdwdata /lhdata

[root@pvcecs-master ~]# sudo mount /dev/xvdb /docker
[root@pvcecs-master ~]# sudo mount /dev/xvdc /cdwdata
[root@pvcecs-master ~]# sudo mount /dev/xvdd /lhdata

[root@pvcecs-master ~]# sudo blkid /dev/xvdb
[root@pvcecs-master ~]# sudo blkid /dev/xvdc
[root@pvcecs-master ~]# sudo blkid /dev/xvdd

[root@pvcecs-master ~]# cat /etc/fstab
UUID=497ad222-04fa-453f-b110-ba8001d38788    /    xfs    defaults    0    0
UUID=2e0d9ec9-a82a-43cc-a8e2-c6db30e7f6a4    /boot    xfs    defaults    0    0
UUID=7B77-95E7 /boot/efi    vfat    defaults,uid=0,gid=0,umask=077,shortname=winnt    0    2
[root@pvcecs-master ~]#

[root@pvcecs-master ~]# cat>> /etc/fstab
UUID=497ad222-04fa-453f-b110-ba8001d38788    /    xfs    defaults    0    0
UUID=2e0d9ec9-a82a-43cc-a8e2-c6db30e7f6a4    /boot    xfs    defaults    0    0
UUID=7B77-95E7 /boot/efi    vfat    defaults,uid=0,gid=0,umask=077,shortname=winnt    0    2
UUID="934fece5-074e-4102-8481-6ed3a3b10931" /docker    xfs    defaults    0    0
UUID="52f4940d-3dca-4b90-a223-3bef6eee2b74" /cdwdata    xfs    defaults    0    0
UUID="d7e9640b-cc12-49ac-b6c2-6eb3136679a1" /lhdata    xfs    defaults    0    0

[root@pvcecs-master ~]# mount -av
[root@pvcecs-master overlay2]# lvs
LV   VG   Attr   LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
home centos -wi-ao---- 183.24g
root centos -wi-ao---- 50.00g
swap centos -wi-ao---- 15.75g
cdwdata vgcdp -wi-ao---- 600.00g
docker vgcdp -wi-ao---- <600.00g
var vgcdp -wi-ao---- <600.00g
lhdata vgcdp -wi-ao---- 1200.00g
[root@pvcecs-master overlay2]#

[root@pvcecs-master overlay2]# pvs
PV   VG   Fmt Attr PSize  PFree
/dev/sda2 centos lvm2 a-- <249.00g 4.00m
/dev/sdb vgcdp lvm2 a-- <600.00g <224.96g
/dev/sdc vgcdp lvm2 a-- <600.00g <224.96g
/dev/sdd vgcdp lvm2 a-- <1200.00g <224.96g
/dev/sde vgcdp lvm2 a-- <600.00g <224.96g
[root@pvcecs-master ~]#

```

Procedure 19. Install httpd on Cloudera-Manager node i.e. cldr-mngr to host a local Parcel repository

Setting up the RHEL repository on the cloudera-manager node requires `httpd`.

Step 1. Install httpd on the cloudera-manager i.e. **cldr-mngr** node to host repositories:

Note: The Red Hat repository is hosted using HTTP on the admin node; this machine is accessible by all the hosts in the cluster.

```
[root@cldr-mngr ~]# dnf install -y httpd mod_ssl createrepo_c
```

Step 2. Generate CA certificate.

[illegible]

Step 3. Create certificate directory to server content from.

```
[root@cldr-mngr ~]# mkdir -p /var/www/https/
[root@cldr-mngr ~]# echo secure content > /var/www/https/index.html
[root@cldr-mngr ~]# cat /var/www/https/index.html
secure content
```

Step 4. Edit `httpd.conf` file; add `ServerName` and make the necessary changes to the server configuration file:

```
[root@cldr-mngr ~]# vi /etc/httpd/conf/httpd.conf
ServerName cldr-mngr.cldrsetup.local:80
```

Step 5. Start httpd service.

```
[root@cldr-mngr ~]# systemctl start httpd
[root@cldr-mngr ~]# systemctl enable httpd
[root@cldr-mngr ~]# systemctl is-enabled httpd
```

Install Cloudera Data Platform Private Cloud (Cloudera on premises)

This chapter contains the following:

- Cloudera Runtime
- Install Cloudera on premises Base **7.3.2 CHF1**
- Install CDP Data Services **1.5.5 SP3 CHF3**

Cloudera Runtime

Cloudera Runtime is the core open-source software distribution within Cloudera on premises Base. Cloudera Runtime includes approximately 50 open-source projects that comprise the core distribution of data management tools within CDP.

For more information review Cloudera Runtime Release notes:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/private-release-notes/topics/rt-whats-new.html>

Please review runtime cluster hosts and role assignments:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-runtime-cluster-hosts-role-assignments.html>

Cloudera Data Platform Private Cloud Installation Requirements (Pre-requisites):

NTP/Chrony

Both Cloudera on premises Base and Cloudera on premises DS cluster should have their time synched with the NTP Clock time from the same NTP source. Also make sure, Active Directory server where Kerberos is setup for data lake and for other services must also be synched with the same NTP source.

JDK 17

The cluster must be configured with JDK 17, as JDK 11 and JDK8 are not supported. You can use Oracle, OpenJDK 17.0.13, or higher. JAVA 17 is a JKS requirement and must be met. In this setup we used OpenJDK 17.0.13.

Kerberos

Kerberos must be configured using an Active Directory (AD), RedHat FreeIPA or MIT KDC. Kerberos will be enabled for all services in the cluster.

Database Requirements

Cloudera Manager and Runtime come packaged with an embedded PostgreSQL database for use in non-production environments. The embedded PostgreSQL database is not supported in production environments. For production environments, you must configure your cluster to use dedicated external databases.

For detailed information about supported database go to: <https://supportmatrix.cloudera.com/>

Configure Cloudera Manager with TLS/SSL

TLS/SSL provides privacy and data integrity between applications communicating over a network by encrypting the packets transmitted between endpoints (ports on a host, for example). Configuring TLS/SSL for any system typically involves creating a private key and public key for use by server and client processes to negotiate an encrypted connection at runtime. In addition, TLS/SSL can use certificates to verify the trustworthiness of keys presented during the negotiation to prevent spoofing and mitigate other potential security issues.

For detailed information on encrypting data in transit, go to:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/security-encrypting-data-in-transit/topics/cm-security-guide-ssl-certs.html>

The Auto-TLS feature automates all the steps required to enable TLS encryption at a cluster level. Using Auto-TLS, you can let Cloudera manage the Certificate Authority (CA) for all the certificates in the cluster or use the company's existing CA. In most cases, all the necessary steps can be enabled easily via the Cloudera Manager UI. This feature automates the following processes when Cloudera Manager is used as a Certificate Authority:

- Creates the root Certificate Authority or a Certificate Signing Request (CSR) for creating an intermediate Certificate Authority to be signed by company's existing Certificate Authority (CA)
- Generates the CSRs for hosts and signs them

Configuring TLS Encryption for Cloudera Manager Using Auto-TLS for detailed information:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/security-encrypting-data-in-transit/topics/cm-security-auto-tls.html>

Manually Configuring TLS Encryption for Cloudera Manager for detailed information:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/security-encrypting-data-in-transit/topics/cm-security-how-to-configure-cm-tls.html>

TLS uses JKS-format (Java KeyStore)

Cloudera Manager Server, Cloudera Management Service, and many other CDP services use JKS formatted key-stores and certificates. Java 11 is required for JKS.

Licensing Requirements

The cluster must be setup with a license with entitlements for installing Cloudera on premises. 60 days evaluation license for Cloudera Data Platform Cloudera on premises Base does not allow you to set up Cloudera on premises Data Services.

Refer to the [Cloudera on premises Base Requirements and Supported Versions](#) for information about hardware, operating system, and database requirements, as well as product compatibility matrices.

Refer Cloudera Manager release note for new feature and support:

<https://docs.cloudera.com/cloudera-manager/7.13.2/manager-release-notes/topics/cm-whats-new-7131.html>

Please review before install steps:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-before-you-install.html>

Please review Cloudera on premises Base requirements and supported versions for information about hardware, operating system, and database requirements, as well as product compatibility matrices:

<https://docs.cloudera.com/cdp-private-cloud-upgrade/latest/upgrade/topics/cdpdc-requirements-supported-versions.html>

Cloudera on premises Cloudera Manager Server Setup

This section outlines the steps needed to set up a 6 node Cloudera on premises Base cluster. Below are the prerequisites which base cluster should have before installing/configuring Data Services.

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-installation.html>

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-prod-installation.html>

Ensure to verify compatibility matrix of Cloudera-Manager, Cloudera RunTime, DataServices/ECS, JDK, Python, PostgreSQL etc. all together:

<https://supportmatrix.cloudera.com/>

Procedure 1. Setup Cloudera Manager Repository

Note: These steps require a Cloudera username and password to access: <https://archive.cloudera.com/p/cm7/>

Step 1: From a host connected to the Internet, download the Cloudera's repositories as shown below and transfer it to the `cldr-mngr` node. We will directly login to `cldr-mngr` and perform below steps::

```
[root@jump ~]# ssh root@cldr-mngr
[root@cldr-mngr ~]# mkdir -p /var/www/html/cloudera-repos/cloudera-manager/
[root@cldr-mngr ~]# cd /var/www/html/cloudera-repos/cloudera-manager/
```

Step 2: Download Cloudera Manager Repository:

```
#!/bin/bash

set -e

# =====
# Cloudera Manager Repository Download Script
# =====

# =====
# 1. Set Cloudera Archive Credentials
# =====

USERNAME="<paywall-username>"
PASSWORD="<paywall-password>"

# =====
# 2. Define Cloudera Manager Version & Build
# =====

CM_VERSION="7.13.2.100"
BUILD_NUMBER="81323056"

# =====
# 3. Define Base URL
# =====

BASE_REPO_URL="https://${USERNAME}:${PASSWORD}@archive.cloudera.com/p/cm7/${CM_VERSION}"
BASE_PARCEL_URL="${BASE_REPO_URL}/redhat9/yum"

# =====
# 4. Prepare Target Local Repo Path
# =====

cd /var/www/html/cloudera-repos/cloudera-manager/

# =====
```

5. Download Repo & GPG Key Files

=====

```
wget ${BASE_PARCEL_URL}/cloudera-manager.repo
wget ${BASE_PARCEL_URL}/cloudera-manager-trial.repo
wget ${BASE_PARCEL_URL}/RPM-GPG-KEY-cloudera
wget ${BASE_REPO_URL}/allkeys.asc
wget ${BASE_REPO_URL}/allkeysha256.asc
```

=====

6. Done

=====

```
echo "✔ Cloudera Manager repo and keys downloaded successfully."
```

=====

7. Verify Downloaded Files

=====

```
ls -lah
```

```
[root@cldr-mngr cloudera-manager]# pwd
/var/www/html/cloudera-repos/cloudera-manager
```

```
[root@cldr-mngr cloudera-manager]# ll
total 36
-rw-r--r--. 1 root root 2464 Apr 23 06:51 RPM-GPG-KEY-cloudera
-rw-r--r--. 1 root root 11019 Apr 23 06:51 allkeys.asc
-rw-r--r--. 1 root root 4901 Apr 23 06:51 allkeysha256.asc
-rw-r--r--. 1 root root 248 Apr 14 11:16 cloudera-manager-trial.repo
-rw-r--r--. 1 root root 501 Apr 23 06:51 cloudera-manager.repo
```

```
[root@cldr-mngr cloudera-manager]#
```

Step 3: Edit cloudera-manager.repo file baseurl and GPG key with username and password provided by Cloudera and edit URL to match repository location **(OR)** Verify, if username and password are already present, so no action needed.

Verify if username and password are already present, so no action needed.

```
[root@cldr-mngr cloudera-manager]# vi cloudera-manager.repo
[cloudera-manager]
name=Cloudera Manager 7.13.2.100
baseurl=https://archive.cloudera.com/p/cm7/7.13.2.100/redhat9/yum/
gpgkey=https://archive.cloudera.com/p/cm7/7.13.2.100/redhat9/yum/RPM-GPG-KEY-cloudera
username=<username>
password=<password>
gpgcheck=1
enabled=1
autorefresh=0
type=rpm-md
```

```
[postgresql10]
name=Postgresql 10
baseurl=https://archive.cloudera.com/postgresql10/redhat9/
gpgkey=https://archive.cloudera.com/postgresql10/redhat9/RPM-GPG-KEY-PGDG-10
enabled=1
gpgcheck=1
module_hotfixes=true
```

If not, update the cloudera-manager.repo to look like below.

```
[root@cldr-mngr cloudera-manager]# vi cloudera-manager.repo
[cloudera-manager]
name=Cloudera Manager 7.13.2.100
baseurl=https://<username>:<password>@archive.cloudera.com/p/cm7/7.13.2.100/redhat9/yum/
gpgkey=https://<username>:<password>@archive.cloudera.com/p/cm7/7.13.2.100/redhat9/yum/RPM-GPG-KEY-cloudera
gpgcheck=1
enabled=1
autorefresh=0
```



```
type=rpm-md
[root@cldr-mngr cloudera-manager]#
```

Step 4: Create directory to download cloudera manager agent, daemon, and server files

```
#!/bin/bash

set -e

# =====
# Cloudera Manager RPM Download Script
# =====

# =====
# 1. Create & Navigate to RPM Directory
# =====

mkdir -p cm${CM_VERSION}/redhat9/yum/RPMS/x86_64/
cd cm${CM_VERSION}/redhat9/yum/RPMS/x86_64/

# =====
# 2. Download Required RPMs
# =====

for pkg in agent daemons server server-db-2; do
  wget ${BASE_PARCEL_URL}/RPMS/x86_64/cloudera-manager-${pkg}-${CM_VERSION}-${BUILD_NUMBER}.el9.x86_64.rpm
done

# =====
# 3. Verify Downloaded Files
# =====

ls -alh

[root@cldr-mngr x86_64]# pwd
/var/www/html/cloudera-repos/cloudera-manager/cm7.13.2/redhat9/yum/RPMS/x86_64
[root@cldr-mngr x86_64]# ll
total 1862092
-rw-r--r--. 1 root root 116384320 Apr 23 06:52 cloudera-manager-agent-7.13.2.100-81323056.el9.x86_64.rpm
-rw-r--r--. 1 root root 1790367515 Apr 23 06:52 cloudera-manager-daemons-7.13.2.100-81323056.el9.x86_64.rpm
-rw-r--r--. 1 root root 20933 Apr 23 06:52 cloudera-manager-server-7.13.2.100-81323056.el9.x86_64.rpm
-rw-r--r--. 1 root root 20933 Apr 23 06:52 cloudera-manager-server-db-7.13.2.100-81323056.el9.x86_64.rpm
[root@cldr-mngr x86_64]#
```

Step 5: Run createrepo command to create a local repository.

```
[root@cldr-mngr ~]# createrepo --baseurl http://${hostname -i}/cloudera-repos/cloudera-manager/ /var/www/html/cloudera-repos/cloudera-manager/
```

Note: In a web browser please check and verify cloudera manager repository created by entering baseurl
<http://13.251.65.11/cloudera-repos/cloudera-manager/>

Step 6: Copy cloudera-manager.repo file to /etc/yum.repos.d/ on all nodes to enable it to find the packages that are locally hosted on the admin node.

```
[root@cldr-mngr ~]# cp /var/www/html/cloudera-repos/cloudera-manager/cloudera-manager.repo /etc/yum.repos.d/cloudera-manager.repo
```

Step 7: Edit cloudera-manager.repo. file as per the customer repository location configuration in the step above. Copy the updated repo file to the ipaserver node so it can be copied to the rest of servers using ansible.

```
[root@cldr-mngr ~]# vi /etc/yum.repos.d/cloudera-manager.repo
[cloudera-manager]
name=Cloudera Manager 7.13.2.100
baseurl=http://<ip\_of\_cldr\_mgr>/cloudera-repos/cloudera-manager/
#Update IP of Repo/cldr-mngr server
gpgcheck=0
```

```
enabled=1
[root@cldr-mngr ~]# scp -r /etc/yum.repos.d/cloudera-manager.repo root@ipaserver:/etc/yum.repos.d/cloudera-manager.repo
```

Step 8: From the ansible control node copy the repo files to /etc/yum.repos.d/ of all the nodes of the cluster:

```
[root@ipaserver ~]# ansible all -m copy -a "src=/etc/yum.repos.d/cloudera-manager.repo dest=/etc/yum.repos.d/cloudera-manager.repo"
```

Procedure 2. Set Up the Local Parcels for Cloudera on premises Base 7.3.2

From a host connected the internet, download Cloudera on premises Base 7.3.2 parcels for RHEL9 from the URL:

Step 1. Create a directory and Download CDH parcels as shown below:

```
#!/bin/bash

set -e

# =====
# CDH Parcels Download Script
# =====

# Credentials
USERNAME="<paywall-username>"
PASSWORD="<paywall-password>"

# CDH Version & Build
CDH_VERSION="7.3.2"
PARCEL_VERSION="7.3.2.100"
BUILD_NUMBER="81274879" # must match build ID from Cloudera archive

# Base URL
BASE_URL="https://${USERNAME}:${PASSWORD}@archive.cloudera.com/p/cdh7/${PARCEL_VERSION}/parcels"

# Local repo directory
TARGET_DIR="/var/www/html/cloudera-repos/cdh${CDH_VERSION}"
mkdir -p "${TARGET_DIR}"
cd "${TARGET_DIR}"

# =====
# Download parcels
# =====
echo "Downloading CDH parcels for version ${CDH_VERSION}..."

# CDH parcel + checksums
wget ${BASE_URL}/CDH-${CDH_VERSION}-1.cdh${CDH_VERSION}.p${PARCEL_VERSION##*}.${BUILD_NUMBER}-el9.parcel
wget ${BASE_URL}/CDH-${CDH_VERSION}-1.cdh${CDH_VERSION}.p${PARCEL_VERSION##*}.${BUILD_NUMBER}-el9.parcel.sha1
wget ${BASE_URL}/CDH-${CDH_VERSION}-1.cdh${CDH_VERSION}.p${PARCEL_VERSION##*}.${BUILD_NUMBER}-el9.parcel.sha256

# Key Trustee parcel + checksums
#wget -q ${BASE_URL}/KEYTRUSTEE_SERVER-${PARCEL_VERSION}-1.keytrustee${PARCEL_VERSION}.p0.${BUILD_NUMBER}-el9.parcel
#wget -q ${BASE_URL}/KEYTRUSTEE_SERVER-${PARCEL_VERSION}-1.keytrustee${PARCEL_VERSION}.p0.${BUILD_NUMBER}-el9.parcel.sha
#wget -q ${BASE_URL}/KEYTRUSTEE_SERVER-${PARCEL_VERSION}-1.keytrustee${PARCEL_VERSION}.p0.${BUILD_NUMBER}-el9.parcel.sha1
#wget -q ${BASE_URL}/KEYTRUSTEE_SERVER-${PARCEL_VERSION}-1.keytrustee${PARCEL_VERSION}.p0.${BUILD_NUMBER}-el9.parcel.sha256

# Manifest
wget ${BASE_URL}/manifest.json

# =====
# Set permissions
# =====
chmod -R ugo+rX "${TARGET_DIR}"

# =====
# Verify downloaded files
```

```
# =====
echo "Downloaded parcels:"
ls -lah "${TARGET_DIR}"

[root@cldr-mngr ~]# ll /var/www/html/cloudera-repos/cdh7/7.3.2.100/
total 14234192
-rw-r--r--. 1 root root 14575729622 Aug  6 09:15 CDH-7.3.2-1.cdh7.3.2.p100.81274879-el9.parcel
-rw-r--r--. 1 root root      40 Aug  6 09:29 CDH-7.3.2-1.cdh7.3.2.p100.81274879-el9.parcel.sha1
-rw-r--r--. 1 root root      64 Aug  6 09:25 CDH-7.3.2-1.cdh7.3.2.p100.81274879-el9.parcel.sha256
-rw-r--r--. 1 root root  73091 Aug  6 09:10 manifest.json

[root@cldr-mngr ~]#
```

Note: In a web browser please check and verify cloudera manager repository created by entering baseurl:
<http://13.251.65.11/cloudera-repos/cdh7.3.2/> (IP is of Cloudera-Manager)

Procedure 3. Set Up the Local Parcels for CDS 3.3 powered by Apache Spark

Note: This section could be skipped, as Spark3 is now bundled with the Cloudera RunTime parcel itself. (applicable for Spark 3.5 <)

Step 1. From a host connected the internet, download CDS 3.3 Powered by Apache Spark parcels for RHEL9 from the URL: <https://archive.cloudera.com/p/spark3/3.3.7191000.4/parcels/>

Note: Although Spark 2 and Spark 3 can coexist in the same Cloudera on premises Base cluster, you cannot use multiple Spark 3 versions simultaneously. All clusters managed by the same Cloudera Manager Server must use exactly the same version of CDS 3.3 Powered by Apache Spark.

Step 2. Create a directory and download CDS parcels as shown below:

```
#!/bin/bash

set -e

# =====
# CDS Parcels Download Script
# =====

# =====
# Set Variables
# =====
USERNAME="<paywall-username>"
PASSWORD="<paywall-password>"
SPARK_VERSION="3.5.7191000.0"
SPARK_UPSTREAM="3.5.4"
BUILD_NUMBER="68499982"
BASE_URL="https://${USERNAME}.${PASSWORD}@archive.cloudera.com/p/spark3/${SPARK_VERSION}/parcels"

# =====
# Create Local Repo Directory
# =====
mkdir -p /var/www/html/cloudera-repos/spark3/${SPARK_VERSION}
cd /var/www/html/cloudera-repos/spark3/${SPARK_VERSION}

# =====
# Download Spark3 Parcels and Manifest
# =====
wget "${BASE_URL}/SPARK3-${SPARK_UPSTREAM}.${SPARK_VERSION}-30-1.p0.${BUILD_NUMBER}-el9.parcel"
wget "${BASE_URL}/SPARK3-${SPARK_UPSTREAM}.${SPARK_VERSION}-30-1.p0.${BUILD_NUMBER}-el9.parcel.sha1"
wget "${BASE_URL}/manifest.json"

# =====
```

```
# Set Permissions
# =====
chmod -R ugo+rX /var/www/html/cloudera-repos/spark3/

# =====
# Verify
# =====
ls -lh /var/www/html/cloudera-repos/spark3/${SPARK_VERSION}

[root@cldr-mngr ~]# ll /var/www/html/cloudera-repos/spark3/3.5.7191000.0/
total 1999000
-rw-r--r--. 1 root root 2046842555 Aug 21 15:20 SPARK3-3.5.4.3.5.7191000.0-30-1.p0.68499982-el9.parcel
-rw-r--r--. 1 root root    41 Aug 21 15:20 SPARK3-3.5.4.3.5.7191000.0-30-1.p0.68499982-el9.parcel.sha1
-rw-r--r--. 1 root root    64 Aug 21 15:20 SPARK3-3.5.4.3.5.7191000.0-30-1.p0.68499982-el9.parcel.sha256
-rw-r--r--. 1 root root  8962 Aug 21 15:20 manifest.json

[root@cldr-mngr ~]#
```

Step 4. In a web browser please check and verify cloudera manager repository created by entering baseurl:
<http://13.251.65.11/cloudera-repos/spark3>

Procedure 4. Install and Configure Database for Cloudera Manager

Cloudera Manager uses various databases and datastores to store information about the Cloudera Manager configuration, as well as information such as the health of the system, or task progress.

Please review [Database Requirement for Cloudera on premises Base](#).

This procedure highlights the installation and configuration steps with PostgreSQL. Please review Install and Configure Databases for Cloudera on premises Base for more details:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-install-config-postgresql-for-cdp.html>

Note: If you already have a PostgreSQL database set up, you can skip to the section Configuring and Starting the PostgreSQL Server to verify that your PostgreSQL configurations meet the requirements for Cloudera Manager.

Note: We will be installing the external PostgreSQL DB server on the cldr-mngr host.

Step 1. Login on *cldr-mngr* server and Install PostgreSQL as shown in the steps below.

```
##### Install and configure POSTGRES DB on cldr-mngr server :
*** When you restart any process in future, the configuration for each of the services is redeployed using information saved in the Cloudera Manager database.
If this information is not available, your cluster cannot start or function correctly. So, you must schedule and maintain regular backups of the Cloudera Manager
database to recover the cluster in the event of the loss of this database.

##### Install the repository RPM:
[root@cldr-mngr ~]# sudo dnf install -y https://download.postgresql.org/pub/repos/yum/repopms/EL-9-x86_64/pgdg-redhat-repo-latest.noarch.rpm

##### Disable the built-in PostgreSQL module:
[root@cldr-mngr ~]# sudo dnf -qy module disable postgresql

##### Install PostgreSQL:
[root@cldr-mngr ~]# sudo dnf install -y postgresql17 postgresql17-server postgresql17-libs
[root@cldr-mngr ~]#
```

Step 2. Install the PostgreSQL JDBC driver by running the following command on ansible-controller/ipaserver node. Rename the Postgres JDBC driver .jar file to postgresql-connector-java.jar and copy it to the /usr/share/java directory. The following copy command can be used if the Postgres JDBC driver .jar file is installed from the OS repositories: *(Skip this, as we already performed it in prior steps)*

```
##### Install JDBC Connector
[root@jump ~]# wget https://jdbc.postgresql.org/download/postgresql-42.7.13.jar
```

```
[root@ipaserver ~]# mv -v postgresql-42.7.13.jar postgresql-connector-java.jar

[root@jump ~]# ansible all -m copy -a "src=postgresql-connector-java.jar dest=/usr/share/java/postgresql-connector-java.jar"

[root@jump ~]# ansible all -m shell -a "sudo chmod 644 /usr/share/java/postgresql-connector-java.jar"

[root@jump ~]# ansible all -m shell -a "sudo ls -l /usr/share/java/postgresql*.jar"
ipaserver.cldrsetup.local | CHANGED | rc=0 >>
-rw-r--r-- 1 root root 1089312 Jun  3 08:06 /usr/share/java/postgresql-connector-java.jar

##### Alternate way (Not recommended)
[root@jump ~]# ansible all -m shell -a "sudo dnf install postgresql-jdbc -y"
[root@jump ~]# ansible all -m shell -a "java -version"
```

Step 4. Make sure that the data directory, which by default is `/var/lib/pgsql/17/data/`, is on a partition that has sufficient free space.

Note: Cloudera Manager supports the use of a custom schema name for the Cloudera Manager Server database. By default, PostgreSQL only accepts connections on the loopback interface. You must reconfigure PostgreSQL to accept connections from external hosts.

Step 5. Installing the psycopg2 Python package for PostgreSQL-backed Hue. *(Skip this, as we already performed it in prior steps)*

Note: If you are installing Runtime 7 and using PostgreSQL as a backend database for Hue, then you must install the 2.9.3 version (or greater) of the psycopg2 package on all Hue hosts. The psycopg2 package is automatically installed as a dependency of Cloudera Manager Agent, but the version installed is often lower than 2.9.3

Step 6. Make sure the psycopg2 package dependencies for RHEL 9 is installed on all required hosts, by running the following commands:

```
##### Install the psycopg2-binary package as follows:
[root@jump ~]# ansible all -m shell -a "pip3 install psycopg2-binary && pip3 list | grep psyc"

ipaserver.cldrsetup.local | CHANGED | rc=0 >>
Requirement already satisfied: psycopg2-binary in /usr/local/lib64/python3.9/site-packages (2.9.12)
psycopg2-binary 2.9.12
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a
virtual environment instead: https://pip.pypa.io/warnings/venv

[root@jump ~]#
```

Step 7. Initialize the database:

```
# Initialize the DB
[root@cldr-mngr ~]# sudo /usr/pgsql-17/bin/postgresql-17-setup initdb

# Verify PG Version
[root@cldr-mngr ~]# cat /var/lib/pgsql/17/data/PG_VERSION
17

# Verify Psycpg Version
[root@cldr-mngr ~]# pip3 list |grep psycpg
psycpg2-binary 2.9.12

# data directory is very critical, if you want to cleanup postgres simply rename or remove /var/lib/pgsql/17/data directory
```

Step 8. Make sure that `LC_ALL` is set to `C.UTF-8` to enable UTF-8 CHARSET and initialize the database as follows:

```
[root@cldr-mngr ~]# echo 'LC_ALL="C.UTF-8"' >> /etc/locale.conf
```

Step 9. To enable MD5 authentication, edit `/var/lib/pgsql/17/data/pg_hba.conf` by adding the following lines, to enable connection from all outside hosts:

(Enable md5 auth to serve password authentication and TLS/SSL encryption from outside world)

```
[root@cldr-mngr ~]# vi /var/lib/pgsql/17/data/pg_hba.conf
host all all 0.0.0.0/0 md5 # Enable md5 authentication
host ranger rangeradmin 0.0.0.0/0 md5 # Allow ranger database connection from any host
hostssl all all 0.0.0.0/0 md5 # Allow SSL connection from client(s)
# replace 127.0.0.1 with host IP if PostgreSQL access from a different host is required.
# Edit section for replication privilege. HA not documented in this solution.
```

Backup the config so you can use it in case of re-setup.

```
[root@cldr-mngr ~]# cp /var/lib/pgsql/17/data/pg_hba.conf ~
```

If you have the file backed up, copy it (only applicable in case of resetup)

```
[root@cldr-mngr ~]# cp /var/lib/pgsql/17/data/pg_hba.conf /var/lib/pgsql/17/data/pg_hba.conf_orig
```

```
[root@cldr-mngr ~]# cp ~/pg_hba.conf /var/lib/pgsql/17/data/pg_hba.conf
```

Step 10. Configure settings to ensure your system performs as expected. Update these settings in the `/var/lib/pgsql/17/data/postgresql.conf` file. Settings may vary based on cluster size and resources:

```
[root@cldr-mngr ~]#
# =====
# 1. Define PostgreSQL Variables
# =====

PG_VERSION="17"
PG_DATA_DIR="/var/lib/pgsql/${PG_VERSION}/data"
PG_CONF="${PG_DATA_DIR}/postgresql.conf"

# =====
# 2. Define PostgreSQL Parameters
# =====

PG_PORT="5432"
PG_LISTEN_ADDRESSES=""
PG_MAX_CONNECTIONS="1000"
PG_SHARED_BUFFERS="1024MB"
PG_WAL_BUFFERS="16MB"
PG_MAX_WAL_SIZE="6GB"
PG_MIN_WAL_SIZE="512MB"
PG_CHECKPOINT_COMPLETION_TARGET="0.9"
PG_STANDARD_CONFORMING_STRINGS="off"
PG_JIT="off"

# =====
# 3. Verify PostgreSQL Config
# =====

cd "${PG_DATA_DIR}" || exit 1
cp "${PG_CONF}" "${PG_CONF}.bak.${date +%Y%m%d_%H%M%S}"

# =====
# 4. Update postgresql.conf
# =====

sed -i -E \
-e "s|^([[:space:]]*#?([[:space:]]*port[[:space:]]*=.*|port = ${PG_PORT})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*listen_addresses[[:space:]]*=.*|listen_addresses = '${PG_LISTEN_ADDRESSES}')|\" \
-e "s|^([[:space:]]*#?([[:space:]]*max_connections[[:space:]]*=.*|max_connections = ${PG_MAX_CONNECTIONS})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*shared_buffers[[:space:]]*=.*|shared_buffers = ${PG_SHARED_BUFFERS})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*wal_buffers[[:space:]]*=.*|wal_buffers = ${PG_WAL_BUFFERS})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*max_wal_size[[:space:]]*=.*|max_wal_size = ${PG_MAX_WAL_SIZE})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*min_wal_size[[:space:]]*=.*|min_wal_size = ${PG_MIN_WAL_SIZE})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*checkpoint_completion_target[[:space:]]*=.*|checkpoint_completion_target = ${PG_CHECKPOINT_COMPLETION_TARGET})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*standard_conforming_strings[[:space:]]*=.*|standard_conforming_strings = ${PG_STANDARD_CONFORMING_STRINGS})|\" \
-e "s|^([[:space:]]*#?([[:space:]]*jit[[:space:]]*=.*|jit = ${PG_JIT})|\" \
"${PG_CONF}"
```

```
# =====
# 5. Verify Updated Parameters
# =====

grep -E \
'^[[:space:]]*(port|listen_addresses|max_connections|shared_buffers|wal_buffers|max_wal_size|min_wal_size|checkpoint_completion_target|standard_conforming_strings|jit)[[:space:]]*=' \
"${PG_CONF}"
cd -

#### Below vi based part is already covered in above, hence commented – ignore / skip this.
# [root@cldr-mngr ~]# vi /var/lib/pgsql/17/data/postgresql.conf
# port = 5432                # (change requires restart) ##### uncomment
# listen_addresses = '*'     # what IP address(es) to listen on;
# max_connections = 1000     # (change requires restart)
# shared_buffers = 1024MB    # min 128kB
# wal_buffers = 16MB         # min 32kB, -1 sets based on shared_buffers
# max_wal_size = 6GB
# min_wal_size = 512MB
# checkpoint_completion_target = 0.9 # checkpoint target duration, 0.0 - 1.0 ##### uncomment
# standard_conforming_strings = off
# jit = off

##### Backup the config so you can use it in case of re-setup.
[root@cldr-mngr ~]# cp /var/lib/pgsql/17/data/postgresql.conf ~

##### If you have the file backed up, copy it (only applicable in case of resetup)
[root@cldr-mngr ~]# cp /var/lib/pgsql/17/data/postgresql.conf /var/lib/pgsql/17/data/postgresql.conf_orig
[root@cldr-mngr ~]# cp ~/postgresql.conf /var/lib/pgsql/17/data/postgresql.conf
```

Note: Settings vary based on cluster size and resources.

Step 11. Start the PostgreSQL Server and configure it to start at boot.

```
[root@cldr-mngr ~]# systemctl start postgresql-17.service
[root@cldr-mngr ~]# systemctl enable postgresql-17.service
[root@cldr-mngr ~]# systemctl status postgresql-17.service -l
[root@cldr-mngr ~]# netstat -ltnupa | grep LIST | grep -E '5432|postgres'
```

Step 12. Create or verify login

```
[root@cldr-mngr ~]# sudo -u postgres psql
could not change directory to "/root": Permission denied
psql (17.13)
Type "help" for help.

postgres=# ALTER USER postgres PASSWORD 'postgres';
ALTER ROLE
postgres=# \q

[root@cldr-mngr ~]# psql -h cldr-mngr.cldrsetup.local -d postgres -U postgres
Password for user postgres:
psql (17.13)
Type "help" for help.

postgres=# \q
```

Step 13. Enable TLS 1.2 for PostgreSQL database before setting up Cloudera Manager.

```
##### Verify TLS is enabled or not:
[root@cldr-mngr ~]# sudo -u postgres psql
could not change directory to "/root": Permission denied
psql (17.13)
```

Type "help" for help.

```
postgres=# SELECT * FROM pg_stat_ssl;
 pid | ssl | version | cipher | bits | client_dn | client_serial | issuer_dn
```

```
-----+-----+-----+-----+-----+-----+-----+-----
41275 | f   |         |        |      |           |              |
```

(1 row)

```
postgres=# SHOW ssl;
```

```
ssl
----
off
(1 row)
```

```
postgres=# \q
```

```
[root@cldr-mngr ~]# sudo dnf install -y mod_ssl
```

```
##### Stop Postgres DB service.
```

```
[root@cldr-mngr ~]# systemctl stop postgresql-17
```

```
[root@cldr-mngr ~]# cd /var/lib/pgsql/17/data/
```

```
##### Generate CA-signed certificates for clients to verify with openssl command line tool.
```

```
##### Update value for "-days 3650". Currently set for 3650 days = 10 years.
```

```
##### create a certificate signing request (CSR) and a public/private key file
```

```
[root@cldr-mngr data]# openssl req -new -nodes -text -out root.csr -keyout root.key -subj '/C=US/ST=California/L=Santa Clara/O=Cloudera Inc/OU=CLDR/CN=cldr-mngr.cldrsetup.local'
```

```
##### Output for above command
```

```
[root@cldr-mngr data]# ls -ltr root*
```

```
total 8
-rw----- 1 root root 1704 Jun  3 07:43 root.key
-rw-r--r-- 1 root root 3589 Jun  3 07:43 root.csr
[root@cldr-mngr data]#
```

```
[root@cldr-mngr data]# chmod 400 root.key
```

```
[root@cldr-mngr data]# ls -ltr root*
```

```
total 8
-r----- 1 root root 1704 Jun  3 07:43 root.key
-rw-r--r-- 1 root root 3589 Jun  3 07:43 root.csr
[root@cldr-mngr data]#
```

```
##### create a root certificate authority
```

```
[root@cldr-mngr data]# openssl x509 -req -in root.csr -text -days 3650 -extfile /etc/ssl/openssl.cnf -extensions v3_ca -signkey root.key -out root.crt
```

```
Certificate request self-signature ok
```

```
subject=C = US, ST = California, L = Santa Clara, O = Cloudera Inc, OU = CLDR, CN = cldr-mngr.cldrsetup.local
```

```
[root@cldr-mngr data]# ls -l
```

```
total 16
-r----- 1 root root 1704 Jun  3 07:43 root.key
-rw-r--r-- 1 root root 3589 Jun  3 07:43 root.csr
-rw-r--r-- 1 root root 4592 Jun  3 07:46 root.crt
[root@cldr-mngr data]#
```

```
# create a server certificate signed by the new root certificate authority
```

```
[root@cldr-mngr data]# openssl req -new -nodes -text -out server.csr -keyout server.key -subj "/CN=cldr-mngr.cldrsetup.local"
```

```
[root@cldr-mngr data]# ls -ltr root* server*
```

```
total 24
-r----- 1 root root 1704 Jun  3 07:43 root.key
-rw-r--r-- 1 root root 3589 Jun  3 07:43 root.csr
-rw-r--r-- 1 root root 4592 Jun  3 07:46 root.crt
-rw----- 1 root root 1704 Jun  3 07:47 server.key
```



```
-rw-r--r-- 1 root root 3388 Jun  3 07:47 server.csr
```

```
[root@cldr-mngr data]#
```

```
[root@cldr-mngr data]# chmod 400 server.key
```

```
[root@cldr-mngr data]# ls -ltr root* server*
```

```
total 24
```

```
-r----- 1 root root 1704 Jun  3 07:43 root.key
```

```
-rw-r--r-- 1 root root 3589 Jun  3 07:43 root.csr
```

```
-rw-r--r-- 1 root root 4592 Jun  3 07:46 root.crt
```

```
-r----- 1 root root 1704 Jun  3 07:47 server.key
```

```
-rw-r--r-- 1 root root 3388 Jun  3 07:47 server.csr
```

```
[root@cldr-mngr data]#
```

```
[root@cldr-mngr data]# openssl x509 -req -in server.csr -text -days 3650 -CA root.crt -CAkey root.key -CAcreateserial -out server.crt
```

```
Certificate request self-signature ok
```

```
subject=CN = cldr-mngr.cldrsetup.local
```

```
[root@cldr-mngr data]# ls -ltr root* server*
```

```
total 32
```

```
-r----- 1 root root 1704 Jun  3 07:43 root.key
```

```
-rw-r--r-- 1 root root 3589 Jun  3 07:43 root.csr
```

```
-rw-r--r-- 1 root root 4592 Jun  3 07:46 root.crt
```

```
-r----- 1 root root 1704 Jun  3 07:47 server.key
```

```
-rw-r--r-- 1 root root 3388 Jun  3 07:47 server.csr
```

```
-rw-r--r-- 1 root root  41 Jun  3 07:51 root.srl
```

```
-rw-r--r-- 1 root root 3933 Jun  3 07:51 server.crt
```

```
[root@cldr-mngr data]#
```

```
##### The above steps will create a server.crt and server.key file in that location.
```

```
[root@cldr-mngr data]# chown postgres:postgres server.crt server.key root.crt
```

```
##### Artifacts generated from above command:
```

```
[root@cldr-mngr data]# ls -l server\.* root\.*
```

```
-rw-r--r-- 1 postgres postgres 4586 Aug  9 14:34 root.crt
```

```
-rw-r--r-- 1 root root 3584 Aug  9 14:33 root.csr
```

```
-r----- 1 root root 1704 Aug  9 14:33 root.key
```

```
-rw-r--r-- 1 root root 41 Aug  9 14:37 root.srl
```

```
-rw-r--r-- 1 postgres postgres 3928 Aug  9 14:37 server.crt
```

```
-rw-r--r-- 1 root root 3388 Aug  9 14:35 server.csr
```

```
-r----- 1 postgres postgres 1704 Aug  9 14:35 server.key
```

```
[root@cldr-mngr data]#
```

```
##### Verify Key and Certs generated fine
```

```
[root@cldr-mngr data]# openssl rsa -noout -text -in server.key
```

```
Private-Key: (2048 bit, 2 primes)
```

```
modulus:
```

```
00:b3:30:86:66:49:8d:c4:de:62:c6:17:e2:50:6c:
```

```
88:91:10:49:26:6a:7f:a7:1d:6a:33:3a:71:0d:2c:
```

```
f0:08:1b:3d:88:bc:73:43:b9:82:00:1a:a3:15:0f:
```

```
08:ed:53:94:be:1e:25:7b:dd:99:66:c0:f5:2d:42:
```

```
92:f0:d6:52:67:18:80:ab:a1:86:e1:aa:5c:53:47:
```

```
41:3c:e2:2e:e1:dd:f8:5d:b7:e0:d0:39:26:f4:23:
```

```
3d:78:71:9f:75:66:a0:0e:c7:9a:bc:c2:fb:db:1b:
```

```
d1:fe:b2:2e:5d:a5:72:54:5f:04:54:1a:d8:76:77:
```

```
a8:04:9d:05:9a:f6:25:5b:ed:73:88:6b:1a:e6:0f:
```

```
09:62:d3:19:07:7c:2b:77:d0:5d:af:c3:bd:ff:44:
```

```
7f:a9:08:b9:b2:e3:8c:5a:fd:90:dd:c7:bf:db:1e:
```

```
c9:fe:72:16:e2:09:c2:0c:90:de:31:8b:06:58:e8:
```

```
6c:37:7a:a4:bf:91:7e:ca:d4:15:60:d8:6f:b7:0b:
```

```
e5:a1:5c:a2:30:98:d4:34:9c:69:88:57:f4:d1:b8:
```

```
2a:1d:a1:c6:1f:5c:1d:10:56:5a:80:b5:5d:f3:f1:
```

```
59:7f:4b:42:2c:82:3d:96:6d:5d:91:88:2a:de:12:
```

```
6b:b4:65:f3:9d:c0:b8:02:4b:a6:21:bc:3b:5c:3f:
```

```
32:3b
```

```

publicExponent: 65537 (0x10001)
privateExponent:
  12:78:80:8a:1f:af:dc:e8:bd:8e:c4:dc:7f:c4:c8:
  49:07:c0:3a:95:04:c6:91:aa:26:50:b2:61:94:cd:
  c3:50:27:86:26:42:cd:6a:dc:63:2d:5b:bd:2a:79:
  15:99:a5:7d:f9:76:8c:af:99:85:f5:82:f0:60:e9:
  eb:a8:74:03:0b:8c:0b:e5:11:15:c6:ed:50:6a:4a:
-----
-----
-----

[root@cldr-mngr data]# openssl x509 -noout -text -in server.crt
Certificate:
  Data:
    Version: 1 (0x0)
    Serial Number:
      40:26:f6:7b:84:d1:ad:30:65:2a:07:df:20:f8:4f:a3:91:0e:09:c7
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C = US, ST = California, L = Santa Clara, O = Cloudera Inc, OU = CLDR, CN = cldr-mngr.cldrsetup.local
    Validity
      Not Before: May 14 06:33:09 2024 GMT
      Not After : May 12 06:33:09 2034 GMT
    Subject: CN = cldr-mngr.cldrsetup.local
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:b3:30:86:66:49:8d:c4:de:62:c6:17:e2:50:6c:
        88:91:10:49:26:6a:7f:a7:1d:6a:33:3a:71:0d:2c:
        f0:08:1b:3d:88:bc:73:43:b9:82:00:1a:a3:15:0f:
        08:ed:53:94:be:1e:25:7b:dd:99:66:c0:f5:2d:42:
-----
-----
-----

```

```

# Correct permissions for the private key file
[root@cldr-mngr data]# chmod 644 server.crt root.crt
[root@cldr-mngr data]# chmod 0600 /var/lib/pgsql/17/data/server.key

##### Edit Configuration file for PostgreSQL (postgresql.conf) to enable SSL
[root@cldr-mngr data]# cat <<EOF >> /var/lib/pgsql/17/data/postgresql.conf
ssl = on
ssl_ca_file = 'root.crt'
ssl_cert_file = 'server.crt'
ssl_key_file = 'server.key'
EOF

```

```

# Find the location of the private key file, typically in the data directory
[root@cldr-mngr data]# ls -l /var/lib/pgsql/17/data/server.key

##### Restart PostgreSQL database service to pick up the SSL related configuration changes and verify login with SSL
[root@cldr-mngr data]# systemctl restart postgresql-17.service
[root@cldr-mngr data]# systemctl status postgresql-17.service -l

```

```

[root@cldr-mngr data]# psql -h cldr-mngr.cldrsetup.local -d postgres -U postgres
Password for user postgres: <postgres>
psql (17.13)
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

postgres=# SELECT * FROM pg_stat_ssl;
 pid | ssl | version |   cipher   | bits | client_dn | client_serial | issuer_dn
-----+-----+-----+-----+-----+-----+-----+-----
43895 | t   | TLSv1.3 | TLS_AES_256_GCM_SHA384 | 256 |          |              |
(1 row)

```

```

postgres=# SHOW ssl;
ssl
-----
on
(1 row)

postgres=#
##### Verify SSL is actually applied for DB
postgres=# SELECT name, setting FROM pg_settings WHERE name LIKE '%ssl%';
postgres=#
postgres-#
      name          |      setting
-----+-----
ssl                 | on
ssl_ca_file         | root.crt
ssl_cert_file       | server.crt
ssl_ciphers          | HIGH:MEDIUM:+3DES:!aNULL
ssl_crl_dir         |
ssl_crl_file        |
ssl_dh_params_file  |
ssl_ecdh_curve       | prime256v1
ssl_key_file        | server.key
ssl_library          | OpenSSL
ssl_max_protocol_version |
ssl_min_protocol_version | TLSv1.2
ssl_passphrase_command |
ssl_passphrase_command_supports_reload | off
ssl_prefer_server_ciphers | on
(15 rows)
postgres-# \q

##### Copy /var/lib/pgsql/17/data/root.crt to /root/.postgresql/root.crt
[root@cldr-mngr data]# mkdir -p /root/.postgresql/
[root@cldr-mngr data]# cp /var/lib/pgsql/17/data/root.crt /root/.postgresql/root.crt

##### Copy the root.crt to all other hosts with the help of ansible, for this copy the root.crt file to ipaserver/ansible control node
[root@cldr-mngr data]# scp -r /root/.postgresql/root.crt root@jump:~/

##### Login to ipaserver and copy the root.crt Postgres DB certificate file to all other nodes at location /root/.postgresql/ with the help of ansible.

[root@jump ~]# ls -l
[root@jump ~]# chmod 644 root.crt
[root@jump ~]# ansible all -m shell -a "mkdir -p /root/.postgresql/ && chmod -R 755 /root/.postgresql/"
[root@jump ~]# ansible all -m copy -a "src=root.crt dest=/root/.postgresql/root.crt"

[root@cldr-mngr data]# psql -h cldr-mngr.cldrsetup.local -p 5432 -U postgres "dbname=postgres sslmode=verify-full"
Password for user postgres:
psql (17.13)
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.
postgres=#
postgres=# \q

[root@cldr-mngr data]# psql -h cldr-mngr.cldrsetup.local -p 5432 -U postgres "dbname=postgres sslmode=verify-ca"
Password for user postgres:
psql (17.13)
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.
postgres=#
postgres=# \q
[root@cldr-mngr data]#

```

Step 14. Create databases and service accounts for components that require databases. Following components requires databases:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-required-databases.html>
<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-configuring-starting-postgresql-server.html>

Note: The databases must be configured to support the PostgreSQL UTF8 character set encoding.

Note: Record the values you enter for database names, usernames, and passwords. The Cloudera Manager installation wizard requires this information to correctly connect to these databases.

Create CM DB and USERS

```
[root@cldr-mngr data]# sudo -u postgres psql
```

```
CREATE ROLE scm LOGIN PASSWORD 'scm';
CREATE DATABASE scm OWNER scm ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE scm TO scm;
```

```
CREATE ROLE rman LOGIN PASSWORD 'rman';
CREATE DATABASE rman OWNER rman ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE rman TO rman;
```

```
CREATE ROLE hue LOGIN PASSWORD 'hue';
CREATE DATABASE hue OWNER hue ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE hue TO hue;
```

```
CREATE ROLE hive LOGIN PASSWORD 'hive';
CREATE DATABASE hive OWNER hive ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE hive TO hive;
```

```
CREATE ROLE oozie LOGIN PASSWORD 'oozie';
CREATE DATABASE oozie OWNER oozie ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE oozie TO oozie;
```

```
CREATE ROLE rangeradmin LOGIN PASSWORD 'rangeradmin';
CREATE DATABASE ranger OWNER rangeradmin ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE ranger TO rangeradmin;
```

```
/*For Ranger KMS, use rangerkms rather than rangeradmin user.*/
CREATE ROLE rangerkms LOGIN PASSWORD 'rangerkms';
CREATE DATABASE rangerkms OWNER rangerkms ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE rangerkms TO rangerkms;
```

```
CREATE ROLE schemaregistry LOGIN PASSWORD 'schemaregistry';
CREATE DATABASE schemaregistry OWNER schemaregistry ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE schemaregistry TO schemaregistry;
```

```
CREATE ROLE yqm LOGIN PASSWORD 'yqm';
CREATE DATABASE yqm OWNER yqm ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE yqm TO yqm;
```

```
CREATE ROLE nifiregistry LOGIN PASSWORD 'nifiregistry';
CREATE DATABASE nifiregistry OWNER nifiregistry ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE nifiregistry TO nifiregistry;
```

```
CREATE ROLE impala LOGIN PASSWORD 'impala';
CREATE DATABASE impala OWNER impala ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE impala TO impala;
```

```
CREATE ROLE dataviz LOGIN PASSWORD 'dataviz';
CREATE DATABASE dataviz OWNER dataviz ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE dataviz TO dataviz;
```

```
CREATE ROLE Knox LOGIN PASSWORD 'knox';
CREATE DATABASE Knox OWNER Knox ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE Knox TO Knox;
```

```

/*For the SMM metadata store, create a database called smm with the password smm:*/
CREATE ROLE smm LOGIN PASSWORD 'smm';
CREATE DATABASE smm OWNER smm ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE smm TO smm;

```

```

CREATE ROLE das LOGIN PASSWORD 'das';
CREATE DATABASE das OWNER das ENCODING 'UTF8';
GRANT ALL PRIVILEGES ON DATABASE das TO das;

```

```

ALTER DATABASE hive SET standard_conforming_strings=off;
ALTER DATABASE oozie SET standard_conforming_strings=off;
SELECT 1;
SHOW ssl;
\q

```

--- Alternate commands

```

--- CREATE USER registry WITH PASSWORD 'registry';
--- GRANT ALL PRIVILEGES ON DATABASE "registry" to registry;

```

Note: If you plan to use Apache Ranger, please visit [Configuring a PostgreSQL Database for Ranger or Ranger KMS](#) for instructions on creating and configuring the Ranger database. included above- install JDBC driver, create DB for ranger etc.

Note: If you plan to use Schema Registry or Streams Messaging Manager, please visit [Configuring the Database for Streaming Components](#) for instructions on configuring the database. included above- create smm and registry db etc.

The following procedure describes how to install Cloudera Manager and then using Cloudera Manager to install Cloudera Data Platform Cloudera on premises Base 7.3.2.

Procedure 5. Install Cloudera Manager Server (CM-UI)

Cloudera Manager, an end-to-end management application, is used to install and configure Cloudera on premises Base. During CDP Installation, Cloudera Manager's Wizard will help to install Hadoop services and any other role(s)/service(s) on all nodes using the following procedure:

- Discovery of the cluster nodes
- Configure the Cloudera parcel or package repositories
- Install Hadoop, Cloudera Manager Agent (CMA) and Impala on all the cluster nodes.
- Install the Oracle JDK or OpenJDK if it is not already installed across all the cluster nodes.
- Assign various services to nodes.
- Start the Hadoop services

Note: Please see the [JAVA requirements](#) for Cloudera on premises Base.

Step 1. Install the Cloudera Manager Server packages by running following command:

```
[root@cldr-mngr data]# dnf install -y cloudera-manager-agent cloudera-manager-daemons cloudera-manager-server
```

Recommendation: Always install agents via CM-UI only. Never install manually as it generates agent config as localhost and leads to heartbeat error. If HeartBeat error comes up, then run below command to update agent config (before start scm-server)

```
[root@cldr-mngr data]# sed -i 's/server_host=localhost/server_host=cldr-mngr.cldrsetup.local/g' /etc/cloudera-scm-agent/config.ini
```

Step 2. Enable TLS 1.2 on Cloudera Manager Server.

<https://docs.cloudera.com/cloudera-manager/7.13.2/cloudera-manager-installation/topics/cdpdc-enable-tls-12-cm-server.html>

Step 3. Import the PostgreSQL root certificate in Step 5.

Step 4. If the Database host and Cloudera Manager Server host are located on the same machine, then perform the following steps to import the PostgreSQL database root certificate, as mentioned below in Step 5:

Step 5. Go to the path where root certificates are stored. By default it is `/var/lib/pgsql/17/data/`.

Configure CDP to use SSL Enabled DB

Create a new directory in the following path by running the following command:

```
[root@cldr-mngr data]# mkdir -p /var/lib/cloudera-scm-server/.postgresql
[root@cldr-mngr data]# chmod 755 /var/lib/cloudera-scm-server/.postgresql
[root@cldr-mngr data]# cd /var/lib/cloudera-scm-server/.postgresql
```

Copy the PostgreSQL root certificate to the new directory on the Cloudera Manager server host by running the following command:

```
[root@cldr-mngr data]# cp /var/lib/pgsql/17/data/root.crt root.crt
```

Change the ownership of the root certificate by running the following command:

```
[root@cldr-mngr .postgresql]# chown cloudera-scm: root.crt
[root@cldr-mngr .postgresql]# ls -lt
total 8
-rw-r--r-- 1 cloudera-scm cloudera-scm 4639 Mar  5 16:59 root.crt
```

Include this root certificate path in the JDBC URL as follows:

```
# jdbc:postgresql://<DB HOSTNAME>:<DB-PORT>/<DB NAME>?ssl=true&sslmode=verify-ca&sslrootcert=<PATH_TO_ROOT_CERTIFICATE>
# jdbc:postgresql://cldr-mngr.cldrsetup.local:5432/scm?ssl=true&sslmode=verify-ca&sslrootcert=/var/lib/cloudera-scm-server/.postgresql/root.crt
```

Changes required for Ranger SSL

```
[root@cldr-mngr data]# cp /usr/share/java/postgresql-connector-java.jar /opt/cloudera/cm/lib/postgresql-connector.jar
[root@cldr-mngr data]# unlink /opt/cloudera/cm/lib/postgresql-42.*.jar
[root@cldr-mngr data]# ls -ltr /opt/cloudera/cm/lib/*postgres*
[root@cldr-mngr data]# chmod 755 /var/lib/cloudera-scm-server/
[root@cldr-mngr data]# ls -ltr /var/lib/cloudera-scm-server/.postgresql/*.*.crt
[root@cldr-mngr data]#
```

Step 6. Run the `scm_prepare_database.sh` script to check and generate database configuration file for cloudera-manager i.e. `db.properties` and test the database connection between cloudera-manager and database server:

```
# Run the script to configure PostgreSQL with TLS 1.2 enabled
##### sudo /opt/cloudera/cm/schema/scm_prepare_database.sh -h<DB HOSTNAME> --jdbc-url
"jdbc:postgresql://db_server_host:db_port/db_name?ssl=true&sslmode=verify-ca&sslrootcert=<PATH_TO_DB_ROOT_CERTIFICATE>" <db_type:postgresql>
<db_name> <db_role_user> <dn_user_password> --ssl
[root@cldr-mngr ~]# sudo /opt/cloudera/cm/schema/scm_prepare_database.sh -h cldr-mngr.cldrsetup.local --jdbc-url
"jdbc:postgresql://cldr-mngr.cldrsetup.local:5432/scm?ssl=true&sslmode=verify-ca&sslrootcert=/var/lib/cloudera-scm-server/.postgresql/root.crt" postgresql
scm scm scm --ssl
JAVA_HOME=/usr/lib/jvm/java-17-openjdk-17.0.14.0.7-1.el9.x86_64
Verifying that we can write to /etc/cloudera-scm-server
Creating SCM configuration file in /etc/cloudera-scm-server
Executing: /usr/lib/jvm/java-17-openjdk-17.0.14.0.7-1.el9.x86_64/bin/java -cp
/usr/share/java/mysql-connector-java.jar:/usr/share/java/oracle-connector-java.jar:/usr/share/java/postgresql-connector-java.jar:/opt/cloudera/cm/schema/./lib
/* com.cloudera.enterprise.dbutil.DbCommandExecutor /etc/cloudera-scm-server/db.properties com.cloudera.cmf.db.
[main] DbCommandExecutor INFO A JDBC URL override was specified. Using this as the URL to connect to the database and overriding all other values.
[main] DbCommandExecutor INFO Successfully connected to database.
All done, your SCM database is configured correctly!
[root@cldr-mngr ~]#
```

Step 7. Upon successful connection, the `scm_prepare_database.sh` script writes the content of `/etc/cloudera-scm-server/db.properties` file as shown below, verify the content, should look like below:

```
[root@cldr-mngr ~]# cat /etc/cloudera-scm-server/db.properties
# Auto-generated by scm_prepare_database.sh on Tue Mar 5 08:02:56 PM PST 2024
#
# For information describing how to configure the Cloudera Manager Server
# to connect to databases, see the "Cloudera Manager Installation Guide."
#
com.cloudera.cmf.db.type=postgresql
com.cloudera.cmf.db.host=cldr-mngr.cldrsetup.local
com.cloudera.cmf.db.name=scm
com.cloudera.cmf.db.user=scm
com.cloudera.cmf.db.setupType=EXTERNAL
```

```
com.cloudera.cmf.db.password=scm
com.cloudera.cmf.orm.hibernate.connection.url=jdbc:postgresql://cldr-mngr.cldrsetup.local:5432/scm?ssl=true&sslmode=verify-ca&sslrootcert=/var/lib/cloudera
-scm-server/.postgresql/root.crt
[root@cldr-mngr ~]#
```

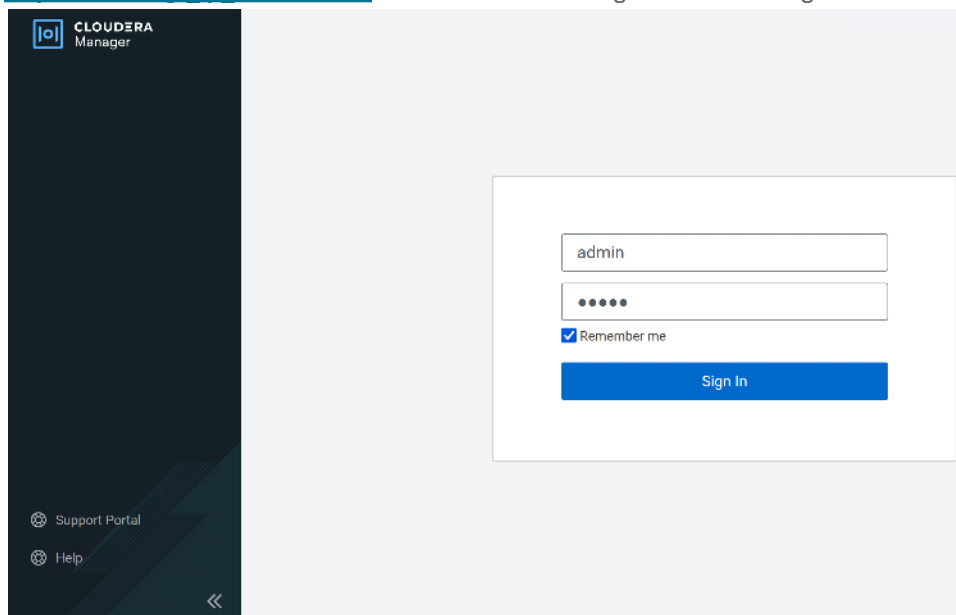
Step 8. Start the Cloudera Manager Server:

```
[root@cldr-mngr ~]# systemctl start cloudera-scm-server
[root@cldr-mngr ~]# systemctl enable cloudera-scm-server cloudera-scm-agent
[root@cldr-mngr ~]# systemctl status cloudera-scm-server -l
##### Run the below command to check the logs of cloudera-scm-server starting up. Wait until you see the Started Jetty server message on the screen.
[root@cldr-mngr ~]# sudo tail -f /var/log/cloudera-scm-server/cloudera-scm-server.log
# Start the cloudera manager agent
[root@cldr-mngr ~]# systemctl start cloudera-scm-agent
```

Step 9. The Cloudera Manager should show the below logs before the UI actually comes up.

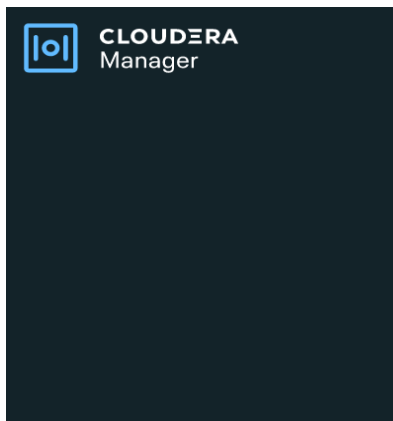
```
INFO WebServerImpl:org.eclipse.jetty.server.Server: Started @89711ms
INFO WebServerImpl:com.cloudera.server.cmf.WebServerImpl: Started Jetty server.
```

Step 10. Once the Cloudera-Manager(CM) installation is completed, open the endpoint URL http://<cldr-mngr_ip_addr>:7180/ of the Cloudera Manager WebUI and login to the CM using default credentials.



Note: The default username and password for Cloudera Manager is **admin/admin**.

Step 11. The Welcome to Cloudera Manager page appears. Since you would have received the CDP license before, select **Upload Cloudera Data Platform License** and upload the downloaded .txt or .zip file with the license information.



Welcome to Cloudera Manager 7.11.3

Upload License File

☒ Upload Cloudera Data Platform License

Cloudera Data Platform provides important features that help you manage and monitor capabilities and support. [Contact Cloudera Sales](#)

[Upload License File](#) (Accept .txt or .zip)

Step 12. Activate your license for Cloudera Data Platform by clicking the **Continue** button. Click Continue.

Step 13. The **Add Private Cloud Base Cluster** page appears. Next, we will enable AutoTLS for CM.

Step 14. As a prerequisite step to **enabling AutoTLS**, login to the cldr-mngr node as user root, and verify **cloudera-manager-agent** software is installed and running successfully. Verify the logs in below file:

```
[root@cldr-mngr ~]# tail -f /var/log/cloudera-scm-agent/cloudera-scm-agent.log
```

Verify the same by running the below command. This should return the output stating the service is active and in running state.

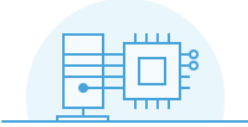
```
[root@cldr-mngr ~]# systemctl status cloudera-scm-agent -l
```

```
[root@cdpbase cloudera-scm-server]# systemctl status cloudera-scm-agent
● cloudera-scm-agent.service - Cloudera Manager Agent Service
   Loaded: loaded (/usr/lib/systemd/system/cloudera-scm-agent.service; enabled; vendor preset: disabled)
   Active: active (running) since Sat 2023-04-08 02:24:22 UTC; 9s ago
     Main PID: 3062 (cmagent)
       CGroup: /system.slice/cloudera-scm-agent.service
               └─3062 /usr/bin/python2 /opt/cloudera/cm-agent/bin/cm agent
```

Procedure 6. Enable AutoTLS

Auto-TLS is managed using the certmanager utility, which is included in the Cloudera Manager Agent software, and not the Cloudera Manager Server software. You must install the Cloudera Manager Agent software on the Cloudera Manager Server host to be able to use the utility. You can use certmanager to manage auto-TLS on a new installation. For more information, go to: [Configuring TLS Encryption for Cloudera Manager Using Auto-TLS](#)

Step 1. Click on the link *here to setup Enable AutoTLS* to set up AutoTLS through Cloudera Manager on the **Add Private Cloud Base Cluster** page.



Private Cloud Base Cluster
Add a cluster to provide storage and metadata for a compute cluster or to run workloads that benefit from data locality.
[Selected](#)

AutoTLS is currently not enabled. This means the over-the-wire communication is insecure. Click [here to setup Enable AutoTLS.](#)

A KDC is currently not configured. This means you cannot create Kerberized clusters. Kerberized clusters are required for Ranger, Atlas, and services that depend on them. Click [here to setup a KDC.](#)

Adding a cluster in Cloudera Manager consists of two steps.

1. Add a set of hosts to form a cluster and install Cloudera Runtime and the Cloudera Manager Agent software.
2. Select and configure the services to run on this cluster.

Quick Links

- [Installation Guide](#)
- [Operating System Requirements](#)
- [Database Requirements](#)
- [JDK Requirements](#)

Step 2. Below screen will appear. Enter the values for the parameters as shown below. (**We will be using the private key approach**, you can use password option as well, both options should considerably work)

Component	Value
Enable TLS for	All existing and future clusters
SSH username	root
Authentication method	All hosts accept same private key / All hosts accept same password
Private Key (If using Key approach)	Choose the private key created and downloaded in earlier section
Password (If using Password approach)	Enter VM's root users' password
Confirm Password	Enter VM's root users' password (again)

1 Generate CA

2 Remaining Steps

Generate CA

This wizard helps you enable Auto-TLS. Ensure that you have installed the Cloudera Manager Agent package on the Cloudera Manager Server host.

Note: You will need to restart The Cloudera Manager Server, the Cloudera Management service, and all clusters to complete this process.

Trusted CA Certificates Location

Enable TLS for

☒ All existing and future clusters
☐ Future clusters only

Cloudera Manager needs to distribute the certificates to all the hosts over ssh.

SSH Username

Authentication Method

☒ All hosts accept same password
☐ All hosts accept same private key

Password

Confirm Password

SSH Port

Cancel

← Back

Next →

a. Screenshot for using the Password based authentication method.

Add Private Cloud Base Cluster

Cluster Basics

Specify Hosts

Select Repository

Select JDK

5 Enter Login Credentials

6 Install Agents

7 Install Parcels

8 Inspect Cluster

Enter Login Credentials

Root access to your hosts is required to install the Cloudera packages. This installer will connect to your hosts via SSH and log in either directly as root or as another user with password-less sudo/pbrun privileges to become root.

SSH Username

Authentication Method

☐ All hosts accept same password
☒ All hosts accept same private key

Private Key

Choose File

Passphrase

Confirm Passphrase

SSH Port

Simultaneous Installations
(Running a large number of installations at once can consume large amounts of network bandwidth and other system resources)

Cancel

← Back

Continue →

b. Screenshot for using the Private Key based authentication method.

Step 3. Click **Next** to continue. Below screen will appear, if all the values are entered properly.

Remaining Steps

Now you must **restart** the Cloudera Manager server from the command line manually.

```
$ ssh my_cloudera_manager_server_host
```

```
$ systemctl restart cloudera-scm-server
```

```
$ tail -f /var/log/cloudera-scm-server/cloudera-scm-server.log
```

Wait until the cloudera-manager-server.log shows the message **Started Jetty server** and then click **Finish**
Afterwards, you must **restart** the Cloudera Management Service and finally **restart** any clusters that are stale.

Step 4. Click on **Finish**.

Step 5. After enabling the AutoTLS for the CM-UI through the browser, login to cldr-mngr node at backend as user root and restart Cloudera Manager Server, suggested in the previous screenshot.

```
[root@cldr-mngr ~]# systemctl restart cloudera-scm-server  
[root@cldr-mngr ~]# systemctl status cloudera-scm-server -l
```

Run the below command to check the logs of cloudera-scm-server starting up. Wait until you see the **Started Jetty server** message on the screen.
[root@cldr-mngr ~]# sudo tail -f /var/log/cloudera-scm-server/cloudera-scm-server.log

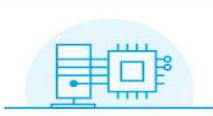
Step 6. Once you see the message **Started Jetty server** in the logs, Login to Cloudera Manager using URL endpoint, <http://<IP for CM Server>:7180>, in a new incognito window.

Step 7. The URL should get redirected to https at 7183 port i.e. https://<CM_SRVR_IP_ADDR>:7183/ This means that the AutoTLS configuration is successful. You might get a warning message on the browser related to the certificate. You can ignore the warning and visit the website as this is not a signed certificate.

Step 8. Enter the default credentials (admin/admin) and click on **Login**. You should see AutoTLS enabled as shown in the image below.

Add Cluster

Select Cluster Type



Private Cloud Base Cluster
Add a cluster to provide storage and metadata for a compute cluster or to run workloads that benefit from data locality.
Selected



Private Cloud Containerized Cluster New
Add a Private Cloud Containerized Cluster to access our latest data analytic data services on a container cloud with separated compute and storage.

✔ AutoTLS has already been enabled.

⚠ A KDC is currently not configured. This means you cannot create Kerberized clusters. Kerberized clusters are required for Ranger, Atlas, and services that depend on them. Click [here](#) to setup a KDC.

Adding a cluster in Cloudera Manager consists of two steps.

1. Add a set of hosts to form a cluster and install Cloudera Runtime and the Cloudera Manager Agent software.
2. Select and configure the services to run on this cluster.

Quick Links

- [Installation Guide](#)
- [Operating System Requirements](#)
- [Database Requirements](#)
- [JDK Requirements](#)

[← Back](#) [Continue →](#)

Procedure 7. Enable Kerberos:- Kerberos Integration with CDP

Cloudera Manager provides a wizard for integrating your organization's Kerberos with your cluster to provide authentication services. Cloudera Manager clusters can be integrated with MIT Kerberos, Red Hat Identity Management (or the upstream FreeIPA), or Microsoft Active Directory. For more information, see [Enable Kerberos Authentication for CDP](#).

Note: In our lab, we configured AD based Kerberos authentication. We presume that AD is pre-configured with user(s) and proper authentication is set up for Kerberos Authentication.

Note: Before integrating Kerberos with your cluster, configure TLS encryption between Cloudera Manager Server and all Cloudera Manager Agent host systems in the cluster. During the Kerberos integration process, Cloudera Manager Server sends keytab files to the Cloudera Manager Agent hosts, and TLS encrypts the network communication, so these files are protected.

Note: For FreeIPA, you must have administrative privileges to the ipaserver instance for initial setup and for on-going management, or you will need to have the help of your LDAP administrator prior to and during the integration process. For example, administrative access is needed to access the FreeIPA Kerberos KDC, create principals, and troubleshoot Kerberos TGT/TGS-ticket-renewal and take care of any other issues that may arise.

Note: In case, you configure **Active-Directory** based Kerberos authentication. We presume that Active Directory is pre-configured with OU, user(s) and proper authentication is setup for Kerberos Authentication. LDAP users and bind users are expected to be in the same branch/OU.

Note: For **Active Directory**, you must have administrative privileges to the Active Directory instance for initial setup and for on-going management, or you will need to have the help of your AD administrator prior to and during the integration process. For example, administrative access is needed to access the Active Directory KDC, create principals, and troubleshoot Kerberos TGT/TGS-ticket-renewal and take care of any other issues that may arise.

Step 1. Before proceeding further with KDC setup, we need to ensure that the changes to **krb5.conf** related to the default cache is not reversed. View the contents of the file **/etc/krb5.conf** after logging in to both **ipaserver** node and **cldr-mngr** node and check whether the property **default_ccache_name** is commented out. If not, then open the file and comment it out.

```
[root@ipaserver ~]# sudo vi /etc/krb5.conf
#default_ccache_name = KEYRING:persistent:%{uid}
```

```
[libdefaults]
default_realm = CDPPVCDS.COM
dns_lookup_realm = false
dns_lookup_kdc = true
rdns = false
ticket_lifetime = 24h
forwardable = true
udp_preference_limit = 0
# default_ccache_name = KEYRING:persistent:%{uid}
```

If you have made any changes, only then run the below commands to restart all the IPA services. If not, skip to the next step.

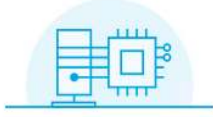
```
[root@ipaserver ~]# ipactl restart
```

Step 2. Now, move to CM-UI on your browser.

Step 3. In the Cloudera manager console click on **here to set up a KDC**, on the same TLS page, to enable the kerberos authentication on the cluster.

Add Cluster

Select Cluster Type



Private Cloud Base Cluster
Add a cluster to provide storage and metadata for a compute cluster or to run workloads that benefit from data locality.
Selected



Private Cloud Containerized Cluster New
Add a Private Cloud Containerized Cluster to access our latest data analytic data services on a container cloud with separated compute and storage.

✔ AutoTLS has already been enabled.

⚠ A KDC is currently not configured. This means you cannot create Kerberized clusters. Kerberized clusters are required for Ranger, Atlas, and services that depend on them. Click [here](#) to setup a KDC.

Adding a cluster in Cloudera Manager consists of two steps.

1. Add a set of hosts to form a cluster and install Cloudera Runtime and the Cloudera Manager Agent software.
2. Select and configure the services to run on this cluster.

Quick Links

- [Installation Guide](#)
- [Operating System Requirements](#)
- [Database Requirements](#)
- [JDK Requirements](#)

← Back Continue →

Step 4. Click Continue.

Step 5. Select **RedHat IPA** as shown below and check the box for **I have completed all the above steps**.

We have already installed the required RedHat FreeIPA/ Kerberos dependency packages openldap-clients, krb5-workstation and krb5-libs in previous steps.



4 Enter Account Credentials

5 Command Details

1. Read the [documentation](#) about enabling Kerberos.

2. Set up a working KDC (Key Distribution Center) and specify the **KDC Type**:

KDC Type

☐ MIT KDC

☐ Active Directory

☒ Red Hat IPA

[Undo](#)

3. Configure the KDC to have **non-zero ticket lifetime and renewal lifetime**. Clusters will not work properly if tickets are not renewable.

4. Configure the KDC to have an account that has **permissions to create other accounts**.

5. Install OpenLdap client libraries on the **Cloudera Manager Server host** if you want to use Active Directory.

6.

```
# RHEL / CentOS
$ yum install openldap-clients krb5-workstation krb5-libs

# if Red Hat IPA is used as the KDC
$ yum install freeipa-client

# SUSE
$ zypper install openldap2-client krb5-client

# if Red Hat IPA is used as the KDC
$ zypper install freeipa-client

# Ubuntu
$ apt-get install ldap-utils krb5-user

# if Red Hat IPA is used as the KDC
$ apt-get install freeipa-client
```

7. The Cloudera Manager principal must be authorized to add services and hosts. If the IPA server is on a host that is part of the cluster, the principal Cloudera Manager is going to use must have the permission to retrieve the keytab for the HTTP principal used by the IPA.

✔ ☒ I have completed all the above steps.

Cancel ← Back Continue →

Step 6. Select **Active Directory** as shown below, **if you're proceeding with AD based Kerberos integration** and check the box for **I have completed all the above steps**. We have already installed the required RedHat FreeIPA/ Kerberos dependency packages openldap-clients, krb5-workstation and krb5-libs in previous steps. **(Skip this step, in our case)**

Getting Started

This wizard walks you through the steps to configure Cloudera Manager for Kerberos authentication.

Before using the wizard, ensure that you have performed the following steps:

1. Read the [documentation](#) about enabling Kerberos.
2. Set up a working KDC (Key Distribution Center) and specify the **KDC Type**:

KDC Type

kdc_type

☐ MIT KDC
 ☒ Active Directory
 ☐ Red Hat IPA
 [Undo](#)

3. Configure the KDC to have **non-zero ticket lifetime and renewal lifetime**. Clusters will not work properly if tickets are not renewable.
4. Configure the KDC to have an account that has **permissions to create other accounts**.
5. Install OpenLdap client libraries on the **Cloudera Manager Server host** if you want to use Active Directory.
6.


```
# RHEL / CentOS
$ yum install openldap-clients krb5-workstation krb5-libs

# if Red Hat IPA is used as the KDC
$ yum install freeipa-client
```

```
# SUSE
$ zypper install openldap2-client krb5-client

# if Red Hat IPA is used as the KDC
$ zypper install freeipa-client
```

```
# Ubuntu
$ apt-get install ldap-utils krb5-user

# if Red Hat IPA is used as the KDC
$ apt-get install freeipa-client
```

☒ I have completed all the above steps.

Step 7. As recommended, install the following in all Cloudera Manager hosts by running the following command. Once completed, click the checkbox **"I have completed all the above steps"** and click **Continue**. **(Skip the below command execution step, as we already installed the dependencies in prior steps)**

```
[root@jump ~]# ansible all -m command -a "dnf install -y openldap-clients krb5-workstation krb5-libs"
```

Step 8. For enabling Kerberos, under the Enter KDC Information page, provide below inputs for **Enter KDC information** for this Cloudera Manager. Use [Table 6](#) as an example to fill-in the KDC setup information, provide below inputs and click Next.

Table 4. KDC Setup components and their corresponding value

Component	Value
Kerberos Encryption Types	aes256-cts-hmac-sha1-96
Kerberos Security Realm	CLDRSETUP.LOCAL

Component	Value
KDC Server Host	ipaserver.cldrsetup.local
KDC Admin Server Host	ipaserver.cldrsetup.local
Domain Name(s)	cldrsetup.local
Base DN	dc=cldrsetup,dc=local
Active Directory Suffix (Only for AD based Kerberos)	OU=cloudera,DC=cldrsetup,DC=local Credentials: (Administrator / Cloudera@123)
Active Directory Delete Accounts on Credential Regeneration (Only for AD based Kerberos)	Select (Check)

Check the picture below on where to populate the above mentioned fields:

Note: In this setup, we used Kerberos authentication with *RedHat FreeIPA*.

Step 9. On the Next Page, check the box for *Manage “krb5.conf”* to enable it through Cloudera Manager. This will install the krb5.conf file in all the hosts selected for the data lake.

Manage krb5.conf

Specify the properties needed for generating the krb5.conf file for the cluster. You can use the Advanced Configuration Snippet to specify configuration of an advanced KDC setup; for example, with cross-realm authentication.

krb5.conf file path

Requires Server Restart

krb_krb5_conf_path

Manage krb5.conf through Cloudera Manager

☒

Undo

Requires Server Restart

krb_manage_krb5_conf

Step 10. Next, enter the details as per the configuration of FreeIPA you did before. i.e., provide the domain and password of the admin user configured earlier in the FreeIPA setup. Enter account credentials for the admin which you have created. This credential will be used to generate the keytabs. In our lab setup, “admin” user is created during the IPA server installation. Click Continue. In case of **AD based Kerberos**, we used *Administrator* as the username

Enter the REALM portion of the principal in upper-case only to confirm to Kerberos convention.

Enter Account Credentials

Enter the credentials for the account that has permissions to **create** other users. Cloudera Manager will store the credentials in encrypted form and use them whenever new principals need to be generated.

Username

@

Password

Step 11. Click Finish to complete the KDC setup.

Step 12. KDC Account manager credentials should get imported successfully as shown below.

Setup KDC for this Cloudera Manager

Getting Started

Enter KDC Information

Manage krb5.conf

Enter Account Credentials

5 Command Details

Command Details

Import KDC Account Manager Credentials Command

Status **Finished**

Mar 5, 8:34:05 PM

5.01s

Successfully imported KDC Account Manager credentials.


```
##### For Red Hat IdM, make sure that all cluster hosts are joined to the IPA domain, after freeipa-client is installed.
## Kerberos client OS-specific packages must be installed on all cluster hosts and client hosts that will authenticate using Kerberos.
```

```
[root@ipaserver ~]# rpm -qa |grep ipa-client
ipa-client-common-4.13.1-3.el9_8.2.noarch
ipa-client-4.13.1-3.el9_8.2.x86_64
[root@ipaserver ~]#
```

```
##### If keytab error come up during kerberos configuration -- go to ipa server and run -
[root@ipaserver ~]# ipactl restart && ipactl status
```

Step 13. Once the KDC setup is completed, the Cloudera Manager wizard for adding a cluster will reflect the following:

Add Cluster

Select Cluster Type


Private Cloud Base Cluster
Add a cluster to provide storage and metadata for a compute cluster or to run workloads that benefit from data locality.
Selected


Private Cloud Containerized Cluster New
Add a Private Cloud Containerized Cluster to access our latest data analytic data services on a container cloud with separated compute and storage.

✓ AutoTLS has already been enabled.

✓ The KDC is already set up. You can now create Kerberized clusters.

Adding a cluster in Cloudera Manager consists of two steps.

1. Add a set of hosts to form a cluster and install Cloudera Runtime and the Cloudera Manager Agent software.
2. Select and configure the services to run on this cluster.

Quick Links

- [Installation Guide](#)
- [Operating System Requirements](#)
- [Database Requirements](#)
- [JDK Requirements](#)

[← Back](#) [Continue →](#)

Step 14. Verify Kerberos configuration. (You can run this on any server to verify)

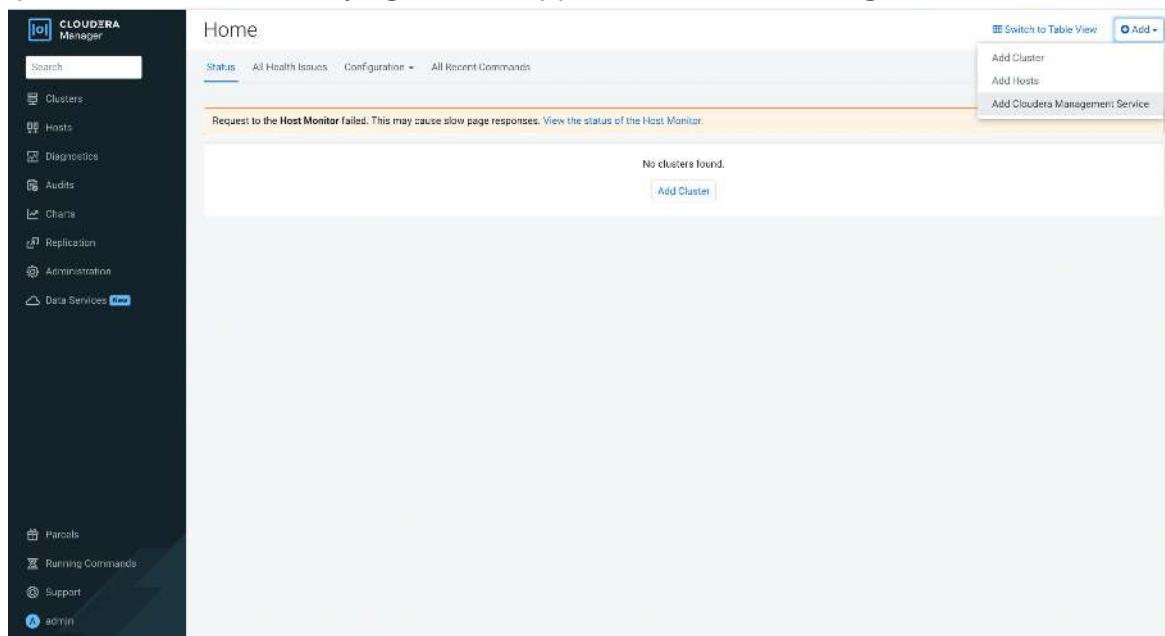
```
[root@ipaserver ~]# kinit admin@CLDRSETUP.LOCAL
Password for admin@CLDRSETUP.LOCAL: <cloudera123>
[root@ipaserver ~]# klist -e
Ticket cache: KCM:0
Default principal: admin@CLDRSETUP.LOCAL

Valid starting    Expires          Service principal
03/05/2024 20:35:11 03/06/2024 20:35:07 krbtgt/CLDRSETUP.LOCAL@CLDRSETUP.LOCAL
    renew until 03/12/2024 21:35:07
[root@ipaserver ~]#
```

Step 15. Setup the *Cloudera Management Services*: (Only if the status of hosts/services/charts are not visible / visible as (?) / or showing the errors on console)

Setup the *Cloudera Management Services* (need rman DB details), it will start service monitor and other services and enable charts view. If *Cloudera Management Services* are not installed/ enabled or not working properly, status of hosts, or installed services will not be updated on CM-UI.

a) Go to —> WebUI -> Top Right Corner -> (+)Add -> Add Cloudera Management Service



b) Assign Roles for different Cloudera Management Services to CLDR-MNGR Host (e.g. cldr-mngr.cldrsetup.local)

Add Cloudera Management Service Service

Select Dependencies

2 Assign Roles

3 Review Changes

4 Command Details

5 Summary

Assign Roles

You can customize the role assignments for your new service here, but note that if assignments are made incorrectly, such as assigning too many roles to a single host, performance will suffer.

You can also view the role assignments by host: [View By Host](#)

Service Monitor × 1 New	Host Monitor × 1 New	Reports Manager × 1 New
cldr-mngr.cdppvcds.com	cldr-mngr.cdppvcds.com	cldr-mngr.cdppvcds.com
Event Server × 1 New	Alert Publisher × 1 New	Telemetry Publisher
cldr-mngr.cdppvcds.com	cldr-mngr.cdppvcds.com	Select a host

[Cancel](#)[← Back](#)[Continue →](#)

c) Setup Report Manager Database integration by providing the **DBHostname**, **DBNAME**, **DBUser** and **DBPassword**. After entering the details, click on **Test Connection**. After the successful connection test, click **Continue**.

Add Cloudera Management Service Service

✓ Select Dependencies

✓ Assign Roles

3 Setup Database

4 Review Changes

5 Command Details

6 Summary

Setup Database

Configure and test database connections. If using custom databases, create the databases first according to the [Installing and Configuring an External Database](#) section of the [Installation Guide](#).

Reports Manager

✓ Successful

Currently assigned to run on **cldr-mngr.cdppvcds.com**.

Type	Database Hostname	Database Name
PostgreSQL	cldr-mngr.cdppvcds.com	rman
Username	Password	
rman	****	

☐ Show Password

Test Connection

Notes:

- The value in the **Database Hostname** field must match the value you used for the hostname when creating the database.
- If the database is not running on its default port, specify the port number using **host:port** in the **Database Hostname** field.
- It is highly recommended that each database is on the same host as the corresponding role instance.
- If a value in the **JDBC URL** field is provided, it will be used when establishing a connection to the database. This customized connection URL will override **Database Hostname**, **Type**, and **Database Name**. Only some services currently support this.

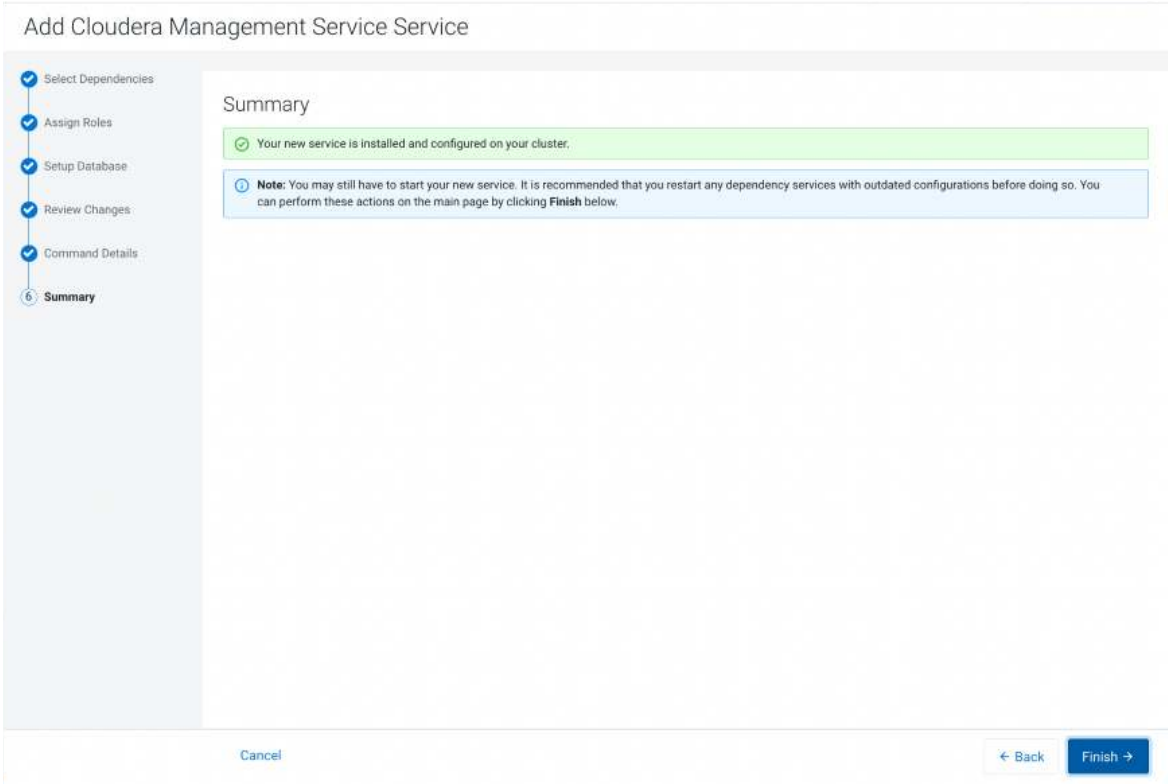
[Learn more](#)

Cancel

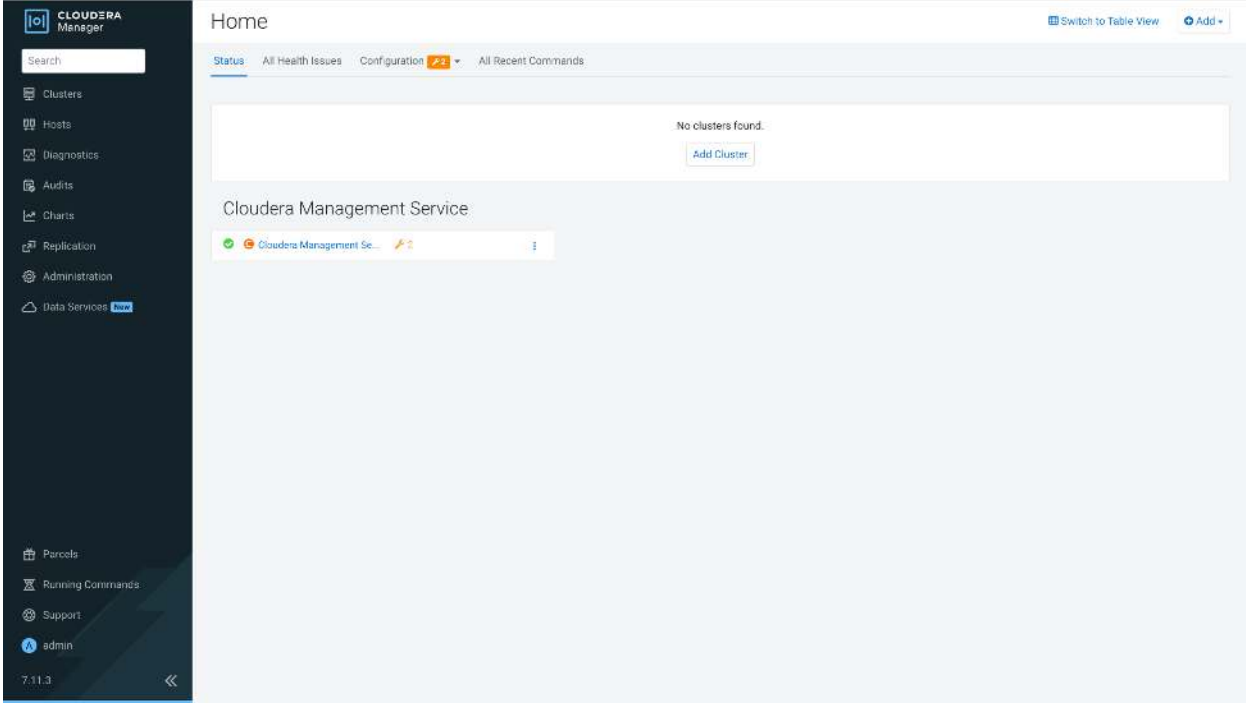
← Back

Continue →

d) **Summary page** will come up. Click on **Finish**.



e) Now, Cloudera Management Service is visible as installed and status of different components is also visible.



Configure Cloudera Manager for external authentication using LDAP (LDAP integration):

An LDAP-compliant identity/directory service, such as OpenLDAP/FreeIPA/AD, provides different options for enabling Cloudera Manager to look-up user accounts and groups in the directory:

- Use a single Distinguished Name (DN) as a base for matching usernames in the directory, or
- Search filter options let you search for a particular user based on somewhat broader search criteria – for example Cloudera Manager users could be members of different groups or organizational units (OUs), so a single pattern does not find all those users. Search filter options also let you find all the groups to which a user belongs, to help determine if that user should have login or admin access.

Note: The *LDAP Distinguished Name Pattern* property is deprecated. Leave this field empty while configuring authentication using LDAP in Cloudera Manager.

Step1. Obtain CA certificate from active directory and copy it as for example. **(Only applied, in case you are going with an AD based setup)**

```
# cp ad.cert.cer /etc/pki/ca-trust/source/anchors/ad.cert.pem
# update-ca-trust force-enable
# update-ca-trust extract
# update-ca-trust check
```

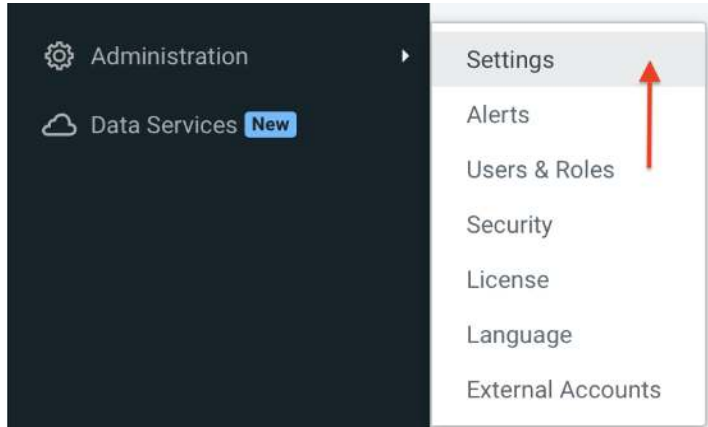
Step2. Update AutoTLS configuration by rotating Auto-TLS certificate with new CA certificate obtained. **(Only applied, in case you are going with an AD based setup).**

Rotate Auto-TLS Certificates

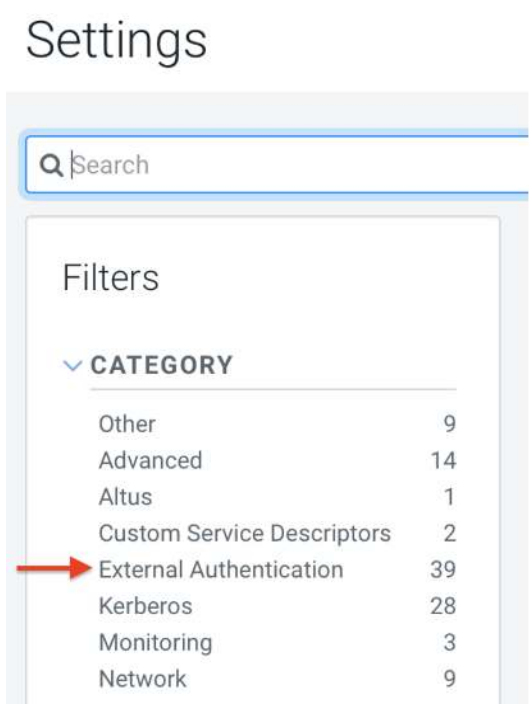
Step3. Restart cloudera server configuration and restart cluster role/services and deploy client configuration. **(Only applied, in case you are going with AD based setup)**

```
# systemctl restart cloudera-scm-server
# systemctl status cloudera-scm-server.service -l
# tail -f /var/log/cloudera-scm-server/cloudera-scm-server.log
```

Step4. From Cloudera Manager, navigate to Administration→Settings.



Step5. In the filters section, click on External Authentication as shown in the screenshot below:



Step6. Search for “ldap” and enter values for ldap authentication, please refer to the sample values as mentioned in the below table and update according to your actual setup.

Table 7. FreeIPA LDAP Integration

Component	Value
Authentication Backend Order:	Database then EXTERNAL
Authorization Backend Order:	Database and EXTERNAL

Component	Value
External Authentication Type:	LDAP
LDAP URL:	ldap://ipaserver.cldrsetup.local:389
LDAP Bind User Distinguished Name:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Bind Password:	<cloudera123> (password for KDC admin, configured earlier)
LDAP User Search filter:	(&(uid={0})(objectClass=person))
LDAP User Search Base:	cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Group Search filter:	(&(member={1})(objectClass=posixgroup))
LDAP Group Search Base:	cn=groups,cn=accounts,dc=cldrsetup,dc=local
LDAP DistName Pattern:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local

Table 8. AD LDAP Integration

Component	Value
Authentication Backend Order:	Database then EXTERNAL
Authorization Backend Order:	Database and EXTERNAL
External Authentication Type:	LDAP
LDAP URL:	ldap://ipaserver.cldrsetup.local:389
LDAP Bind User Distinguished Name:	CN=Administrator, CN=users,DC=cldrsetup,DC=local
LDAP Bind Password:	<cloudera123> (password for KDC admin, configured earlier)
Active Directory Domain	<AD DOMAIN> (e.g CLDRSETUP.LOCAL)
LDAP User Search filter	sAMAccountName={0}
LDAP User Search Base:	DC=cldrsetup,DC=local
LDAP Group Search filter	member={0}
LDAP Group Search Base:	DC=cldrsetup,DC=local

Step7. Below page will get open, select the appropriate options as mentioned below in the screenshot:



Authentication Backend Order

Requires Server Restart

auth_backend_order

☐ Database Only

☐ External then Database

☒ Database then External

☐ External Only (with emergency Administrator access)

☐ External Only (without emergency Administrator access)

Show All Descriptions

Authorization Backend Order

Requires Server Restart

authorization_backend_order

☐ Database Only

☒ Database and External

☐ External Only

External Authentication Type

Requires Server Restart

ldap_type

☐ Active Directory

☒ LDAP

☐ External Program

☐ SAML

☐ PAM

LDAP URL

Requires Server Restart

ldap_url

ldap://ipaserver.cdppvcds.com:389

LDAP Bind User Distinguished Name

Requires Server Restart

ldap_bind_dn

uid=admin,cn=users,cn=accounts,dc=cdppvcds,dc=com

LDAP Bind Password

Requires Server Restart

ldap_bind_pw

.....

Active Directory Domain

Requires Server Restart

nt_domain

LDAP User Search Filter

Requires Server Restart

ldap_user_search_filter

(&(uid={0})(objectClass=person))

Undo

LDAP User Search Base

Requires Server Restart

ldap_user_search_base

cn=users,cn=accounts,dc=cdppvcds,dc=com

Undo

LDAP Group Search Filter

Requires Server Restart

ldap_group_search_filter

(&(member={1})(objectClass=posixgroup))

Undo

LDAP Group Search Base

Requires Server Restart

ldap_group_search_base

cn=groups,cn=accounts,dc=cdppvcds,dc=com

Undo

Additional Parameters for AD Based Setup:

Active Directory Domain

△ Requires Server Restart

⚙️ nt_domain

cdip.cisco.local



LDAP User Search Filter

△ Requires Server Restart

⚙️ ldap_user_search_filter

sAMAccountName={0}



LDAP User Search Base

△ Requires Server Restart

⚙️ ldap_user_search_base

OU=cloudera,DC=cdip,DC=cisco,DC=local



LDAP Group Search Filter

△ Requires Server Restart

⚙️ ldap_group_search_filter

member={0}



LDAP Group Search Base

△ Requires Server Restart

⚙️ ldap_group_search_base

DC=cdip,DC=cisco,DC=local



LDAP Distinguished Name Pattern

△ Requires Server Restart

⚙️ ldap_dn_pattern



Allowed Groups for Knox Proxy

△ Requires Server Restart

⚙️ proxyuser_knox_groups

Active Directory LDAP Port

⚙️ ad_ldap_port

389

Active Directory LDAPS Port

⚙️ ad_ldaps_port

636

Step8.

Click **Save**. Once you click on the Save button, it will tell you to restart the CM Server, in order to bring the changes in effect. When you restart the CM Server from Backend. You will see the below entries in the logs.

```
[root@cldr-mngr ~]# systemctl restart cloudera-scm-server
[root@cldr-mngr ~]# systemctl status cloudera-scm-server -l
```

Run the below command to check the logs of cloudera-scm-server starting up. Wait until you see the **Started Jetty server** message on the screen.

```
[root@cldr-mngr ~]# sudo tail -f /var/log/cloudera-scm-server/cloudera-scm-server.log
```

```
2024-08-21 03:14:10,007 INFO WebServerImpl:com.cloudera.server.cmf.ExternalAuthenticationHelper: Using LDAP authentication with properties: DN
pattern=(uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local) user search base=(cn=users,cn=accounts,dc=cldrsetup,dc=local) user search
filter=((&(uid={0}))(objectClass=person))) group search base=(cn=groups,cn=accounts,dc=cldrsetup,dc=local) group search
filter=((&(member={1}))(objectClass=posixgroup)))
```

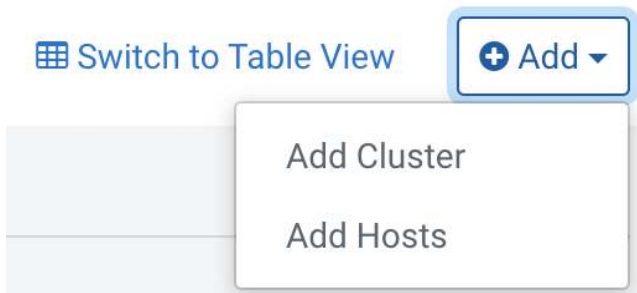
```
2024-08-21 03:14:10,009 INFO WebServerImpl:com.cloudera.server.cmf.WebServerImpl: Authenticating against database, then LDAP
```

Setup Cloudera on premises (PvC) Base Cluster

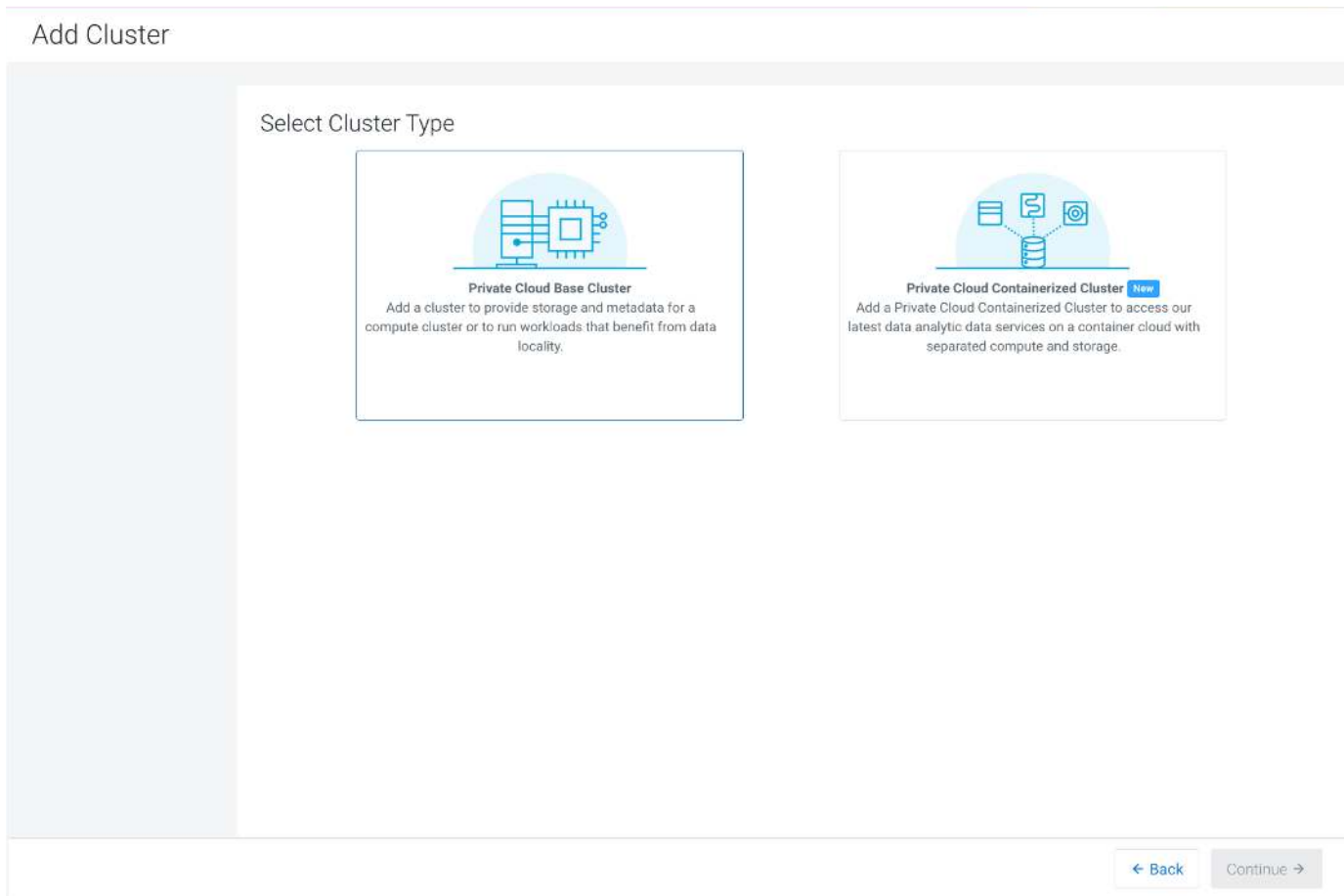
In this step, we will setup the Base cluster which will serve as the DataLake for the CDP Data Services that need the SDK capabilities for the cluster wide features like lineage, governance, security etc.,

Procedure 8. Install Cloudera on premises Base using the Cloudera Manager WebUI

Step 1. In Cloudera Manager, on the top right corner, click **(+) Add > Add Cluster**. The Select Cluster Type page appears.



Step 2. On the **Select Cluster Type** page, select the cluster type as **Cloudera on premises Base Cluster** (first option) and enter a cluster name. Click **Continue**.



Add Private Cloud Base Cluster

1 Cluster Basics

2 Specify Hosts

3 Select Repository

4 Install Parcels

5 Inspect Cluster

Cluster Basics

Cluster Name



Base Cluster

A Base Cluster contains storage nodes, compute nodes, and other services such as metadata and security collocated in a single cluster.

Cancel

← Back

Continue →

Step 3. Enter the cluster host names or IP addresses in the Hostnames field. You can Provide the host pattern pvcbase-master, pvcbase-worker[1-3] or pvcbase-worker[1-3].cldrsetup.local etc separated with a new line and Click on **Search**.

Note: Host names must be in lowercase. If you use uppercase letters in any host name, the cluster services will not start after enabling Kerberos.

Step 4. Specify the hosts that are part of the cluster using their IP addresses or hostname. The figure below shows a pattern that specifies the IP addresses range. Cloudera Manager will "discover" the hosts based on matching the pattern provided by you to add in the cluster. Verify that all desired nodes have been found and **selected for installation**. Verify host entries, **deselect** any that you do not want to install services on, and click **Continue**.

```
pvcbase-master.cldrsetup.local
pvcbase-worker[1-3].cldrsetup.local
```

Cluster Basics

Specify Hosts

Select Repository

Select JDK

Enter Login Credentials

Install Agents

Install Parcels

Inspect Cluster

Parcels

Running Commands

Support

admin

7.11.3

Add Private Cloud Base Cluster

Cluster Basics

Specify Hosts

Select Repository

Select JDK

Enter Login Credentials

Install Agents

Install Parcels

Inspect Cluster

Currently Managed Hosts (0/1 Selected)

New Hosts (6 Selected)

Hosts should be specified using the same hostname (FQDN) that they will identify themselves with.

Hostname

pvcbase-master
pvcbase-worker[1-5]

Hint: Search for hostnames or IP addresses using [pattern](#)

SSH Port

22

Search

6 hosts scanned, 6 running SSH.

	Expanded Query	Hostname (FQDN)	IP Address	Currently Managed	Result
☒	pvcbase-master	pvcbase-master.cdppvcds.com	192.168.1.34	No	Host was successfully scanned.
☒	pvcbase-worker1	pvcbase-worker1.cdppvcds.com	192.168.1.35	No	Host was successfully scanned.
☒	pvcbase-worker2	pvcbase-worker2.cdppvcds.com	192.168.1.36	No	Host was successfully scanned.
☒	pvcbase-worker3	pvcbase-worker3.cdppvcds.com	192.168.1.37	No	Host was successfully scanned.
☒	pvcbase-worker4	pvcbase-worker4.cdppvcds.com	192.168.1.30	No	Host was successfully scanned.
☒	pvcbase-worker5	pvcbase-worker5.cdppvcds.com	192.168.1.31	No	Host was successfully scanned.

Cancel

Back

Continue

Step 5. The Select Repository section appears. Select Cloudera Repository option as mentioned. Enter **Custom Repository** or Cloudera Repository to install Cloudera Manager Agent on all nodes in the cluster. We have earlier configured the private yum repository on **cldr-mngr** node. Please provide the path here:
i.e. <http://13.251.65.11/cloudera-repos/cloudera-manager/> and click on **Continue**.

Cluster Basics

Specify Hosts

Select Repository

Select JDK

Enter Login Credentials

Install Agents

Install Parcels

Select Repository

Cloudera Manager Agent

Cloudera Manager Agent 7.11.3 (#56530239) needs to be installed on all new hosts.

Repository Location

Cloudera Repository (Requires direct Internet access on all hosts.)

Custom Repository

http://192.168.1.38/cloudera-repos/cloudera-manager/

Example: http://LOCAL_SERVER/cloudera-repos/cm/7.11.3

Do not include operating system-specific paths in the URL. The path will be automatically derived.

Learn more at [How to set up a custom repository.](#)

Step 6. In the other software section, select **Use Parcels (Recommended)** and click **Parcel Repository & Network Settings** to provide a custom Parcels location to be installed (*in a new tab in the same browser window*).

Other Software

Cloudera recommends the use of parcels for installation over packages deployment and upgrade of service binaries. Electing not to use parcels will prevent you from using Cloudera Manager's rolling upgrade capability.

Install Method

Use Packages

Use Parcels (Recommended)

Parcel Repositories & Network Settings

Step 7. Enter custom repository URL for CDH7 and CDS 3.3 parcels. Click on **Save and Verify Configuration**. Close the Parcel Repository & Network Settings wizard.

i.e. <http://13.251.65.11/cloudera-repos/spark3/3.3.7191000.4/>

<http://13.251.65.11/cloudera-repos/cdh7.3.2/>

Parcel Repository & Network Settings

Cloudera Manager checks the connection to the configured parcel repository URLs. A valid license is required to access most Cloudera parcel repositories.

Last Updated: Aug 19, 6:29:26 AM EDT

> 9/9 URL(s) - The repository was successfully accessed and the manifest downloaded and validated. (HTTP Status: 200)

Remote Parcel Repository URLs

remote_parcel_repo_urls

https://archive.cloudera.com/p/cdh7/{latest_supported}/parcels/

http://192.168.1.38/cloudera-repos/spark3/3.3.7190.7

http://192.168.1.38/cloudera-repos/cdh7.1.9/

https://parcels.repos.intel.com/mkl/latest

https://archive.cloudera.com/accumulo-c5/parcels/latest/

https://archive.cloudera.com/p/accumulo6/6.1.0/parcels/

http://archive.cloudera.com/kudu/parcels/latest/

https://archive.cloudera.com/p/spark3/3.3.7180.0/parcels/

https://archive.cloudera.com/p/spark3/3.3.7190.0/parcels/

Enable Automatic Authentication for Cloudera Repositories

remote_repo_auth

☒

Close

Save & Verify Configuration

Step 8. Select the parcels for installation.

Other Software

Cloudera recommends the use of parcels for installation over packages, because parcels enable Cloudera Manager to easily manage the software on your cluster, automating the deployment and upgrade of service binaries. Electing not to use parcels will require you to manually upgrade packages on all hosts in your cluster when software updates are available, and will prevent you from using Cloudera Manager's rolling upgrade capabilities.

Install Method ☐ Use Packages

☒ Use Parcels (Recommended)

☒ Parcel Repositories & Network Settings ☐ Other Parcel Configurations

CDH Version Versions that are too new for this version of Cloudera Manager (7.11.3) will not be shown.

☒ CDH 7.1.9-1.cdh7.1.9.p1000.55406660

Additional Parcels ☐ ACCUMULO 1.9.2-1.ACCUMULO6.1.0.p0.908695

☐ ACCUMULO 1.7.2-5.5.0.ACCUMULO5.5.0.p0.8

☒ None

☒ SPARK3 3.3.2.3.3.7190.7-2-1.p0.55847114

☐ SPARK3 3.3.2.3.3.7190.0-91-1.p0.45265883

☐ SPARK3 3.3.0.3.3.7180.0-274-1.p0.31212967

☐ None

☐ mkl 2024.0.0.49671

☒ None

Cancel

← Back

Continue →

Step 9. Click **Continue**.

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Step 10. Select the appropriate option for JDK. (manual installation for JDK11 with CDH 7.1.x and JDK17 with 7.3.2 and above): (Select **Manually manage JDK** here, as we have already installed a System-provided version of OpenJDK11 manually on all servers. Click on **Continue**.

Add Private Cloud Base Cluster

Cluster Basics

Specify Hosts

Select Repository

Select JDK

Enter Login Credentials

Install Agents

Install Parcels

Inspect Cluster

Select JDK

CDH Version	Supported JDK Version
7.1.9 and above	OpenJDK 8, 11, 17 or Oracle JDK 8, 11, 17
7.1.1 to 7.1.8	OpenJDK 8, 11 or Oracle JDK 8, 11
7.0 and above	OpenJDK 8 or Oracle JDK 8
6.3 and above	OpenJDK 8 or Oracle JDK 8
6.2	OpenJDK 8 or Oracle JDK 8
6.1 or 6.0	Oracle JDK 8
5.16 and above	OpenJDK 8 or Oracle JDK 8
5.7 to 5.15	Oracle JDK 8

If you plan to use JDK 11 with CDH 7.1.x and above or JDK 17 with CDH 7.1.9 and above , you will need to install JDK 11 or 17 on all hosts below.

☒ Manually manage JDK

Please ensure that a supported JDK is **already installed** on all hosts. You will need to manage its installation.

Step 11. The **Enter Login Credentials** section appears. On this page, provide the required **details** as mentioned **in the below table**, for the hosts to install Cloudera packages. Click **Continue**.

Component	Value
SSH Username	root
Authentication Method	All hosts accept same password / All hosts accept same private key
Password (If selected password based auth)	<password_for_vm_root_user> (e.g. cloudera123)
Confirm Password (If selected password based auth)	<password_for_vm_root_user> (e.g. cloudera123) (again)
Private Key (If selected private key based auth)	Upload the private key e.g. id_rsa generated in earlier steps
Passphrase (If selected private key based auth)	If Applicable
Repeat Passphrase (If selected private key based auth)	If Applicable

Add Private Cloud Base Cluster

✓ Cluster Basics

✓ Specify Hosts

✓ Select Repository

✓ Select JDK

5 Enter Login Credentials

6 Install Agents

7 Install Parcels

8 Inspect Cluster

Enter Login Credentials

Root access to your hosts is required to install the Cloudera packages. This installer will require sudo/pbrun privileges to become root.

SSH Username ⓘ root

Authentication Method

☒ All hosts accept same password

☐ All hosts accept same private key

Password

••••••••

Confirm Password

••••••••

SSH Port

22

Simultaneous Installations

10

(Running a large number of installations at once can consume a lot of system resources.)

a. If Selected the **Authentication method** as **All hosts accept same password**

✓ Cluster Basics

✓ Specify Hosts

✓ Select Repository

✓ Select JDK

5 Enter Login Credentials

6 Install Agents

7 Install Parcels

8 Inspect Cluster

Enter Login Credentials

Root access to your hosts is required to install the Cloudera packages. This installer will require password-less sudo/pbrun privileges to become root.

SSH Username ⓘ root

Authentication Method

☐ All hosts accept same password

☒ All hosts accept same private key

Private Key

Choose File

id_rsa

Passphrase

Confirm Passphrase

SSH Port

22

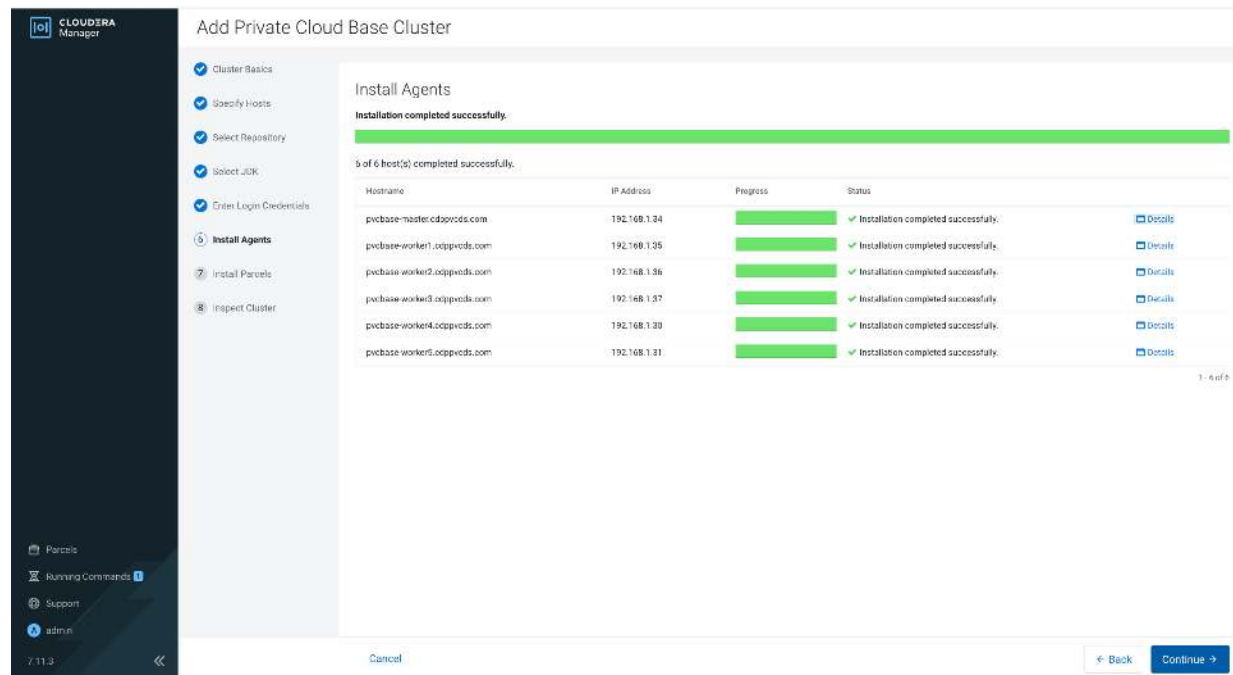
Simultaneous Installations

10

(Running a large number of installations at once can consume a lot of system resources.)

b. If Selected the **Authentication method** as **All hosts accept the same private key**.

Step 12. The **Install Agents** section appears showing the progress of the installation. It will check for and install the **JDK** (If not already there) and **cloudera-scm-agent** software on all the Base Cluster nodes. Click **Continue** after the Cloudera Agent **Installation completed successfully** on all hosts.



Step 13. Stop at this stage. Create a directory on base cluster nodes (for handling of a bug associated with p1000)

```
[root@jump ~]# ansible namenodes,datanodes -m shell -a "mkdir /var/lib/hadoop-hdfs"
[root@jump ~]# ansible namenodes,datanodes -m shell -a "ls -lart /var/lib/hadoop-hdfs"
```

Step 14. After the agents are installed, the **Install Parcels** section appears showing the progress of the parcels distribution, activation and installation on all hosts part of the cluster creation. Once the parcels are installed successfully for all hosts, click on **Continue**.

Add Private Cloud Base Cluster

Cluster Basics

Specify Hosts

Select Repository

Select JDK

Enter Login Credentials

Install Agents

7 Install Parcels

8 Inspect Cluster

Install Parcels

The selected parcels are being downloaded and installed on all the hosts in the cluster.

Cloudera Runtime 7.1.9

Downloaded: 100%

Distributed: 6/6 (223.7 MiB/s)

Unpacked: 6/6

Activated: 6/6

☒ All (6)
 ☐ Running (0)
 ☐ Failed (0)
 ☐ Completed (6)

Hostname	Throughput	Status	Errors
pvcbase-worker3.cdpprvcds.com	42.7 MiB/s	NONE	
pvcbase-worker4.cdpprvcds.com	38 MiB/s	NONE	
pvcbase-worker5.cdpprvcds.com	42.3 MiB/s	NONE	
pvcbase-master.cdpprvcds.com	42.8 MiB/s	NONE	
pvcbase-worker2.cdpprvcds.com	37.3 MiB/s	NONE	
pvcbase-worker1.cdpprvcds.com	42.7 MiB/s	NONE	

1 - 6 of 6

SPARK3 3.3.2.3.3.7190.7

Downloaded: 100%

Distributed: 6/6 (232.2 MiB/s)

Unpacked: 6/6

Activated: 6/6

☐ All (6)
 ☐ Running (0)
 ☐ Failed (0)
 ☐ Completed (6)

Hostname	Throughput	Status	Errors
pvcbase-worker3.cdpprvcds.com	42 MiB/s	DISTRIBUTING	
pvcbase-worker4.cdpprvcds.com	38.8 MiB/s	DISTRIBUTING	
pvcbase-worker5.cdpprvcds.com	46 MiB/s	DISTRIBUTING	
pvcbase-master.cdpprvcds.com	48.4 MiB/s	DISTRIBUTING	
pvcbase-worker2.cdpprvcds.com	38.7 MiB/s	DISTRIBUTING	
pvcbase-worker1.cdpprvcds.com	42 MiB/s	DISTRIBUTING	

1 - 6 of 6

Cancel

← Back

Continue →

Step 15. Stop at this stage. On base cluster nodes (for handling of a bug associated with p1000). Once Parcels are activated, follow below steps: **(Failed to execute command Install YARN MapReduce Framework JARs on service YARN)**

```
[root@jump ~]# ansible namenodes,datanodes -m shell -a "chown hdfs:hadoop /var/lib/hadoop-hdfs"
[root@jump ~]# ansible namenodes,datanodes -m shell -a "ls -lart /var/lib/hadoop-hdfs"
```

Verify if links are good for the CM version for hadoop-hdfs filesystem JAR.

```
[root@jump ~]# ansible namenodes,datanodes -m shell -a "namei -om /var/lib/hadoop-hdfs/ozone-file-system-hadoop3.jar"
```

If jar not found on the host, you can download using the link below:

```
wget
```

```
https://repository.cloudera.com/repository/cloudera-repos/org/apache/hadoop/hadoop-ozone-file-system-hadoop3/1.1.0.7.2.15.0-147/hadoop-ozone-file-system-hadoop3-1.1.0.7.2.15.0-147.jar
```

If links are not working fine, deactivate and activate the parcel from CM Parcel Manager. Then Perform Below steps:

```
[root@cldr-mngr ~]# ansible namenodes,datanodes -m shell -a "unlink /etc/alternatives/ozone-file-system-hadoop3.jar; unlink /var/lib/hadoop-hdfs/ozone-file-system-hadoop3.jar"
```

[Notes]

Here are the notes from product/engineering team on the issue:

Verify if /var/lib/hadoop-hdfs exists on all nodes.

Check if hdfs user exists on all nodes. (grep hdfs /etc/passwd; grep hdfs /etc/group)

Create the /var/lib/hadoop-hdfs directory on all nodes.

Deactivate and activate the Cloudera Runtime parcel.

Alternatively, if there is a separate Ozone parcel installed on the cluster, deactivate and activate the Ozone parcel instead.

Ensure that the directory has at least the following permissions:

```
/var/lib/hadoop-hdfs/ozone-file-system-hadoop3.jar
```

```
lrwxrwxrwx [owner: root, group: root]
```

```
/var/lib/hadoop-hdfs
```

```
drwxr-xr-x [owner: hdfs, group: hadoop]
```

Output for permissions should look similar to:

```
$ namei -om /var/lib/hadoop-hdfs/ozone-filessystem-hadoop3.jar
```

```
f: /var/lib/hadoop-hdfs/ozone-filessystem-hadoop3.jar
```

```
drwxr-xr-x root    root    /
```

```
drwxr-xr-x root    root    var
```

```
drwxr-xr-x root    root    lib
```

```
drwxr-xr-x hdfs    hadoop   hadoop-hdfs
```

```
lrwxrwxrwx root    root     ozone-filessystem-hadoop3.jar -> /etc/alternatives/ozone-filessystem-hadoop3.jar
```

The issue can also be found in the Ozone Known issues documentation.

Create the alternatives link manually for CDP binaries.

This step would have failed during the activation of the CDP parcel due to the missing folder.

```
$ alternatives --install /var/lib/hadoop-hdfs/ozone-filessystem-hadoop3.jar ozone-filessystem-hadoop3.jar
```

```
/opt/cloudera/parcels/CDH-<path_to_active_parcel>/lib/hadoop-ozone/ozone-filessystem-hadoop3.jar 5
```

If an Ozone parcel is installed, create the symbolic links for the Ozone parcel.

```
$ alternatives --install /var/lib/hadoop-hdfs/ozone-filessystem-hadoop3.jar ozone-filessystem-hadoop3.jar
```

```
/opt/cloudera/parcels/OZONE-<path_to_activated Ozone_parcel>/lib/hadoop-ozone/ozone-filessystem-hadoop3.jar 10
```

Upgrade:

Upgrade to a fixed release of CDP/Cloudera Manager once available.

For the latest update on this issue see the corresponding Knowledge Article:

TSB 2024-775: FileNotFoundException for the Ozone FS JAR during or after installation or upgrade

Step 16. After the parcels are installed the Inspect Cluster section appears.

Inspect Cluster by running **Inspect Network Performance**. After the network inspector completes, click **Show Inspector Results** to view the results in a new tab. Address any reported issues (If there), and click **Run Again**.

Click Inspect Hosts. After the host inspector completes, click **Show Inspector Results** to view the results in a new tab. Address any reported issues (If there), and click **Run Again**.

Both the inspection tests should run successfully. Review inspector summary. Click **Finish**.

Add Private Cloud Base Cluster

Cluster Basics

Specify Hosts

Select Repository

Select JDK

Enter Login Credentials

Install Agents

Install Parcels

8 Inspect Cluster

Inspect Cluster

You have created a new empty cluster. Cloudera recommends that you run the following inspections. For accurate measurements, Cloudera recommends that they are performed sequentially.

Inspect Network Performance

> Advanced Options

Status **Finished** Last Run a few seconds ago Duration 14.88s [Show Inspector Results](#) [Run Again](#) [More](#)

Host Inspector

No issues were detected, review the inspector results to see what checks were performed.

Status **Finished** Last Run a few seconds ago Duration 14.21s [Show Inspector Results](#) [Run Again](#) [More](#)

☐ I understand the risks of not running the inspections or the detected issues, let me continue with cluster setup.

[Cancel](#) [Back](#) [Finish](#)

If java path mismatch error occurs in network/host inspection:

Take - Java Path from the host (eg. `/usr/lib/jvm/java-17-openjdk-17.0.13.0.11-4.el9.x86_64/`), goto CM, open a new tab of CM-UI on browser and search for JAVA PATH in search bar present at left hand side:

Override JAVA PATH in CM-UI> Search JAVA PATH> Override the value> Save Changes> Restart Cloudera-SCM-Server
Re-run the inspect tools, this time all checks should be green.

Step 17. Click *Continue*.

Cloudera on premises Base Cluster (Data Lake) Creation

Step 18. After the Cloudera on premises Base Cluster (runtime) setup is complete, if successful, it will automatically move to add services (**Add Cluster -Configuration**) wizard. It will ask to **Select Services** i.e. (a) Data Engineering, (b) Data Warehouse/ Data Mart, © Operational Database specific or (d) Custom control plane/base cluster services i.e. HDFS, YARN, Hive, Tez, HiveOnTez, Ozone, Zookeeper, Kafka, SOLR, Ranger, Atlas, HBase, Phoenix, Impala, HUE, Spark2, Spark3, YARN Queue Manager etc. Choose from a combination of services defined or select custom services. Services required based on selection will be automatically added.

Step 19. Select **Custom Services** option to install.

Add Cluster - Configuration

1

Select Services

2

Assign Roles

3

Setup Database

4

Enter Required Parameters

5

Review Changes

6

Configure Kerberos

7

Command Details

8

Command Details

9

Summary

Select Services

Choose a combination of services to install.

☐ Data Engineering

Process, develop, and serve predictive models.
Services: HDFS, YARN, YARN Queue Manager, Ranger, Atlas, Hive, Hive on Tez, !

☐ Data Mart

Browse, query, and explore your data in an interactive way.
Services: HDFS, Ranger, Atlas, Hive, Hive on Tez, Impala, and Hue

☐ Operational Database

Real-time insights for modern data-driven business.
Services: HDFS, Ranger, Atlas, and HBase

☒ Custom Services

Choose your own services. Services required by chosen services will automatic

Note: It is important to select host(s) to deploy services based on the role intended for it. For detailed information, go to: [Runtime Cluster Hosts and Role Assignments](#)

Step 20. Under the **Custom Services**, select the below custom Cloudera DataLake/Control Plane management services to be installed on the cluster. The Selection would look similar to below, Select services and Click **Continue**.

Custom Services

Choose your own services. Services required by chosen services will automatically be included.

Service Type	Description
☒  Atlas	Apache Atlas provides a set of metadata management and governance services that enable you to find, organize, and manage data assets. This service requires Kerberos.
☐  Cruise Control	Cruise Control simplifies the operation of Kafka clusters automating workload rebalancing and self-healing.
☒  HBase	Apache HBase is a highly scalable, highly resilient NoSQL OLTP database that enables applications to leverage big data.
☒  HDFS	Apache Hadoop Distributed File System (HDFS) is the primary storage system used by Hadoop applications. HDFS creates multiple replicas of data blocks and distributes them on compute hosts throughout a cluster to enable reliable, extremely rapid computations.
☒  Hive	Apache Hive is a SQL based data warehouse system. In CDH 6 and earlier, this service includes Hive Metastore and HiveServer2. In Cloudera Runtime 7.0 and later, this service only includes the Hive Metastore; HiveServer2 and other components of the Hive execution engines are part of the Hive on Tez service.
☒  Hive on Tez	Hive on Tez is a SQL query engine using Apache Tez.
☒  Hue	Hue is the leading SQL Workbench for optimized, interactive query design and data exploration.
☒  Iceberg Replication	Iceberg Replication facilitates the replication of Iceberg tables across clusters.
☒  Impala	Apache Impala provides a real-time SQL query interface for data stored in HDFS and HBase. Impala requires the Hive service and shares the Hive Metastore with Hue.
☒  Kafka	Apache Kafka is publish-subscribe messaging rethought as a highly scalable distributed commit log.
☐  Key-Value Store Indexer	Key-Value Store Indexer listens for changes in data inside tables contained in HBase and indexes them using Solr.
☒  Knox	The Apache Knox Gateway is an Application Gateway for interacting with the REST APIs and UIs of Apache Hadoop deployments. This service requires Kerberos.
☐  Kudu	Apache Kudu is a data store that enables real-time analytics on fast changing data.
☐  Livy	Apache Livy is a REST service for deploying Spark applications.
☐  Livy for Spark 3	Apache Livy for Spark 3 is a REST service used for deploying Spark3 applications Before adding this service, ensure that you have installed the Spark3 binaries, which are not included in CDH.
☒  Ozone	Apache Ozone is a Scalable, S3 Compatible, Distributed object store for Big Data.
☒  Phoenix	Apache Phoenix is a scale-out relational database that supports OLTP workloads and provides secondary indexes, materialized views, star schema support, and common HBase optimizations. Phoenix uses Apache HBase as the underlying data store.
☐  Query Processor	Query Processor indexes Hive & Tez events and provides APIs to access them



☒	Ranger	Apache Ranger is a framework to enable, monitor and manage comprehensive data security across the Hadoop platform. This service requires Kerberos.
☐	SQOOP_CLIENT	Apache Sqoop is a CLI-based tool for efficient and reliable bulk transfers of data between relational databases and HDFS, or cloud object stores including Amazon S3 and Microsoft ADLS.
☐	Schema Registry	Schema Registry is a shared repository of schemas that allows applications to flexibly interact with each other. A common Schema Registry provides end-to-end data governance and introduces operational efficiency by saving and retrieving reusable schema, defining relationships between schemas and enabling data providers and consumers to evolve at different speeds.
☒	Solr	Apache Solr is a highly scalable, distributed service for indexing and relevance-based exploring of all forms of data.
☒	Spark	Apache Spark is an open source cluster computing system. This service runs Spark as an application on YARN.
☒	Spark 3	Apache Spark is an open source cluster computing system. This service runs Spark 3 as an application on YARN. Before adding this service, ensure that you have installed the Spark3 binaries, which are not included in CDH.
☐	Streams Messaging Manager	Streams Messaging Manager (SMM) is an operations monitoring and management tool that provides end-to-end visibility in an enterprise Apache Kafka environment.
☐	Streams Replication Manager	Streams Replication Manager (SRM) is an enterprise-grade replication solution that enables fault tolerant, scalable, and robust cross-cluster Kafka topic replication.
☐	Stub DFS	Stub DFS is a storage-less service for clusters where services have a mandatory DFS dependency.
☒	Tez	Apache Tez is the next generation Hadoop Query Processing framework written on top of YARN.
☒	YARN	Apache Hadoop MapReduce 2.0 (MRv2), or YARN, is a data computation framework that supports MapReduce applications (requires HDFS).
☐	YARN Queue Manager	YARN Queue Manager is the queue management user interface for Apache Hadoop YARN Capacity Scheduler.
☐	Zeppelin	Apache Zeppelin is a web-based notebook that enables data-driven, interactive data analytics and collaborative documents with SQL, Scala and more.
☒	ZooKeeper	Apache ZooKeeper is a centralized service for maintaining and synchronizing configuration data.

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This wizard will also install the **Cloudera Management Service**. These are a set of components that enable monitoring, reporting, events, and alerts; these components require databases to store information, which will be configured on the next page.

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Step 21. Select host assignment for different services in the **Add cluster - configuration** wizard. You need to assign hosts to different roles across all the selected services. Use the below table as a reference to assign the roles.

Table: Cloudera Data Platform Cloudera on premises Base host and Role assignment example

Service Name (Role Instance)	Host
HDFS	NameNode : pvcbase-master SecondaryNameNode : pvcbase-master Balancer : pvcbase-master DataNode : pvcbase-worker[1-3]
YARN	ResourceManager : pvcbase-master NodeManager : pvcbase-worker[1-3] (Same as DataNode) JobHistoryServer : pvcbase-master
Core Configuration	Gateway : pvcbase-master
Iceberg Replication	Gateway : pvcbase-master

Service Name (Role Instance)	Host
Knox	Gateway : pvcbase-master
Cloudera Management Service This is already configured in previous steps	Service Monitor : cldr-mngr Host Monitor : cldr-mngr Reports Manager : cldr-mngr Event Server : cldr-mngr Alert Publisher : cldr-mngr
Spark2 (i.e. SparkOnYARN)/ Spark3	Spark History Server : pvcbase-master Spark Gateway : pvcbase-worker[1-3]
Hive	Hive Metastore Server (HMS) : pvcbase-master Gateway : pvcbase-worker[1-3]
Tez	Gateway : pvcbase-worker[1-3]
Hive on Tez	HiveServer2 : pvcbase-master Gateway : pvcbase-worker[1-3]
Impala	Impala Catalog Server : pvcbase-master Impala State Store : pvcbase-master Impala Daemon : pvcbase-worker[1-3] (Same as DataNode)
HUE	HUE Server : pvcbase-master LoadBalancer : pvcbase-master
HBase	HBase Master : pvcbase-master RegionServer : pvcbase-worker[1-3] (Same as DataNode)
Phoenix	Query Server : pvcbase-master
Ozone	Storage Container Manager : pvcbase-master Ozone Manager : pvcbase-master Ozone Recon : pvcbase-master S3 Gateway : pvcbase-master Gateway : pvcbase-worker[1-3] OzoneDataNode : pvcbase-worker[1-3] (Same as DataNode)
CDP-INFRA-SOLR	Solr Server : pvcbase-master (can be installed on all hosts if needed if there is a search use case)
Kafka	Kafka Broker : base-master, pvcbaseworker[1-5] (Same as DataNode)
ZooKeeper (must be >3)	Zookeeper Server : pvcbase-master, pvcbase-worker[1-2]
Ranger	Ranger Admin : pvcbase-master UserSync : pvcbase-master Ranger Tagsync : pvcbase-master
Atlas	Atlas Server : pvcbase-master
Oozie Server (Optional)	pvcbase-master

Service Name (Role Instance)	Host
YARN Queue Manager (Optional)	

Step 22. Assign roles as updated above and shown as below.

Assign Roles

You can customize the role assignments for your new cluster here, but if assignments are made incorrectly, such as assigning too many roles to a single host, this can impact the performance of your services. Cloudera does not recommend altering assignments unless you have specific requirements, such as having pre-selected a specific host for a specific role.

You can also view the role assignments by host. [View By Host](#)

Kafka

MirrorMaker

Select hosts

Kafka Connect

Select hosts

KRaft Controller

Select hosts

Gateway

Select hosts

Kafka Broker × 5 New

Same as DataNode

Atlas

Atlas Server × 1 New

pvcbase-master.cdppvcds.com

Gateway

Select hosts

Core Configuration

Gateway × 1 New

pvcbase-master.cdppvcds.com

HBase

Master × 1 New

pvcbase-master.cdppvcds.com

RegionServer × 5 New

Same as DataNode

REST Server

Select hosts

Thrift Server

Select hosts

HDFS

NameNode × 1 New

pvcbase-master.cdppvcds.com

HttpFS

Select hosts

SecondaryNameNode × 1 New

pvcbase-master.cdppvcds.com

NFS Gateway

Select hosts

Balancer × 1 New

pvcbase-master.cdppvcds.com

DataNode × 5 New

pvcbase-worker[1-5].cdppvcds.com

Hive

Gateway × 5 New

pvcbase-worker[1-5].cdppvcds.com

HiveServer2

Select hosts

Metastore Server × 1 New

pvcbase-master.cdppvcds.com

WebHCat Server

Select hosts

Hive on Tez

Gateway

Select hosts

HiveServer2 × 1 New

pvcbase-master.cdppvcds.com

Hue

Hue Server × 1 New

pvcbase-master.cdppvcds.com

Load Balancer × 1 New

pvcbase-master.cdppvcds.com

Iceberg Replication

Admin Server × 1 New

pvcbase-master.cdppvcds.com

Impala

StateStore × 1 New

pvcbase-master.cdppvcds.com

Catalog Server × 1 New

pvcbase-master.cdppvcds.com

Impala Daemon × 5 New

Same as DataNode

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Ozone

Storage Container Manager × 1 New

pvcbase-master.cdppvcds.com ▾

Ozone Manager × 1 New

pvcbase-master.cdppvcds.com ▾

Ozone Recon × 1 New

pvcbase-master.cdppvcds.com ▾

S3 Gateway × 1 New

pvcbase-worker4.cdppvcds.com

HTTPFS Gateway

Select hosts

Prometheus

Select a host

Gateway × 5 New

pvcbase-worker[1-5].cdppvcds.com ▾

Ozone DataNode × 5 New

Same as DataNode ▾

Phoenix

Query Server × 1 New

pvcbase-master.cdppvcds.com ▾

Ranger

Ranger Admin × 1 New

pvcbase-master.cdppvcds.com ▾

Usersync × 1 New

pvcbase-master.cdppvcds.com ▾

Ranger Tagsync × 1 New

pvcbase-master.cdppvcds.com ▾

Solr

Solr Server × 1 New

pvcbase-master.cdppvcds.com

Spark 3

History Server × 1 New

pvcbase-master.cdppvcds.com ▾

Gateway

Select hosts

Spark

History Server × 1 New

pvcbase-master.cdppvcds.com

Gateway × 5 New

pvcbase-worker[1-5].cdppvcds.com ▾

←

→

↶

↷

Ranger Admin × 1 New

pvcbase-master.cdppvcds.com ▾

Usersync × 1 New

pvcbase-master.cdppvcds.com ▾

Ranger Tagsync × 1 New

pvcbase-master.cdppvcds.com ▾

Solr

Solr Server × 1 New

pvcbase-master.cdppvcds.com

Spark 3

History Server × 1 New

pvcbase-master.cdppvcds.com ▾

Gateway × 5 New

pvcbase-worker[1-5].cdppvcds.com ▾

Spark

History Server × 1 New

pvcbase-master.cdppvcds.com

Gateway × 5 New

pvcbase-worker[1-5].cdppvcds.com ▾

Tez

Gateway × 6 New

pvcbase-master.cdppvcds.com; pvcbase-worker[1-5].cdppvcds.com

YARN

ResourceManager × 1 New

pvcbase-master.cdppvcds.com

JobHistory Server × 1 New

pvcbase-master.cdppvcds.com

NodeManager × 5 New

Same as DataNode ▾

ZooKeeper

Server × 1 New

pvcbase-master.cdppvcds.com

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Continue →

Step 23. Click **Continue**. When you're doing the set-up for the first time on this CM Sever. **Cloudera Management Services** will be installed only once along with other control plane services.

Step 24. On the **Setup Databases** page, Select **database type** as **Use Custom Database**. Provide database hostname, username, and password (created in earlier steps) for different services and click on **Test Connection**. After a successful connection test, click **Continue** to install, configure and start services sequentially.

- Reports Manager (For Cloudera Management Service)
- Oozie Server (If selected to install, as optional)
- Ranger
- Hive
- YARN Queue Manager
- Hue

Add Cluster - Configuration

- Select Services
- Assign Roles
- Setup Database**
- Enter Required Parameters
- Review Changes
- Configure Kerberos
- Command Details
- Command Details
- Summary

Setup Database

Configure and test database connections. If using custom databases, create the databases first according to the [Installing and Configuring an External Database](#) section of the [Installation Guide](#).

Reports Manager

Currently assigned to run on **pvcbase-master.cdppvcds.com**. ✓ Successful

Type	Database Hostname	Database Name
PostgreSQL	cldr-mngr.cdppvcds.com	rman
Username	Password	
rman	rman	

Ranger

✓ Successful

Type	Use JDBC URL Override	Database Hostname
PostgreSQL	No	cldr-mngr.cdppvcds.com
Database Name	Username	Password
ranger	rangeradmin	rangeradmin

Hive

✓ Successful

Type	Use JDBC URL Override	Database Hostname
PostgreSQL	No	cldr-mngr.cdppvcds.com
Database Name	Username	Password
hive	hive	hive

Hue

✓ Successful

Type	Database Hostname	Database Name
PostgreSQL	cldr-mngr.cdppvcds.com	hue

[← Back](#) [Continue →](#)

Step 25. Next, the **Enter Required Parameters** page appears.

Step 26. Wait at the parameter screen, enter the required parameters. For all the remaining parameters, set a common password so that it becomes easier while using those services. This password must have 1 lowercase, 1 uppercase, and 1 numeric value. Failing to adhere to this, the final step in the cluster setup would fail. Please see the required inputs below:

- Knox Master Secret: **Cloudera@123**
- Knox IDBroker Master Secret: **Cloudera@123**
- Enter a suitable name for Ozone Service ID. i.e.: **ozone11**
- Ranger Admin: **Cloudera@123**
- Ranger Usersync: **Cloudera@123**
- Ranger Tagsync: **Cloudera@123**
- Ranger KMS Keyadmin: **Cloudera@123**

Enter Required Parameters

Knox Master Secret gateway_master_secret gateway_master_secret	Knox Gateway Default Group Undo		?
Ozone Service ID ozone.service.id ozone.service.id	Ozone (Service-Wide) Undo	ozone11	?
Ranger Admin User Initial Password (Use strong password as per updated 7.1.8+ password criteria). rangeradmin_user_password rangeradmin_user_password	Ranger (Service-Wide) Undo		?
Ranger Usersync User Initial Password (Use strong password as per updated 7.1.8+ password criteria). rangerusersync_user_password rangerusersync_user_password	Ranger (Service-Wide) Undo		?
Ranger Tagsync User Initial Password (Use strong password as per updated 7.1.8+ password criteria). rangertagsync_user_password rangertagsync_user_password	Ranger (Service-Wide) Undo		?
Ranger KMS Keyadmin User Initial Password (Use strong password as per updated 7.1.8+ password criteria). keyadmin_user_password keyadmin_user_password	Ranger (Service-Wide) Undo		?

Step 27. Click *Continue*.

Step 28. The *Review Changes* page appears. Set a *password* for *Atlas*. You can set it to the same value set for Ranger above.

- Atlas Admin password: **Cloudera@123**

Admin Password atlas.admin.password atlas_admin_password	Cluster 1 > Atlas Server Default Group Undo		?
--	---	--------------------------	---

Step 29. Scroll down and verify/update the HDFS disks configuration according to the below.

- DataNode Data Directory: **/hdfs/dfs/dn**
- NameNode Data Directories: **/hdfs/dfs/nn**
- HDFS Checkpoint Directories: **/hdfs/dfs/snn**

DataNode Data Directory dfs.datanode.data.dir dfs_data_dir_list	PvCBBaseCluster1 > DataNode Default Group Undo	/hdfs/dfs/dn	?
NameNode Data Directories dfs.namenode.name.dir dfs_name_dir_list	PvCBBaseCluster1 > NameNode Default Group Undo	/hdfs/dfs/nn	?
HDFS Checkpoint Directories dfs.namenode.checkpoint.dir fs_checkpoint_dir_list	PvCBBaseCluster1 > SecondaryNameNode Default Group Undo	/hdfs/dfs/snn	?

Step 30. Review the changes for all the services on the *Review Changes* page and verify/edit the configuration parameters as per your requirements. Click *Continue*.

Step 31. A few sets of commands are running in the background. Wait for them to get executed successfully. Once done, Click *Continue*.

Step 32. Configure Kerberos and Keep Review and customize the configuration changes based on your requirements. Check the box for *Enable Kerberos for this cluster*. **Required libraries i.e. krb5-workstation, krb5-libs and freeipa-client are already installed on all servers in prior steps.**

✓ Select Services

✓ Assign Roles

✓ Setup Database

✓ Enter Required Parameters

✓ Review Changes

6 **Configure Kerberos**

7 Command Details

8 Command Details

9 Summary

Configure Kerberos

☒ Enable Kerberos for this cluster
Kerberos is a network authentication protocol that provides security for your cluster.

Install Kerberos client libraries on all hosts before proceeding.

```
# RHEL / CentOS
$ yum install krb5-workstation krb5-libs

# if Red Hat IPA is used as the KDC
$ yum install freeipa-client
```

```
# SUSE
$ zypper install krb5-client

# if Red Hat IPA is used as the KDC
$ zypper install freeipa-client
```

```
# Ubuntu
$ apt-get install krb5-user

# if Red Hat IPA is used as the KDC
$ apt-get install freeipa-client
```

Configure DataNode Ports

Configure the privileged ports required by DataNodes in a secure HDFS service.

DataNode Transceiver Port ⓘ

1004 ▾

DataNode HTTP Web UI Port ⓘ

1006 ▾

Step 33. Click **Continue** after Cloudera Manager successfully runs the **Enable Kerberos** command.

Add Cluster - Configuration

The screenshot shows the 'Command Details' page for the 'Enable Kerberos Command'. The status is 'Finished' at 102.95s. The command was successfully executed on 'PvCBaseCluster1' on Aug 10, 3:36:46 AM. The command details list the following steps:

- Stop cluster (1ms)
- Stop Cloudera Management Services (0ms)
- Deploy krb5.conf (15.17s)
- Configure all services to use Kerberos (73ms)
- Wait for credentials to be generated (27.53s)
- Deploy client configuration (36.54s)
- Start Cloudera Management Services (22.77s)

Step 34. Installation wizard will run the first command to start cluster roles and services. Click **Continue**.

Add Cluster - Configuration

The screenshot shows the 'Command Details' page for the 'First Run Command'. The status is 'Finished' at 6.9m. The command was successfully executed on 'PvCBaseCluster1' on Aug 10, 3:38:39 AM. The command details list the following steps:

- Run a set of services for the first time (6.9m)
- Execute 12 steps in sequence (6.9m)

Step 35. If you still face any issue while making the services up or during the installation or start of any Control Plane services please refer to troubleshooting PvC Base Cluster part at the end of this document. Though, some of the major issues during installation, their cause and their resolution is listed as below:

Zookeeper SASL error:

Solution:resolved by regenerate key tab

Kafka error:

Solution:

Update clusterid and broker id from Role logs in meta.properties

```
rm -vf /var/local/kafka/data/meta.properties
```

```
rm -vf /tmp/kafka-logs/*
```

<https://gautambangalore.medium.com/resolved-error-fatal-error-during-kafkaserver-startup-37f638c2c00c>

```
ansible datanodes -m shell -a "sed -i 's#aHRUaTqoQzOpGf8qA5bXlQ#rxsV4DwNRvWtlcl5ejG-IA#g'
```

Service NodeManager failed in state INITED

```
org.apache.hadoop.yarn.exceptions.YarnRuntimeException: Failed NodeManager login
    at org.apache.hadoop.yarn.server.nodemanager.NodeManager.serviceInit(NodeManager.java:511)
    at org.apache.hadoop.service.AbstractService.init(AbstractService.java:164)
    at org.apache.hadoop.yarn.server.nodemanager.NodeManager.initAndStartNodeManager(NodeManager.java:974)
    at org.apache.hadoop.yarn.server.nodemanager.NodeManager.main(NodeManager.java:1054)
```

Caused by: org.apache.hadoop.security.KerberosAuthException: failure to login: for principal: yarn/pvcbase-worker3.cldrsetup.local@CLDRSETUP.LOCAL from keytab yarn.keytab javax.security.auth.login.LoginException: Client not found in Kerberos database (6) - CLIENT_NOT_FOUND

Solution:

Regenerate kerberos creds from administration>security for yarn

YARN issue

Solution:

<https://community.cloudera.com/t5/Support-Questions/Error-CM-Server-guid-updated-CDH-5-9-0/m-p/47221>

<https://community.cloudera.com/t5/Support-Questions/CDH-6-1-Installation-Issues-Unable-to-obtain-CM-release/td-p/88238>

Fixed it by deleting /var/lib/cloudera-scm-agent/cm_guid on each node.

Chrony issue on during ipaserver/ipaclient installation

Solution:

Stop chrnoyd and remove chrony from all hosts, then install ipa-server and then ipa-client. It will work.

Rman db error

Exception while executing ddl scripts.

```
org.postgresql.util.PSQLException: ERROR: relation "rman_usergrouphistory_seq" already exists
    at org.postgresql.core.v3.QueryExecutorImpl.receiveErrorResponse(QueryExecutorImpl.java:2725)
    at org.postgresql.core.v3.QueryExecutorImpl.processResults(QueryExecutorImpl.java:2412)
```

Solution:

drop and recreate db solved issues.

Ozone Error:

Found SOLR_SERVICE: "

solution:

ozone install error

include solr in service dependency and restart services

Issue: IPASERVER failed to resolve DNS ipaservices not working

Solution:

Port 53 was not open on AWS SecGrp: ipaserver was not working on aws due to it, updated secgrp added 53 rule for dns fixed issue.

Ranger Error:

Repo cm_kafka already exists ->

Solution:

delete cm_kafka from ranger

CM Not able to login

2024-05-14 04:41:02,375 INFO CommandPusher-1:com.cloudera.server.cmf.CommandPusherThread: Acquired lease lock on DbCommand:1546336335

2024-05-14 04:41:02,378 INFO CommandPusher-1:com.cloudera.cmf.service.AbstractOneOffHostCommand: Unsuccessful 'RepMgrTestDatabaseConnection'

2024-05-14 04:41:02,379 INFO CommandPusher-1:com.cloudera.cmf.service.AbstractDbConnectionTestCommand: Command exited with code: 1

```

2024-05-14 04:41:02,379 INFO CommandPusher-1:com.cloudera.cmf.service.AbstractDbConnectionTestCommand: +
MGMT_JAVA_OPTS='-Djava.net.preferIPv4Stack=true '
+ exec /usr/lib/jvm/java-17-openjdk-17.0.11.0.9-2.el9.x86_64/bin/java -Djava.net.preferIPv4Stack=true -Djava.security.egd=file:///dev/urandom -cp
'/run/cloudera-scm-agent/process/1546336334-MGMT.REPORTSMANAGER-test-db-connection:/usr/share/java/mysql-connector-java.jar:/usr/share/java/postgre
sql-connector-java.jar:/usr/share/java/oracle-connector-java.jar:/opt/cloudera/cm/lib/*' com.cloudera.enterprise.dbutil.DbCommandExecutor db.properties
Exception in thread "main" java.lang.NoClassDefFoundError: com/ongres/scram/common/stringprep/StringPreparation
    at org.postgresql.core.v3.ConnectionFactoryImpl.doAuthentication(ConnectionFactoryImpl.java:759)
    at org.postgresql.core.v3.ConnectionFactoryImpl.tryConnect(ConnectionFactoryImpl.java:161)
    at org.postgresql.core.v3.ConnectionFactoryImpl.openConnectionImpl(ConnectionFactoryImpl.java:213)
    at org.postgresql.core.ConnectionFactory.openConnection(ConnectionFactory.java:51)
    at org.postgresql.jdbc.PgConnection.<init>(PgConnection.java:225)
    at org.postgresql.Driver.makeConnection(Driver.java:465)
    at org.postgresql.Driver.connect(Driver.java:264)
    at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:681)
    at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:229)
    at com.cloudera.enterprise.dbutil.DbCommandExecutor.testDbConnection(DbCommandExecutor.java:265)
    at com.cloudera.enterprise.dbutil.DbCommandExecutor.main(DbCommandExecutor.java:140)
Caused by: java.lang.ClassNotFoundException: com.ongres.scram.common.stringprep.StringPreparation
    at java.base/jdk.internal.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:641)
    at java.base/jdk.internal.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)
    at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:525)
    ... 11 more
2024-05-14 04:41:02,379 ERROR CommandPusher-1:com.cloudera.cmf.model.DbCommand: Command 1546336335(RepMgrTestDatabaseConnection) has
completed. finalstate:FINISHED, success:false, msg:Unexpected error. Unable to verify database connection.
Caused by: java.lang.ClassNotFoundException: com.ongres.scram.common.stringprep.StringPreparation
    at java.base/jdk.internal.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:641)
    at java.base/jdk.internal.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)
    at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:525)
    ... 11 more
2024-05-14 04:50:38,519 ERROR CommandPusher-1:com.cloudera.cmf.model.DbCommand: Command 1546336734(OozieTestDatabaseConnection) has
completed. finalstate:FINISHED, success:false, msg:Unexpected error. Unable to verify database connection.
2024-05-14 04:50:38,519 INFO CommandPusher-1:com.cloudera.cmf.command.components.CommandStorage: Invoked delete temp files for
command:DbCommand{id=1546336734, name=OozieTestDatabaseConnection, host=pvcbase-master.cldrsetup.local} at
dir:/var/lib/cloudera-scm-server/temp/commands/1546336734
Solution:
Caused after change in hostssl parameter for Postgres (suspected)
Delete scm db and recreated db and restart scm server fixed the issue. This lead to reinstall entire base and ecs clusters as metadata deleted from scm db

=====

Other:
Configure Ozone with other data services before env creation, else it will lead to CDE installation error
Configure thrift server role in hbase for hue
Knox and Atlas works with local Linux Users and Password credentials i.e. PAM
For accessing the WebUI and fixing issues for web based authentication is not working for some of the services including Knox, Atlas, HDFS (Namenode UI),
Yarn (History server UI) etc.Disable Kerberos Authentication for WebUI under each service configuration section which are showing 403 or 401 error.
In case of Cleanup and re-installation (end-to-end), make sure cleanup steps are performed properly and no control plane services left user and groups created
in /etc/passwd and /etc/group on all nodes of the cluster include cldr-mngr.

```

Step 36. Next, all the Cloudera services would get started and their prerequisite operations would also be run. These processes will run in a combination of serial and parallel processes. Wait for them to complete. Once completed, you will see a Green tick next to all the steps, as shown in the screenshot above.

Step 37. Once completed above step, click **Continue**. You will see a summary page like below.

Add Cluster - Configuration

Step 38. Click **Finish** on the Summary page.

Step 39. This will navigate to the main page where you can see all the services installed.

Step 40. All the services should be in Healthy state. If there are any instances in Bad Health, troubleshoot the same and fix it.

Step 41. This completes the **Cloudera on premises Base cluster** setup.

Note: You might need to adjust configuration parameters of the cluster after successful first run command execution. Apply the changes and restart the cluster.

Step 42. We will perform the adjustment of the configuration parameters of the cluster to fix some of the issues with the services on the base cluster, in the next steps below.

Step 43. Optionally, we can also perform the prerequisites and compatibility tests for Hardware using the script provided below by Cloudera.

https://github.com/cloudera-labs/toolkits/tree/main/data_services-toolkit/DS_Pre-Install_Check

Additional requirements and details for Cloudera on premises Base Cluster services:

Note: Common Data Lake Services' URLs:

CM-UI:

HTTP: <http://cldr-mngr.cldrsetup.local:7180/cmf/>

HTTPS: <https://cldr-mngr.cldrsetup.local:7183/cmf/>

HDFS:

HDFS-NAMENODE UI: <https://pvcbase-master.cldrsetup.local:9871/dfshealth.html>

YARN:

HDFS-YARN JobHistory UI: <https://pvcbase-master.cldrsetup.local:19890/jobhistory>

YARN RM UI: <https://pvcbase-master.cldrsetup.local:8090/ui2/#/cluster-overview>

Ranger: <https://pvcbase-master.cldrsetup.local:6182/index.html#/policymanager/resource>

Atlas: <https://pvcbase-master.cldrsetup.local:31443/login.jsp>

(Login works with PAM-Linux Server Local User Credentials)

Knox:

<https://pvcbase-master.cldrsetup.local:8443/gateway/knoxsso/knoxauth/login.html?originalUrl=https://pvcbase-master.cldrsetup.local:8443/gateway/homepage/home/?profile=token>

(Login works with PAM-Linux Server Local User Credentials)

HiveServer2 UI: <https://pvcbase-master.cldrsetup.local:10002/>

HUE: <https://pvcbase-master.cldrsetup.local:8889/hue/editor/?type=hive>

HBASE: <https://pvcbase-master.cldrsetup.local:16010/master-status>

Ozone:

Ozone Recon: <https://pvcbase-master.cldrsetup.local:9889/#/Overview>

Ozone SCM: <https://pvcbase-master.cldrsetup.local:9877/#/>

Ozone Manager: <https://pvcbase-master.cldrsetup.local:>

S3 Gateway: <https://pvcbase-master.cldrsetup.local:>

Gateway: <https://pvcbase-worker1.cldrsetup.local:>

OzoneDataNode: <https://pvcbase-worker1.cldrsetup.local:>

Spark JobHistory Server:

Spark2: <https://pvcbase-master.cldrsetup.local:18488/>

Spark3: <https://pvcbase-master.cldrsetup.local:18489/>

Impala:

Impala Catalog: <https://pvcbase-master.cldrsetup.local:25020/>

Impala Statestore: <https://pvcbase-master.cldrsetup.local:25010/>

Job History Server: <https://pvcbase-master.cldrsetup.local:9991/>

Step 1. Port requirements for different services on PvC Base Cluster/Data Services (ECS) Cluster:

Please whitelist the below ports or make sure , firewall is disabled in the internal network. (Not required in on-premise Private datacenter based network)

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-ports-used-by-runtime.html>

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-ports-third-party-components.html>

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-ports-used-by-cm.html>

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-service-dependencies.html>

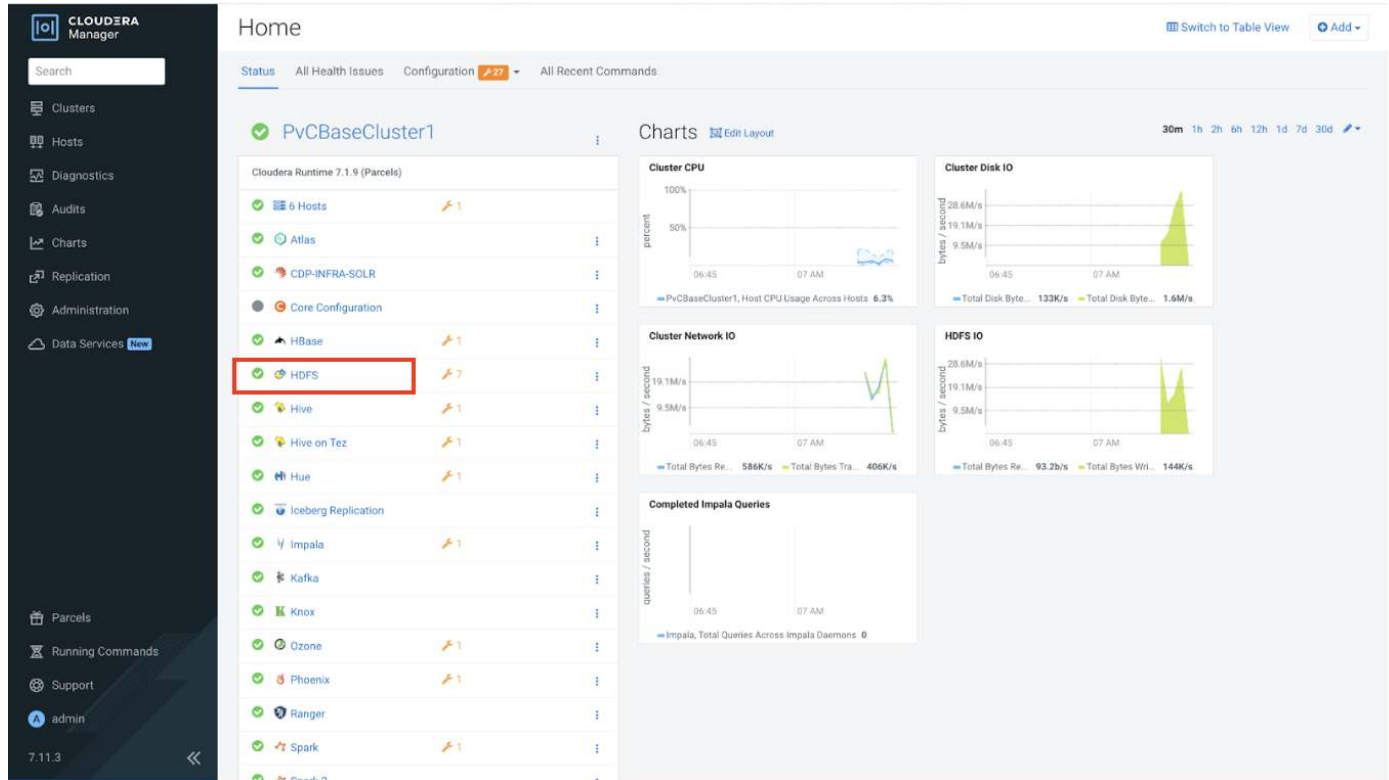
<https://docs.cloudera.com/cdp-private-cloud-data-services/latest/installation-ecs/topics/cdppvc-installation-ecs-steps.html>

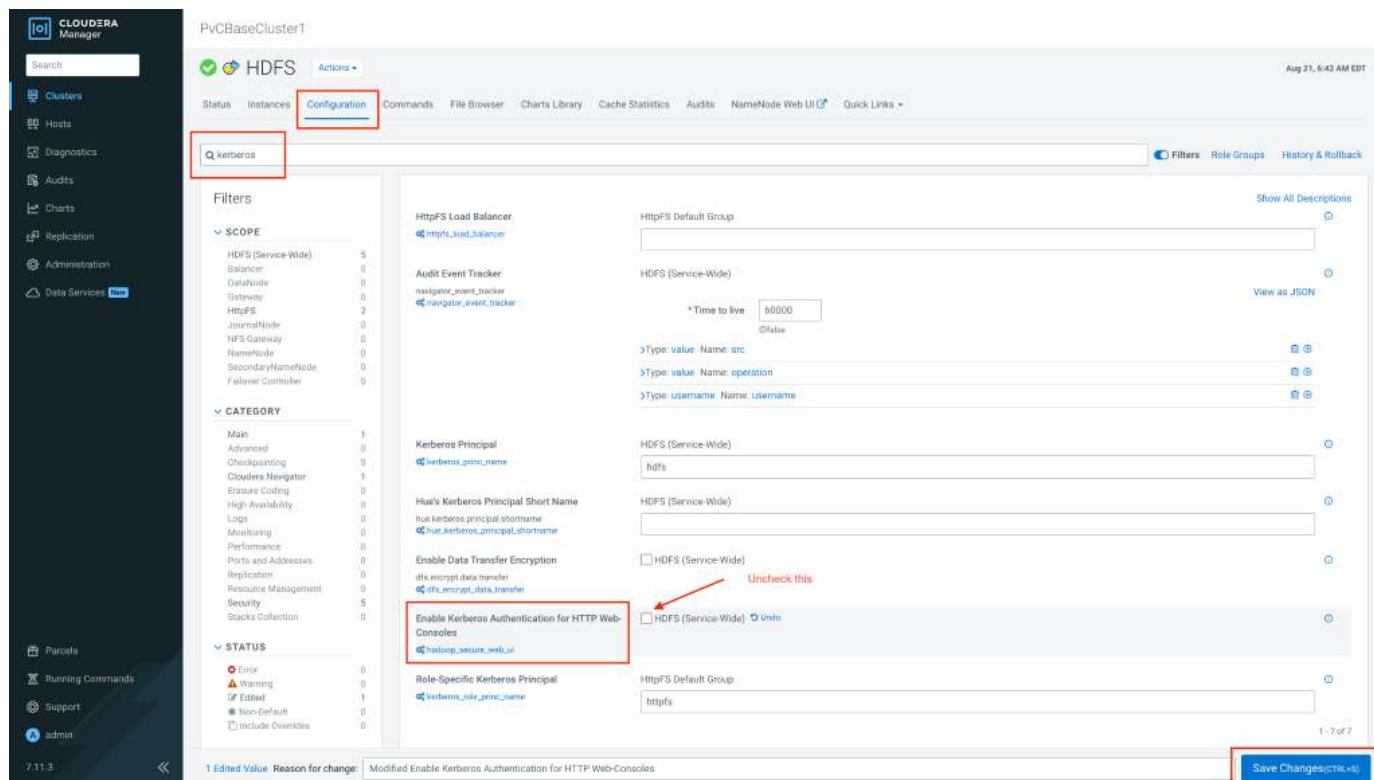
Step 2. Disable Kerberized Web-UI for YARN, Spark, HBase, HDFS, etc. from configuration:

While accessing Web UIs from web browsers, for HDFS-Namenode, YARN services, HiveServer2, Impala Services, etc. If gets 401 unauthorized error: We need to enable non-kerberized webUI for those services, for this we need to:

- Login to CM-UI> Go to individual services> Go to the Configuration section for individual services> Search for **Kerberos**.
- Disable the option, by unchecking the checkbox, for Kerberos authentication for WebUI. Save changes. Restart the Stale Services to update the backend configuration files.

Example Screenshots for HDFS. You can follow the same for other services, which requires WebUI access.





1. For Impala, Hive, Hive on Tez edit value for Ranger Plugin URL Auth Filesystem Schemes - file:,wasb:,adl:

2. Disable ~~Enable~~ Kerberos Authentication for HTTP Web-Consoles - HBase (Service-Wide), YARN, Spark2, Spark3, HiveServer2, Impala, HDFS, etc. ~~Click on Generate missing credentials for Kerberos.~~

3. For TLS/SSL enabled HDFS configuration you might see a warning as "DataNode configuration is valid, but not recommended. There are two recommended configurations:

- (1) DataNode Transceiver Port and Secure DataNode Web UI Port (TLS/SSL) both ≥ 1024 , DataNode Data Transfer Protection set, Hadoop TLS/SSL enabled;
- (2) DataNode Transceiver Port and DataNode HTTP Web UI Port both < 1024 , DataNode Data Transfer Protection not set, Hadoop TLS/SSL disabled."

DataNode Transceiver Port (dfs_datanode_port)- 9866

DataNode HTTP Web UI Port (dfs.datanode.http.address) - 9864

DataNode Data Transfer Protection (dfs.data.transfer.protection) – Authentication

4. Install Ranger Plugin for services, add service dependency i.e. Hive etc. and restart the cluster.

Ensure that the Ranger Solr and Ranger HDFS plugins are enabled. See [Additional Steps for Apache Ranger](#) for more details on Configuration Steps.

5. Make Sure HDFS, Ozone services are Installed and Running Successfully (critical services – if not working properly, then no other service will work properly)

1.HDFS 2.Zookeeper 3.

6. Atlas and Knox work with PAM authentication i.e. Local (Non-LDAP) users created on the base-master node where your Atlas server is running, unless Atlas is explicitly configured (integrated) to use LDAP. So you may need to create a local user on the base-master node, if it does not already exist.

7. Atlas is having dependencies on some additional services i.e. HBase, SOLR and Kafka

Step 3. Optionally, Update the /etc/hosts file on your working machine/JumpHost where you are trying to access your CM-UI to work with the URLs smoothly:

Open C:\Windows\System32\drivers\etc (on Windows) or /etc/hosts (on MAC/Linux), with sudo privileges.

```
ksahu@Kuldeeps-MacBook-Air ~ % sudo vi /etc/hosts
##
```

```

# Host Database
#
# localhost is used to configure the loopback interface
# when the system is booting. Do not change this entry.
##
127.0.0.1    localhost
255.255.255.255 broadcasthost
::1         localhost

52.221.202.246 pvcecs-master.cldrsetup.local
52.221.202.246 hue-kd-hive-vw1.apps.cldrsetup.local
52.221.202.246 cml-task-bo6klv.kuldeep-cml.apps.cldrsetup.local
52.221.202.246 kuldeep-cml.apps.cldrsetup.local
18.139.222.78 pvcbase-master.cldrsetup.local
13.251.65.11 cldr-mngr.cldrsetup.local

# ECS Links
52.221.202.246 console-cdp.apps.cldrsetup.local prometheus-cp.apps.cldrsetup.local infra-prometheus.apps.cldrsetup.local validation-cdp.apps.cldrsetup.local
kube-dashboard.apps.cldrsetup.local longhorn.apps.cldrsetup.local fluent-console-cdp.apps.cldrsetup.local vault.localhost.localdomain

# PvC Base Cluster Nodes
18.139.222.78 pvcbase-master.cldrsetup.local pvcbase-master
13.215.202.164 pvcbase-worker1.cldrsetup.local pvcbase-worker1
172.31.23.0 pvcbase-worker2.cldrsetup.local pvcbase-worker2
18.141.13.157 pvcbase-worker3.cldrsetup.local pvcbase-worker3

# PvC Data Services Cluster Nodes
172.31.30.239 pvcecs-master.cldrsetup.local pvcecs-master
172.31.22.43 pvcecs-worker1.cldrsetup.local pvcecs-worker1
172.31.30.249 pvcecs-worker2.cldrsetup.local pvcecs-worker2
172.31.26.24 pvcecs-worker3.cldrsetup.local pvcecs-worker3
172.31.24.198 pvcecs-worker4.cldrsetup.local pvcecs-worker4
172.31.24.53 pvcecs-worker5.cldrsetup.local pvcecs-worker5

```

Step 4. Optionally, Update the /etc/hosts file on your working machine/JumpHost (The /etc/hosts entries required for ECS Data Services)

<https://docs.cloudera.com/management-console/1.5.5/private-cloud-security-overview/mc-private-cloud-security-overview.pdf>

Embedded Container Service (ECS) :

- console-cdp.apps.**APPDOMAIN**
- prometheus-cp.apps.**APPDOMAIN**
- infra-prometheus.apps.**APPDOMAIN**
- validation-cdp.apps.**APPDOMAIN**
- kube-dashboard.apps.**APPDOMAIN**
- longhorn.apps.**APPDOMAIN**
- fluent-console-cdp.apps.**APPDOMAIN**

Entries required by CDW

Let **APPDOMAIN** be the base app domain for the ECS cluster. For example, if your console URL is "console-cdp.apps.cldrsetup.local", then the APPDOMAIN is "cldrsetup.local". Let **VWHNAME** be the name of the CDW Virtual Warehouse. This must match the name the user provides when creating a new Virtual Warehouse (VW).

Endpoints of Hive VW:

- hue-**VWHNAME**.apps.**APPDOMAIN**
- hs2-**VWHNAME**.apps.**APPDOMAIN**

Endpoints of Impala VW:

- hue-**VWHNAME**.apps.**APPDOMAIN**
- coordinator-**VWHNAME**.apps.**APPDOMAIN**
- admissiond-web-**VWHNAME**.apps.**APPDOMAIN**
- catalogd-web-**VWHNAME**.apps.**APPDOMAIN**
- coordinator-web-**VWHNAME**.apps.**APPDOMAIN**
- statestored-web-**VWHNAME**.apps.**APPDOMAIN**
- impala-proxy-**VWHNAME**.apps.**APPDOMAIN**
- impala-autoscaler-web-**VWHNAME**.apps.**APPDOMAIN**

Endpoints of Viz:

- viz-VWHNAME.apps.APPDOMAIN

Configure Ranger with SSL/TLS enabled PostgreSQL Database

Login to Cloudera Manager Web Console. Go to **Ranger > Configuration**.

Note: Make sure that:

- The database and database user for Ranger service are created in the required postgresQL.
- A database server certificate is issued by a trusted certificate authority.
- The server host name matches the host name in the database server certificate.

From CDPDC-7.1.5 onwards, Ranger service requires postgres JDBC driver **version >= 42.2.5**. The Ranger code also constructs the JDBC connection string to have **sslmode=verify-full**, if Ranger Database SSL configurations are set in case of postgresql database type.

For more details:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/installation/topics/cdpdc-enable-ssl-tls-ranger-postgres-db.html>

Copy the database server certificate to **/var/lib/ranger/** path, or use any custom path.

```
[root@pvcbase-master ~]# cp -rv /root/.postgresql/root.crt /var/lib/ranger/root.crt
```

- In Review Config, search for SSL and update the following configurations:

Component	Value
Ranger DB SSL Enabled: (ranger.db.ssl.enabled)	true (Checked)
Ranger DB SSL Required: (ranger.db.ssl.required)	true (Checked)
Ranger DB SSL Verify Server Certificate: (ranger.db.ssl.verifyServerCertificate)	true (Checked)
Ranger DB Auth Type: (ranger.db.ssl.auth.type)	1-way
Ranger Admin Database SSL Certificate File: (ranger.db.ssl.certificateFile)	<path-to-db-server-certificate>: /var/lib/ranger/root.crt or custom path /var/lib/cloudera-scm-server/.postgresql/root.crt
Ranger Database JDBC URL Override:	jdbc:postgresql://<db_host>:<db_port>/<db_name>?sslmode=verify-full& sslrootcert=<server_certificate_path> jdbc:postgresql://cldr-mngr.cldrsetup.local:5432/ranger?ssl=true&sslmode=verify-full&ssl =/var/lib/ranger/root.crt
Set Load Balancer Address (Optional)	http://<ranger_host>:6080 http://pvcbase-master.cldrsetup.local:6080 https://<ranger_host>:6182 https://pvcbase-master.cldrsetup.local:6182

- After updating the configurations, click on **Save Changes**.
- Ranger Service restart is required for rangeradmin after updating Ranger configuration.
- Click on **Actions -> Restart** under Ranger Service.
- Run below command on pvcbase-master node to check the Ranger logs, during restart.

```
[root@pvcbase-master ~]# tail -f /var/log/ranger/admin/catalina.out
```

PvCBaseCluster1

Ranger Actions

Aug 21, 7:49 AM EDT

Status Instances **Configuration** Commands Charts Library Audits Ranger Admin Web UI Quick Links

Q ssl Filters Role Groups History & Rollback

Filters

SCOPE

- Ranger (Service-Wide) 1
- Ranger Admin 14
- Ranger Tagsync 8
- Ranger Usersync 7

CATEGORY

- Main 11
- Advanced 1
- Database 0
- Logs 0
- Monitoring 0
- Performance 0
- Ports and Addresses 3
- Resource Management 0
- Security 15
- Stacks Collection 0

STATUS

- Error 0
- Warning 0
- Edited 4
- Non Default 19
- Include Overrides 0

Exclude Users from Audit Access Tab

ranger.accesslogs.exclude.users.list
ranger.accesslogs.exclude.users.list

Ranger DB SSL Enabled

ranger.db.ssl.enabled
ranger.db.ssl.enabled

Ranger DB SSL Required

ranger.db.ssl.required
ranger.db.ssl.required

Ranger DB SSL Verify Server Certificate

ranger.db.ssl.verifyServerCertificate
ranger.db.ssl.verifyServerCertificate

Ranger DB Auth Type

ranger.db.ssl.auth.type
ranger.db.ssl.auth.type

Ranger Admin Keystore File

ranger.keystore.file
ranger.keystore.file

Ranger Admin Database SSL Certificate File

ranger.db.ssl.certificateFile
ranger.db.ssl.certificateFile

Ranger Admin TLS/SSL Keystore File Alias

ranger.service.https.attrb.keystore.keyalias
ranger.service.https.attrb.keystore.keyalias

Ranger Admin Access Log Rotation Max Days

ranger.accesslog.rotate.max.days
ranger.accesslog.rotate.max.days

Ranger Admin Default Group

ranger.tagsync,mapred,spark,oozie,hue,streamsmgmt,streamrepmgr,cruisecontrol,impala,zeppelin

☒ Ranger Admin Default Group Undo

☒ Ranger Admin Default Group Undo

☒ Ranger Admin Default Group Undo

☒ 1-way
☐ 2-way

Ranger Admin Default Group

jdbc:postgresql://cld-mngr.cdppvcds.com:5432/ranger?ssl=true&sslmode=verify-full&sslrootcert=/var/lib/ranger/root.crt

Ranger Admin Default Group

Ranger Admin Default Group

15

4 Edited Values Reason for change: Modified Ranger DB SSL Enabled, Ranger DB SSL Required, Ranger DB SSL Verify Server Certificate, Ranger Admin Database SSL Certificate File

Save Changes(ctrl+s)

Configure Hive metastore with SSL/TLS enabled PostgreSQL Database (Mandatory Step for CDW)

In the Cloudera Manager Web console; go to *Hive > Configuration > Hive Metastore Database JDBC URL Override*.

Copy the database server certificate to */var/lib/hive/* path, or use any custom path.

```
[root@pvcbase-master ~]# cp -rv /root/.postgresql/root.crt /var/lib/hive/root.crt
```

Edit value as: jdbc:postgresql://<db_host>:<db_port>/<db_name>?sslmode=verify-full&sslrootcert=<server_certificate_path>
jdbc:postgresql://cldr-mngr.cldrssetup.local:5432/hive?ssl=true&sslmode=verify-full&sslrootcert=/var/lib/hive/root.crt



Note: Restart required for Hive Metastore Server and HiveServer2 after updating Hive configuration.

Note: Click on *Actions -> Restart* under *Hive* and *Hive-on-Tez* Services.

Scale the Cluster (Optional– Skip this step)

The role assignment recommendation above is for clusters with at least 64 servers and in High Availability. For smaller clusters running without High Availability the recommendation is to dedicate one server for Name Node and a second server for secondary name node and YARN Resource Manager. For larger clusters larger than 16 nodes the recommendation is to dedicate one server each for name node, YARN Resource Manager and one more for running both Name Node (High Availability) and Resource Manager (High Availability) as in the table (no Secondary Name Node when in High Availability).

Note: For production clusters, it is recommended to set up Name Node and Resource manager in High Availability mode.

This implies that there will be at least 3 master nodes, running the Name Node, YARN Resource manager, the failover counterpart being designated to run on another node and a third node that would have similar capacity as the other two nodes.

All the three nodes will also need to run zookeeper and quorum journal node services. It is also recommended to have a minimum of 8 Data Nodes in a cluster. Please refer to the next section for details on how to enable HA.

Enable High Availability (Optional– Skip this step)

Note: Setting up High Availability is done after the Cloudera Installation is completed.

<https://docs.cloudera.com/cloudera-manager/7.13.2/cloudera-manager-managing-clusters/topics/cm-high-availability.html>

Configure Browsers for Kerberos Authentication

Note: To enable specific web browsers to use SPNEGO to negotiate Kerberos authentication, please visit:

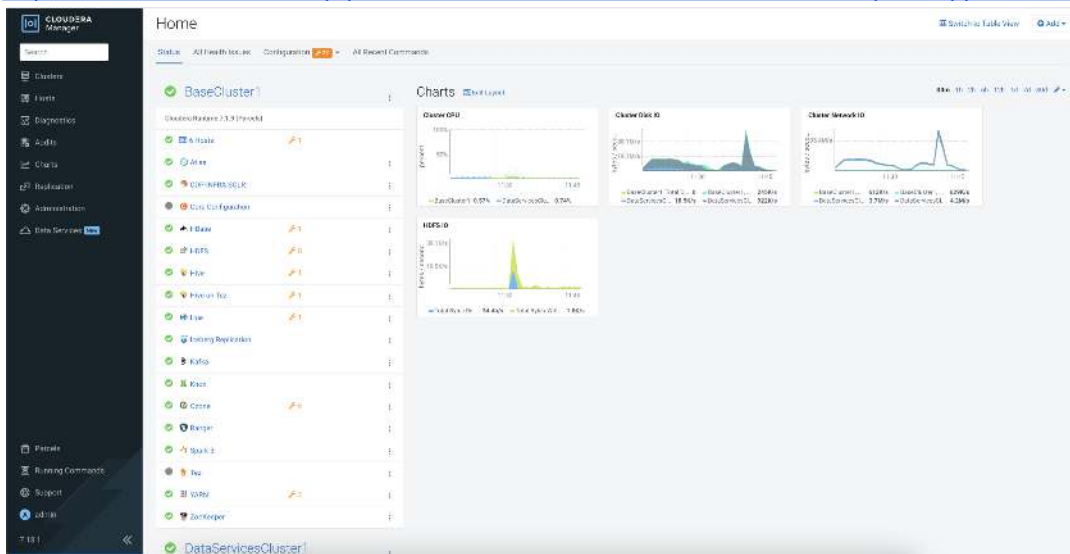
<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/security-how-to-guides/topics/cm-security-browser-access-kerberos-protected-url.html>

Cloudera on premises Base checklist

[Cloudera support matrix](#) lists the supported software for the Cloudera on premises Base cluster and the Cloudera on premises Data Services containerized cluster.

Please review Cloudera on premises Base Checklist:

<https://docs.cloudera.com/cdp-private-cloud-data-services/latest/installation/topics/cdppvc-installation-pvcbase-checklist.html>



Configure Ranger authentication for LDAP (Optional– Skip this Step)

Follow steps below to configure Ranger for LDAP authentication.

1. In Cloudera Manager, select **Ranger**, then click the **Configuration** tab.
2. To display the authentication settings, type "**authentication**" in the Search box. Scroll down to see all of the **LDAP** settings.
3. Select LDAP for "Admin Authentication Method".

PvCBaseCluster1

The screenshot shows the Ranger configuration interface in Cloudera Manager. The top navigation bar includes Status, Instances, Configuration (selected), Commands, Charts Library, Audits, Ranger Admin Web UI, and Quick Links. A search bar at the top left contains the text 'authentication'. On the left side, there are two filter sections: 'SCOPE' and 'CATEGORY'. The 'SCOPE' section lists Ranger (Service-Wide) with 0 items, Ranger Admin with 21 items, Ranger Tagsync with 1 item, and Ranger Usersync with 2 items. The 'CATEGORY' section lists Main with 23 items, Advanced with 0 items, Database with 0 items, and Error with 0 items. The main configuration area is divided into two columns. The left column is titled 'Admin Authentication Method' and shows the configuration for 'ranger.authentication.method' with a link to 'ranger_authentication_method'. The right column is titled 'Ranger Admin Default Group' and shows the configuration for 'ranger.admin.default.group' with a link to 'ranger_admin_default_group'. In the 'Admin Authentication Method' section, the 'LDAP' radio button is selected, and the 'Ranger Admin Default Group' checkbox is checked.

4. Configure the following settings for LDAP authentication as shown below, the details depends on your configuration, based on existing LDAP/AD setup:

Table 9. User LDAP Integration

Component	Value
Admin LDAP/AD Auth URL:	ldap://ipaserver.cldrsetup.local:389/ (Give LDAP or AD Server LDAP ADDR)
Admin LDAP/AD Auth Bind User/ Bind DN:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
Admin LDAP/AD Auth Bind User Password:	<cloudera123> (password for KDC admin, configured earlier)
Admin LDAP/AD Auth User DN Pattern:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
Admin LDAP/AD Auth User Search Filter:	For AD: (&(objectClass=user)(sAMAccountName={0})) For LDAP: (&(objectClass=person)(uid={0}))
Admin LDAP/AD Auth Group Search Base:	cn=groups,cn=accounts,dc=cldrsetup,dc=local
Admin LDAP/AD Auth Group Search Filter:	For AD: (&(objectClass=group)(member={0})) For LDAP: (&(objectClass=posixGroup)(memberUid={0}))
Admin LDAP/AD Auth Group Role Attribute:	cn

Component	Value
Admin LDAP /AD Auth Base DN:	dc=cldrsetup,dc=local
Admin LDAP /AD Auth Referral:	follow
Admin AD Auth Domain Name: (For AD Setup)	cldrsetup.local

(Search under Configuration for- *ranger.idap*)

Filters Role Groups History & Rollback

Filters

SCOPE

Ranger (Service-Wide) 0
Ranger Admin 17
Ranger Toolsync 0
Ranger Usersync 1

CATEGORY

Main 18
Advanced 0
Database 0
Logs 0
Monitoring 0
Performance 0
Ports and Addresses 0
Resource Management 0
Security 0
Stacks Collection 0

STATUS

Error 0
Warning 0
Edited 3
Non Default 10
Include Overrides 0

Admin LDAP Auth URL

ranger.idap.url
🔗 ranger.idap.url

Admin LDAP Auth Bind User

ranger.idap.bind.cn
🔗 ranger.idap.bind.cn

Admin LDAP Auth Bind User Password

ranger.idap.bind.password
🔗 ranger.idap.bind.password

Admin LDAP Auth User DN Pattern

ranger.idap.user.dn.pattern
🔗 ranger.idap.user.dn.pattern

Admin LDAP Auth User Search Filter

ranger.idap.user.search.filter
🔗 ranger.idap.user.search.filter

Admin LDAP Auth Group Search Base

ranger.idap.group.search.base
🔗 ranger.idap.group.search.base

Admin LDAP Auth Group Search Filter

ranger.idap.group.search.filter
🔗 ranger.idap.group.search.filter

Admin LDAP Auth Group Role Attribute

ranger.idap.group.role.attribute
🔗 ranger.idap.group.role.attribute

Admin LDAP Auth Base DN

ranger.idap.base.dn
🔗 ranger.idap.base.dn

Admin LDAP Auth Referral

ranger.idap.referral
🔗 ranger.idap.referral

Ranger Admin Default Group

🔄 undo
ldap://ipaserver.cdppvcds.com:389/

Ranger Admin Default Group

🔄 undo
admin

Ranger Admin Default Group

🔄 undo

Ranger Admin Default Group

uid=admin,cn=users,cn=accounts,dc=cdppvcds,dc=com

Ranger Admin Default Group

(&(objectClass=person)(uid=*))

Ranger Admin Default Group

cn=groups,cn=accounts,dc=cdppvcds,dc=com

Ranger Admin Default Group

(&(objectClass=posixGroup)(memberUid=*))

Ranger Admin Default Group

cn

Ranger Admin Default Group

dc=cdppvcds,dc=com

Ranger Admin Default Group

☐ ignore
☒ follow
☐ throw

Additional parameters required for AD Based integration: (Search under Configuration for- *ranger.idap.ad*)

Admin AD Auth URL

ranger.idap.ad.url
🔗 ranger.idap.ad.url

Admin AD Auth Bind DN

ranger.idap.ad.bind.dn
🔗 ranger.idap.ad.bind.dn

Admin AD Auth Bind Password

ranger.idap.ad.bind.password
🔗 ranger.idap.ad.bind.password

Admin AD Auth Domain Name

ranger.idap.ad.domain
🔗 ranger.idap.ad.domain

Admin AD Auth Base DN

ranger.idap.ad.base.dn
🔗 ranger.idap.ad.base.dn

Admin AD Auth Referral

ranger.idap.ad.referral
🔗 ranger.idap.ad.referral

Ranger Admin Default Group

🔗 Show All Description

Ranger Admin Default Group

Ranger Admin Default Group

Ranger Admin Default Group

Ranger Admin Default Group

Ranger Admin Default Group

☐ ignore
☒ follow
☐ throw

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Admin AD Auth User Search Filter
ranger ldap.ad.user.searchfilter
ranger ldap.ad.user.searchfilter

Ranger Admin Default Group

1 - 7 of

5. Edit Usersync configuration. Example values set are shown in the screenshot below:

Source for Syncing User and Groups

ranger.usersync.source.impl.class

ranger.usersync.source.impl.class

Ranger Usersync Default Group

- ☐ org.apache.ranger.unixusersync.process.UnixUserGroupBuilder
- ☐ org.apache.ranger.unixusersync.process.FileSourceUserGroupBuilder
- ☒ org.apache.ranger.ldapusersync.process.LdapUserGroupBuilder

Usersync LDAP/AD URL

ranger.usersync.ldap.url

ranger.usersync.ldap.url

Ranger Usersync Default Group

ldaps://winjb-ucsg16.cdip.cisco.local:636

Usersync Bind User

ranger.usersync.ldap.binddn

ranger.usersync.ldap.binddn

Ranger Usersync Default Group

CN=cdpbind,OU=cloudera,DC=cdip,DC=cisco,DC=local

Usersync Bind User Password

ranger.usersync.ldap.ldapbindpassword

ranger_usersync_ldap_ldapbindpassword

Ranger Usersync Default Group

.....

Usersync Incremental Sync

ranger.usersync.ldap.deltasync

ranger.usersync.ldap.deltasync

☒ Ranger Usersync Default Group

Usersync Enable STARTTLS

ranger.usersync.ldap.starttls

ranger.usersync.ldap.starttls

☐ Ranger Usersync Default Group

Usersync User Search Base

ranger.usersync.ldap.user.searchbase

ranger.usersync.ldap.user.searchbase

Ranger Usersync Default Group

CN=cdipadmin,OU=cloudera,DC=cdip,DC=cisco,DC=local

Usersync User Search Scope

ranger.usersync.ldap.user.searchscope
[ranger.usersync.ldap.user.searchscope](#)

Ranger Usersync Default Group

- ☒ sub
☐ base
☐ one

Usersync User Object Class

ranger.usersync.ldap.user.objectclass
[ranger.usersync.ldap.user.objectclass](#)

Ranger Usersync Default Group [↗](#)

user

Usersync User Search Filter

ranger.usersync.ldap.user.searchfilter
[ranger.usersync.ldap.user.searchfilter](#)

Ranger Usersync Default Group [↗](#)

(&(objectClass=user)(sAMAccountName={0}))

Usersync User Name Attribute

ranger.usersync.ldap.user.nameattribute
[ranger.usersync.ldap.user.nameattribute](#)

Ranger Usersync Default Group [↗](#)

sAMAccountName

Usersync Referral

ranger.usersync.ldap.referral
[ranger.usersync.ldap.referral](#)

Ranger Usersync Default Group [↗](#)

- ☐ ignore
☒ follow
☐ throw

Usersync Username Case Conversion

ranger.usersync.ldap.username.caseconversion
[ranger.usersync.ldap.username.caseconversion](#)

Ranger Usersync Default Group [↗](#)

- ☐ none
☒ lower
☐ upper

Usersync Groupname Case Conversion

ranger.usersync.ldap.groupname.caseconversion
[ranger.usersync.ldap.groupname.caseconversion](#)

Ranger Usersync Default Group [↗](#)

- ☐ none
☒ lower
☐ upper

Usersync Enable User Search

ranger.usersync.user.searchenabled

 ranger.usersync.user.searchenabled

☒ Ranger Usersync Default Group

Usersync Group Search Base

ranger.usersync.group.searchbase

 ranger.usersync.group.searchbase

Ranger Usersync Default Group 

CN=cdipadmin,OU=cloudera,DC=cdip,DC=cisco,DC=local

Usersync Group Object Class

ranger.usersync.group.objectclass

 ranger.usersync.group.objectclass

Ranger Usersync Default Group 

group

Usersync Group Name Attribute

ranger.usersync.group.nameattribute

 ranger.usersync.group.nameattribute

Ranger Usersync Default Group 

cn

Usersync Group Member Attribute

ranger.usersync.group.memberattributename

 ranger.usersync.group.memberattributename

Ranger Usersync Default Group 

member

Usersync Group Hierarchy Levels

ranger.usersync.ldap.grouphierarchylevels

 ranger.usersync.ldap.grouphierarchylevels

Ranger Usersync Default Group

0  

Usersync Ldap Group Names

ranger.usersync.ldap.groupnames

 ranger.usersync.ldap.groupnames

Ranger Usersync Default Group



Table 10. UserSync LDAP Integration

Component	Value
Source for Syncing User and Groups:	org.apache.ranger.ldapusersync.process.LdapUserGroupBuilder
Ranger Usersync Unix Backend:	nss
Usersync LDAP/AD URL:	ldap://ipaserver.cldrsetup.local:389/
Usersync Bind User:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
Usersync Bind User Password:	<cloudera123> (password for KDC admin, configured earlier)
Usersync User Search Base:	cn=users,cn=accounts,dc=cldrsetup,dc=local
Usersync User Search Scope:	sub
Usersync User Object Class:	person
Usersync User Search Filter:	uid=*
Usersync User Name Attribute:	uid
Usersync Referral:	follow
Usersync Username Case Conversion:	none
Usersync Groupname Case Conversion:	none
Usersync Enable User Search:	Ranger Usersync Default Group
Usersync Group Search Base:	cn=groups,cn=accounts,dc=cldrsetup,dc=local
Usersync Group Search Filter:	For AD: (&(objectClass=group)(member={0}))

Component	Value
	For LDAP: (&(objectClass=posixGroup)(memberUid={0}))
Usersync Group Search Scope:	sub
Usersync Group Object Class:	ipausergroup
Usersync Group Name Attribute:	cn
Usersync Group Member Attribute:	member

Table 11. UserSync AD Integration

Component	Value
Source for Syncing User and Groups:	org.apache.ranger.Idapusersync.process.LdapUserGroupBuilder
Ranger Usersync Unix Backend:	nss
Usersync LDAP/AD URL:	ldap://ipaserver.cldrsetup.local:389/
Usersync Bind User:	Administrator@cldrsetup.local
Usersync Bind User Password:	<cloudera123> (password for KDC admin, configured earlier)
Usersync User Search Base:	OU=cloudera,DC=cldrsetup,DC=local
Usersync User Search Scope:	sub
Usersync User Object Class:	user
Usersync User Search Filter:	(sAMAccountName=*)
Usersync User Name Attribute:	sAMAccountName
Usersync Referral:	follow
Usersync Username Case Conversion:	none
Usersync Groupname Case Conversion:	none
Usersync Enable User Search:	Ranger Usersync Default Group
Usersync Group Search Base:	OU=cloudera,DC=cldrsetup,DC=local * An OU would need to be mentioned to configure ldap if you are using AD based setup
Usersync Group Search Scope:	sub
Usersync Group Object Class:	group
Usersync Group Name Attribute:	cn
Usersync Group Member Attribute:	(member={1})

6. Click on save changes.
7. Restart Ranger service.

8. Login to Ranger Admin WebUI with ldap authentication

For more details:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/security-ranger-authentication-unix-ldap-ad/topics/security-ranger-authentication-ldap-settings.html>

Configure Hue for LDAP Authentication (Optional– Skip this Step)

Configuring Hue for Lightweight Directory Access Protocol (LDAP) enables you to import users and groups from a directory service, synchronize group membership manually or automatically at login, and authenticate with an LDAP server. Hue supports Microsoft Active Directory (AD) and open standard LDAP such as OpenLDAP and Forgerock OpenDJ Directory Services.

1. Login to **Cloudera Manager**. Go to **Cluster > Hue > Configuration**.
2. Change value for Authentication Backend –
desktop.auth.backend.LdapBackend,desktop.auth.backend.AllowFirstUserDjangoBackend

Authentication Backend	Hue (Service-Wide) ↶
backend	
⚙️ auth_backend	desktop.auth.backend.LdapBackend,desktop.auth.backend.AllowFirstUserDjangoBackend

3. Edit value for LDAP configuration. Example values set are shown in the screenshot below:

LDAP URL	Hue (Service-Wide) ↶
ldap_url	
⚙️ ldap_url	ldaps://winjb-ucsg16.cdip.cisco.local:636
LDAP Server CA Certificate	Hue (Service-Wide) ↶
ldap_cert	
⚙️ ldap_cert	/etc/pki/ca-trust/source/anchors/ad.cert.pem
Enable LDAP TLS	☒ Hue (Service-Wide)
use_start_tls	
⚙️ use_start_tls	
Active Directory Domain	Hue (Service-Wide) ↶
nt_domain	
⚙️ nt_domain	cdip.cisco.local
LDAP Username Pattern	Hue (Service-Wide)
ldap_username_pattern	
⚙️ ldap_username_pattern	
Use Search Bind Authentication	☒ Hue (Service-Wide) ↶
search_bind_authentication	
⚙️ search_bind_authentication	
Create LDAP users on login	☒ Hue (Service-Wide)
create_users_on_login	
⚙️ create_users_on_login	
LDAP Search Base	Hue (Service-Wide) ↶
base_dn	
⚙️ base_dn	DC=cdip,DC=cisco,DC=local

LDAP Bind User Distinguished Name

bind_dn

bind_dn

Hue (Service-Wide)

CN=cdpbind,OU=cloudera,DC=cdip,DC=cisco,DC=local

LDAP Bind Password

bind_password

bind_password

Hue (Service-Wide)

••••••

LDAP Username for Test LDAP Configuration

test_ldap_user

test_ldap_user

Hue (Service-Wide)

cdpbind

LDAP Group Name for Test LDAP Configuration

test_ldap_group

test_ldap_group

Hue (Service-Wide)

cdipadmin

LDAP User Filter

user_filter

user_filter

Hue (Service-Wide)

(objectClass=user)

LDAP Username Attribute

user_name_attr

user_name_attr

Hue (Service-Wide)

sAMAccountName

LDAP Group Filter

group_filter

group_filter

Hue (Service-Wide)

(objectClass=group)

LDAP Group Name Attribute

group_name_attr

group_name_attr

Hue (Service-Wide)

cn

LDAP Group Membership Attribute

group_member_attr

group_member_attr

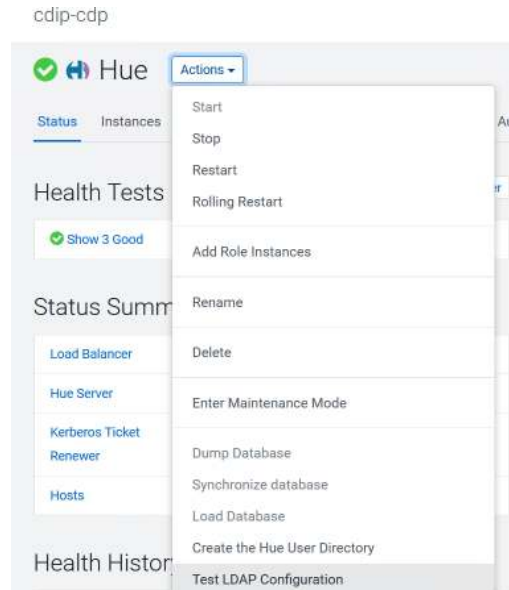
Hue (Service-Wide)

member

Table: LDAP Integration

Component	Value
LDAP URL:	ldap://ipaserver.cldrsetup.local:389/
LDAP Server CA Certificate (Optional):	Leave it blank
Enable LDAP TLS (Hue):	True (Checked)
LDAP Search Base:	cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Bind User Distinguished Name:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Bind Password:	<cloudera123> (password for KDC admin, configured earlier)
LDAP Username for Test LDAP Config:	admin
LDAP Group Name for Test LDAP Config:	Leave it blank
LDAP User filter:	Leave it blank
LDAP Group filter:	Leave it blank
LDAP Group Name Attribute:	cn
LDAP Group Membership Attribute:	member

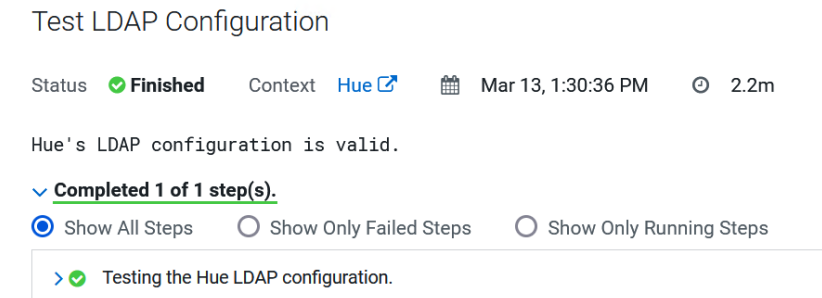
4. Click on save changes
5. Restart HUE service.
6. Click on Actions next to Hue. Click on Test LDAP Configuration.



7. Click on Test LDAP Configuration.



8. Click on Finish.



For more details:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/securing-hue/topics/hue-authenticate-users-with-ldap.html>

Configure Atlas for LDAP authentication (Optional– Skip this Step)

Follow steps below to configure Atlas authentication for LDAP. (If LDAP is not integrated for Atlas, we need to use local OS users present on the node where Atlas server is installed i.e. base-master)

1. Login to Cloudera Manager WebUI. Go to Cluster > Atlas > Configuration.
2. Edit LDAP configuration. Sample configuration is shown in the screenshot below:

Enable LDAP Authentication

atlas.authentication.method.ldap
[atlas_authentication_method_ldap](#)

☒ Atlas Server Default Group ↗

LDAP Server URL

atlas.authentication.method.ldap.url
[atlas_authentication_method_ldap_url](#)

Atlas Server Default Group ↗

ldaps://winjb-ucsg16.cdip.cisco.local:636

User DN Pattern

atlas.authentication.method.ldap.userDNpattern
[atlas_authentication_method_ldap_userDNpattern](#)

Atlas Server Default Group ↗

CN=\$USER\$,CN=cdipadmin,OU=cloudera,DC=cdip,DC=cisco,DC=local

LDAP Group-Search Base

atlas.authentication.method.ldap.groupSearchBase
[atlas_authentication_method_ldap_groupSearchBase](#)

Atlas Server Default Group ↗

CN=cdipadmin,OU=cloudera,DC=cdip,DC=cisco,DC=local

LDAP Group-Search Filter

atlas.authentication.method.ldap.groupSearchFilter
[atlas_authentication_method_ldap_groupSearchFilter](#)

Atlas Server Default Group ↗

(&(objectClass=Group)(sAMAccountName={0}))

LDAP Group-Role Attribute

atlas.authentication.method.ldap.groupRoleAttribute
[atlas_authentication_method_ldap_groupRoleAttribute](#)

Atlas Server Default Group

cn

LDAP DN

atlas.authentication.method.ldap.base.dn
[atlas_authentication_method_ldap_base_dn](#)

Atlas Server Default Group ↗

DC=cdip,DC=cisco,DC=local

LDAP Bind DN Username

atlas.authentication.method.ldap.bind.dn
[atlas_authentication_method_ldap_bind_dn](#)

Atlas Server Default Group ↗

CN=cdpbind,OU=cloudera,DC=cdip,DC=cisco,DC=local

LDAP Bind DN Password

atlas.authentication.method.ldap.bind.password
[atlas_authentication_method_ldap_bind_password](#)

Atlas Server Default Group ↗

•••••

LDAP Referral

atlas.authentication.method.ldap.referral
[atlas_authentication_method_ldap_referral](#)

Atlas Server Default Group ↗

☒ follow
☐ throw
☐ ignore

LDAP User Search Filter

atlas.authentication.method.ldap.user.searchfilter
[atlas_authentication_method_ldap_user_searchfilter](#)

Atlas Server Default Group ↗

(&(objectClass=user)(sAMAccountName={0}))

AD Referral

atlas.authentication.method.ldap.ad.referral
 [atlas_authentication_method_ldap_ad_referral](#)

Atlas Server Default Group 

- ☒ follow
☐ throw
☐ ignore

AD User Search Filter

atlas.authentication.method.ldap.ad.user.searchfilter
 [atlas_authentication_method_ldap_ad_user_searchfilter](#)

Atlas Server Default Group

(sAMAccountName={0})

AD User Default Role

atlas.authentication.method.ldap.ad.default.role
 [atlas_authentication_method_ldap_ad_default_role](#)

Atlas Server Default Group

ROLE_USER

LDAP Authentication Type

atlas.authentication.method.ldap.type
 [atlas_authentication_method_ldap_type](#)

Atlas Server Default Group 

- ☐ none
☒ ldap
☐ ad

Table 12. Atlas LDAP Integration

Component	Value
Enable LDAP Authentication (Atlas):	True (Checked)
LDAP Server URL:	ldap://ipaserver.cldrsetup.local:389/
User DN Pattern:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Group Search filter:	(&(member={1})(objectClass=posixgroup))
LDAP Group Search Base:	cn=groups,cn=accounts,dc=cldrsetup,dc=local
LDAP Group-Role Attribute:	cn
LDAP DN:	dc=cldrsetup,dc=local
LDAP Bind DN Username:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Bind DN Password:	<cloudera123> (password for KDC admin, configured earlier)
LDAP Referral:	follow
LDAP User filter:	(&(uid={0})(objectClass=person))
LDAP Authentication Type:	LDAP
AD Referral: (Only for AD Setup)	follow
AD User Search Filter: (Only for AD Setup)	(sAMAccountName={0})
AD User Default Role: (Only for AD Setup)	ROLE_USER

- 
3. Click on save changes.
 4. Restart Atlas service.

For more details:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/atlas-securing/topics/atlas-configure-ldap-authentication.html>

Configure Hive for LDAP Authentication (Optional– Skip this Step)

LDAP username

hiveserver2_ldap_replication_user

Hive (Service-Wide)

cdpbind

LDAP password

hiveserver2_ldap_replication_password

Hive (Service-Wide)

•••••

Enable LDAP Authentication for HiveServer2

hiveserver2_enable_ldap_auth

☒ Hive (Service-Wide)

LDAP URL

hive.server2.authentication.ldap.url

hiveserver2_ldap_uri

Hive (Service-Wide)

ldaps://winjb-ucsg16.c DIP.cisco.local:636

Active Directory Domain

hive.server2.authentication.ldap.Domain

hiveserver2_ldap_domain

Hive (Service-Wide)

cdip.cisco.local

LDAP BaseDN

hive.server2.authentication.ldap.baseDN

hiveserver2_ldap_basedn

Hive (Service-Wide)

Enable LDAP Authentication for Hive Metastore

hive_metastore_enable_ldap_auth

☒ Hive (Service-Wide)

LDAP URL

hive.metastore.authentication.ldap.url

hive_metastore_ldap_uri

Hive (Service-Wide)

ldaps://winjb-ucsg16.c DIP.cisco.local:636

Active Directory Domain

hive.metastore.authentication.ldap.Domain

hive_metastore_ldap_domain

Hive (Service-Wide)

cdip.cisco.local

LDAP BaseDN

hive.metastore.authentication.ldap.baseDN

hive_metastore_ldap_basedn

Hive (Service-Wide)

Table 13. LDAP Integration-Hive

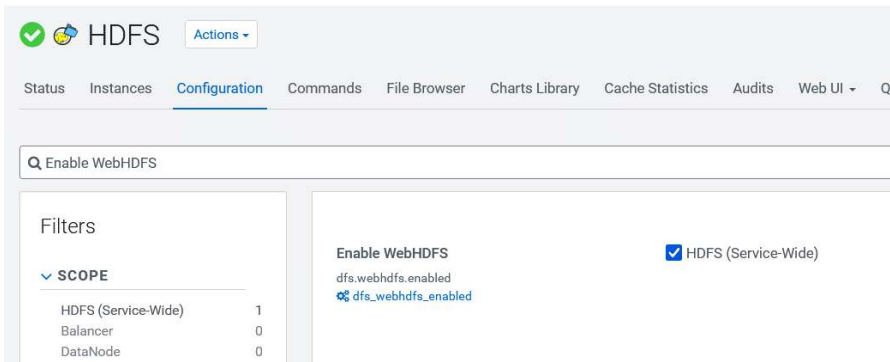
Component	Value
LDAP Username:	admin
LDAP Password:	<cloudera123> (password for KDC admin, configured earlier)
Enable LDAP Authentication for HiveS2:	True (Checked)
LDAP URL:	ldap://ipaserver.cldrsetup.local:389/
LDAP Base DN:	dc=cldrsetup,dc=local
Enable LDAP Authentication for HMS:	True (Checked)
LDAP URL:	ldap://ipaserver.cldrsetup.local:389/
LDAP Base DN:	dc=cldrsetup,dc=local

Configure HDFS properties to optimize log collection (Optional– Skip this Step)

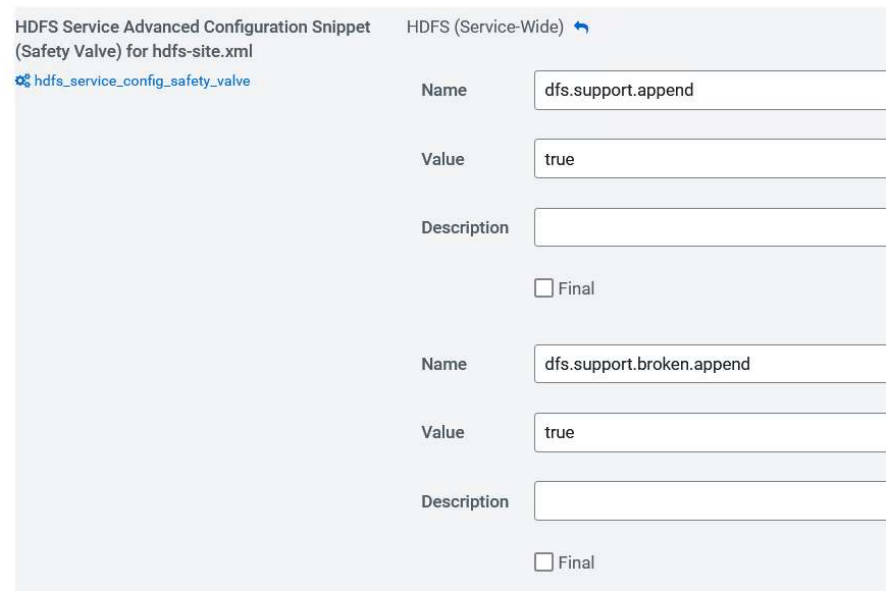
CDP uses “out_webhdfs” Fluentd output plugin to write records into HDFS, in the form of log files, which are then used by different Data Services to generate diagnostic bundles. Over time, these log files can grow in size. To optimize the size of logs that are captured and stored on HDFS, you must update certain HDFS configurations in the hdfs-site.xml file using Cloudera Manager.

- 1. Login to **Cloudera Manager WebUI**.
- 2. Go to **Cluster > HDFS Service > Configuration**.
- 3. Enable **WebHDFS**.

cdip-cdp



- 4. Edit value for HDFS Service Advanced Configuration Snippet (Safety Valve) for hdfs-site.xml as shown in the screenshot below:



- 5. Click Save Changes.
- 6. Restart the HDFS service.
- 7. Restart Cloudera on premises Base cluster.

Cloudera on premises (PvC) Data Services (DS) Installation

Cloudera on premises Data Services lets you deploy and use the Cloudera Data Warehouse (CDW), Cloudera AI a.k.a. Cloudera Machine Learning (CML/CAI), and Cloudera Data Engineering (CDE) Data Services.

This section summarizes Cloudera on premises Data Science v1.5.5 installation through Embedded Container Service on Cloudera on premises Base 7.3.2.

A Cloudera on premises Data Services deployment includes an Environment, a Data Lake, the Management Console, and Data Services (Data Warehouse, Machine Learning, Data Engineering). Other tools and utilities include Replication Manager, Data Recovery Service, CDP CLI, and monitoring using Grafana.

To deploy Cloudera on premises Data Services you need a Cloudera on premises Base cluster, along with container-based clusters that run the Data Services. You can either use a dedicated **RedHat OpenShift container cluster (OCP)** or deploy an **Embedded Container Service (ECS)** container cluster.

The Cloudera on premises deployment process involves configuring Management Console, registering an environment by providing details of the Data Lake configured on the Base cluster, and then creating the workloads.

Platform Managers and Administrators can rapidly provision and deploy the data services through the Management Console, and easily scale them up or down as required.

Cloudera on premises Base provides the following components and services that are used by Cloudera on premises Data Services:

- SDX Data Lake cluster for security, metadata, and governance
- HDFS and Ozone for storage
- Powerful and open-source Cloudera Runtime services such as Ranger, Atlas, Hive Metastore (HMS), etc.
- Networking infrastructure that supports network traffic between storage and compute environments.

Embedded Container Service (ECS) checklist

Use the checklist for Embedded Container Service (ECS) for Cloudera on premises Data Services:

<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/installation-ecs/topics/cdppvc-installation-ds-checklist.html>

Cloudera on premises Data Services software requirements

1. You must have a minimum of one agent node for ECS.
2. Enable TLS on the Cloudera Manager cluster for communication with components and services.
3. Set up Kerberos on these clusters.
4. Ensure that all of the hosts in the ECS cluster have more than **300 GiB** of free space in the each **/var/lib** and **/docker** directories, on each host in a Cloudera on premises Containerized Cluster, at the time of installation. The default docker service uses **/docker** folder. Whether you wish to retain **/docker** or override **/docker** with any other folder, you **must have a minimum of 300 GiB free space**. The hosts in a Cloudera on premises Containerized Cluster that have GPUs are required to have nVidia Drivers and nvidia-container-runtime installed.

Status	Description
❗	A minimum of 16 cores are required for the hosts in a Private Cloud Containerized Cluster. The following hosts do not satisfy the minimum number of cores: > View Details
❗	A minimum of 300 GiB of storage is required in the /var/lib directory and 300 GiB in the /ecs/docker directory for the hosts in a Private Cloud Containerized Cluster. /var/lib/longhorn directory cannot be a symbolic link. The following hosts do not meet these criteria: > View Details

5. **Python 3.8<** is required for Cloudera Manager version 7.11.3.0 and higher versions. Cloudera Manager agents will not start unless Python 3.8 is installed on the cluster nodes.

6. The cluster generates multiple hosts and host-based routing is used in the cluster in order to route it to the right service. You must decide on a domain for the services which Cloudera Manager by default points to one of the host names on the cluster. However, during the installation, you should check the default domain and override the default domain (only if necessary) with what you plan to use as the domain. The default domain must have a [wildcard DNS entry](#). For example, `*.apps.cldrsetup.local`.

7. It is recommended that you leave IPv6 enabled at the OS level on all ECS nodes.

8. Take care that enough disk space is available on each host in the ECS cluster.

9. On each of the ECS hosts, create three partitions of the attached 2T EBS volume (non-root) and mount those partitions as volumes `/lhdata` `/cdwdata` `/docker`. (`/docker` is used for docker cache, `/lhdata` for LongHorn NFS Volume storage, and `/cdwdata` for local storage and CDW)

10. ECS requires JDK, krb5-workstation, krb5-libs, NTP, iptables packages to be installed on all hosts.

11. Enable cgroup v1 in Red Hat Enterprise Linux 9. **(optional, skip this step)**

```
# In RHEL 9 cgroup-v2 is enabled by default, follow steps below to enable cgroup v1:
```

```
# Check if the cgroup-v2 is mounted currently as default.
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "mount | grep cgroup"
```

```
# Add the kernel command line parameter systemd.unified_cgroup_hierarchy=0 & systemd.legacy_systemd_cgroup_controller.
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a 'grubby --update-kernel=/boot/vmlinuz-$(uname -r) --args="systemd.unified_cgroup_hierarchy=0\nsystemd.legacy_systemd_cgroup_controller"'
```

```
# Reboot jump system for changes to take effect.
```

```
[root@ipaserver ~]# ansible ecsmasternodes,ecsnodes -m shell -a "systemctl reboot"
```

```
# Verify the changes after reboot:
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "cat /proc/cmdline"
```

```
BOOT_IMAGE=(hd0,gpt2)/vmlinuz-5.14.0-162.23.1.el9_1.x86_64 root=/dev/mapper/rhel-root ro crashkernel=1G-4G:192M,4G-64G:256M,64G-:512M\nresume=/dev/mapper/rhel-swap rd.lvm.lv=rhel/root rd.lvm.lv=rhel/swap rhgb quiet systemd.unified_cgroup_hierarchy=0\nsystemd.legacy_systemd_cgroup_controller
```

```
# Mount shows legacy cgroup-v1 mounted now.
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "mount | grep cgroup"
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "ll /sys/fs/cgroup"
```

12. For CAI, you must install `nfs-utils` in order to mount longhorn-nfs provisioned mounts. The `nfs-utils` package is required on every node of the ECS cluster. Run this command `dnf install nfs-utils` to install `nfs-utils`.

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "dnf install -y nfs-utils iscsi-initiator-utils"
```

13. Check `iptables` is installed and working fine.

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "iptables -L"
```

14. If `iptables` is not working as expected or giving command not found error then remove the package on all ecs nodes and re-install.

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "dnf remove -y iptables && dnf install -y iptables && iptables -L"
```

15. If `iptables` is not present on ecs cluster nodes, then Install `iptables`, and verify if it is working as expected.

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "dnf install -y iptables && iptables -L"
```

16. For nodes with NVIDIA GPU (if applicable), ensure that the GPU hosts have nVidia Drivers and `nvidia-container-runtime` installed. You must confirm that drivers are properly loaded on the host by executing the command `nvidia-smi`. You must also install the `nvidia-container-toolkit` package. **(skip this step)**

17. You must install **nvidia-container-toolkit** (if applicable). (**nvidia-container-runtime** is migrated to **nvidia-container-toolkit**, see Migration Notice.) The steps for this are shown in the [NVIDIA Installation Guide](#). If using Red Hat Enterprise Linux (RHEL), use dnf to install the package. See Installing the [NVIDIA Container Toolkit](#). **(skip this step)**

Verify linux version:

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "uname -m && cat /etc/*release"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "uname -a"
```

Verify GCC installation and version

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "gcc --version"
```

Verify nodes with NVIDIA GPU installed:

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "lspci -nnv | grep -i nvidia"
```

Set subscription to RHEL9.x

```
[root@jump ~]# ansible all -m shell -a "subscription-manager release --set=9.4"
[root@jump ~]# ansible all -m shell -a "subscription-manager release --show"
[root@jump ~]# ansible all -m shell -a "sudo dnf clean all"
[root@jump ~]# ansible all -m shell -a "sudo rm -rvf /var/cache/dnf"
```

(Optional if not completed prior) Enable optional repos - On RHEL 9 Linux only

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "subscription-manager repos --enable=rhel-9-for-x86_64-appstream-rpms"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "subscription-manager repos --enable=rhel-9-for-x86_64-baseos-rpms"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "subscription-manager repos --enable=codeready-builder-for-rhel-9-x86_64-rpms"
```

Install kernel headers and development packages for the currently running kernel

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo dnf install kernel-devel-$(uname -r) kernel-headers-$(uname -r)"
```

Remove outdated Signing Key:

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo rpm --erase gpg-pubkey-7fa2af80"
```

Download and Install NVIDIA CUDA Toolkit [This exercise documented with CUDA 12.3.2 for RHEL 9 rpm(local) installation]

```
[root@ipaserver ~]# wget
https://developer.download.nvidia.com/compute/cuda/12.2.2/local_installers/cuda-repo-rhel9-12-2-local-12.2.2_535.104.05-1.x86_64.rpm
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m copy -a "src=/root/cuda-repo-rhel9-12-2-local-12.2.2_535.104.05-1.x86_64.rpm
dest=/root/cuda-repo-rhel9-12-2-local-12.2.2_535.104.05-1.x86_64.rpm"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo rpm --install cuda-repo-rhel9-12-2-local-12.2.2_535.104.05-1.x86_64.rpm"
```

From the NVIDIA Driver Downloads page, download NVIDIA Driver <https://www.nvidia.com/download/index.aspx> as per the GPU, OS and NVIDIA CUDA version.

```
[root@jump ~]# wget https://nvidia.github.io/libnvidia-container/stable/rpm/nvidia-container-toolkit.repo
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m copy -a "src=/root/nvidia-driver-local-repo-rhel9-535.161.07-1.0-1.x86_64.rpm
dest=/root/nvidia-driver-local-repo-rhel9-535.161.07-1.0-1.x86_64.rpm"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo rpm --install nvidia-driver-local-repo-rhel9-535.161.07-1.0-1.x86_64.rpm"
```

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo dnf clean all"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo dnf -y module install nvidia-driver:latest-dkms"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo dnf -y install cuda"
```

Enable nvidia-persistenced services:

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo systemctl enable nvidia-persistenced.service"
```

Reboot the machine:

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo reboot"
```

After the machine boots, verify that the NVIDIA drivers are installed properly:

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo nvidia-smi"
```

Installing with dnf

Configure the production repository:

```
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m copy -a "src=/root/nvidia-container-toolkit.repo dest=/etc/yum.repos.d/nvidia-container-toolkit.repo"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo dnf config-manager --enable nvidia-container-toolkit-experimental"
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m shell -a "sudo dnf install -y nvidia-container-toolkit"
```

If you are installing ECS on RHEL 8 or RHEL 9: *(Not Required for Cloud Based Servers, perform only on Datacenter/Bare-metal environments for RKE Based ECS clusters)*

```
##### Run the following command to check to see if the nm-cloud-setup.service and nm-cloud-setup.timer services are enabled:
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m copy -a "systemctl status nm-cloud-setup.service nm-cloud-setup.timer"
##### If the nm-cloud-setup.service and nm-cloud-setup.timer services are enabled, disable them by running the following command on each host you added:
[root@jump ~]# ansible ecsmasternodes,ecsnodes -m copy -a "systemctl disable nm-cloud-setup.service nm-cloud-setup.timer && systemctl stop nm-cloud-setup.service nm-cloud-setup.timer"
##### For more information, see Known issues and limitations.
If you disabled the nm-cloud-setup.service and nm-cloud-setup.timer services, reboot the added hosts.
```

Note: For more information, see: [Known issues and limitations](#).

Note: Prepare Cloudera on premises Base for the Cloudera on premises Data Services installation:
<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/installation-ecs/topics/cdppvc-installation-ecs-prepare-cdp-private-cloud-base.html>

Note: Use this checklist to ensure that your Cloudera on premises Base is configured and ready for installing Cloudera on premises Data Services:
<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/installation-ecs/topics/cdppvc-installation-pvcbase-checklist.html>

Note: Use this checklist to ensure that your Embedded Container Service (ECS) is configured and ready for installing Cloudera on premises Data Services:
<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/installation-ecs/topics/cdppvc-installation-ds-checklist.html>

Checklist Item	Details
Docs Links	Docs - ECS Checklist Docs - ECS System Requirements Support Matrix
ClouderaManager version	7.11.3 - CHF6 (minimum)
LDAP Configuration	CM is configured for LDAP, and you have a copy of the CA trust chain that signed your LDAP server's cert (i.e. root cert & intermediates if any).
LDAP Bind User Account	You have the account and password for LDAP Bind user account.
Base Cluster & CM Agents TLS	TLS: AutoTLS? Yes/No TLS: AutoTLS, signed with customer CA? Yes/No TLS: Manual, signed with customer CA? Yes/No

	TLS: Manual self signed (not recommended) xxx
Kerberos Admin Account	CM is setup with Kerberos admin account (cloudera-scm account & password).
ECS Installer Prerequisites Checker	IMPORTANT! CM ECS installer now has an Enforcing prerequisites checker, install will fail if your infra does not meet requirements outlined in docs. See item 14, in doc.
Runtime Version	7.3.2 (CHF6+) is installed & configured. See support matrix
New Changes with 1.5.5	
Minimum Required Components	Zookeeper, HDFS, Ozone, HBase, Hive Metastore, Kafka, Solr-Infra, Ranger, Atlas, YARN (optional for CDW, but required for Spark pushdown in CDE, CAI).
Base Components TLS	All Base components configured with TLS.
Base Components Kerberos	All Base components configured with Kerberos.
CDW Base Cluster HMS	CDW - Base HMS, mTLS can be used instead of user/password. If using CDW, the Base Cluster HMS is configured to allow TLS connections (set to allow TLS, but not forced to require it).
Wildcard DNS Record	CRITICAL: A wildcard subdomain (called app_domain in cm) has been created in DNS. For AD this usually means a subdomain folder + an "A record" with name = "" <i>pointing to the ECS host and a CNAME record. The "A record" does not need a reverse PTR.</i> AD Example: If corp domain is "company.com", then create a new subdomain folder called "ecs-dev", then within that create AD another called "apps", then create a "A record" inside that folder, its name is "", its IP will be the IP of the host you will install ECS Server onto. Verify with dig utility e.g., <code>\$ dig foobar.apps.cdppvc.com</code> . Look for "ANSWER" section in the result.
DNS Resolver	DNS needs to be the primary resolver, do not use entries in /etc/hosts (except for localhost).

ECS Nodes File System	RHEL 8 & RHEL 9 - disk partition that backs <code>"/var"</code> should be xfs file system (using the default <code>ftype = 1</code>), this is usually the root partition <code>/</code> .
RHEL/Centos 7.x Support	New in 1.5.5: RH/Centos 7.x no longer supported. If running RHEL 7.x, you must upgrade to a higher version before installing Cloudera on premises Data Services.
Hostname Configuration	Hostname must = FQDN, use <code>hostnamectl set-hostname <fqdn></code> .
/etc/resolv.conf Entries	<code>/etc/resolv.conf</code> must NOT have more than 3 "nameserver" entries.
MTU Size	Ensure that net interfaces are configured to a maximum transmission unit size (MTU) not smaller than 1450, (typical MTU is 1500), required by the Calico CNI plugin.
Storage Devices	Validate that Storage devices meet Sys Requirements (see docs) and are mounted using a consistent naming convention. Devices are expected to be SCSI block devices (NFS mounts do not work). Devices for CDW Cache and Block should be separate mounts. Use a Logical Volume Manager if you have multiple devices per type.
Disk Mounts	All disk mounts should be xfs(type1), including root partition. The partition backing <code>/var/lib</code> must have at least 300GB free after install of CM agent. If 300GB free is not available, then must symlink <code>"/var/lib/docker"</code> & <code>"/var/lib/rancher"</code> from another partition (must be xfs).
Required Mounts	3 mounts required per host, name them any you like, DO NOT USE symbolic links. See: https://longhorn.io/docs/1.5.5/volumes-and-nodes/multidisk/#use-an-alternative-path-for-a-disk-on-the-node 1) A xfs mount for Docker storage (used for install staging, CAI model building, and embedded repo storage) at least 300GB 2) A xfs mount for CDW Cache, this is a physical NVMe device, sizing dependent on use case, consumed in 600GB chunks per executor & coordinator 3) A xfs mount for Longhorn Block storage which provides k8s Persistent Volumes (PVs), min 1 TB per node, vendor recommends this mount should be a Logical Volume (from LVM2) (for future expansion needs)

Passwordless Sudo User	You have credentials for a passwordless sudo user for each host. If you don't have these credentials, then you will have to install CMAgents manually (not using wizard) and configure them for TLS manually.
CM Agents Configuration	Nodes are configured with requirements to run CM Agents.
iptables Configuration	Do NOT disable iptables, but ensure iptables is set to defaults. Any customer-set chains/rules must be un-set. (firewalld not supported, nftables not yet certified). Check if <code>/etc/sysconfig/iptables</code> exists and inform the customer to remove it. Ensure no system management software like puppet, ansible, etc., will recreate it on reboot. ECS will attempt to install iptables-services if it's not present.
RHEL8 - iptables	Install iptables like: <code># yum --setopt=tsflags=noscripts install -y iptables</code> . See docs: ECS Installation Steps .
RHEL8 - Net Manager	Services <code>nm-cloud-setup.service</code> and <code>nm-cloud-setup.timer</code> must be stopped (if they exist) and disabled. This is seen on AWS RHEL images, unlikely to see this on-prem. (Risky, as we found instances can get corrupt, due to network connectivity may lost from AWS side)
RHEL8 - VMware	Certain versions of VMware virtual network interface (vmxnet3) can cause what seems like intra-pod packet drops. This is due to the Calico Issue . Workaround: See case Cloudera Support .
RHEL9 - iptables	Check iptables configuration as described for RHEL8.
Time Service	Host has chronyd or NTP running.
DNS Resolving	Each ECS host is forward and reverse DNS resolvable.
DNS Forward and Reverse	Each ECS host can forward and reverse DNS resolve each base host.
SELinux and System Settings	SELinux off or set to Permissive, <code>vm.swappiness = 1</code> , Transparent Huge Pages Off (1.3.4+ support SELinux, see docs).
NFS Requirements	Linux package "nfs-utils" must be installed for CAI & CDE.

GPU Requirements & Config	See docs for GPU requirements & config for CAI.
3rd Party Software	Certain 3rd party network monitors/firewalls may seriously interfere with ECS cluster traffic. Illumio... Illumio runs an agent on each host, uses iptables and can block traffic if it is running in Primary Firewall Mode. That mode must be set to = off. See Illumio Docs. VMware NSX-T can create blocking firewall rules. See NSX-T Jira.
Air Gapped Customers	If a customer is Air Gapped, they must download all the ECS bits and stage them behind an HTTP server. Make sure they do this right away as this is 150GB of content (circa 1.5.0).
License Key	In every case, Customer (or Cloudera employee) MUST have a valid license key that includes an entitlement for PrivateCloud!
Ingress Certificates	It is usually better to have a customer CA signed ECS Ingress domain cert, but not required as ECS will sign using an RKE CA cert (the RKE CA will have to be pre-trusted on a user's machine, or the browser will show "insecure" TLS connection). This ECS generated CA is not managed by CM's auto-TLS, so for JDBC/impala-shell/beeline/other connections, you must add this to a client truststore manually. The ECS generated CA will have a CN=rke2-server-ca@xxxxxxxxxx and is located at /var/lib/rancher/rke2/server/tls/server-ca.crt.
Customer CA Signed Certs	For customer CA signed certs: This cert must include 2 SubjectAltNames. If wildcard format is not allowed, let the installer use the RKE CA. Example: DNS.1 = *.apps.cdppvc.com DNS.2 = apps.cdppvc.com
New in 1.5:	
No Wildcard TLS	There is an option to deploy when wildcard SAN entries are not allowed. See doc for method & limits: No wildcard TLS .
CAI Workbench Certificates	If installing in a network domain that requires strict host checking, HSTS, you will need to use TLS for CAI Workbenches. Fun fact: "cloudera.com" requires HSTS, which is usually not seen with customers but is emerging. This cert must include SubjectAltNames.

	DNS.1 = *.xxxx-xxxx.apps.cdppvc.com (xxxx-xxxx = the CAI workspace ID)
CDE Virtual Cluster Certificates	Same as CAI for HSTS. CDE includes a utility to generate RKE signed certs for each virtual cluster. If you make your own cert for a CDE virtual cluster, this cert must include SubjectAltNames. DNS.1 = *.xxxx-xxxx.apps.cdppvc.com (xxxx-xxxx = the CDE virtual cluster ID)
CDW HMS Configuration	CDW can be configured to allow non-TLS connections for HMS db. See CDW Documentation .
FreeIPA Support	FreeIPA is now supported. "LDAP Group Search Filter" in Mgmt Console must include !(cn=admins)), or group sync will break e.g. (&(member={0})(objectclass=posixgroup)!(cn=admins)). Must alter the /etc/krb5.conf - comment out includedir directives. In CM7.10.1, CM generated "local to auth" krb5 rules include embedded "\"Q" and "\"E" chars, this is legal for base but causes a parsing problem for CDW. Contact for workaround (Should be patched in 1.5.1 - CHF1).
Fixed in 1.5.0:	
/etc/resolv.conf Entries	/etc/resolv.conf must NOT have more than 2 "search" entries, causes a side-effect in Impala Statestore pod. Per Jira .
External DS Metadata DB	New: External DS metadata db is deprecated. This will be a problem for "legacy" customers as we have no migrate from external to embedded db.
Container Repos	Able to use customer-owned container repos is now supported e.g., Artifactory, Nexus, Harbor, Docker dist.
Iceberg Tables	Support for Iceberg Tables.
CDSW to CAI Migration	Tech Preview: CDSW to CAI migration. See release notes.
New in 1.4.1:	

/var/lib Storage Requirements	/var/lib must have 100GB.
Docker Storage Requirements	Docker storage must have 200GB.
vCores Requirements	Min 16 vcores required (on Masters too? Yes).
iptables-save	iptables-save to /var/lib/ecs/iptables.save (we don't need to flush IPtables upon reinstall anymore).
Oracle HMS	Oracle HMS is now GA. This can get complicated as many Oracle customers use TLS + Client cert auth, which we don't support. We support TLS + user password auth only. Contact for workaround.
Data Viz	Data Viz is GA.
Ozone for CDW	Ozone for CDW is Tech Preview.
Impala Custom Pod Sizes	Impala Custom pod sizes now GA. See Impala Pod Configs .
CDE Resource Limits	CDE Resource Limits - Tech Preview.
/etc/resolv.conf Entries (2)	/etc/resolv.conf must NOT have more than 2 "search" entries, causes a side-effect in Impala Statestore pod. Per Jira .
New In 1.3.4:	
Cloudera Manager	Cloudera Manager now prevents ECS Server hosts from running workloads. ECS Servers are masters only. See Cloudera Manager Release Notes .
ECS Hosts Workloads	ECS hosts can now be configured to reserve hosts for workloads that require GPU drivers. See GPU Node Setup .



SELinux Support	SELinux is now supported for ECS clusters. See SELinux Documentation .
-----------------	--

Installing Cloudera on premises Data Services using ECS

Follow the steps here to install Cloudera on premises Data Services with the *Embedded Container Service (ECS)*.

Follow the steps outlined below to add hosts to be part of the Cloudera on premises Data Services cluster and the install ECS (embedded container service) through either internet or air gapped method.

<https://docs.cloudera.com/cdp-private-cloud-data-services/latest/installation-ecs/topics/cdppvc-installation-ecs-steps.html>
<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/index.html>

Note: We will be installing Cloudera on premises Data Services via the internet method.

Note: For more details on dedicating ECS node for specific workload type please visit:

<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/managing-ecs/topics/cm-managing-ecs-dedicating-nodes-for-workloads.html>

Note: If you do not have entitlements to access <https://archive.cloudera.com/p/cdp-pvc-ds/latest/>, contact your Cloudera account team to get the necessary entitlements.

Latest ECS Supported Version Of Cloudera-Manager is 7.13.2 CHF1

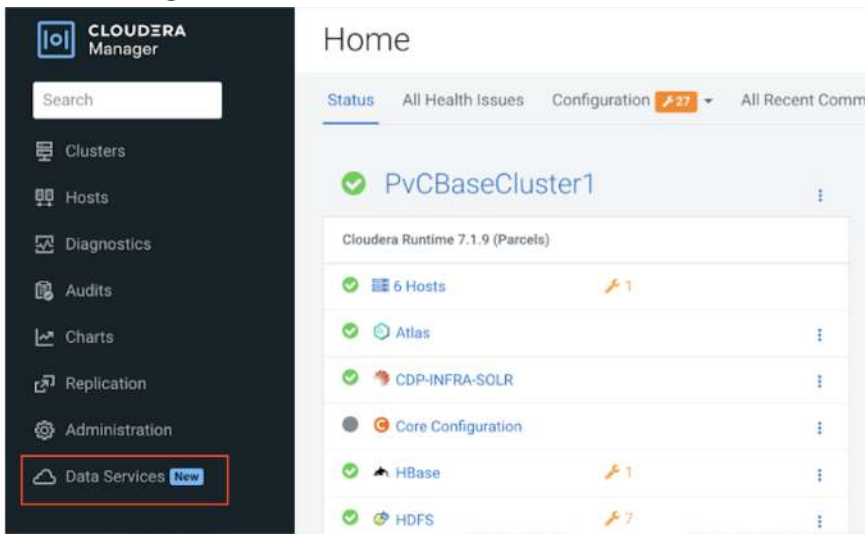
Installing ECS Cluster

Follow the steps in this topic to install Cloudera on premises Data Services with the Embedded Container Service (ECS).

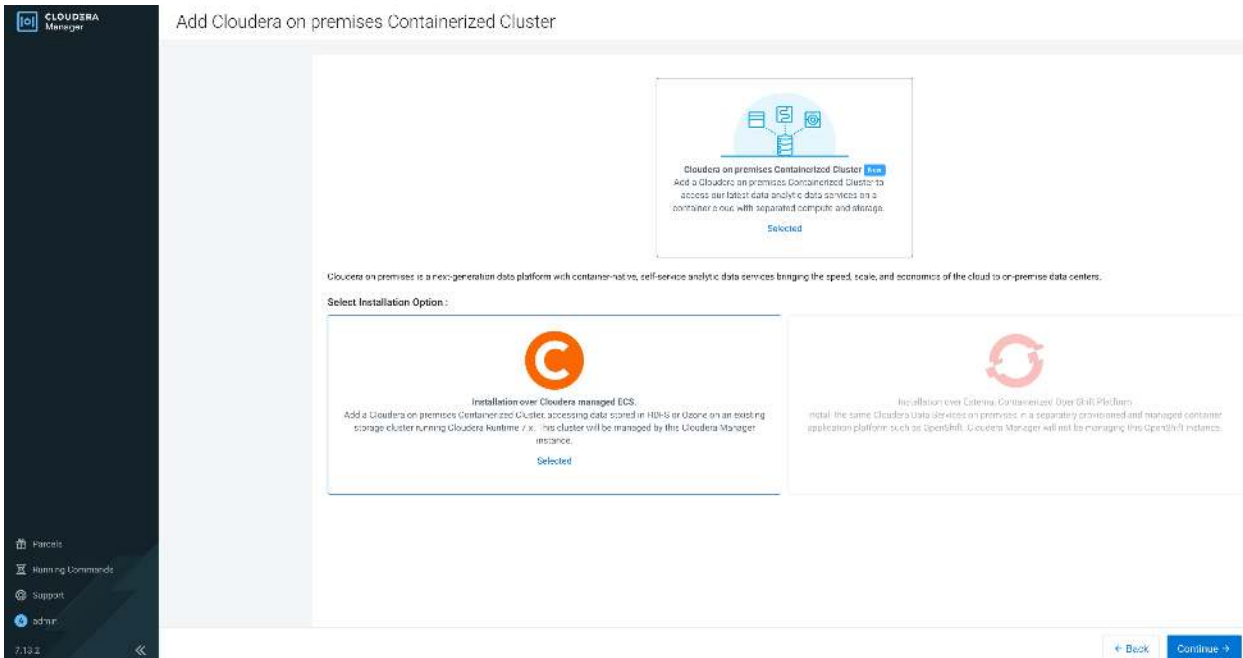
Important:

RHEL 7.x support on ECS has been dropped in Cloudera on premises Data Services 1.5.5 and higher versions. If you are running RHEL 7.x, you must upgrade to a higher version before installing Cloudera on premises Data Services.

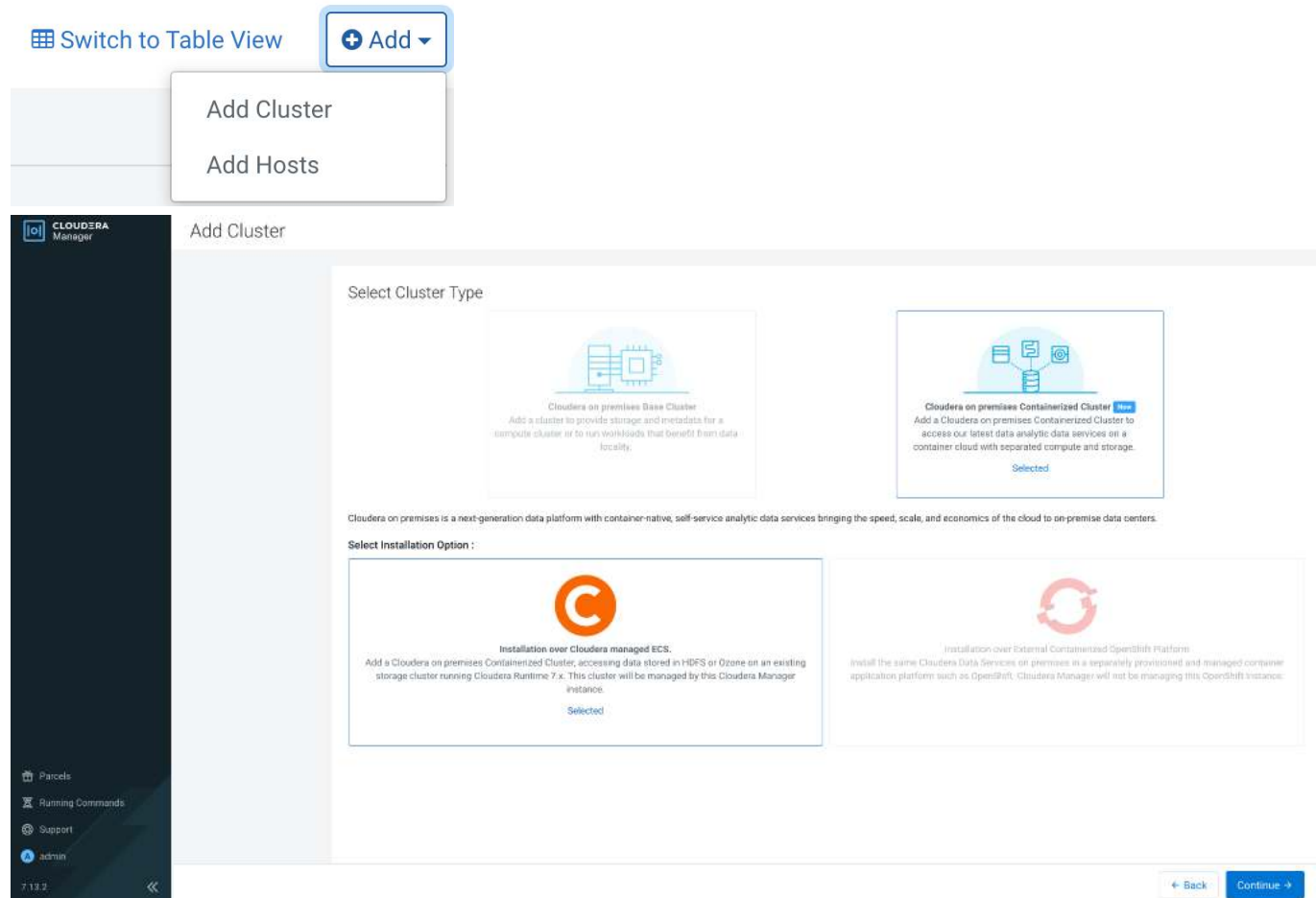
1. In the **Cloudera Manager WebUI** console, go to the **Data Services** page by clicking on the **Data Services** link on the Pane located at the Left Hand side of the browser window. Alternatively, you can also click **(+) Add > Add Cluster** at the top right in Cloudera Manager, then select **Private Cloud Containerized Cluster** as the cluster type.



2. The **Add Private Cloud Containerized Cluster** page appears. Click **Continue** on the page.



Note: Alternatively, you can also click **(+) Add > Add Cluster** at the **top right** in **Cloudera Manager**, then select **Private Cloud Containerized Cluster** as the **cluster type**, then click **Continue**.



Step. Getting started

- On the getting started page of the installation wizard, select either **Internet** or **Air Gapped** as the Install Method. **(We are going here with Internet Installation method only)**
- If you select the **Internet Install** Method option on the **Getting Started** page, images are copied over the internet from the Cloudera repository. For this deployment, we will select the **Internet** as the install method. **Select Repository.** (To use a custom repository link provided to you by Cloudera, click **Custom Repository, instructions for this are mentioned in later steps**):

Add Private Cloud Containerized Cluster

1 Getting Started

2 Cluster Basics

3 Specify Hosts

4 Assign Roles

5 Configure Docker Repository

6 Configure Data Services

7 Configure Databases

8 Install Parcels

9 Check Prerequisites

10 Inspect Cluster

11 Install Data Services

12 Summary

Getting Started

This wizard provides step-by-step guidance for installing CDP Private Cloud Containerized cluster.

Installation of the CDP Private Cloud Data Services components (for trial purposes or for production use) requires an appropriate license key.

Visit the [CDP Private Cloud Installation](#) documentation for more information.

Install Method

☒ Internet ☐ Air Gapped

1. Select Repository

You are about to install CDP Private Cloud Data Services version **1.5.4-h5-b104**.

Before you start, verify the following prerequisites:

- A Cloudera Runtime cluster running 7.1.7 SP3, 7.1.9 or 7.1.9 SP1 that include the required services (Hive, Ranger, Atlas, HDFS).
- Ozone is required for CDE.
- Kerberos has been setup on the cluster using an MIT KDC or Active Directory.
- TLS has been enabled on the cluster.

What's new in version **1.5.4-h5-b104**.

- [Release Notes](#)
- RHEL 7.x support has been removed for CDP Private Cloud Data Services 1.5.4 and above. Please ensure that prior to upgrading the Data Ser and upgrades will fail for CDP Private Cloud Data Services if the OS requirement is not met. Please note that this restriction applies to ECS de
- Cloudera Manager 7.11.3 CHF8 does not support any ECS deployments of CDP Private Cloud Data Services.

Note: Verify if the version shown below is the same as the version that we are willing to install i.e. **1.5.5-h3300-b26** for current setup.

Install Method

☒ Internet ☐ Air Gapped

1. Select Repository

You are about to install CDP Private Cloud Data Services version **1.5.4-h5-b104**.

If not (as shown in below screenshot), then we need to add-up an additional custom URL and configure that to use our specific version.

Custom Repository

Configuration

CDP Private Cloud Repository URLs

+

-

+

-

Undo

Reason for change:

Modified CDP Private Cloud Repository URLs

Cancel

Save Changes

Add the additional URL by (+) and click on **Save Changes**.

Install Method

☒ Internet ☐ Air Gapped

1. Select Repository

<https://archive.cloudera.com/p/cdp-pvc-ds/1.5.4-h4/>

You are about to install CDP Private Cloud Data Services version **1.5.4-h4-b27**.

5. When you select the **Air Gapped** install option, extra steps are displayed. Follow these steps on the *cldr-mngr* node (our bits server), to download and mirror the Cloudera archive URL using a local HTTP server:

6. (For installing via a local mirror with an http server. You will need to set up a full mirror of Cloudera's repositories via a temporary HTTP server within the perimeter network of all hosts.): **(Skip this step, as we will choose Internet method in next steps to Install)**

Getting Started

This wizard provides step-by-step guidance for installing CDP Private Cloud Containerized cluster.

Installation of the CDP Private Cloud Data Services components (for trial purposes or for production use) requires an appropriate network configuration.

Visit the [CDP Private Cloud Installation](#) documentation for more information.

Install Method

☐ Internet ☒ Air Gapped

Installing via a local mirror with an http server. You will need to setup a full mirror of Cloudera's repositories via a temporary HTTP server within the perimeter network of all hosts.

1. Download everything under <https://archive.cloudera.com/p/cdp-pvc-ds/latest>

```
$ wget -l 0 --recursive --no-parent -e robots=off -nH --cut-dirs=2 --reject="index.html*" https://archive.cloudera.com/p/cdp-pvc-ds/latest
```

2. Modify the file `manifest.json` inside the downloaded directory, change `"http_url": "..."` to `"http_url": "http://your_local_repo/cdp-pvc-ds/latest"`

3. Mirror the downloaded directory to your local http server, e.g. `http://your_local_repo/cdp-pvc-ds/latest`

4. Add `http://your_local_repo/cdp-pvc-ds/latest` to your [Custom Repository](#) settings and select it from the drop-down menu.

```
[root@cldr-mngr ~]# mkdir -p /var/www/html/cloudera-repos/cdp-pvc-ds/
[root@cldr-mngr ~]# cd /var/www/html/cloudera-repos/cdp-pvc-ds/
```

Download everything under <https://archive.cloudera.com/p/cdp-pvc-ds/latest/> to your local httpserver, e.g. http://your_local_repo/cdp-pvc-ds/latest/ using the below command

```
[root@cldr-mngr cdp-pvc-ds]# wget -l 0 --recursive --no-parent -e robots=off -nH --cut-dirs=2 --reject="index.html*" -t 10 https://<username>:<password>@archive.cloudera.com/p/cdp-pvc-ds/latest/
[root@cldr-mngr cdp-pvc-ds]#
```

```
[root@cldr-mngr cdp-pvc-ds]# ls -lt 1.5.5-h3300/
total 116300
-rw-r--r-- 1 root root 284820 Mar 15 10:13 manifest.json
-rw-r--r-- 1 root root 118747085 Mar 15 10:13 cdp-private-1.5.5-h3300-b26.tgz
drwxr-xr-x 2 root root 4096 Mar 15 10:13 parcels
drwxr-xr-x 2 root root 4096 Mar 15 10:12 manifests
drwxr-xr-x 2 root root 32768 Mar 15 10:12 images
[root@cldr-mngr ~]#
```

Modify the `manifest.json` file inside the downloaded directory. Change `"http_url": "..."` to `"http_url": "http://your_local_repo/cloudera-repos/cdp-pvc-ds/1.5.5-h3300/"`

```
[root@cldr-mngr ~]# vi manifest.json
"http_url": "http://192.168.1.38/cloudera-repos/cdp-pvc-ds/1.5.5-h3300/"
[root@cldr-mngr ~]#
```

7. Click **Custom Repository**. Add http://your_local_repo/cloudera-repos/cdp-pvc-ds/1.5.5-h3300/ as a custom repository. Click on **Save Changes**. (Skip this step, as we have chosen Internet method steps to Install)

The Configuration dialog box shows two input fields for CDP Private Cloud Repository URLs. The first field contains 'https://archive.cloudera.com/p/cdp-pvc-ds/latest' and the second field contains 'http://192.168.1.38/cloudera-repos/cdp-pvc-ds/1.5.4-h2'. There are icons for adding, removing, and refreshing the list. Below the fields is an 'Undo' button. At the bottom, there is a 'Reason for change' field with the text 'Modified CDP Private Cloud Repository URLs', and 'Cancel' and 'Save Changes' buttons.

8. Click the **Select Repository** drop-down and select http://your_local_repo/cloudera-repos/cdp-pvc-ds/1.5.5-h3300/ (Skip this step, as we will chosen Internet method steps to Install)

The 'Add Private Cloud Containerized Cluster' wizard is shown at the 'Getting Started' step. The left sidebar lists steps from 1 to 12. The main content area provides instructions for installing the CDP Private Cloud Data Services components. It includes a section for 'Install Method' with radio buttons for 'Internet' and 'Air Gapped'. Below this, it states 'Installing via a local mirror with an http server' and lists five steps: 1. Download everything under a specific URL, 2. Modify the file manifest.json, 3. Mirror the downloaded directory to a local http server, 4. Add the local repo URL to the Custom Repository settings, and 5. Select Repository. A dropdown menu for 'Select Repository' shows 'http://192.168.1.38/cloudera-repos/cdp-pvc-ds/1.5.4-h2' selected. At the bottom, there are 'Cancel', 'Back', and 'Continue' buttons.

9. Click **Continue**.

Step. Cluster Basics

10. On the **Cluster Basics- Add Private Cloud Containerized Cluster** page, enter the **Cluster Name** for the **Cloudera on premises Data Services (ECS) cluster** that you want to create in the **Cluster Name** field. From the **Base Cluster** drop-down list,

select the **Cloudera on premises Base Cluster** (which is created earlier i.e. **PvCBaseCluster1**), that has the storage and SDX services that you want this new Cloudera on premises Data Services instance to connect with. Click **Continue**.

Add Private Cloud Containerized Cluster

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Inspect Cluster

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Summary

Cluster Basics

Cluster Name


Private Cloud Containerized Cluster
A Private Cloud Containerized Cluster helps you to install and run CDP Private Cloud Data Services such as Machine Learning and Data Warehouse with data from an existing Base Cluster. Learn more at [CDP Private Cloud Containerized Cluster](#)

Base Cluster

☐ Use Default Configuration
Use embedded Docker Repository, Vault and Database with default settings, and use default configurations for Role Assignments. Not recommended for production.

Cancel

Back

Continue

Step. Specify Hosts

11. On the **Specify Hosts** page, hosts that have already been added to Cloudera Manager are listed on the **Currently Managed Hosts** tab. If the intended ECS nodes are not added previously (in case of a fresh setup) and you can't see them listed under **Currently Managed Hosts** tab, Click on **New Hosts** tab.

Note: To specify the hosts that are part of the cluster, enter Fully Qualified Domain Names (FQDNs)/hostnames or IP addresses in the Hostname field, or provide a list of search patterns/ IP address range of available matching ECS hosts. Host names must be in lowercase. If you use uppercase letters in any host name, the cluster services will not start after enabling Kerberos.

Note: You can Provide host pattern pvcbase-master, pvcbase-worker[1-3] or pvcbase-worker[1-3].cldrsetup.local etc separated with a new line and Click on **Search**. Cloudera Manager will "discover" the hosts based on matching the pattern provided by you to add in the cluster. Verify that all desired nodes have been found and **selected for installation**. Verify host entries, **deselect** any that you do not want to install services on. Select and/or deselect one or more of these hosts to add to the ECS cluster and click **Continue**.

```
pvcecs-master.cldrsetup.local
pvcecs-worker[1-10].cldrsetup.local
```

Note: Click the pattern link under the Hostname box to display more information about allowed **FQDN** patterns.

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Summary

Specify Hosts

Currently Managed Hosts (0/1 Selected) New Hosts (11 Selected)

Hosts should be specified using the same hostname (FQDN) that they will identify themselves with.

Hostname

pvccs-master.cdppvcds.com
pvccs-worker{1-10}.cdppvcds.com

Hint: Search for hostnames or IP addresses using pattern

SSH Port

22

Search

11 hosts scanned, 11 running SSH.

	Expanded Query	Hostname (FQDN) 1	IP Address	Currently Managed	Result
☒	pvccs-master.cdppvcds.com	pvccs-master.cdppvcds.com	172.31.40.79	No	Host was successfully scanned.
☒	pvccs-worker1.cdppvcds.com	pvccs-worker1.cdppvcds.com	172.31.47.98	No	Host was successfully scanned.
☒	pvccs-worker10.cdppvcds.com	pvccs-worker10.cdppvcds.com	172.31.45.62	No	Host was successfully scanned.
☒	pvccs-worker2.cdppvcds.com	pvccs-worker2.cdppvcds.com	172.31.33.114	No	Host was successfully scanned.
☒	pvccs-worker3.cdppvcds.com	pvccs-worker3.cdppvcds.com	172.31.39.35	No	Host was successfully scanned.
☒	pvccs-worker4.cdppvcds.com	pvccs-worker4.cdppvcds.com	172.31.41.212	No	Host was successfully scanned.
☒	pvccs-worker5.cdppvcds.com	pvccs-worker5.cdppvcds.com	172.31.45.196	No	Host was successfully scanned.
☒	pvccs-worker6.cdppvcds.com	pvccs-worker6.cdppvcds.com	172.31.32.137	No	Host was successfully scanned.
☒	pvccs-worker7.cdppvcds.com	pvccs-worker7.cdppvcds.com	172.31.32.61	No	Host was successfully scanned.
☒	pvccs-worker8.cdppvcds.com	pvccs-worker8.cdppvcds.com	172.31.40.109	No	Host was successfully scanned.
☒	pvccs-worker9.cdppvcds.com	pvccs-worker9.cdppvcds.com	172.31.39.141	No	Host was successfully scanned.

Cancel

Back

Continue

Step. Select JDK

12. On the **Select JDK** page, select any one from the below options:
 - **Manually manage JDK (Select this option)** (manual installation of JDK11/17 with CDH 7.3.2+).
 - Install a Cloudera-provided version of OpenJDK
 - Install a system-provided version of OpenJDK

Add Private Cloud Containerized Cluster

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Summary

Select JDK

CDH Version	Supported JDK Version
7.1.9 and above	OpenJDK 8, 11, 17 or Oracle JDK 8, 11, 17
7.1.1 to 7.1.8	OpenJDK 8, 11 or Oracle JDK 8, 11
7.0 and above	OpenJDK 8 or Oracle JDK 8
6.3 and above	OpenJDK 8 or Oracle JDK 8
6.2	OpenJDK 8 or Oracle JDK 8
6.1 or 6.0	Oracle JDK 8
5.16 and above	OpenJDK 8 or Oracle JDK 8
5.7 to 5.15	Oracle JDK 8

1 - 8 of 8
[More details on supported JDK version.](#)

If you plan to use JDK 11 with CDH 7.1.x and above or JDK 17 with CDH 7.1.9 and above, you will need to install it manually on all hosts and then select the **Manually manage JDK** option below.

☒ **Manually manage JDK**

ⓘ Please ensure that a supported JDK is **already installed** on all hosts. You will need to manage installing the unlimited strength JCE policy file, if necessary.

☐ Install a Cloudera-provided version of OpenJDK
By proceeding, Cloudera will install a supported version of OpenJDK version 8.

☐ Install a system-provided version of OpenJDK
By proceeding, Cloudera will install the default version of OpenJDK version 8 provided by the Operating System.

Cancel

← Back

Continue →

Step. Enter Login Credentials

13. On the **Enter Login Credentials** page, '**All hosts accept the same password**' is selected by default. Enter the user name in the SSH Username box, and type in and confirm the password. You can also select the **All hosts accept the same private key** option and provide the Private Key generated in previous steps and passphrase (If applicable).

Enter the values for the parameters as shown below. (**We will be using the private key approach**, you can use password option as well, both options should considerably work)

Component	Value
Enable TLS for	All existing and future clusters
SSH username	root
Authentication method	All hosts accept same private key / All hosts accept same password
Private Key (If using Key approach)	Choose the private key created and downloaded in earlier section
Password (If using Password approach)	Enter VM's root users' password
Confirm Password	Enter VM's root users' password (again)

Add Private Cloud Containerized Cluster

The screenshot shows the 'Enter Login Credentials' step in the Cloudera installation wizard. On the left is a vertical navigation menu with steps 1 through 11. Step 5, 'Enter Login Credentials', is highlighted. The main content area has the title 'Enter Login Credentials' and a paragraph explaining that root access is required. Below this are several form fields: 'SSH Username' with the value 'root', 'Authentication Method' with two radio buttons (the first is selected), 'Password' and 'Confirm Password' fields with masked text, 'SSH Port' with the value '22', and 'Simultaneous Installations' with the value '10'. A note below the last field explains that many simultaneous installations can consume resources.

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Specify Hosts

Select JDK

5 Enter Login Credentials

6 Install Agents

7 Assign Roles

8 Configure Docker Repository

9 Configure Data Services

10 Configure Databases

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Enter Login Credentials

Root access to your hosts is required to install the Cloudera packages. This installer will connect to your hosts via SSH and log in either directly as root or as another user with password-less sudo/pbrun privileges to become root.

SSH Username ⓘ root

Authentication Method

☒ All hosts accept same password

☐ All hosts accept same private key

Password

Confirm Password

SSH Port 22

Simultaneous Installations 10
(Running a large number of installations at once can consume large amounts of network bandwidth and other system resources)

a. Screenshot for using the Password based authentication method.

This screenshot shows the same 'Enter Login Credentials' step, but with the 'All hosts accept same private key' radio button selected. The 'SSH Username' is still 'root'. The 'Private Key' field now shows a 'Choose File' button and the filename 'id_rsa'. The 'Passphrase' and 'Confirm Passphrase' fields are present but empty. The 'SSH Port' is '22' and 'Simultaneous Installations' is '10'.

Enter Login Credentials

Root access to your hosts is required to install the Cloudera packages. This installer will connect to you password-less sudo/pbrun privileges to become root.

SSH Username ⓘ root

Authentication Method

☐ All hosts accept same password

☒ All hosts accept same private key

Private Key Choose File id_rsa

Passphrase

Confirm Passphrase

SSH Port 22

Simultaneous Installations 10
(Running a large number of installations at once can consume large amount

b. Screenshot for using the Private Key based authentication method.

Step. Install Agents

14. The **Install Agents** page appears and displays a progress indicator showing the agent packages getting installed. Click on **Continue** after successful agent installation on hosts to be added in Cloudera Manager.

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Install Agents

Installation in progress.

0 of 11 host(s) completed successfully. [Abort Installation](#)

Hostname	IP Address	Progress	Status
pvcecs-master.cdppvcds.com	172.31.40.79		Installing cloudera-manager-agent package... Details
pvcecs-worker1.cdppvcds.com	172.31.47.98		Installing cloudera-manager-agent package... Details
pvcecs-worker10.cdppvcds.com	172.31.45.62		Installing cloudera-manager-agent package... Details
pvcecs-worker2.cdppvcds.com	172.31.33.114		Installing cloudera-manager-agent package... Details
pvcecs-worker3.cdppvcds.com	172.31.39.35		Installing cloudera-manager-agent package... Details
pvcecs-worker4.cdppvcds.com	172.31.41.212		Installing cloudera-manager-agent package... Details
pvcecs-worker5.cdppvcds.com	172.31.45.196		Installing cloudera-manager-agent package... Details
pvcecs-worker6.cdppvcds.com	172.31.32.137		Installing cloudera-manager-agent package... Details
pvcecs-worker7.cdppvcds.com	172.31.32.61		Installing cloudera-manager-agent package... Details
pvcecs-worker8.cdppvcds.com	172.31.40.109		Installing cloudera-manager-agent package... Details
pvcecs-worker9.cdppvcds.com	172.31.39.141		Pending... Details

Rows per page: 25 1 - 11 of 11

Install Agents

Installation completed successfully.

11 of 11 host(s) completed successfully.

Hostname	IP Address	Progress	Status
pvcecs-master.cdppvcds.com	172.31.40.79		Installation completed successfully. Details
pvcecs-worker1.cdppvcds.com	172.31.47.98		Installation completed successfully. Details
pvcecs-worker10.cdppvcds.com	172.31.45.62		Installation completed successfully. Details
pvcecs-worker2.cdppvcds.com	172.31.33.114		Installation completed successfully. Details
pvcecs-worker3.cdppvcds.com	172.31.39.35		Installation completed successfully. Details
pvcecs-worker4.cdppvcds.com	172.31.41.212		Installation completed successfully. Details
pvcecs-worker5.cdppvcds.com	172.31.45.196		Installation completed successfully. Details
pvcecs-worker6.cdppvcds.com	172.31.32.137		Installation completed successfully. Details
pvcecs-worker7.cdppvcds.com	172.31.32.61		Installation completed successfully. Details
pvcecs-worker8.cdppvcds.com	172.31.40.109		Installation completed successfully. Details
pvcecs-worker9.cdppvcds.com	172.31.39.141		Installation completed successfully. Details

Rows per page: 25 1 - 11 of 11

Cancel

Back

Continue

Step. Assign Roles

15. Next on the **Assign Roles** page, ensure that the roles assignment for your new **Cloudera on premises Containerized cluster** is as follows. You can customize the role assignment for your cluster. But, Cloudera does not recommend altering assignments unless you have specific requirements such as having selected a specific host for a specific role.

Note: Single node ECS installation is supported, but is only intended to enable CDSW to CAI migration. If you are installing ECS on a single node, only the Docker and ECS Server roles are assigned. The ECS Agent role is not required for single node installation.

Note: With 1 mgmt node and 12 worker node for CDP Data Services ECS cluster we select host role assignment as:

Role	ECS Host
Docker Server	All ECS Hosts (i.e., ECS master and worker nodes) pvcecs-master, pvcecs-worker[1-11]
ECS Server	ECS Master Nodes only (pvcecs-master)
ECS Agent	ECS Worker Nodes only (pvcecs-worker[1-11])

Add Private Cloud Containerized Cluster

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✓ Cluster Basics

✓ Specify Hosts

✓ Select JDK

✓ Enter Login Credentials

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Assign Roles

You can customize the role assignments for your new cluster here, but if assignments are made incorrectly, such as assigning too many roles Cloudera does not recommend altering assignments unless you have specific requirements, such as having pre-selected a specific host for a :

You can also view the role assignments by host. [View By Host](#)

 DOCKER

Docker Server × 11 New

pvcecs-master.cdppvcds.com; pvcecs-worker[1-10].cdppvcds....

 ECS

Ecs Server × 1 New

pvcecs-master.cdppvcds.com ▼

Ecs Agent × 10 New

pvcecs-worker[1-10].cdppvcds.com ▼

16. Click **Continue**.

Step. Configure Docker Repository

17. On the **Configure Docker Repository** page, select **Cloudera default Docker Repository** or (**Use an Embedded Docker Repository** option. Then select Default in the below section. There are several options for configuring a Docker Repository. For more information about these options, see [Docker repository access](#).)

The following ports must be opened and allowed no matter which Docker repository option you choose.

- Ports required for Cloudera Manager/Cloudera Manager agent (port 5000 is required for CAI):

Protocol	Port
TCP	7180-7192
TCP	19001
TCP	5000
TCP	9000

- Inbound rules for ECS Server nodes (Kubernetes/RKE2):

Protocol	Port
TCP	9345
TCP	6443
UDP	8472
TCP	10250
TCP	2379
TCP	2380
TCP	30000-32767

- Inbound Rules for the ECS Agent (Kubernetes/RKE2):

Protocol	Port
UDP	4789

Embedded Docker Repository:

Proceed with default selection to deploy all of the default Docker images to the repository, or select **Select the Optional Images** to choose which images to deploy. If you will be deploying Cloudera AI (CAI) a.k.a. Cloudera Machine Learning (CAI), toggle the **Cloudera Machine Learning** switch on to copy the images for CAI.

Add Private Cloud Containerized Cluster

The screenshot shows the 'Configure Docker Repository' step in the Cloudera installation wizard. On the left is a vertical sidebar with 15 steps, each with a numbered circle and a checkmark. Step 8, 'Configure Docker Repository', is highlighted with a blue circle. The main content area has the title 'Configure Docker Repository' and a paragraph: 'Cloudera uses a Docker Repository to deliver CDP Private Cloud Data Services. [Learn more](#) about how to set up custom Docker Repository for CDP Private Cloud Data Services.' Below this are three radio button options: 'Use an embedded Docker Repository' (selected with a blue dot), 'Use Cloudera's default Docker Repository', and 'Use a custom Docker Repository'. Further down, there is a paragraph: 'This release comes with 301 container images that need to be deployed to the Docker repository. Some images are optional and can be skipped by toggling them from the list below. Other images are always installed.' Below this are two radio button options: 'Default' and 'Select the Optional Images' (selected with a blue dot). Then, there is a toggle switch for 'Cloudera Machine Learning', which is currently turned off. Below the toggle is a paragraph: 'Docker images required to create a Cloudera Machine Learning workspace. Without these images, it will not be possible to use Cloudera Machine Learning.' At the bottom of the main content area is a paragraph: 'The system will deploy 301 container images, approximately 173 GiB, to the embedded Docker repository.' At the bottom of the wizard are three buttons: 'Cancel', '← Back', and 'Continue →'.

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Configure Docker Repository

Cloudera uses a Docker Repository to deliver CDP Private Cloud Data Services. [Learn more](#) about how to set up custom Docker Repository for CDP Private Cloud Data Services.

☒ Use an embedded Docker Repository

☐ Use Cloudera's default Docker Repository

☐ Use a custom Docker Repository

This release comes with 301 container images that need to be deployed to the Docker repository. Some images are optional and can be skipped by toggling them from the list below. Other images are always installed.

☐ Default ☒ Select the Optional Images

☐ Cloudera Machine Learning

Docker images required to create a Cloudera Machine Learning workspace. Without these images, it will not be possible to use Cloudera Machine Learning.

The system will deploy 301 container images, approximately 173 GiB, to the embedded Docker repository.

Cancel

← Back

Continue →

a. Screenshot for Use an embedded Docker Repository Option

Cloudera default Docker Repository: (We will setup this way)

This option requires that cluster hosts have access to the internet and you have selected Internet as the install method.

Add Private Cloud Containerized Cluster

The screenshot shows a multi-step installation wizard. On the left, a vertical list of steps is shown, with steps 1 through 7 marked with checkmarks and steps 9 through 15 in grey. Step 8, 'Configure Docker Repository', is highlighted with a blue circle. The main content area is titled 'Configure Docker Repository' and contains explanatory text and three radio button options. The second option, 'Use Cloudera's default Docker Repository', is selected with a blue dot. At the bottom, there are three buttons: 'Cancel', '← Back', and 'Continue →'.

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Configure Docker Repository

Cloudera uses a Docker Repository to deliver CDP Private Cloud Data Services. [Learn more](#) about how to set up custom Docker Repository for CDP Private Cloud Data Services.

☐ Use an embedded Docker Repository

☒ Use Cloudera's default Docker Repository

☐ Use a custom Docker Repository

Cancel

← Back

Continue →

b. Screenshot for Use Cloudera's Docker Repository Option

Custom Docker Repository:

This option requires that you set up a Docker Repository in your environment and that all cluster hosts have connectivity to the repository.

Note: If you are installing ECS on a single node, you should select the Use a Custom Docker Repository option. Single node ECS installation is supported, but is only intended to enable CDSW to CAI migration.

CDEP Deployment from 2024-Apr-22 12:02

Add Private Cloud Containerized Cluster

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Configure Docker Repository

Cloudera uses a Docker Repository to deliver CDP Private Cloud Data Services. [Learn more](#) about how to set up custom Docker Repository for CDP Private Cloud Data Services.

☐ Use an embedded Docker Repository

☐ Use Cloudera's default Docker Repository

☒ Use a custom Docker Repository

Custom Docker Repository ⓘ

Prepare your Docker Repository from a machine that is running Docker locally and has access to all the Docker images either directly from Cloudera or from a local http mirror in your network. If your custom repository already has all the Docker images for this version, this section can be skipped.

1. Generate the copy-docker script

2. Optionally, review the script. The file contains usage information and lists the Docker images that it will download and push.

3. Login to your custom Docker Registry and run the script with the following commands (Note: this downloads 100+ Docker images and it will take a while):

```
docker login <your_custom_registry> -u <user_with_write_access>
bash copy-docker.txt
```

☐ I confirm that I have downloaded all the Docker images to my custom Docker Repository.

Docker Username ⓘ

Docker Password ⓘ

Docker Certificate ⓘ

Choose File

Cancel

← Back

Continue →

c. Screenshot for Use Cloudera's Docker Repository Option

You must enter the following options:

Option	Value
Custom Docker Repository:	Enter the URL for your Docker Repository
Docker Username:	Enter the username for the Docker Repository
Docker Password:	Enter the password for the Docker Repository

Important: Do not use the \$ character for this password.

Docker Certificate – Click the **Choose File** button to upload a TLS certificate to secure communications with the Docker Repository.

Click the **Generate the copy-docker script** button to generate and download a script that copies the Docker images from Cloudera, or (for air-gapped installation) from a local http mirror in your network.

Run the script from a machine that is running Docker locally and has access to the Docker images using the following commands:

```
docker login [***URL for Docker Repository***] -u [***username of user with write access***]
bash copy-docker.txt
```

The copying operation may take 4 - 5 hours.

Note: Embedded Repository can be a single point of failure. If the node that runs the Docker Repository fails or becomes unavailable, some cluster functionalities might become unavailable. Moving the Docker Repository to another node is a complex process and will require engaging Cloudera Professional Services.

Note: Cloudera Repository option is best suited for proof-of-concept, non-production deployments or deployments that do not have security requirements that disallow internet access. This option requires that cluster hosts have access to the internet, and an installation method selected as Internet.

Step. Configure Data Services

18. On the **Configure Data Services** page, modify configuration as appropriate and modify the storage related parameters. Edit Application domain to match "app.example.com". For example in this solution we configure AD Domain Services with "CLDRSETUP.LOCAL" as domain name. Created a wildcard entry "*.apps.cldrsetup.local". Click **Continue**.

Role	ECS Host
Data Storage Directory:	/docker
ECS (Service Wide):	/lhdata
Application Domain:	cldrsetup.local
Local Path Storage Directory:	/cdwdata

On the **Configure Data Services** page, you can modify configuration settings such as the data storage directory, number of replicas, and so on. If you want to specify a custom certificate, place the certificate and the private key in a specific location on the Cloudera Manager server host and specify the paths in the input boxes labeled as Ingress Controller TLS/SSL Server Certificate/Private Key File below. This certificate will be copied to the Control Plane during the installation process.

Note: The "Ingress Controller TLS/SSL Server Certificate File (PEM Format)" must only contain -----BEGIN CERTIFICATE----- through -----END CERTIFICATE----- (inclusive) for the server and CA certs. It cannot include any preamble text and, and must not include a private key.

The "Ingress Controller TLS/SSL Server Private Key File (PEM Format)" must only contain the unencrypted key, and only the header through the footer, with no preamble text.

Both of these files must be readable by the "cloudera-scm" account.

For information on the required entries that must be present in DNS and TLS certificates when not using wildcards, refer to 'No Wildcard DNS/TLS Setup'

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Configure Data Services

The Private Cloud Containerized Cluster needs to act as a TLS/SSL Server. By default, Cloudera Private Cloud Containerized Cluster using TLS.

If you want to specify a custom certificate, place the certificate and the private key in a specific Controller TLS/SSL Server Certificate/Private Key File, below.

This certificate must be valid for the application domain and one level underneath it. For example, The certificate will be copied to the Private Cloud Containerized Cluster during the installation.

Data Storage Directory

defaultDataPath
[Edit Individual Values](#)
⚙️ defaultDataPath

DOCKER (Service-Wide) ↩️

/docker

ECS (Service-Wide) ↩️ Undo

/lhdata

Application Domain

app_domain
⚙️ app_domain

ECS (Service-Wide) ↩️ Undo

cdppvcds.com

Local Path Storage Directory

IsoDataPath
⚙️ IsoDataPath

ECS (Service-Wide) ↩️ Undo

/cdwdata

Number of Replicas

longhorn_replication
⚙️ longhorn_replication

ECS (Service-Wide)

Number of replicas

target_redundancy
⚙️ target_redundancy

ECS (Service-Wide)

Use internal alias for registry

⚙️ internal_mirror

☐ ECS (Service-Wide)

Cluster Signing Duration

⚙️ cluster_signing_duration

ECS (Service-Wide)

Note: Please review the range of cluster IP and service IP as part of the ECS installation. It might conflict existing network configuration. Please adjust the range of IPs to be configured. Consult with the network team to avoid potential conflict.

Cluster IP Range

cluster-cidr
⚙️ cluster_cidr

ECS (Service-Wide)

10.42.0.0/16

IPv4/IPv6 network CIDRs to use for pod IPs. ✕

Service IP Range

service-cidr
⚙️ service_cidr

ECS (Service-Wide)

10.43.0.0/16

IPv4/IPv6 network CIDRs to use for service IPs. ✕

19. Click **Continue**.



Step. Configure Databases

20. On the *Configure Databases* page, edit size for the *Embedded Database Disk Space*. Click *Continue*.
Add Private Cloud Containerized Cluster

✓ Getting Started

✓ Cluster Basics

✓ Specify Hosts

✓ Select JDK

✓ Enter Login Credentials

✓ Install Agents

✓ Assign Roles

✓ Configure Docker Repository

✓ Configure Data Services

10 Configure Databases

11 Install Parcels

12 Check Prerequisites

13 Inspect Cluster

14 Install Data Services

15 Summary

Configure Databases

CDP Private Cloud Control Plane uses an embedded Database to store configuration and other metadata information for the cluster being managed.

Embedded Database Disk Space (GiB) ⓘ

200

Cancel

← Back

Continue →

Step. Install Parcels

21. On the *Install Parcels* page, the selected parcel is downloaded to the Cloudera Manager server host, distributed, unpacked, and activated on the ECS cluster hosts. Click *Continue*.
Add Private Cloud Containerized Cluster

Getting Started

Cluster Basics

Specify Hosts

Select JDK

Enter Login Credentials

Install Agents

Assign Roles

Configure Docker Repository

Configure Data Services

Configure Databases

11 Install Parcels

12 Check Prerequisites

13 Inspect Cluster

14 Install Data Services

15 Summary

Install Parcels

The selected parcels are being downloaded and installed on all the hosts in the cluster.

Embedded Container Service 1.5.4

Downloaded: 100%

Distributed: 11/11 (478.7 MiB/s)

Unpacked: 11/11

Activated: 11/11

All (11)

Running (0)

Failed (0)

Completed (11)

Hostname	Throughput	Status	Errors
pvcacs-worker6.cdppvcds.com	46.6 MiB/s	NONE	
pvcacs-worker2.cdppvcds.com	46.6 MiB/s	NONE	
pvcacs-worker1.cdppvcds.com	46.6 MiB/s	NONE	
pvcacs-worker3.cdppvcds.com	46.6 MiB/s	NONE	
pvcacs-master.cdppvcds.com	46.6 MiB/s	NONE	
pvcacs-worker8.cdppvcds.com	46.6 MiB/s	NONE	
pvcacs-worker4.cdppvcds.com	43.5 MiB/s	NONE	
pvcacs-worker10.cdppvcds.com	43.5 MiB/s	NONE	
pvcacs-worker7.cdppvcds.com	46.6 MiB/s	NONE	
pvcacs-worker9.cdppvcds.com	43.5 MiB/s	NONE	
pvcacs-worker5.cdppvcds.com	43.5 MiB/s	NONE	

Rows per page: 25 1 - 11 of 11 < > >>

Cancel

BackContinue

Step. Check Prerequisites

22. If the hosts do not meet the prerequisites, the *Check Prerequisites* page displays the applicable issues. Correct the issues, then click *Run Again*. After all of the issues have been resolved, click *Continue*. Prerequisites checks are included in the new ECS release version 1.5.5.

Status	Description
	A minimum of 16 cores are required for the hosts in a Private Cloud Containerized Cluster. The following hosts do not satisfy the minimum number of cores: > View Details
	A minimum of 300 GiB of storage is required in the /var/lib directory and 300 GiB in the /ecs/docker directory for the hosts in a Private Cloud Containerized Cluster. /var/lib/longhorn directory cannot be a symbolic link. The following hosts do not meet these criteria: > View Details

The following prerequisites are checked:

Host Prerequisite Inspection	Validation
StorageInspection:	Checks for a minimum of 300GiB space in the /var/lib and docker data directories. Checks if /var/lib/longhorn or its parent directories are symlinked. If they are, this inspection will fail.
CPUInspection:	Checks to make sure the hosts have 16 virtual cores.
PortsInspection:	Checks for the availability of ports 443 and 80.

Host Prerequisite Inspection	Validation
EcsHostDnsInspection:	Checks to make sure there are less than 3 nameserver entries in the <code>/etc/resolv.conf</code> file, and checks the connections to the Cloudera Manager cluster and the CDP console. It also checks to see if <code>vault.localhost.localdomain's</code> ping can be resolved. If not, it is likely that the host <code>/etc/nsswitch.conf</code> file is misconfigured. If this inspection fails: • Check the <code>/etc/resolv.conf</code> and <code>/etc/nsswitch.conf</code> files and ensure that <code>/etc/resolv.conf</code> does not contain 3 or more nameservers, and that <code>/etc/nsswitch.conf</code> does not contain <code>myhostname</code> under the <code>hosts</code> field. • Check to see if the connections were resolved correctly. If connection to the CDP console fails, check to see if your DNS wildcard is configured properly.
VersionInspection:	Checks that Java is installed and consistent among all ECS hosts.
IPTablesInspection:	Checks that if the <code>iptables</code> command exists, rules are cleared. If the <code>iptables</code> command does not exist, <code>iptables</code> gets installed during FirstRun so this inspection passes. If <code>iptables</code> are installed and the rules are not cleared, this inspection will fail.
EcsCleanUpHostInspection:	Checks to make sure that the <code>/var/lib/rancher</code> and <code>docker</code> data directories do not contain any files.

Add Private Cloud Containerized Cluster

CDEP Deployment from 2024-Apr-23 12:43

- Getting Started
- Cluster Basics
- Specify Hosts
- Select JDK
- Enter Login Credentials
- Install Agents
- Assign Roles
- Configure Docker Repository
- Configure Data Services
- Configure Databases
- Install Parcels
- 12 Check Prerequisites**
- 13 Inspect Cluster
- 14 Install Data Services
- 15 Summary

Check Prerequisites

We are verifying if your hosts meet minimum storage, ports, cpu, and network requirements. The minimum requirements must be met before proceeding.

✖ Host Prerequisites

Error(s) were detected, review the inspector results and correct the problems found. Once corrected, please run the inspections again.

Status **Finished** Last Run a few seconds ago Duration 7.56s [Show Inspector Results](#) [Run Again](#)

[More](#)

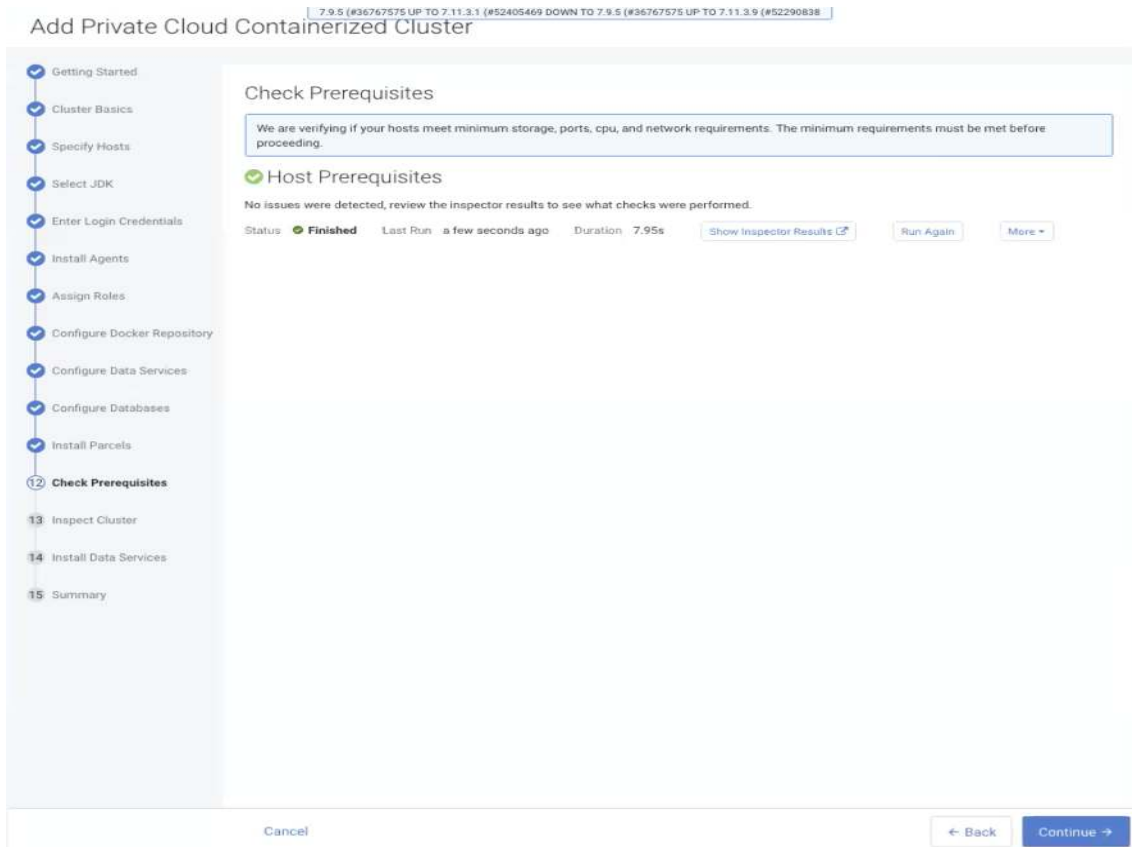
Status	Description
✖	A minimum of 16 cores are required for the hosts in a Private Cloud Containerized Cluster. The following hosts do not satisfy the minimum number of cores: View Details

[Cancel](#)
[Back](#)
[Continue](#)

Note: If the prerequisite check fails due to any reason or you need to skip this step, try the workaround by right-clicking on the screen, select **Inspect**, go to the **Console** tab, and run the below command:

```
document.querySelector('.btn.next').removeAttribute('disabled');
```

This will enable the **Continue** button.



Step. Inspect Cluster

23. On the **Inspect Cluster** page, you can **Inspect your Network Performance and Hosts**. Click on the **Show Inspector Results**. If the inspect tool displays any issues, you can fix those issues and click on **Run Again** to rerun the inspect tool. After all of the issues have been resolved, click on **Continue**.

Note: These inspections are more comprehensive host and network tests that you can **optionally run**. To **skip these tests**, select the **I understand the risks of not running the inspections or the detected issues, let me continue with cluster setup** checkbox.

Note: Safe to ignore unrelated errors in the host inspector result. For example, the hosts in a Cloudera on premises Containerized Cluster that have GPUs are required to have NVidia Drivers and NVidia-container-runtime installed. The following hosts do not satisfy this requirement: pvcecs-worker[1-12].cldrsetup.local
Since all hosts part of the ECS installation might not have NVIDIA GPU installed and NVidia driver and NVidia container-runtime is not installed on non-GPU node(s). It is safe to ignore the warning and click on the checkbox to continue with ECS installation.

Add Private Cloud Containerized Cluster

- Getting Started
- Cluster Basics
- Specify Hosts
- Select JDK
- Enter Login Credentials
- Install Agents
- Assign Roles
- Configure Docker Repository
- Configure Data Services
- Configure Databases
- Install Parcels
- Check Prerequisites
- 13 Inspect Cluster**
- 14 Install Data Services
- 15 Summary

Inspect Cluster

You have created a new empty cluster. Here are additional inspections Cloudera recommends you to run. For accurate measurements, Cloudera recommends that they are performed sequentially.

Host Inspector

Error(s) were detected, review the inspector results and correct the problems found.

Status **Finished** Last Run a few seconds ago Duration 5.74s [Show Inspector Results](#) [Run Again](#) [More](#)

Status	Description
	Starting with CDH 6, Hue requires Python version 2.7. This warning can be ignored if hosts will not be running CDH 6. The following hosts do not satisfy this requirement: View Details tina-rhel89-[1-3].vpc.cloudera.com
	The hosts in a Private Cloud Containerized Cluster that have GPUs are required to have nVidia Drivers and nvidia-container-runtime installed. The following hosts do not satisfy this requirement: View Details tina-rhel89-[1-3].vpc.cloudera.com

Network Performance Inspections

[Advanced Options](#)

Status **Finished** Last Run a few seconds ago Duration 10.18s [Show Inspector Results](#) [Run Again](#) [More](#)

Tested within **Containerized Cluster 1**:


Latency Test

Minimum

Average

Maximum

0.09ms

0.13ms

0.17ms

☒ I understand the risks of not running the inspections or the detected issues, let me continue with cluster setup.

[Cancel](#)

[← Back](#)

[Continue →](#)

Step. Install Data Services

Completed 0 of 1 step(s).

Show All Steps Show Only Failed Steps Show Only Running Steps

Run a set of services for the first time. 0/1 steps completed.	May 7, 5:08:51 PM	
Execute 2 steps in sequence 0/1 steps completed.	May 7, 5:08:51 PM	
Start DOCKER	May 7, 5:08:51 PM	43.92s
Start ECS 0/1 steps completed.	May 7, 5:09:35 PM	
Execute 3 steps in sequence Waiting for process ecs-tolerations-webhook-install (id=1546398465) on host ecsmastercoe.rhdt.com (id=1546397508) to finish	May 7, 5:09:35 PM	
Execute command Save or Restore Iptables on service ECS	May 7, 5:09:35 PM	1.88s
Start ECS	May 7, 5:09:37 PM	2.5m
Execute 15 steps in sequence Waiting for process ecs-tolerations-webhook-install (id=1546398465) on host ecsmastercoe.rhdt.com (id=1546397508) to finish	May 7, 5:12:05 PM	
- Kubectl command. - Setup Storage. - Execute command Unseal Vault on service ECS - Setup infrastructure monitoring - Install Control Plane	- Execute 5 steps in parallel - Execute command Install Longhorn UI on servi... - Execute command Reapply All Settings to Clust... - Install Nvidia device plugin for GPU support. - Execute command Create Environment on servi...	- Install mutating webhook for ECS tolerations. - Initialize embedded Vault. - Setup ECS Web UI - Run additional installation steps from parcel - Execute command Update Ingress Controller C...

24. While the other steps for installing data services cluster (ECS) is running, login to the pvcecs-master node and create kubeconfig file by copying the rke2.yaml file on ecs master node, in order to be able to run the kubectl commands from master node:

```
##### Install K9s for troubleshooting purpose
[root@pvcecs-master ~]# dnf install wget -y
[root@pvcecs-master ~]# rm -rvf ~/.kube && mkdir -p ~/.kube && cp /etc/rancher/rke2/rke2.yaml ~/.kube/config
[root@pvcecs-master ~]# wget https://github.com/derailed/k9s/releases/latest/download/k9s_Linux_amd64.tar.gz
[root@pvcecs-master ~]# tar -xzf k9s_Linux_amd64.tar.gz
[root@pvcecs-master ~]# sudo mv k9s /usr/bin/
[root@pvcecs-master ~]# k9s version

##### The kubectl binary path may vary, do whereis kubectl or use find or locate commands to find the exact path of the kubectl binary on the node.
[root@pvcecs-master ~]# export PATH=/var/lib/rancher/rke2/data/v1.30.12-rke2r1-328c510931ca/bin/:$PATH
[root@pvcecs-master ~]# kubectl get pods
No resources found in the default namespace.
[root@pvcecs-master ~]# kubectl get all
NAME                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes  ClusterIP     10.43.0.1    <none>        443/TCP   13h
[root@pvcecs-master ~]#

##### Create alias on node, to run the kubectl command
[root@pvcecs-master ~]# echo "export PATH=/var/lib/rancher/rke2/data/v1.30.12-rke2r1-328c510931ca/bin/:$PATH" >> ~/.bashrc && source ~/.bashrc
[root@pvcecs-master ~]# echo "alias k=kubectl" >> ~/.bashrc && source ~/.bashrc
##### You can copy the config file and set the above ~/.bashrc paths on any of ECS nodes to run kubectl commands from there.
[root@pvcecs-master ~]#
```

Note: Run `# kubectl get pods -A` to review all pods and their status as either running or completed.

25. After the **RKE2 installation step is completed**, keep the remaining steps running but immediately log in to **pvcecs-master** node and **update the config for coredns to point to private DNS** (ipaserver or AD server). The coredns config should look like similar to below one:

```
##### Test the nslookup and dig from ubuntu image if it is able to resolve the IP from Private DNS
[root@pvcecs-master ~]# kubectl run -it ubuntu1 --image=ubuntu bash
```

If not able to resolve the IP from Private DNS, we need to point coredns to PrivateDNS/ FreeIPA server

```
[root@pvcecs-master ~]# kubectl get configmap -n kube-system | grep coredns
```

```
[root@pvcecs-master ~]# kubectl edit configmap rke2-coredns-rke2-coredns -n kube-system
```

```
# Please edit the object below. Lines beginning with a '#' will be ignored,  
# and an empty file will abort the edit. If an error occurs while saving this file will be  
# reopened with the relevant failures.
```

```
#
```

```
apiVersion: v1
```

```
data:
```

```
Corefile: |
```

```
.:53 {
```

```
  errors
```

```
  health {
```

```
    lameduck 5s
```

```
  }
```

```
  ready
```

```
  kubernetes cluster.local cluster.local in-addr.arpa ip6.arpa {
```

```
    pods insecure
```

```
    fallthrough in-addr.arpa ip6.arpa
```

```
    ttl 30
```

```
  }
```

```
  prometheus :9153
```

```
  forward . 172.31.24.240 {
```

```
    max_concurrent 1000
```

```
  }
```

```
  cache 30
```

```
  loop
```

```
  reload
```

```
  loadbalance
```

```
}
```

```
kind: ConfigMap
```

```
metadata:
```

```
  annotations:
```

```
    meta.helm.sh/release-name: rke2-coredns
```

```
    meta.helm.sh/release-namespace: kube-system
```

```
  creationTimestamp: "2024-05-17T07:12:19Z"
```

```
  labels:
```

```
    app.kubernetes.io/instance: rke2-coredns
```

```
    app.kubernetes.io/managed-by: Helm
```

```
    app.kubernetes.io/name: rke2-coredns
```

```
    helm.sh/chart: rke2-coredns-1.24.006
```

```
    k8s-app: kube-dns
```

```
    kubernetes.io/cluster-service: "true"
```

```
    kubernetes.io/name: CoreDNS
```

```
  name: rke2-coredns-rke2-coredns
```

```
  namespace: kube-system
```

```
  resourceVersion: "39715"
```

```
  uid: dcb8f600-e107-4c1b-9db7-701060fe063a
```

```
[root@pvcecs-master ~]# kubectl rollout restart deployment rke2-coredns-rke2-coredns -n kube-system
```

Verify the status that deployment of coredns is updated with configmap changes

```
[root@pvcecs-master ~]# kubectl get pod -A |grep dns
```

Test the nslookup and dig from ubuntu image if it is actually able to resolve the IP from Private DNS

```
[root@pvcecs-master ~]# kubectl run -it ubuntu1 --image=ubuntu bash
```

```
root@ubuntu2:/# apt-get update && apt-get install net-tools dnsutils -y
```

```
root@ubuntu2:/# nslookup console-cdp.apps.cldrsetup.local
```

26. **Install Data Services** step will run a set of first run commands and report status on Data Services installation. The installation progress is displayed on the **Install Data Services** page. This step will take **nearly an hour to complete**. When the installation is complete, click **Continue**.

Note: Installing Data Services can take several hours. The copying operation for Docker repository may take 4 - 5 hours.

Add Private Cloud Containerized Cluster

Install Data Services

First Run Command

Status: ✔ **Finished** Context: [Containerized Cluster 1.5.4](#) Apr 24, 6:46:03 PM 17.8m

Finished First Run of the following services successfully: DOCKER, ECS.

Completed 1 of 1 step(s).

☒ Show All Steps ☐ Show Only Failed Steps ☐ Show Only Running Steps

✔ Run a set of services for the first time. Successfully completed 1 steps.	Apr 24, 6:46:03 PM	17.8m
✔ Execute 2 steps in sequence Successfully completed 1 steps.	Apr 24, 6:46:03 PM	17.8m
> ✔ Start DOCKER	Apr 24, 6:46:03 PM	47.21s
> ✔ Start ECS	Apr 24, 6:46:51 PM	17m

Cancel ← Back **Continue →**

27. If you still face any issue while making the services up or during the installation or start of any ECS services please refer to troubleshooting PvC Data Services Cluster part at the end of this document. Though, some of the major issues during installation, their cause and their resolution is listed as below:

ksahu@Kuldeeps-MacBook-Air ~ %

Name resolution not working inside pod for console-cdp leading cli pod to fail and env not getting created

Cluster name CM url not able to connect from ECS cluster

Solution:

Update coredns pod by edit and redeploy and mention private dns server ip in config
make sure wildcard dns is set up properly.

After issue is fixed and env get created for ECS: it will show like below:

Fri May 17 01:23:55 AM PDT 2024

Running on: pvccs-master.cldrsetup.local (172.31.30.239)

Fetching session token...

Creating environment cldrsetup

secret "cm.args" deleted

secret/cm.args created

```

job.batch "cli" deleted
job.batch/cli created
job.batch/cli condition met
/opt/app-root/lib64/python3.9/site-packages/urllib3/connectionpool.py:1103: InsecureRequestWarning: Unverified HTTPS request is being made to host
'console-cdp.apps.cldrsetup.local'. Adding certificate verification is strongly advised. See:
https://urllib3.readthedocs.io/en/latest/advanced-usage.html#tls-warnings
warnings.warn(
{
  "environment": {
    "environmentName": "cldrsetup",
    "crn": "crn:altus:environments:us-west-1:d750ff5a-791d-46c8-a95a-f1efaf6185ef:environment:cldrsetup/ad47b514-a135-47dc-a8dd-8d11f33145f7",
    "cloudPlatform": "standard"
  }
}
Waiting for environment cldrsetup to be available
The http response code is 200
...
The http response code is 200
...
The http response code is 200
...
The http response code is 200
...
The http response code is 200
Environment was successfully created

```

Failed to reconcile with temporary etcd:

Solution:

```
bootstrap data already found and encrypted with different token rke error
rm -vf /var/lib/rancher/k3s/server/node-token
```

Update k8s cordons and rollout

Solution:

https://www.reddit.com/r/k3s/comments/p3rdap/lost_bootstrap_data_already_found_and_encrypted/
[root@pvcecs-master bin]# rm -vf /var/lib/rancher/k3s/server/node-token
rm -rvf /var/lib/rancher/ /etc/rancher

X509 error-

Solution:

do cleanup for ECS properly

Unable to connect 6443

Solution:

```
mkdir -p ~/.kube && cp /etc/rancher/rke2/rke2.yaml ~/.kube/config
```

Aws_key_id error in cde cluster creation

Error Logs:

```

Moving from CadenceInstallCompleted to ClusterChartInstallationFailed
{"StartState": "CadenceInstallCompleted", "EndState": "ClusterChartInstallationFailed", "Attempt": 3}
Chart installation failed, dex base overrides: unable to retrieve aws_key_id from the env service
Error fetching fluent log config: unable to retrieve aws_key_id from the env service
Generating Webhook cert and key
Setting up TGT generator
Installing dex-base charts for CDE 1.24.0
{"StartState": "CadenceInstallCompleted", "EndState": "ClusterChartInstallationFailed", "Attempt": 2}
Chart installation failed, dex base overrides: unable to retrieve aws_key_id from the env service

```

Error fetching fluent log config: unable to retrieve aws_key_id from the env service

Cause:

Configure ozone with other base cluster services before env creation

Solution:

Install ozone on base cluster, restart base cluster services after applying the stale configurations and recreate the ECS environment.

<https://community.cloudera.com/t5/Support-Questions/Unable-to-create-Data-Engineering-Cluster-in-CDP-Private/m-p/377969>

Note: Ozone is a necessary service if you want to install CDE

This error typically occurs when required services like Ozone are not present in the base cluster at the time of CDE environment creation. In such cases, the cdp-services namespace remains empty and lacks critical components like the s3proxy, which is essential for Fluent Bit to forward logs to Ozone. The s3proxy deployment is triggered automatically during the environment creation process and cannot be manually deployed via Helm or other means. Therefore, if Ozone is added after the environment has already been created, the only solution is to delete and recreate the entire Data Services installation. This ensures that the s3proxy is properly deployed, allowing the environment to function correctly and resolving the AWS credential-related error during CDE Virtual Cluster setup.

Other:

<https://repost.aws/knowledge-center/create-lv-on-ebs-partition>

UNRESOLVED: CDE Error

```
[root@pvcecs-master ~]# k get po -A | grep dex | grep -v -E 'Run|Comp'
```

dex-base-j95kx9f	cdp-cde-embedded-db-0	0/1	Pending	0	8m54s
dex-base-j95kx9f	dex-base-data-connectors-746fdcdc8-jvbsx	0/1	CrashLoopBackOff	6 (3m22s ago)	8m54s
dex-base-j95kx9f	dex-base-management-api-6c5d48fd96-dgrnt	0/1	CrashLoopBackOff	6 (2m21s ago)	8m54s

```
[root@pvcecs-master ~]# kubectl logs -n dex-base-j95kx9f dex-base-data-connectors-746fdcdc8-jvbsx
```

Defaulted container "data-connectors" out of: data-connectors, k8tz (init)

```
{"level":"INFO","timestamp":"2024-05-27T01:37:23.834-0700","caller":"cmd/data-connectors.go:92","message":"Configuration file used: /etc/dc/data-connectors.yaml"}
```

```
{"level":"INFO","timestamp":"2024-05-27T01:37:23.834-0700","caller":"cmd/start.go:112","message":"Data store init","store type":"sql"}
```

2024/05/27 01:37:23 /grid/0/jenkins/workspace/workspace/App_builds/SOURCES/data-connectors/pkg/store/sqlstore/utls.go:79

[error] failed to initialize database, got error dial tcp 10.43.139.205:3306: connect: connection refused

```
{"level":"FATAL","timestamp":"2024-05-27T01:37:23.840-0700","caller":"cmd/start.go:115","message":"store initialisation failure","error":"dcObject persistent store init failure: DB session creation error: error creating DB session for 'mysql', error: dial tcp 10.43.139.205:3306: connect: connection refused","stacktrace":"github.com/infra.cloudera.com/CDH/data-connectors/pkg/cmd.startServer\n\tgrid/0/jenkins/workspace/workspace/App_builds/SOURCES/data-connectors/pkg/cmd/start.go:115\ngithub.com/infra.cloudera.com/CDH/data-connectors/pkg/cmd.glob..func3\n\tgrid/0/jenkins/workspace/workspace/App_builds/SOURCES/data-connectors/pkg/cmd/start.go:68\ngithub.com/spf13/cobra.(*Command).execute\n\tgrid/0/jenkins/.asdf/installs/golang/1.17.6/packages/pkg/mod/github.com/spf13/cobra@v1.1.3/command.go:856\ngithub.com/spf13/cobra.(*Command).ExecuteC\n\tgrid/0/jenkins/.asdf/installs/golang/1.17.6/packages/pkg/mod/github.com/spf13/cobra@v1.1.3/command.go:960\ngithub.com/spf13/cobra.(*Command).Execute\n\tgrid/0/jenkins/.asdf/installs/golang/1.17.6/packages/pkg/mod/github.com/spf13/cobra@v1.1.3/command.go:897\ngithub.com/infra.cloudera.com/CDH/data-connectors/pkg/cmd.Execute\n\tgrid/0/jenkins/workspace/workspace/App_builds/SOURCES/data-connectors/pkg/cmd/data-connectors.go:105\nmain.main\n\tgrid/0/jenkins/workspace/workspace/App_builds/SOURCES/data-connectors/cmd/data-connectors/main.go:32\nruntime.main\n\tgrid/0/jenkins/.asdf/installs/golang/1.17.6/go/src/runtime/proc.go:255"}
```

```
[root@pvcecs-master ~]# kubectl logs -n dex-base-j95kx9f dex-base-management-api-6c5d48fd96-dgrnt
```

Defaulted container "dex-base-management-api" out of: dex-base-management-api, k8tz (init)

No Envoy proxy, skip waiting for its readiness

Skip fetching DB certificates

Running main binary: /dex/bin/runtime-management-server

2024/05/27 01:38:24 INFO config.go:61 Loaded config from: /etc/dex/conf/dex.yaml

2024/05/27 01:38:24 INFO db.go:204 Opening database connection, driver: mysql

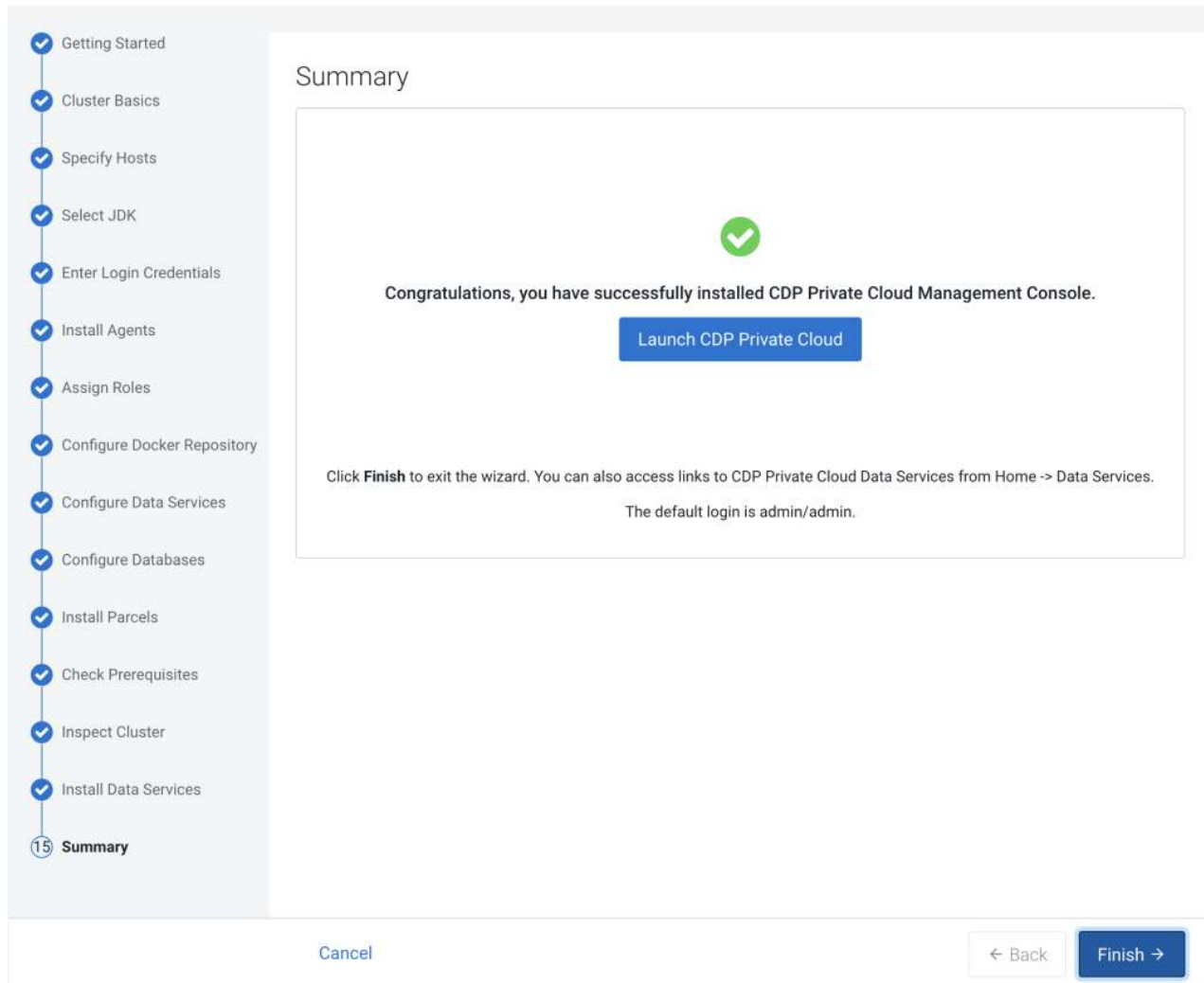
2024/05/27 01:38:24 FATAL db.go:226 dbConn.BeginTx failed: dial tcp 10.43.139.205:3306: connect: connection refused

```
[root@pvcecs-master ~]# kubectl logs -n dex-base-j95kx9f cdp-cde-embedded-db-0
```

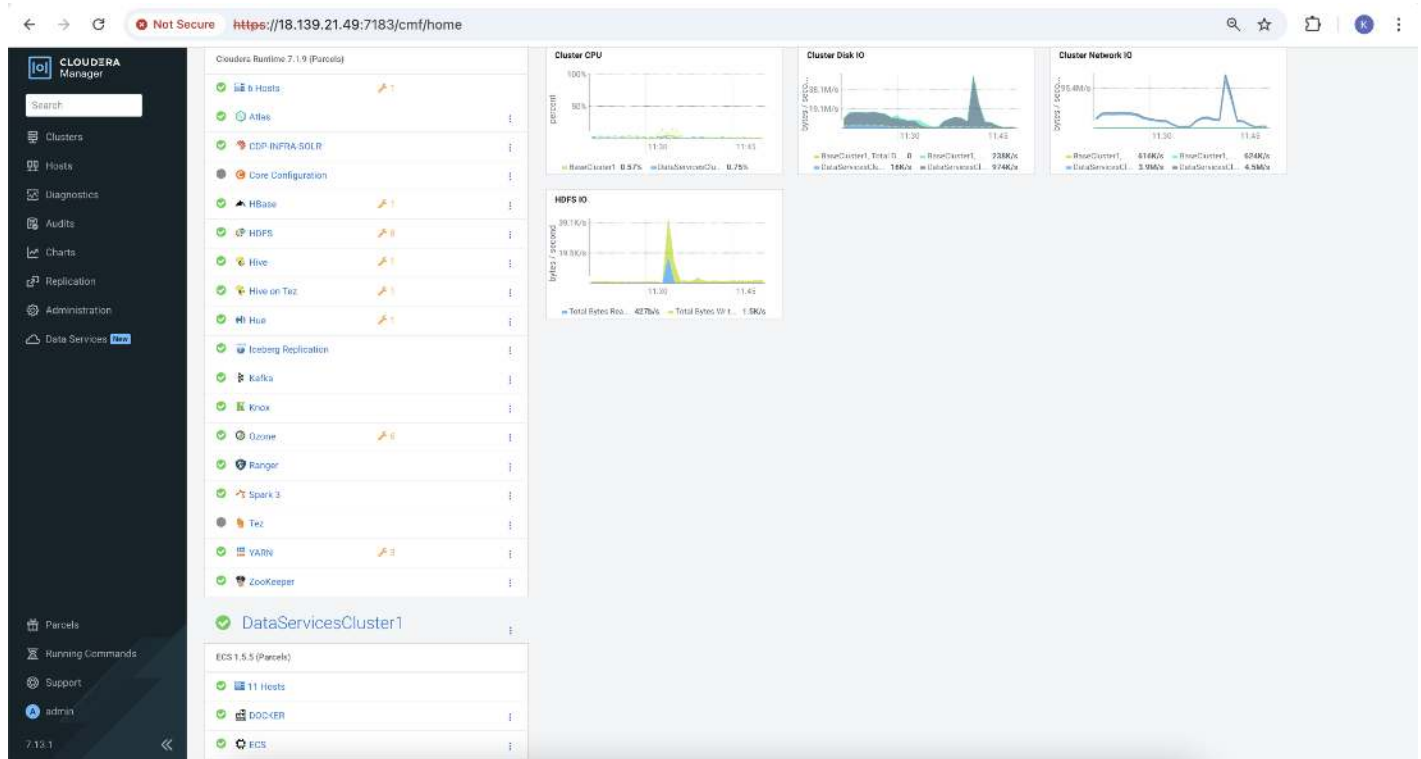
Defaulted container "cdp-cde-embedded-db" out of: cdp-cde-embedded-db, k8tz (init)

```
[root@pvcecs-master ~]#
```


28. When the installation is complete, the **Summary page** appears. Click **Launch Cloudera on premises**.
Add Private Cloud Containerized Cluster



29. Alternatively, you can also click **Finish** and then **access the Private Cloud Data Services** instance from **Cloudera Manager**. Click **Data Services**, then click **Open Private Cloud Data Services** for the applicable Data Services cluster.



Additional Steps for ECS Cluster Setup: *(Optional, Skip this step)*

Note: If `nvgsd-gpu-feature-discovery-xxxx` pods remain in `crashloopbackoff` please apply patch to fix the issue.

```
[root@pvcecs-master ~]# kubectl patch clusterrolebinding gpu-feature-discovery -p
'{"subjects":[{"kind":"ServiceAccount","name":"gpu-feature-discovery","namespace":"kube-system"}]}'
```

The screenshot shows the Cloudera Manager interface. On the left is a sidebar with navigation options: Clusters, Hosts, Diagnostics, Audits, Charts, Replication, Administration, and Data Services (marked as 'New'). The main panel displays the 'ContainerizedCluster1' status. It includes tabs for Status, Health Issues, and Configuration. Under the 'Status' tab, there's a section for 'ECS 1.5.5 (Parcels)' showing 11 Hosts, DOCKER, and ECS components, all with green checkmarks indicating they are healthy. Below this is a 'Tags' section with an 'Edit Tags' button.

To reserve a GPU node in Cloudera on premises Data Services ECS cluster, assign a taint to the node. Set the node taint “`nvidia.com/gpu: true:NoSchedule`” For more details on setting up GPU node:

<https://docs.cloudera.com/machine-learning/1.5.5/private-cloud-requirements/topics/ml-gpu-node-setup.html>

Step 1. To *set up a GPU node* for ECS, go to **Hosts > Configuration**.

The screenshot shows the Cloudera Manager 'All Hosts' page. The left sidebar is the same as in the previous screenshot. The main panel shows a list of hosts filtered by 'ecs'. On the left, there are filter sections for STATUS (Good Health, 11), CLUSTERS, CORES, COMMISSION STATE, LAST HEARTBEAT, LOAD (1 MINUTE), LOAD (5 MINUTES), LOAD (15 MINUTES), MAINTENANCE MODE, UPGRADE DOMAIN, RACK, and SERVICE. On the right, there's a table of hosts with columns for Status, Name, IP, Roles, and Tags. The table lists 10 hosts, all with a green checkmark in the Status column and '2 Roles' in the Roles column.

Status	Name	IP	Roles	Tags
✓	pvcecs-master.cldrsetup.local	172.31.25.60	2 Roles	
✓	pvcecs-worker1.cldrsetup.local	172.31.24.243	2 Roles	
✓	pvcecs-worker10.cldrsetup.local	172.31.20.154	2 Roles	
✓	pvcecs-worker2.cldrsetup.local	172.31.19.45	2 Roles	
✓	pvcecs-worker3.cldrsetup.local	172.31.19.232	2 Roles	
✓	pvcecs-worker4.cldrsetup.local	172.31.18.53	2 Roles	
✓	pvcecs-worker5.cldrsetup.local	172.31.30.54	2 Roles	
✓	pvcecs-worker6.cldrsetup.local	172.31.28.100	2 Roles	
✓	pvcecs-worker7.cldrsetup.local	172.31.17.199	2 Roles	
✓	pvcecs-worker8.cldrsetup.local	172.31.27.59	2 Roles	
✓	pvcecs-worker9.cldrsetup.local	172.31.18.67	2 Roles	

Step 2. Edit value for Data Services: Restrict workload types (node_taint) by clicking on Add Host Overrides.

Hosts Configuration

Q gpu

Filters

SCOPE

All Hosts1

CATEGORY

Advanced1

Monitoring0

Data Services: Restrict workloads types

node_taint

Dedicated GPU Node

Dedicated NVME Node

☒None

Add Host Overrides

Step 3. Add Host Overrides for the ECS nodes as per the requirement. For example, we selected two of the four nodes as Dedicated GPU Nodes.

Add Host Overrides - Data Services: Restrict workloads types

Specify a new override value for the selected hosts below.

☒Dedicated GPU Node

☐Dedicated NVME Node

☐None

Q ecs

☐	Hostname	IP Address	Rack	Cores	Physical Memory
☒	cdip-ecs1.cdip.cisco.local	10.29.148.164	/default	128	1007 GiB
☒	cdip-ecs2.cdip.cisco.local	10.29.148.165	/default	128	1007 GiB
☐	cdip-ecs3.cdip.cisco.local	10.29.148.166	/default	128	1007 GiB
☐	cdip-ecs4.cdip.cisco.local	10.29.148.167	/default	128	1007 GiB

1 - 4 of 4

Cancel

Add (2)

Step 4. Click on **Add** and click **Save Changes**.

Dedicating ECS nodes for specific workloads *(Optional, Skip this step)*

You use Cloudera Manager to dedicate Embedded Container Service (ECS) cluster nodes for specific workloads. You can dedicate GPU nodes for CAI workloads, and NVME nodes for CDW workloads.

Dedicating ECS nodes when creating a new cluster

1. Check the ECS installation requirements.
2. Add the new hosts to **Cloudera Manager**.
3. In Cloudera Manager, click **Hosts > All Hosts**, then select one or more of the **new ECS hosts**.
4. Click the **Configuration** tab, then use the **Search box** to locate the **node_taint** configuration property.
5. **Select Dedicated GPU Node** to dedicate the node for CAI workloads, or select **Dedicated NVME node** to dedicate the node for **CDW** workloads. When either of these options are selected, no other workload pods will be allowed to run on the dedicated node.
6. Click **Save Changes**.
7. **Repeat the previous steps** to add the **other ECS hosts** to **Cloudera Manager** and **assign** workload types.
8. Follow the ECS installation procedure. When you reach the **Specify Hosts** page in the installation wizard, the hosts you added to **Cloudera Manager** appear. Select the hosts, click **Continue**, then proceed through the rest of the installation wizard.
9. After the installation is complete, the applicable workloads will only run on the specified dedicated nodes.

The screenshot shows the Cloudera Manager interface. On the left is a dark sidebar with navigation links: Clusters, Hosts (selected), Diagnostics, Audits, Charts, Replication, Administration, Data Services (with a 'New' button), Parcels, Running Commands, Support, and a user profile 'admin'. The main content area is titled 'Hosts Configuration' and has a search bar containing 'node_taint'. Below the search bar are filter panels. The 'SCOPE' filter shows 'All Hosts' with a count of 1. The 'CATEGORY' filter shows 'Advanced' (1), 'Monitoring' (0), 'Parcels' (0), and 'Resource Management' (0). The 'STATUS' filter shows 'Error' (0), 'Warning' (0), 'Edited' (1), 'Non-Default' (1), and 'Include Overrides' (0). The main configuration area shows 'Data Services: Restrict workloads types' with a search icon and a 'Show All Descriptions' link. Below this, there are three radio button options: 'Dedicated GPU Node' (selected), 'Dedicated NVME Node', and 'None'. There are also links for 'Undo' and 'Add Host Overrides'. At the bottom, a status bar indicates '1 Edited Value' and 'Reason for change: Modified Data Services: Restrict workloads types', followed by a 'Save Changes(CTRL+S)' button.

Dedicating ECS nodes in an existing cluster

1. Open the **Cloudera Manager Admin Console**.
2. On the **Home page**, click the **ECS Cluster**.
3. Click **Hosts**, select one or more of the **ECS hosts**, then click the **Configuration** tab.
4. Click the **Configuration** tab, then use the **Search box** to locate the **node_taint** configuration property.
5. Select **Dedicated GPU Node** to dedicate the node for **CAI** workloads, or select **Dedicated NVME node** to dedicate the node for **CDW** workloads. When either of these options are selected, no other workload pods will be allowed to run on the dedicated node.

Search

Clusters

Hosts

Diagnostics

Audits

Charts

Replication

Administration

Data Services New

Parcels

Running Commands

Support

admin

7.11.3

Hosts Configuration

CDP Deployment from 2023-Sep-28 05:29

node_taint

Filters History & Rollback

Filters

SCOPE

All Hosts 1

CATEGORY

Advanced 1

Monitoring 0

Parcels 0

Resource Management 0

STATUS

Error 0

Warning 0

Edited 1

Non-Default 1

Include Overrides 0

Data Services: Restrict workloads types

node_taint

Dedicated GPU Node

Dedicated NVME Node

None

Undo

Add Host Overrides

Show All Descriptions

1 - 1 of 1

1 Edited Value

Reason for change:

Modified Data Services: Restrict workloads types

Save Changes(CTRL+S)

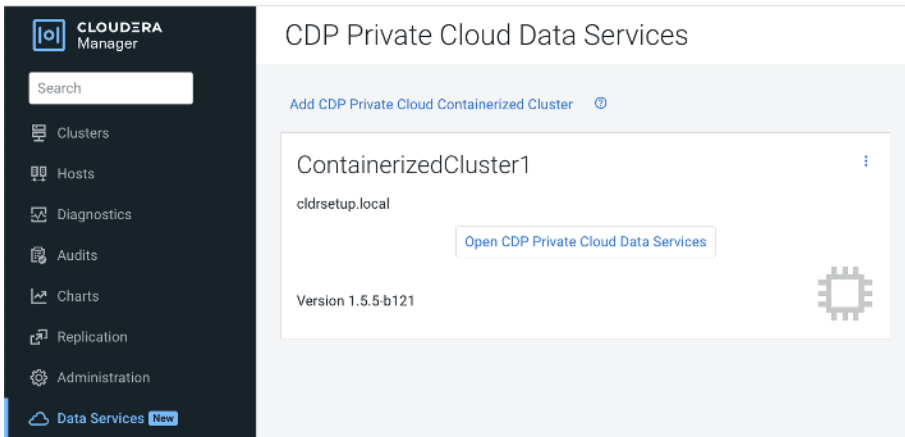
- Click **Save Changes**.
- Repeat the previous steps** to assign workload types to the other **ECS** hosts.
- On the **ECS Cluster** landing page, click **Actions > Refresh Cluster**.
- After the Refresh is complete, click **Actions > Rolling Restart**.

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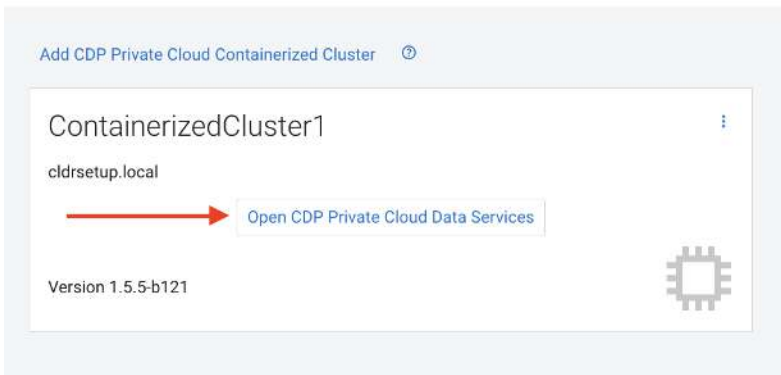
Accessing Cloudera on premises

Step 1. From the **Cloudera Manager** screen, click on **Data Services(New)** in the left pane.

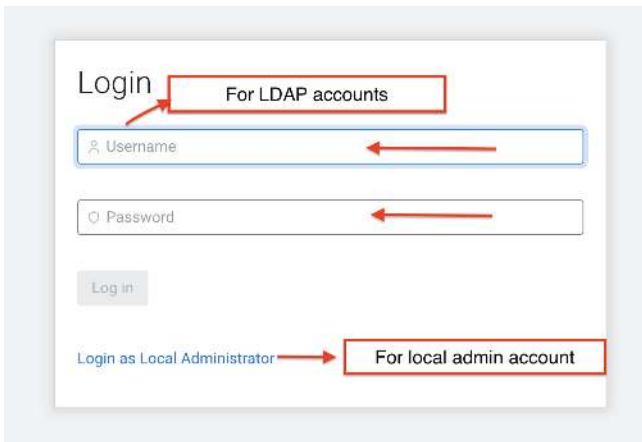


Step 2. On the **CDP Private Cloud Containerized services** page, click on the **Open CDP Private Cloud Data Services** button. This will open the **Cloudera on premises authentication** page.

CDP Private Cloud Data Services



Step 3. On the PvC DS Authentication Page, if you have **LDAP account** credentials, enter its username and password and then click on **Login**. Else, you can click on **Login as Local Administrator**, and enter the default credentials. (**admin/admin**)



Note: From DS 1.5.5 SP1 onwards, it is recommended to use an LDAP user to log in to the DS console and run workloads, as admin users have certain restrictions when executing workloads that were introduced in this release.

Step 4. Login to *Cloudera on premises Data Services* as *local administrator: (admin/admin)*

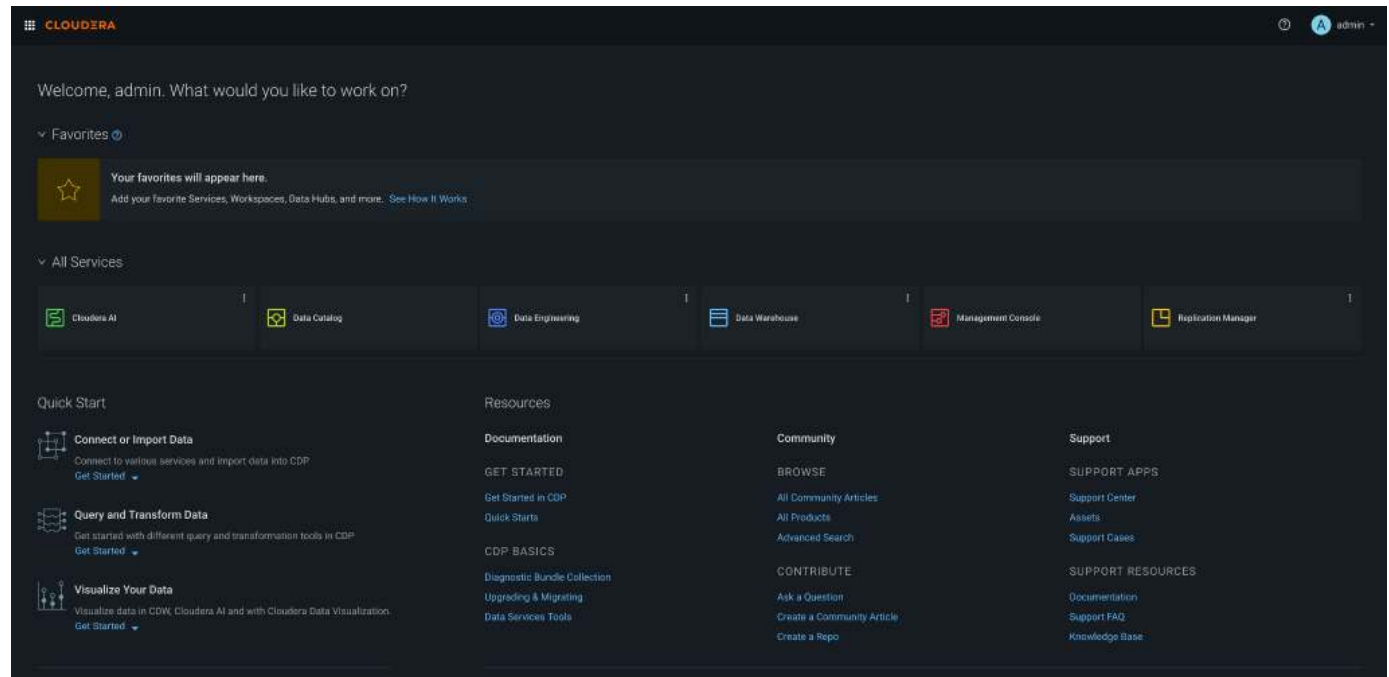
Login as Local Administrator

 admin

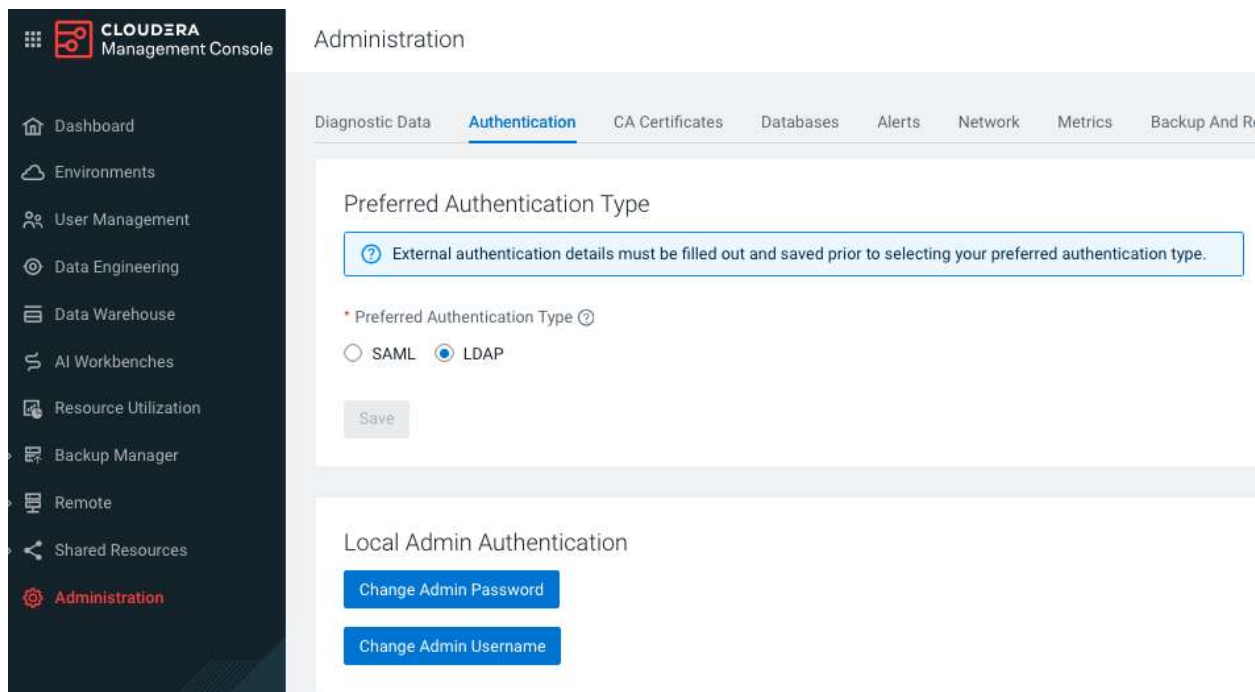
 ●●●●●

Log in

Step 5. After authenticating successfully, you will land at the *CDP console/Data Services* page. From this page, you can navigate to different data services and the management services. Click on the *Management Console*.



Step 6. On the *Welcome to Cloudera on premises* page, click *Reset Password* to change the *Local Administrator Account password*. **(OR)** On the *Management Console* page, navigate to *Administration > Authentication*, and then click *Reset Password* to change the *Local Administrator Account password*.



Welcome to CDP Private Cloud

Local Admin Account

We recommend you to reset your default admin password.

[Reset Password](#)

Step 7. *Set up external authentication* using the URL of the LDAP server and a CA certificate (If using prior existing LDAP server and not proceeding with FreeIPA setup) of your secure LDAP (*e.g. ldap://<ipa_or_ldap_server_fqdn>:389/*). Learn more about [LDAP user authentication for Cloudera on premises](#). Enter values for ldap authentication, as mentioned in the below table.

Table 14. LDAP FreeIPA Integration

Component	Value
Authentication Backend Order:	Database then EXTERNAL
Authorization Backend Order:	Database and EXTERNAL
External Authentication Type:	LDAP
LDAP URL:	ldap://ipaserver.cldrsetup.local:389/
LDAP Bind User Distinguished Name:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Bind Password:	<cloudera123> (password for KDC admin, configured earlier)

Component	Value
LDAP User Search filter:	(&(uid={0})(objectClass=person))
LDAP User Search Base:	cn=users,cn=accounts,dc=cldrsetup,dc=local
LDAP Group Search filter:	(&(member={1})(objectClass=posixgroup))
LDAP Group Search Base:	cn=groups,cn=accounts,dc=cldrsetup,dc=local
LDAP DistName Pattern:	uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local

Table 15. LDAP AD Integration

Component	Value
External Authentication Type:	LDAP
LDAP URL:	ldap://dc.ntnslab.local:389/
LDAP Bind User Distinguished Name:	Administrator@ntnslab.local
LDAP Bind Password:	<cloudera123> (password for KDC admin, configured earlier)
LDAP User Search filter:	(sAMAccountName={0})
LDAP User Search Base:	OU=cloudera,DC=ntnslab,DC=local
LDAP Group Search filter:	(&(member={1})(objectClass=group))
LDAP Group Search Base:	OU=cloudera,DC=ntnslab,DC=local
Username Mapping Attribute	sAMAccountName
Email Mapping Attribute	userPrincipalName

Administration

Do you wish to sync user groups at login? ⓘ
☒ **Sync Groups on Login**

Do you wish to generate workload username by email? ⓘ
☒ **Generate Workload Username by Email**

LDAP
 Configure LDAP settings.

* LDAP URL ⓘ

CA Certificate for Secure LDAP ⓘ
 CA Certificate is not needed

Do you wish to sync user groups at login? ⓘ
☒ **Sync Groups on Login**

Bind Settings
 * Bind Type
☒ **Use Bind DN and Password** ☐ Use Anonymous Bind

* LDAP Bind DN ⓘ

* LDAP Bind Password ⓘ

Search Base Settings
 * LDAP User Search Base ⓘ

* LDAP User Search Filter ⓘ

* LDAP Group Search Base ⓘ

* LDAP Group Search Filter ⓘ

Email Mapping
 Email Mapping Attribute ⓘ

Additional LDAP Settings
 These are additional optional settings that you may want to configure.
 Do you wish to configure advanced settings?
☐ Yes, show me advanced settings

[Test LDAP Connection](#)

Step 8. Follow the instructions on the *Welcome to CDP Private Cloud* page to complete this step.

Step 9. Click **Test Connection** to ensure that you are able to connect to the *configured LDAP server*.

[Test Connection](#)

☒ **Connection Successful!**

[Save](#)

Step 10. The **User Management** tab allows users to add or update roles on existing users. **Groups** tab allows users to sync user groups from the active directory to access *CDP Data Services*.

CLOUDERA
Management Console

Dashboard
Environments
User Management
Data Warehouse
ML Workspaces
Resource Utilization
Clusters
Administration

User Management

Users

Groups

Type: All

Actions

Create Machine User

Upload Users

Update Account Messages

Type	Name	Email	Workload User Name	Password Expiring
★	admin@cdp.example	admin@cdp.example	admin	
	cdpbind@cdp.example	cdpbind@cdp.example	cdpbind	
	dp_profiler_user		dp_profiler_user	
	hardipat@cdp.example	hardipat@cdp.example	hardipat	
	machineuser		machineuser	

Displaying 1 - 5 of 5

1

25 / page

CLOUDERA
Management Console

Dashboard
Environments
User Management
Data Warehouse
ML Workspaces
Resource Utilization
Clusters
Administration

User Management

Users

Groups

Type: All

Actions

Update Roles

Generate Access Key

Delete User

Type	Name	Email	Workload User Name	Password Expiring
★	admin@cdp.example	admin@cdp.example	admin	
	ldapuser1@cdpkvm.cldr	ldapuser1@cdpkvm.cldr	ldapuser1	

Displaying 1 - 2 of

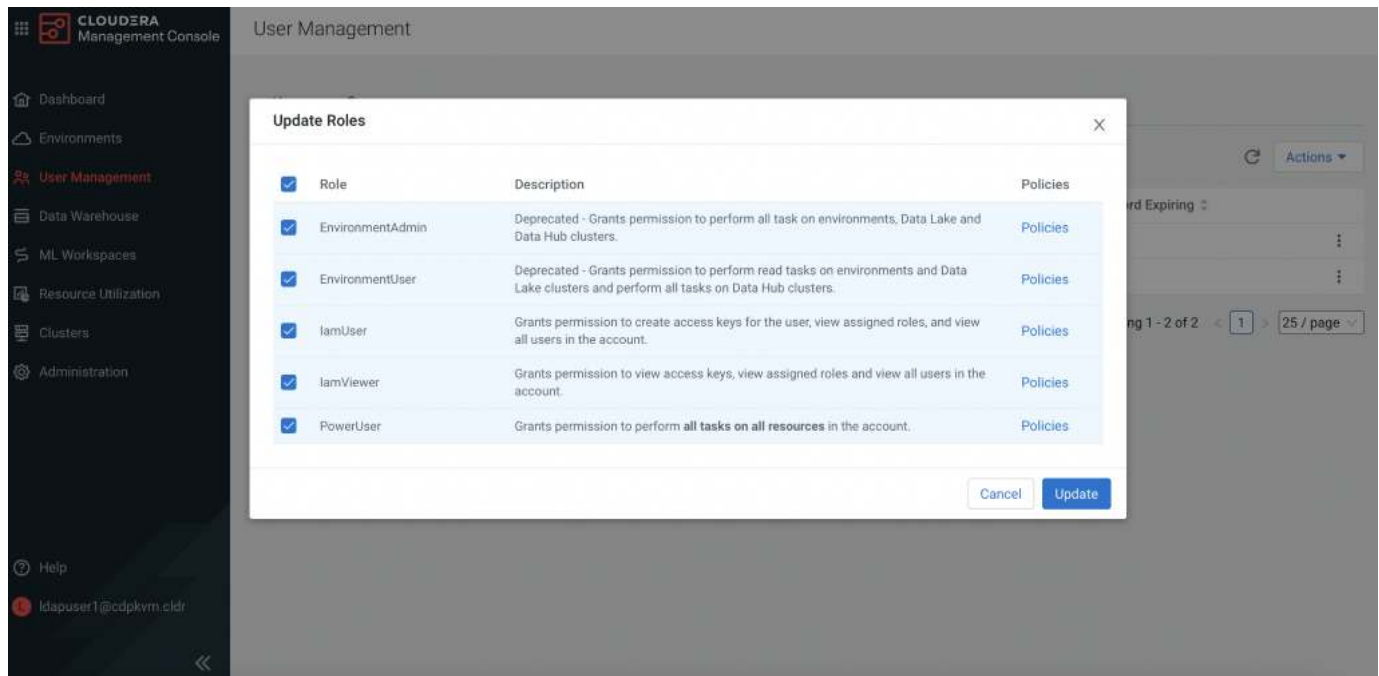
Help
admin@cdp.example

For more details on Cloudera on premises Management console please visit:

<https://docs.cloudera.com/management-console/1.5.5/index.html>

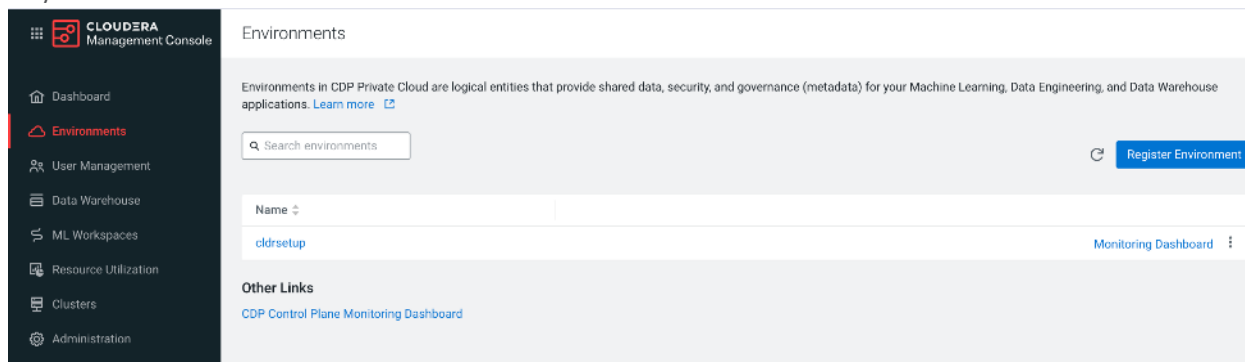
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Step 11. After successfully configuring and testing the setup for **LDAP integration**, the page will auto-redirect for the Environments Page, if not, navigate to the [Environments](#) page, by clicking into the menu in the left pane of your screen.

Step 12. If everything was correctly setup previously, then you will be able to see the environment registered by default for your cluster as shown below:



Step 13. If environment is not created somehow due to some issue during the installation, page will auto-redirect for **Register Environment** Page, if not, navigate to the [Environments](#) page, and select **Register Environment** where you will provide the Cloudera Manager details and credentials to register the **PvC Base Cluster DataLake/Control Plane environment** to DS. Click **Choose Cluster**, select the PvC Base cluster from the populated list and click on **Register**.

 **CLOUDERA**
Management Console

Dashboard

Environments

User Management

Data Warehouse

ML Workspaces

Resource Utilization

Clusters

Administration

Environments

Environments in CDP Private Cloud are logical entities that provide shared data, security, and governance (metadata) for your Machine Learning, Data Engineering, and Data Warehouse applications. [Learn more](#)

Register Environment

Name

No data

Other Links
[CDP Control Plane Monitoring Dashboard](#)

 **CLOUDERA**
Management Console

Dashboard

Environments

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ML Workspaces

Resource Utilization

Clusters

Administration

Environments : Register Environment

Register Environment

Register an environment to share data, security, and governance (metadata) for your machine learning and data warehouse applications

* Environment Name 

Data Lake

* Cloudera Manager 

* Cloudera Manager Admin Username 

* Cloudera Manager Admin Password 

Choose Cluster 

Connected to https://cm.cdpkvm.cldr:7183 with 1 cluster(s) found.

Choose Cluster

 Ozone

 Ranger

 Atlas

 Hive Metastore

 HDFS

Cancel

Register

 **CLOUDERA**
Management Console

Dashboard

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User Management

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ML Workspaces

Resource Utilization

Clusters

Administration

Environments

Environments in CDP Private Cloud are logical entities that provide shared data, security, and governance (metadata) for your Machine Learning, Data Engineering, and Data Warehouse applications. [Learn more](#)

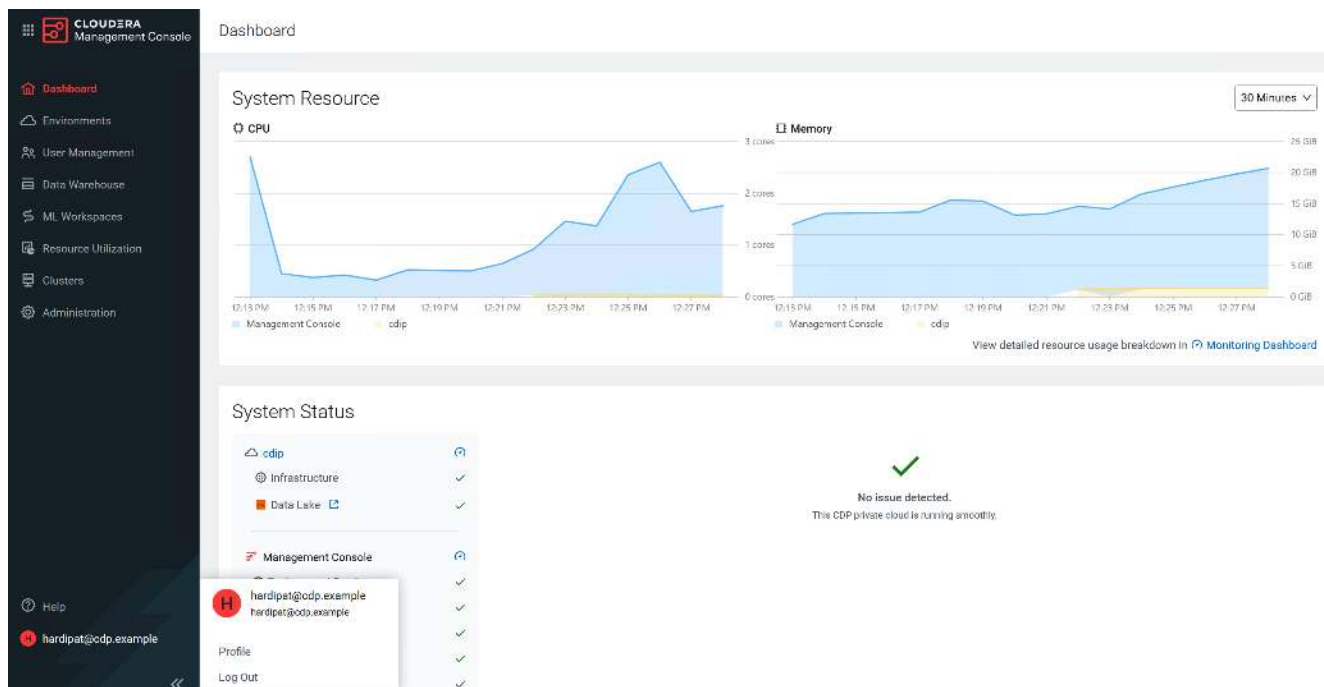
Register Environment

Name

cdp-env-1

Monitoring Dashboard

Other Links
[CDP Control Plane Monitoring Dashboard](#)



Step 14. To come to this section further, from the *Cloudera Manager* screen, click on **Data Services(New)** in the left pane and then click on **Open Private Cloud Data Services** to launch your **CDP Private Cloud Data Services instance**. Log in using the default username and password **admin**.

- Click **Launch CDP** to launch your CDP Cloudera on premises.
- Log in using the default username and password admin.
- In the **Welcome to CDP Private Cloud** page, click **Change Password** to change the Local Administrator Account password.
- Set up external authentication using the URL of the LDAP server and a CA certificate of your secure LDAP. Follow the instructions on the **Welcome to CDP Private Cloud** page to complete this step.
- Click **Test Connection** to ensure that you are able to connect to the configured LDAP server.
- [Register a CDP Cloudera on premises environment](#)
- [Create your first Virtual Warehouse in the CDW Data Service](#)
- [Provision an AI Workbench in the CAI Data Service](#)
- <https://docs.cloudera.com/data-engineering/1.5.5/enable-data-engineering/topics/cde-private-cloud-add-cde-service.html>

Cloudera on premises Machine Learning (CAI)

Please review [Cloudera on premises Machine Learning](#) for more details.

Please review requirements page for ECS and get started with CAI on Cloudera on premises:

<https://docs.cloudera.com/machine-learning/1.5.5/private-cloud-requirements/topics/ml-pvc-intro.html>

For more details on CAI workspace and how to steps, visit:

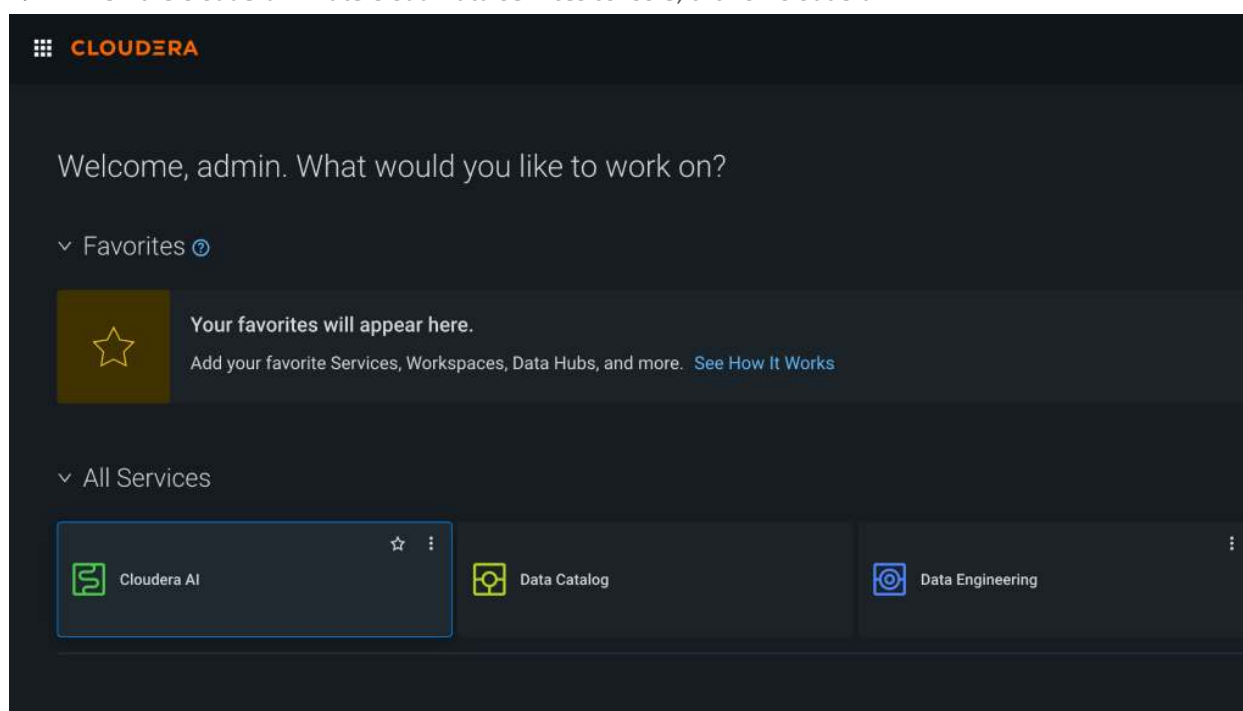
<https://docs.cloudera.com/machine-learning/cloud/workspaces/topics/ml-provision-workspaces.html>

<https://docs.cloudera.com/machine-learning/1.5.5/workspaces-privatecloud/topics/ml-pvc-provision-ml-workspace.html>

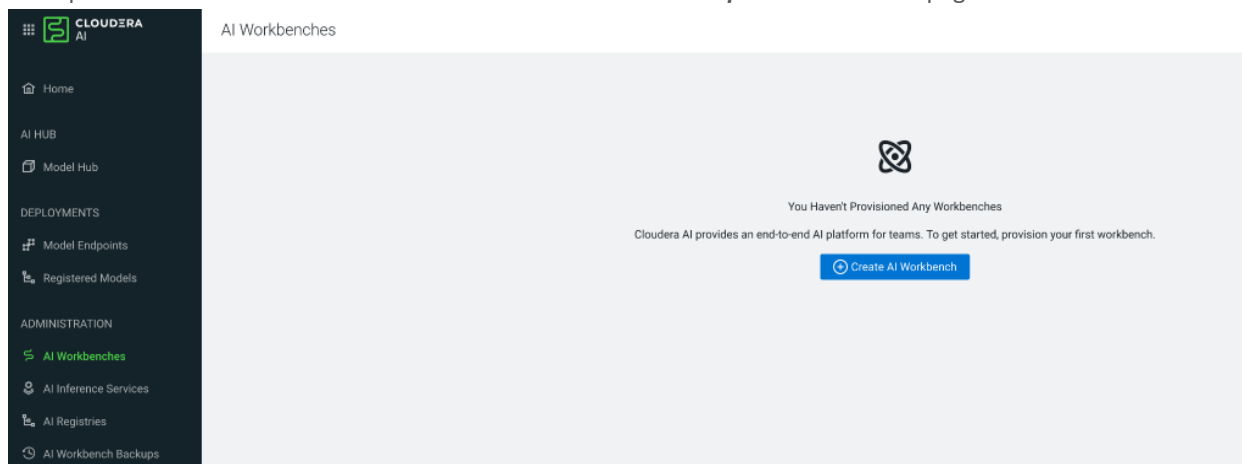
AI Workbench Creation:

To get started with CAI follow steps below:

1. On the **Cloudera Private Cloud Data Services console**, click on Cloudera **AI**.



2. First time login requires provision of a workspace. Since this will be the first time you open CAI, there will be no CAI workspace. You will see the screen below. Click on **Provision Workspace** on the same page.



3. Provide input required to provision machine learning workspace. Click on **Create AI Workbench**.

- Enter the configuration values for the workspace as described below.
 - **Workbench name:** A suitable name for the workbench.
 - **Environment:** Select the default environment from the drop down.
 - **Namespace:** This will be the kubernetes namespace under which the pods would be spinned up. By default, it is set to CAI. You can change it if you wish to.
 - **NFS server:** Select **Internal**.
 - If you choose **External NFS Server**, perform on **all ECS nodes**. (Skip this, as we are not using it in current setup)

```
#### nfs://172.31.30.239:/lndata/nfs_storage/kuldeep-test-cml-w1
[root@pvcecs-master ~]# mkdir -p /lndata/nfs_storage/kuldeep-test-cml-w1
[root@pvcecs-master ~]# chown 8536:8536 /lndata/nfs_storage/kuldeep-test-cml-w1
```

- Under Production Learning, the below parameters need to be updated.
 - **Enable Governance:** This provided advanced lineage and governance features. For simple demos or POCs, you may choose to disable it.
 - **Enable Model Metrics:** Keep it enabled. It provides you with the metrics.
 - **Enable TLS:** You can keep it disabled.
 - **Enable Monitoring:** This helps in monitoring the resource usage for the provisioned workspace. Enable it.
 - **CAI Static Subdomain:** Enter any short name for this parameter that helps in monitoring the resource usage for the provisioned workspace.

Provision Cloudera AI Workbench

Provision an on-demand Cloudera AI Workbench.

* Workbench Name

* Select Environment

Environment type: **None selected**

* Namespace ⓘ

NFS Server ⓘ

☐ Internal ☒ External

ⓘ This selection uses an external NFS export path (or a subdirectory within it).

* Existing NFS ⓘ

⚠ Note: An administrator must run **chown 8536:8536** on the NFS directory.

The directory must be empty and not used by another workbench.

NFS Protocol version ⓘ

Production Cloudera AI

☐ Enable Governance ⓘ

☒ Enable Model Metrics ⓘ

Quota Allocation

CPU (Cores)

50

Memory (Gi)

46

GPU

0

Other Settings

☐ Enable TLS ⓘ

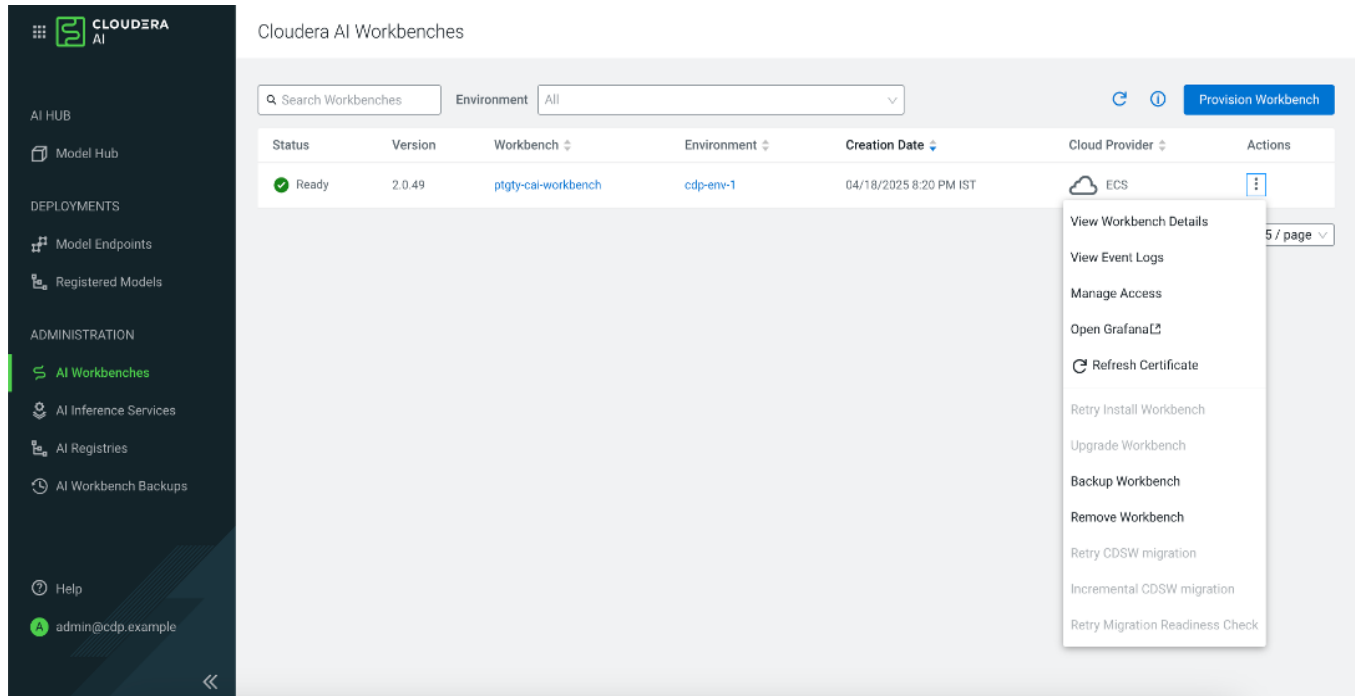
☒ Enable Monitoring ⓘ

☐ Skip Validation ⓘ

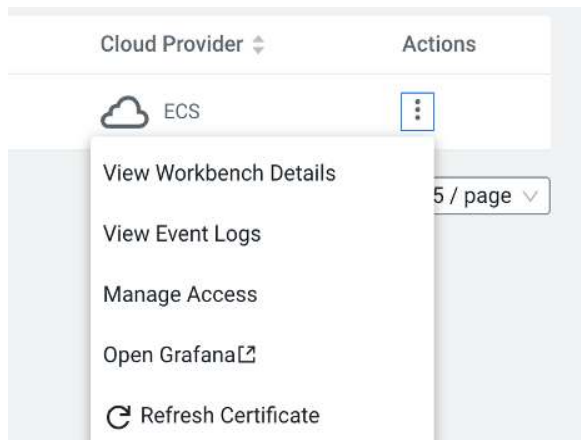
Cloudera AI Static Subdomain ⓘ

Note: Click on ⓘ icon to get more information on the field.

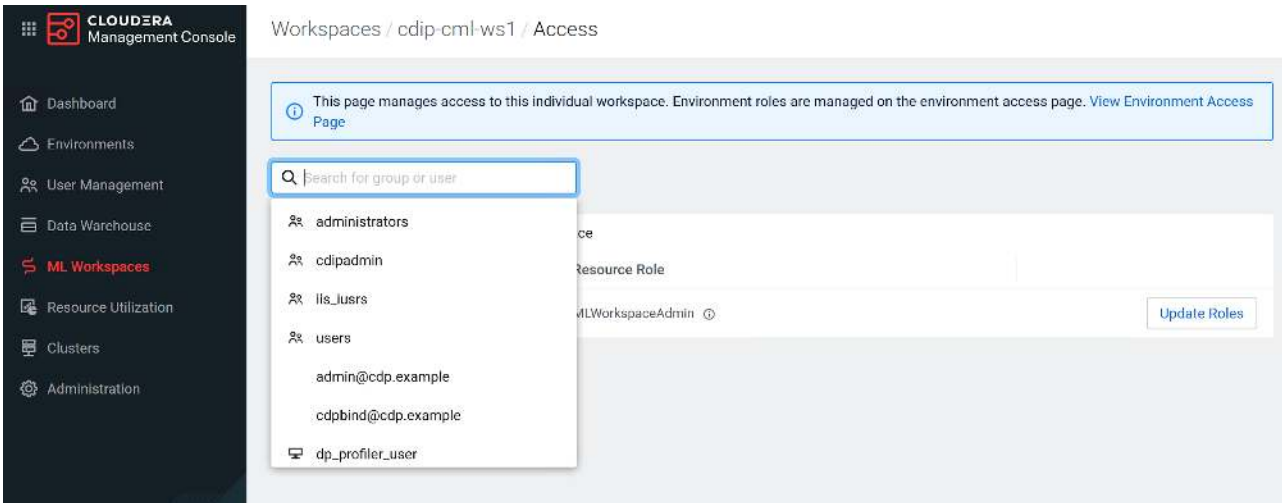
4. When provisioning of the workbench is completed the status reports as **Ready**. Once it is created, it appears on the AI Workbenches page as shown below.



5. Click on Manage Access.



6. In the search field search for a user or group to be able to access AI Workbench.



7. Update Resource role for user or group selected to manage access to workbench provisioned in Cloudera AI.

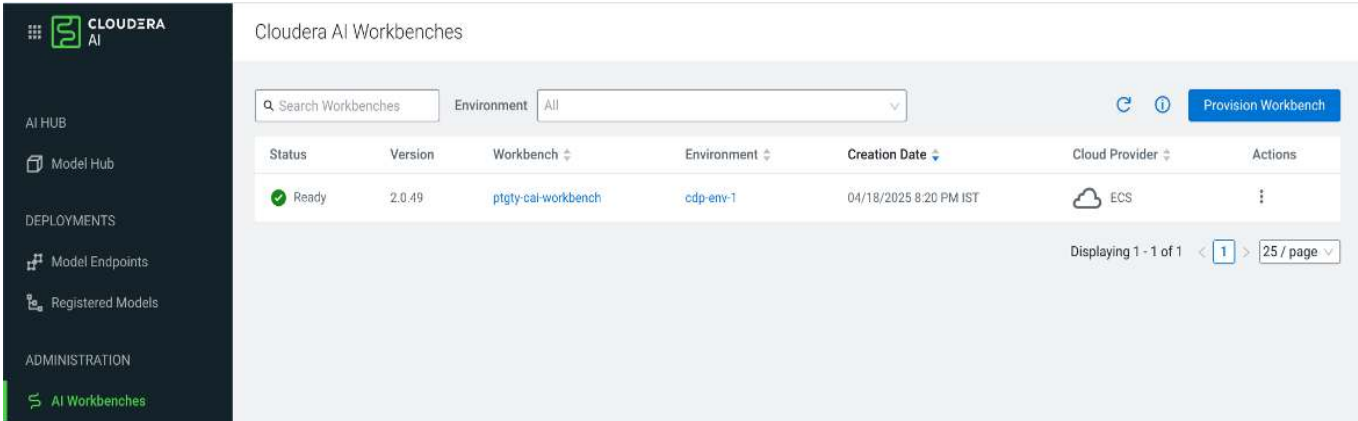
Update Resource Roles for cdipadmin ✕

Resource Roles

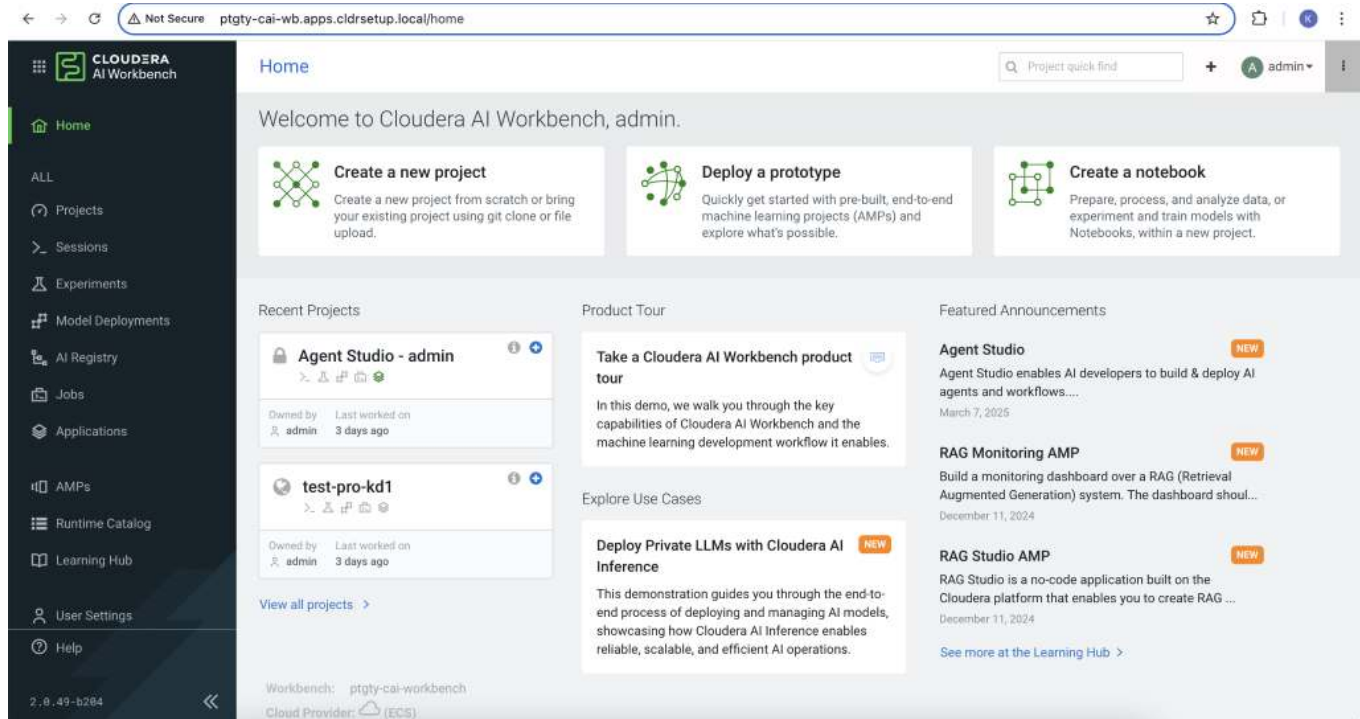
☒	Role	Description
☒	MLWorkspaceAdmin ⓘ	Grants permission to manage all machine learning workloads and settings inside a specific workspace.
☒	MLWorkspaceBusinessUser ⓘ	Grants permission to view shared machine learning applications inside a specific workspace.
☒	MLWorkspaceUser ⓘ	Grants permission to run machine learning workloads inside a specific workspace.
☒	Owner ⓘ	Grants all permissions on the resource.

Cancel Update Roles

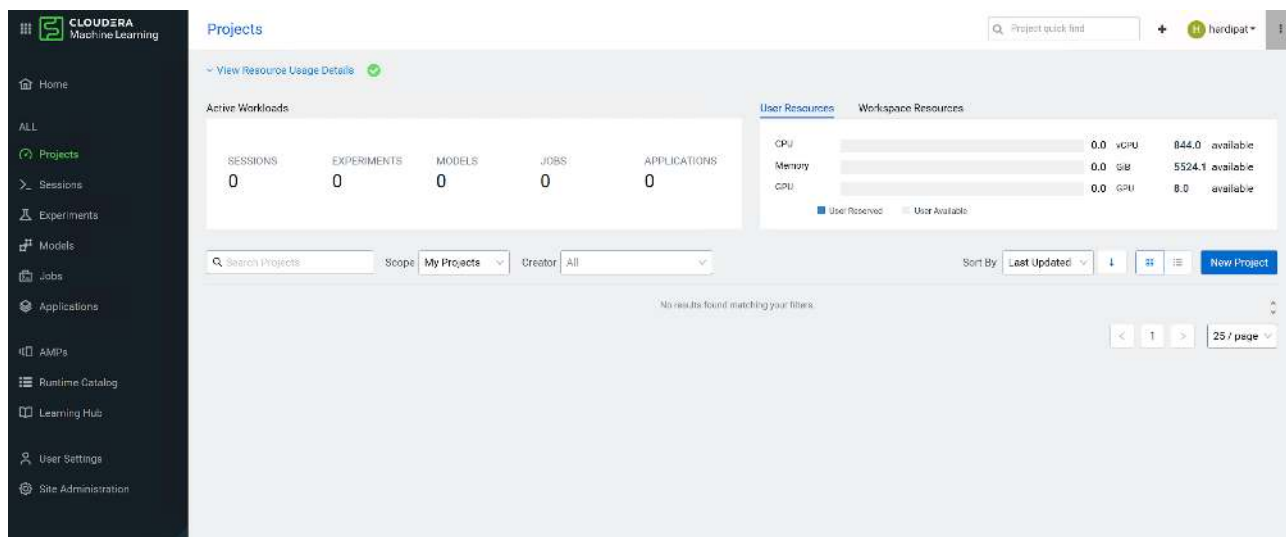
8. Click on the workbench name created.



9. AI Workbench **WebUI** overview.



10. Click on Projects tab, expand View Resource Usage Details to review available resources.



11. For more details and how to review projects section in AI Workbench:
<https://docs.cloudera.com/machine-learning/cloud/projects/index.html>

Creation of Project in AI Workbench:

12. At the middle right, you will find the New Project button. Click on it. New Project page appears. Enter the details as described below. Enter project name and select type of initial setup.

- **Project Name:** Enter a suitable name for your project.
- **Project Description:** Enter a description for the project.

- **Project Visibility:** Keep it Public for any demos or PoC's. If you are creating this in a multi-tenant environment, choose Private.

The Initial Setup Section for the New Project has five options as described below. Choose any of these based on your requirement.

- **Blank:** Choose this if you want to start from scratch.
- **Template:** Template projects contain example code that can help you get started with Cloudera AI. They are available in R, Python, PySpark, and Scala. Using a template project is not required, but it helps you start using Cloudera AI right away.
 - **AMPs:** Applied ML Prototypes provide components to create a complete project. They may include jobs, models and experiments.
- **Local Files:** Choose this if you have all the necessary files in a folder or in a zip.
- **Git:** Choose this if all the resources are stored in a github project.

The screenshot shows the 'Initial Setup' section of the Cloudera AI interface. It includes fields for 'Project Owner' (set to 'admin') and '* Project Name' (set to 'Agent Studio - admin'). Below these is a 'Project Description' text area containing 'Deploy AMP for Agent Studio'. The 'Project Visibility' section has two radio buttons: 'Private - Only added collaborators can view the project' (selected) and 'Public - All authenticated users can view this project'. Under the 'Initial Setup' heading, there are five buttons: 'Blank', 'Template' (highlighted), 'AMPs', 'Local Files', and 'Git'. Below the buttons, a message states 'Templates include example code to help you get started.' followed by a search bar containing 'Python'. A dropdown menu is open, showing the following options: 'R', 'Python' (highlighted), 'PySpark', and 'Scala'.

13. Select **Runtime setup**. For initial exploration, select Basic and keep the kernel to **Python3.9** (enable checkbox to add GPU enabled Runtime variant, if applicable — **We are not using GPU in our current setup**).

Runtime setup

Basic Advanced

Basic configuration adds the most commonly used Editors for the Kernel of your choice. To fine-tune the Editors available in the project, choose the Advanced tab.

Kernel

Cloudera Data Visualization

Python 3.7

Python 3.8

Python 3.9

R 3.6

R 4.0

R 4.1

Runtime setup

Basic Advanced

Basic configuration adds the most commonly used Editors for the Kernel of your choice. To fine-tune the Editors available in the project, choose the Advanced tab.

Kernel

☒ Add GPU enabled Runtime variant

These runtimes will be added to the project:

JupyterLab - Python 3.9 - Nvidia GPU - 2023.08
JupyterLab - Python 3.9 - Standard - 2023.08
PBJ Workbench - Python 3.9 - Nvidia GPU - 2023.08
PBJ Workbench - Python 3.9 - Standard - 2023.08
Workbench - Python 3.9 - Nvidia GPU - 2023.08
Workbench - Python 3.9 - Standard - 2023.08

- Click on **Create Project**. After some time, a new project will be created and will be available on the Projects page.
- Click on the sessions tab and enter details for the new session. You will see a warning like below:

Start A New Session

⚠ Not authenticated to Hadoop
 Before you can connect to your secure Hadoop cluster, you must enter your credentials under [User Settings > Hadoop Authentication](#)

Session Name

Runtime

Editor ⓘ	Kernel ⓘ	Edition ⓘ	Version
JupyterLab	Python 3.10	Nvidia GPU	2023.08

Configure additional runtime options in [Project Settings](#).

☐ Enable Spark ⓘ

Runtime Image -

Resource Profile

2 vCPU / 8 GiB Memory	1 GPU
--	------------------------------------

16. Before starting any new session in the recently created Project, you must complete the hadoop authentication part as the cluster setup is kerberized.

So, to be able to access data from Hadoop clusters **go to user > user settings > Hadoop authentication**.

Open a **new tab in the same browser** window, by duplicating the existing tab. Go to the **CAI home page** and click on **User Settings** in the left pane. Click on **Hadoop Authentication**.

CLOUDERA
Machine Learning

- Projects
- Sessions
- Experiments
- Models
- Jobs
- Applications
- User Settings**
- AMPs
- Runtime Catalog
- Site Administration
- Learning Hub

pkatti / [User Settings](#) / Hadoop Authentication

User Settings

[Profile](#)
[Outbound SSH](#)
[Hadoop Authentication](#)
[API Keys](#)
[Remote Editing](#)

Kerberos

To authenticate to Kerberos, enter your principal and **either** enter your password **or** upload a keytab file.

Principal

Credentials

Password

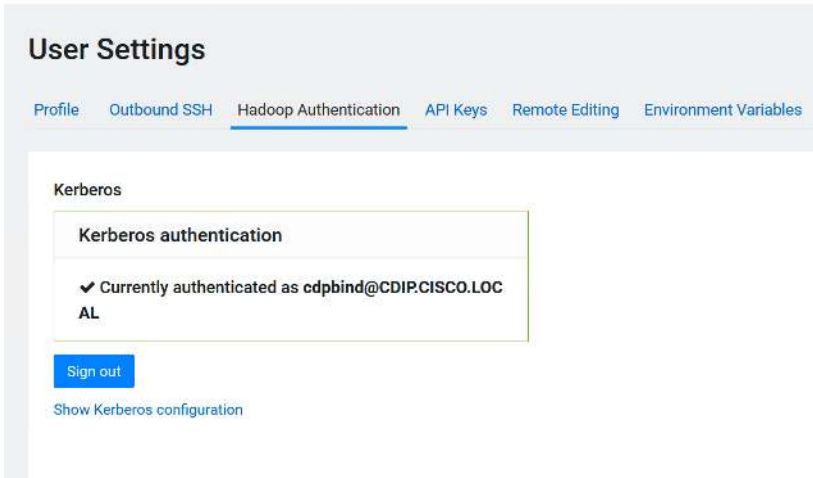
Enter Password

[Keytab](#)

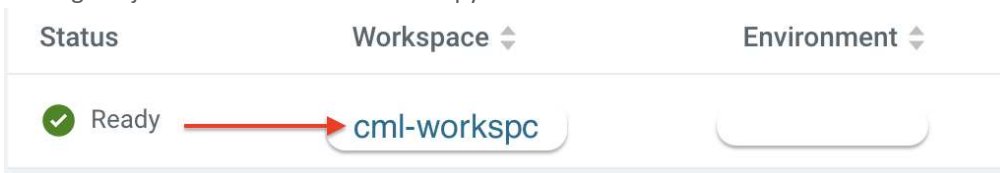
[Show Kerberos configuration](#)

- Enter **Principal** e.g. `username@DOMAIN.LOCAL` i.e. `admin@CLDRSETUP.LOCAL`
- Under the **Credentials** and password (i.e. `cloudera123`) or keytab details of the LDAP user and click on Authenticate.
- Once the authentication is successful, proceed to the next step. You will see the output similar to below screenshot, after the successful authentication and integration to Kerberized Hadoop Cluster.

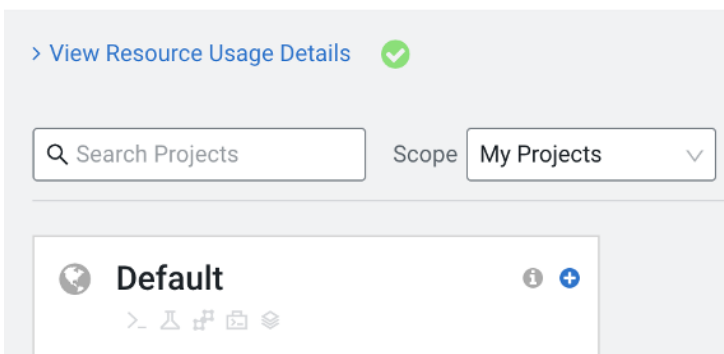
[hardipat](#) / [User Settings](#) / [Hadoop Authentication](#)



17. Now, to explore the CAI IDEs, click on the newly created workspace. It will open the Projects screen of CAI. Go to the **Project page** and click on the newly created project and then click on the **New Session** button on the top right. Explore CAI by running the jobs with different IDEs like Jupyterlab and Workbench.



Projects



18. Go to **Site Administration** to edit **Resource profile** and **GPU per session/ Job**.

Resource Profiles

vCPU is expressed in fractional virtual cores and allows bursting by default. Memory is expressed in fractional GiB and is enforced by memory killer. GPU indicates the number of GPUs that need to be used by the engine. Configurations larger than the maximum allocatable CPU, memory and GPU per node will be unschedulable.

Description	vCPU	Memory (GiB)	Actions
2 vCPU / 4 GiB Memory	2	4	Edit Delete
2 vCPU / 8 GiB Memory	2	8	Edit Delete
2 vCPU / 16 GiB Memory	2	16	Edit Delete
4 vCPU / 16 GiB Memory	4	16	Edit Delete
4 vCPU / 32 GiB Memory	4	32	Edit Delete
8 vCPU / 64 GiB Memory	8	64	Edit Delete
			Add

Maximum GPUs per Session/Job

2

☒ Enable CPU bursting

By default, Resource Profiles are using burstable CPU settings to help better resource utilization. To use the resource profile as a hard limit on vCPU consumption, disable CPU bursting.

Engine Images

☒ Disable Engines

Checking this checkbox will automatically disable Legacy Engine, and set default engine to ML Runtime for all the Projects.

19. Go to the **AMPs** tab to get started with pre-built models.

The screenshot shows the Cloudera AI Workbench interface. The left sidebar contains navigation links: Home, Projects, Sessions, Experiments, Model Deployments, AI Registry, Jobs, Applications, AMPs (highlighted), Runtime Catalog, Learning Hub, User Settings, and Help. The main panel is titled 'Accelerators for ML Projects' and includes a search bar and filters for 'Source' (Cloudera, Hugging Face) and 'Tags'. It lists 38 AMPs. Four featured AMPs are shown in a grid: Synthetic Data Studio (Generate high quality synthetic data for Supervised Fine-Tuning, Model Alignment, Knowledge Distillation and data curation), Agent Studio (Cloudera AI Agent Studio is a platform for building, testing, and deploying AI agents and workflows), RAG Studio (Build chatbots powered by your data - no code RAG studio for a quick start, easy access and fast innovation), and RAG Monitoring (Monitor real-time metrics for a RAG pipeline and gather insights into the pipeline's behavior). Each AMP card displays resource requirements (e.g., LLM +6, Agents ~2) and a 'Deploy' button with a 'Cloudera AMP' badge. The bottom of the page shows a grid of other AMPs like Fine Tuning and Prompt Engineering.

20. Select **AMP** and click on **Configure Project**.

Configure Project: Agent Studio - admin 1

AMP Name: Agent Studio (v1)

Cloudera AI Agent Studio is a workspace for developing and deploying AI agentic workflows.

Environment Variables

The settings below were defined by the AMP:

Name	Value	Description
* AGENT_STUDIO_NUM_WORKFLOW_RUNNERS	5	Number of workflow runners to spawn for testing workflows within Agent Studio. If multiple concurrent users of Agent Studio are expected, you can increase this number accordingly.

Runtime

Editor ⓘ

Kernel ⓘ

Edition ⓘ

Version

JupyterLab

Python 3.10

Standard

2025.01

22. **Agent Studio - admin – AMP** project overview.

Not Secure

ptgty-cai-wb.apps.cirsetup.local/admin/agent-studio-admin

CloudERA AI Workbench

Home

All Projects

PROJECT

Overview

Sessions

Experiments

Model Deployments

Jobs

Applications

Files

Collaborators

Project Settings

AMPs

Runtime Catalog

Learning Hub

Help

2.9.49-b204

admin / Agent Studio - admin

Project quick find

admin

Agent Studio - admin

Cloudera AI Agent Studio is a platform for building, testing, and deploying AI agents and workflows.

0 Fork New Session

Models

This project has no models yet. [Create a new model.](#)

Jobs

Name	Runs / Failures	Duration	Status	Latest Run	Actions
Agent Studio - Upgrade	0 / 0	00:00	Not Yet Run		Run

Files

[Download](#) [New](#) [Upload](#)

Name	Size	Last Modified
alerbic	-	3 days ago
app	-	3 days ago
bin	-	3 days ago
components	-	3 days ago
data	-	3 days ago
docs	-	3 days ago
examples	-	3 days ago
images	-	3 days ago

Workbench: ptgty-cai-workbench

Cloud Provider: ECS

23. Create a new session by **Start A New Session** with desired resources, editor, kernel, and number of GPUs.

Start A New Session

Session Name

llama2-test1

Runtime

Editor ⓘ

Kernel ⓘ

Edition ⓘ

Version

JupyterLab

Python 3.10

Standard

2023.08

Configure additional runtime options in [Project Settings](#).

☐ Enable Spark ⓘ

☒ Spark 3.2.3 - CDP 7.1.7.2035

Runtime Image

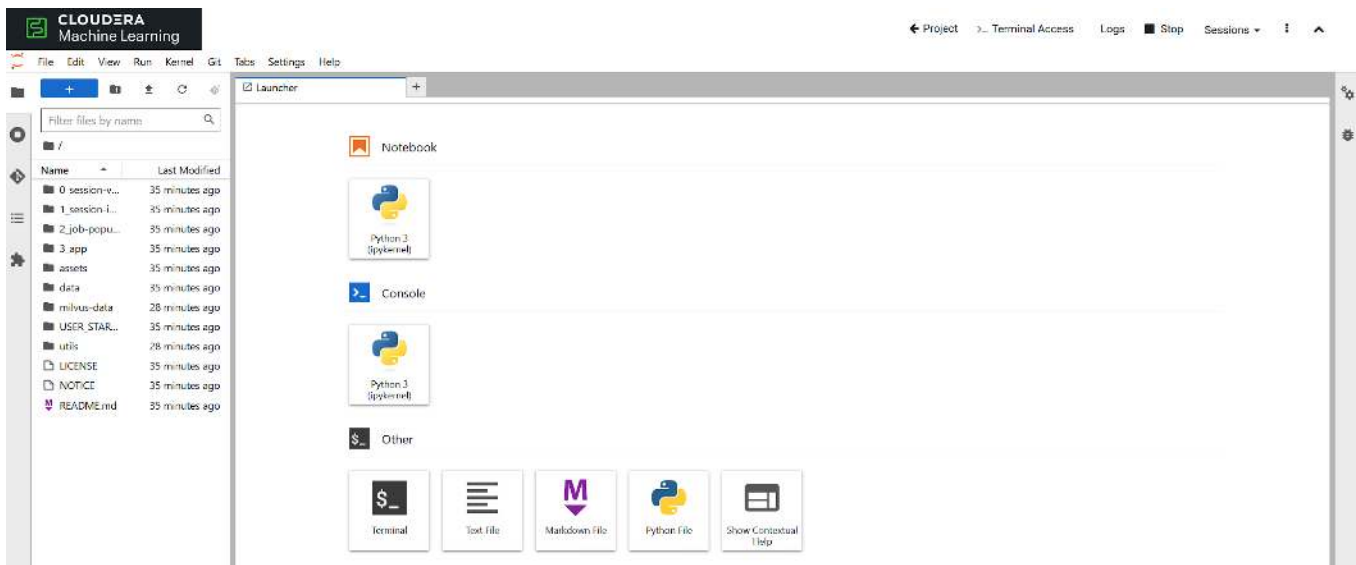
- cdip-ecs1.cdip.cisco.local:5000/cloudera/cdsw/ml-runtime-jupyterlab-python3.10-standard:2023.08.2-b8

Resource Profile

8 vCPU / 64 GiB Memory

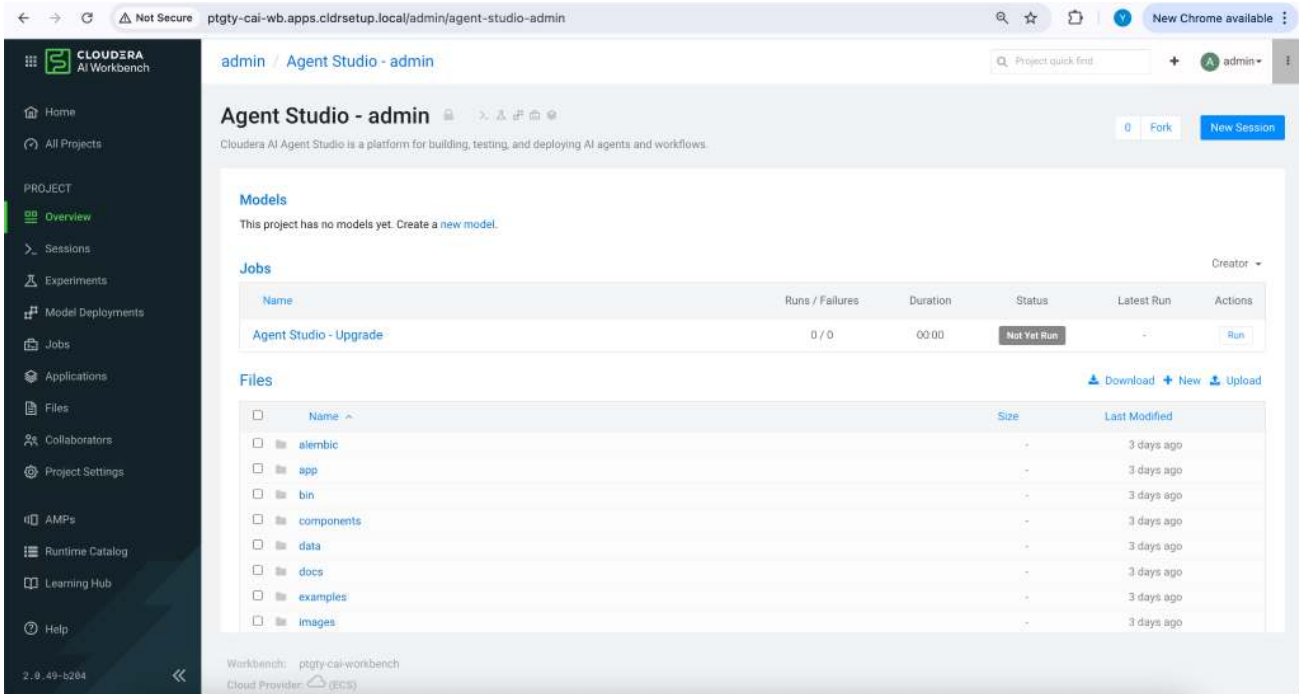
2 GPUs

24. *Jupyter notebook* session in *CAI*.

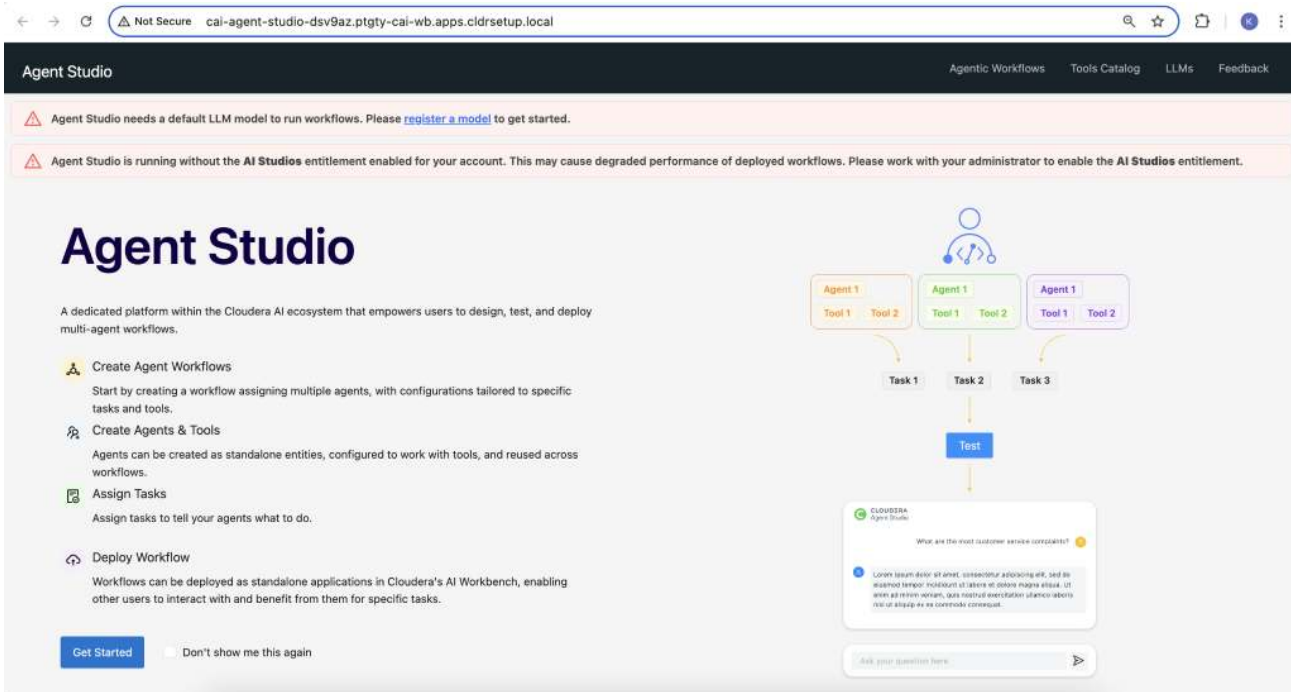


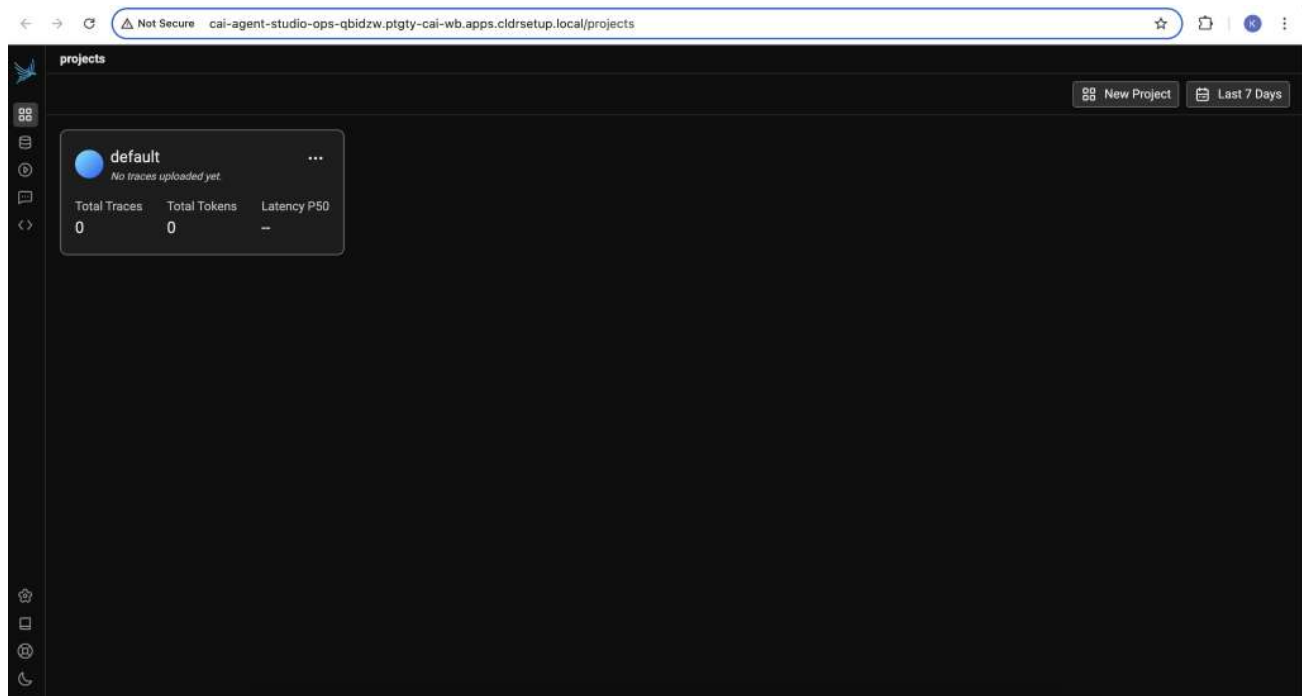
25. Open **AMP** created project. To access **WebUI** click on **Open**.

Note: If you come across any new URLs for accessing a service's Web UI, make sure to add a hostname-to-IP mapping in your laptop's `/etc/hosts` file. Use the ECS master's IP if you're mapping for Data Services. For example: `ptgty-cai-wb.apps.cldrsetup.local` should map to the ECS master IP to be accessible via your browser.

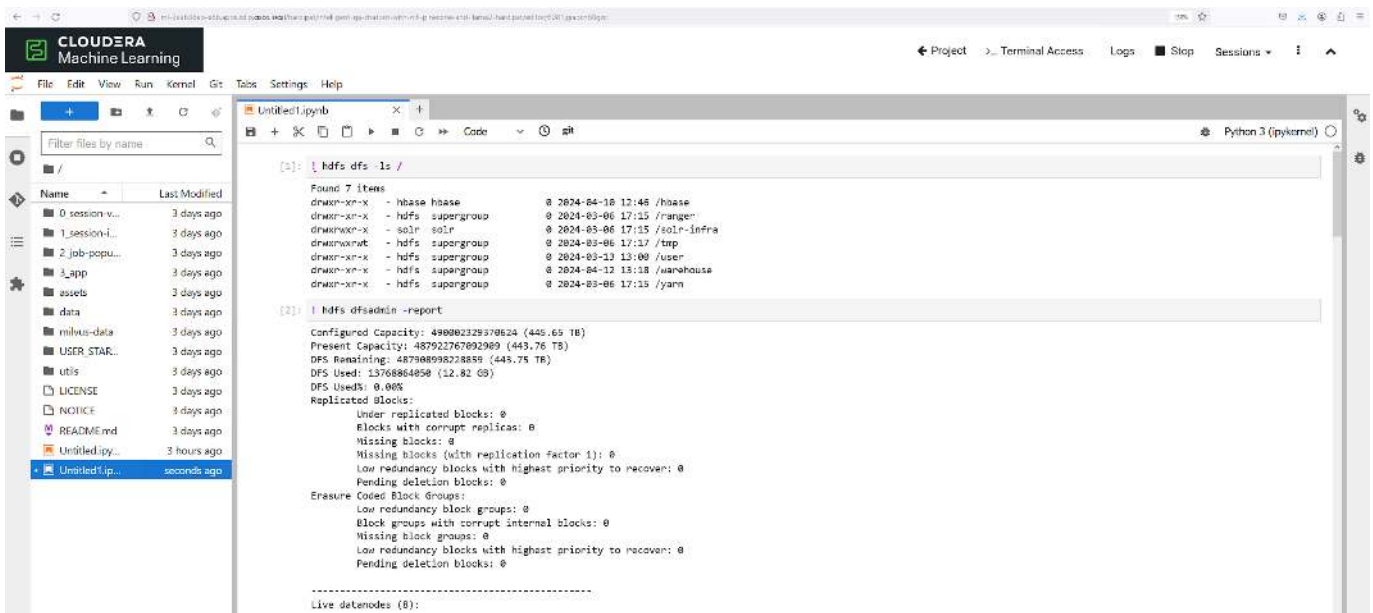


26. WebUI for **Agent Studio - admin** with pre-trained data is now available. **Change settings** on the WebUI or use them out of the box. For example, we questioned “**what is Cloudera Data Platform?**”.





27. Access HDFS data from the *jupyter notebook session* in CAI.



The screenshot shows the Cloudera Machine Learning (CML) interface. On the left is a file explorer showing a directory structure with files like '0_session_v...', '1_session-1...', '2_job-popu...', '3_app', 'assets', 'data', 'milvus-data', 'USER STAR...', 'utils', 'LICENSE', 'NOTICE', 'README.md', and 'Untitled.ipynb'. The main area displays a Jupyter Notebook with two code cells. The first cell runs 'hdfs dfs -ls /' and lists 7 items: 'drwxr-xr-x - hbase hbase', 'drwxr-xr-x - hdfs supergroup', 'drwxr-xr-x - solr solr', 'drwxr-xr-x - hdfs supergroup', 'drwxr-xr-x - hdfs supergroup', 'drwxr-xr-x - hdfs supergroup', and 'drwxr-xr-x - hdfs supergroup'. The second cell runs 'hdfs dfsadmin -report' and shows HDFS statistics: 'Configured Capacity: 490802329378624 (445.65 TB)', 'Present Capacity: 487922767092969 (443.76 TB)', 'DFS Remaining: 487908998278889 (443.75 TB)', 'DFS Used: 13768864850 (12.82 GB)', 'DFS Used%: 0.00%', 'Replicated blocks: 0', 'Under replicated blocks: 0', 'Blocks with corrupt replicas: 0', 'Missing blocks: 0', 'Missing blocks (with replication factor 1): 0', 'Low redundancy blocks with highest priority to recover: 0', 'Pending deletion blocks: 0', 'Erasure Coded Block Groups: 0', 'Low redundancy block groups: 0', 'Block groups with corrupt internal blocks: 0', 'Missing block groups: 0', 'Low redundancy blocks with highest priority to recover: 0', 'Pending deletion blocks: 0', and 'Live datanodes (8):'.

Note: Deploying and documentation of every aspect of *AI Workbench, project, and user management* is not covered here. Please refer to the related *Cloudera documentation* on *Cloudera AI How to section* for more details:

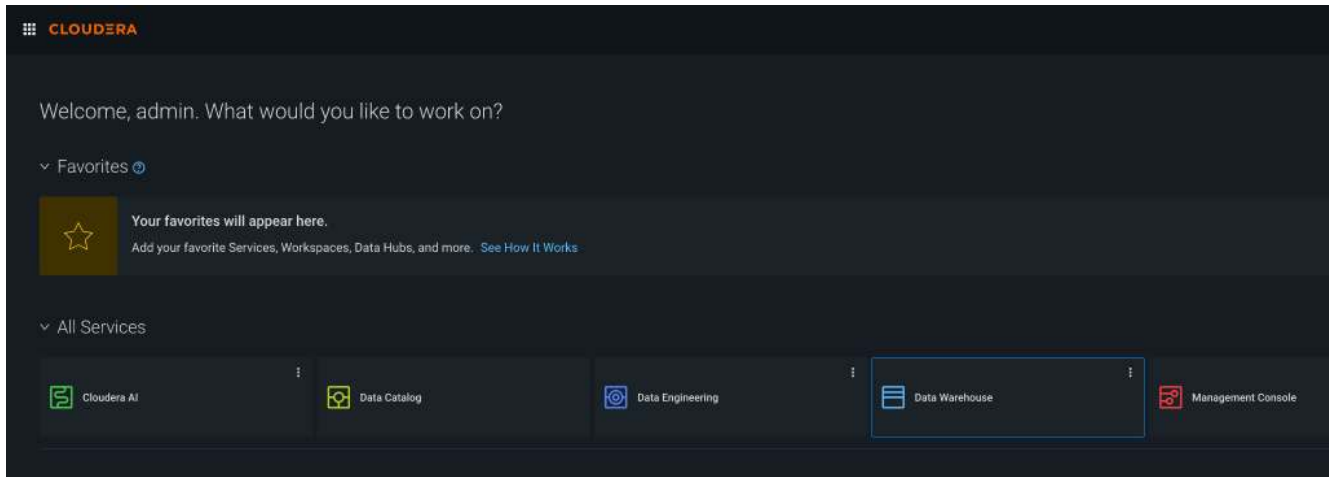
<https://docs.cloudera.com/machine-learning/cloud/product/topics/ml-product-overview.html>

Cloudera on premises Data Warehouse (CDW)

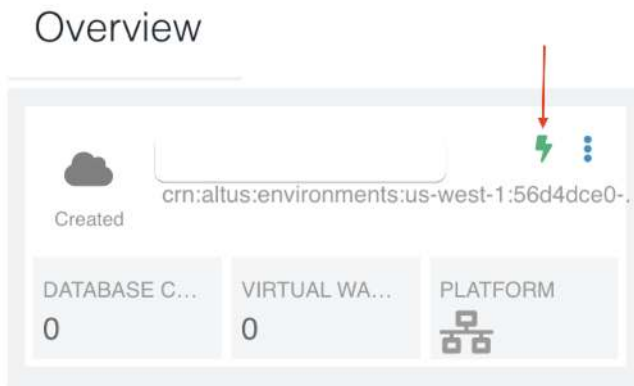
<https://docs.cloudera.com/data-warehouse/1.5.5/private-cloud-getting-started/topics/dw-private-cloud-create-virtual-warehouse-e-openshift-overview.html>

Enable CDW environment and creation of Database Catalog

- Open **CDP Data Services** page.
- Click on **Data Warehouse**.



- On the **Overview** page, click on the **Activate** icon as shown below.



- On the **Activate Environment** page, enter the **LDAP username and password**. Enable **Low resource mode** and click on **Activate**.

Activate Environment

Do you want to activate the environment "cdp-env-temp"?

Delegation Username*

Delegation Password*

Delegation Username

Delegation Password

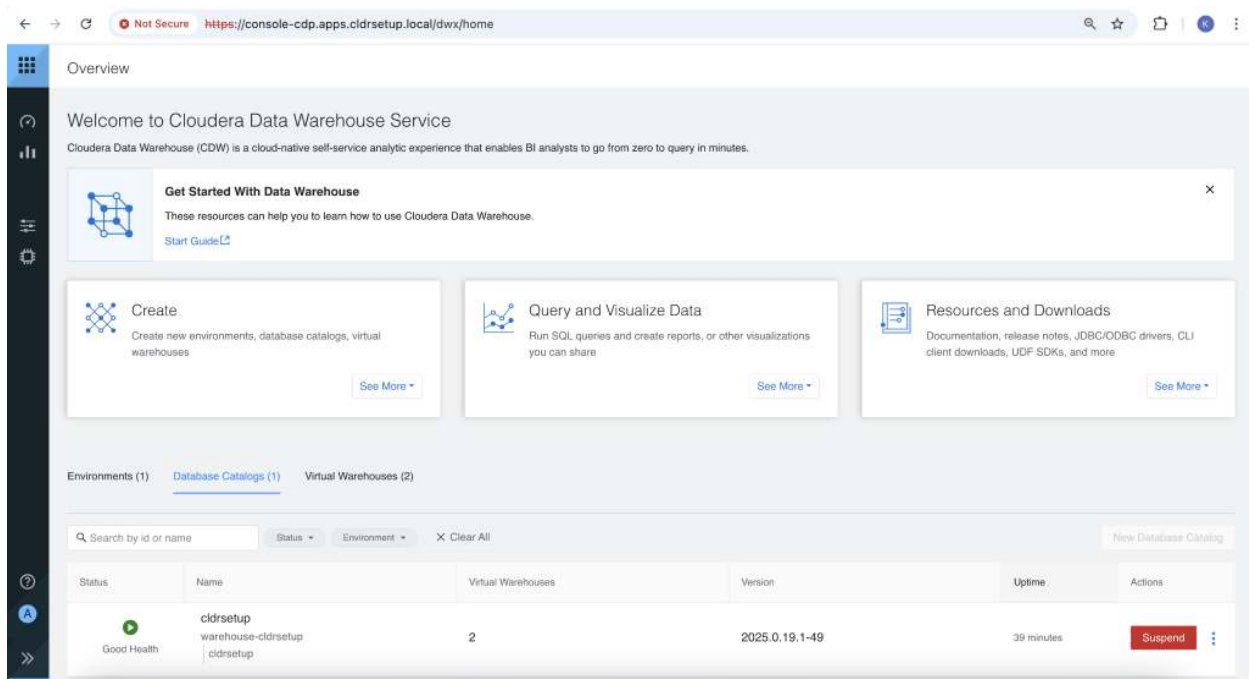
☒ Enable Low Resource Mode

Cancel

ACTIVATE

Create Virtual Warehouse

- Once the **environment** is **activated**, a **default Database Catalog** gets created automatically.



- Once the **database catalog** is created, click on + icon next to **Virtual Warehouses**.



- A **New Virtual Warehouse** tab appears on the same page.
- Enter the **name** for the **new virtual warehouse(VW)**.
- Choose the **type** of VW, i.e. **Hive** or **Impala**.
- Choose the **default Database catalog** that appears in the dropdown.
- Choose **Size** as **xsmall-2 Executors**.

- **AutoSuspend:** If you want the VW to keep running all the time, you can **Disable** it.
- Keep the remaining parameters **default** and click on **Create**.

New Virtual Warehouse X

Name *

Enter Virtual Warehouse Name

Type *

HIVE

IMPALA

Database Catalog *

cdp-pse-development-datalake-default

Size *

xsmall - 2 Executors

☐ Disable AutoSuspend

AutoSuspend Timeout (in seconds): 300



Concurrency Autoscaling ⓘ

Executors: Min:2, Max:6



WaitTime Seconds: 60



☐ Query Isolation ⓘ

Create

- A new **Virtual Warehouse** will be created. You can use **Hue** to submit queries to the underlying engine of the Virtual Warehouse.

CLUSTERA
Data Warehouse

Overview

State of Data Explorer Service

Free Trial Cluster Up

Full Data Visualization

Admin Features

Advanced Configuration

Resource Allocation

Overview

Welcome to Cloudera Data Warehouse Service

Cloudera Data Warehouse (CDW) is a cloud-native self-service analytic experience that enables BI analysts to go from zero to query in minutes.

Get Started With Data Warehouse

These resources can help you to learn how to use Cloudera Data Warehouse.

Start Guide

Create

Create new environments, database catalogs, virtual warehouses.

See More

Query and Visualize Data

Run SQL queries and create reports, or other visualizations you can share.

See More

Resources and Downloads

Documentation, release notes, JDBC/ODBC drivers, CLI client downloads, UDF SDKs, and more.

See More

Environments (1)

Database Catalogs (1)

Virtual Warehouses (2)

Search by ID or name

State

Type

Environment

Database Catalog

OK: All

New Virtual Warehouse

Status	Name	Type	Version	CPU	Director	Agent	Uptime	Actions
Good Health	hiveql1 impala-hq-ml-1 cd-setup	MLQL	2025.0.21.100-7	8		Data Explorer	an hour	Suspend
Stopped	hive2 compute-hq-2 cd-setup	MLQL	2025.0.21.100-7	12		Data Explorer	an hour	Start
Good Health	Trino1 trino-hq-1 cd-setup	MLQL	2025.0.21.1-45	18		Data Explorer	5 days	Suspend

Cloudera on premises Data Engineering (CDE)

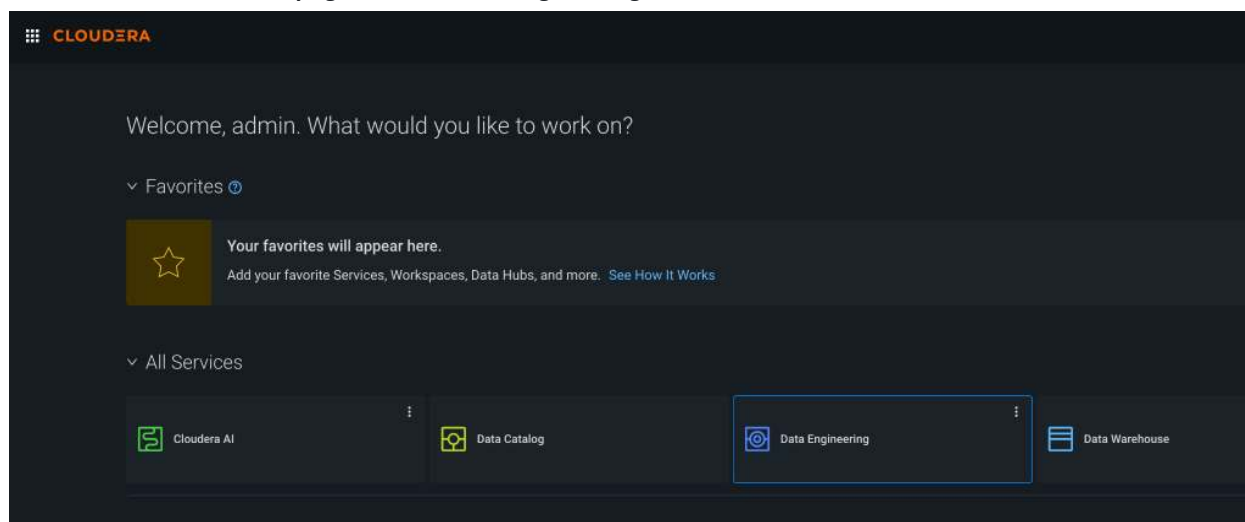
<https://docs.cloudera.com/data-engineering/1.5.5/enable-data-engineering/topics/cde-private-cloud-add-cde-service.html>

CDP Base cluster requirements:

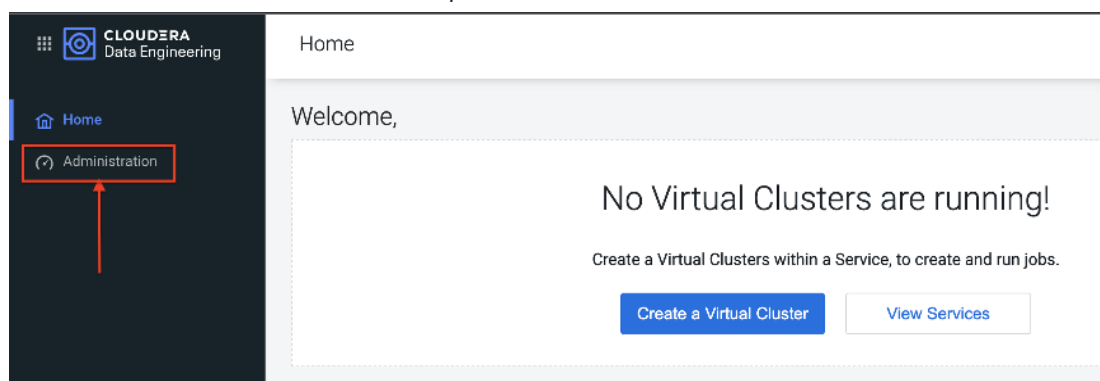
The **Cloudera Data Engineering (CDE)** service requires proper configuration of **Ozone** service in the Base cluster. Ensure that **Ozone** is running properly otherwise you will end up with issues while enabling CDE.

Enabling CDE Service:

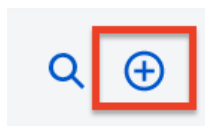
- From the **CDP console page**, click on **Data Engineering**.



- This will open the **CDE home page**. Since this will be the first time you will be opening CDE, you will not see any virtual clusters. Click on **Administration** in the left pane.



- Click on + icon as shown below which will allow you to **enable CDE service** post which you can **create Virtual Clusters**.



- On the **Enable a Service** page, enter the **values** as shown below and then click on **Enable**.

Administration / Enable a Service

Name *

cde-service-name

Environment *

choose the default environment

Resource Quota

Resource Pool *

default keep default

Capacity

CPU

16 999999999999 16

Memory (GB)

48 1000000000 48

Additional Configurations

NFS Storage Class

NFS Storage Class

Default Virtual Cluster ☒

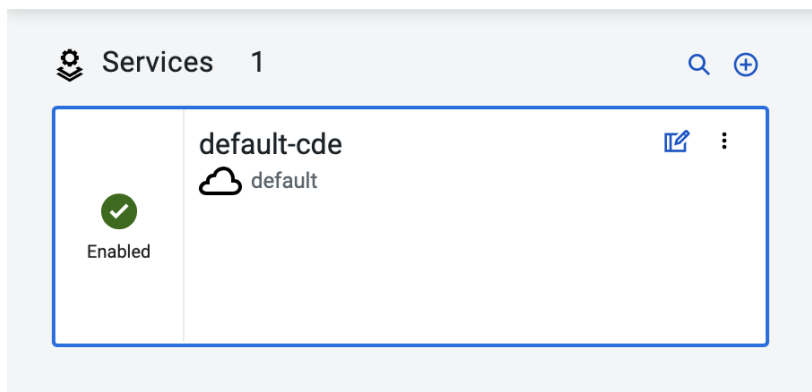
Create a Virtual Cluster by default once this CDE Service is running.

Cancel Enable

Please note that the cpu and memory config chosen here are **minimum values**. You can choose to increase it.

- This will take **approximately 30 mins** after which you will be able to see a **CDE service** on the **CDE** home page.

Administration

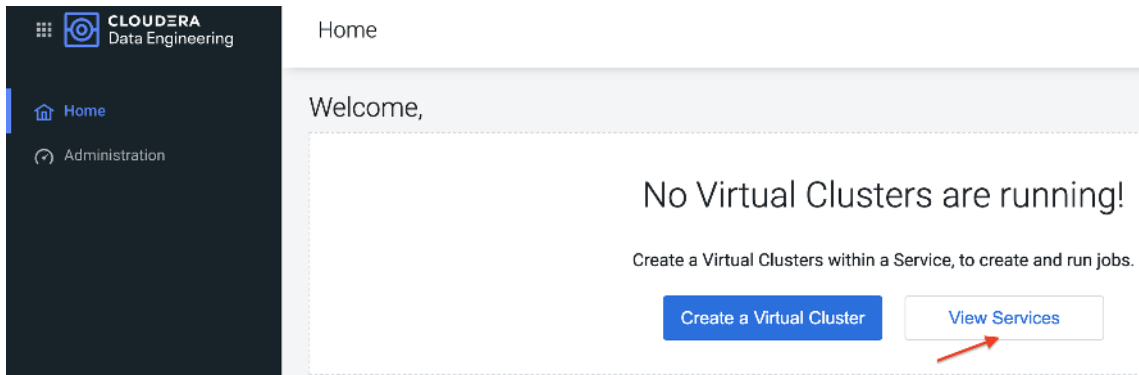


(default **CDE** is the name given as an example. You will see as per the value you entered in the previous step.)

- The **CDE Home page** displays the status of the **CDE** service initialization. You can view logs for the service by clicking on the service **vertical ellipsis (three dots) menu**, and then clicking **View Logs**.

<https://docs.cloudera.com/management-console/1.5.5/private-cloud-environments/topics/mc-private-cloud-environment-register-ui.html>

If you are unable to see the service, then the chances are that the default virtual cluster would not have been created properly. In this case, click on the **View Services** button and then you will be able to see the **CDE** service **enabled**.



Create Virtual Cluster:

When you **enable CDE service**, by default a new **Virtual cluster with Spark2.4** will be created. If you have not enabled this option earlier, then you need to create a virtual cluster again.

- On the **CDE Home page**, click on the **+** icon next to **Virtual clusters** as shown below.



- On the **Create a Virtual Cluster** page, enter the below **values** and click on **Create**.
 - Cluster Name:** Cluster Name should adhere to the below conditions.
 - Begin with a letter
 - Be between 3 and 30 characters (inclusive)
 - Contain only alphanumeric characters and hyphens
 - Service:** Select the CDE service created earlier.
 - Spark Version:** Select the Spark version as per your requirement. If you need both **Spark2.4** and **Spark3.7**, you can create two virtual clusters provided you have sufficient resources.

Cluster Name *

Service *

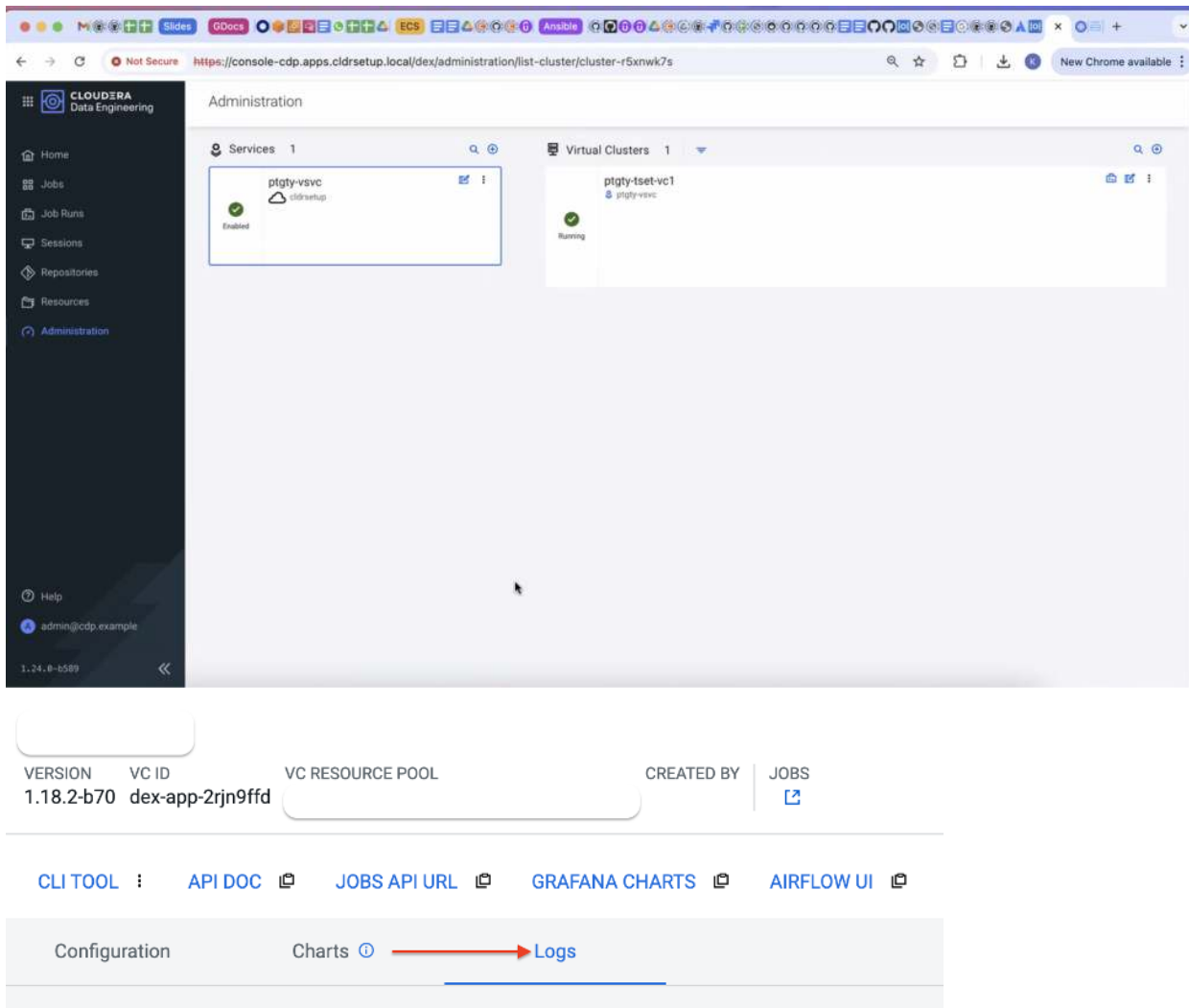
▼

Spark Version

▼

This will take approximately 20 minutes.

- You can check the logs of the cluster creation by clicking on the **pencil** icon and selecting the **Logs** section on the cluster page as shown below.



Initializing Virtual Cluster

Every time a **new virtual cluster** is created, there are a few **manual steps** that must be performed.

- Log in to the **ECS master** and run the next set of **commands** as per the instructions.
- Run the below command to create a temporary directory and navigate to the same.

```
[root@pvcecs-master ~]# mkdir -p /tmp/cde-latest && cd /tmp/cde-latest
```

- Download the script [cdp-cde-utils](https://docs.cloudera.com/data-engineering/1.5.5/cdp-cde-utils.sh) using wget.

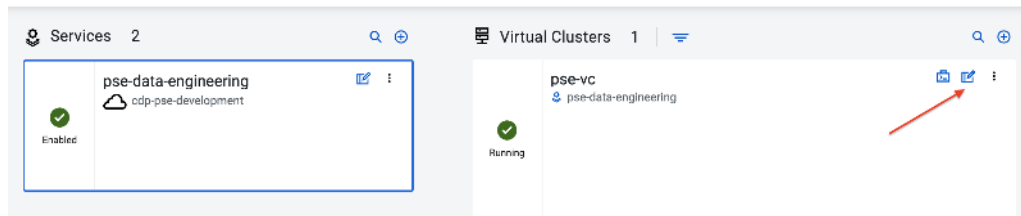
```
[root@pvcecs-master ~]# wget https://docs.cloudera.com/data-engineering/1.5.5/cdp-cde-utils.sh
```

- Add execute permission to this script.


```
[root@pvcecs-master ~]# chmod +x /tmp/cde-latest/cdp-cde-utils.sh
```

- Identify the **virtual cluster endpoint**:

On the **CDE homepage**, select the **CDE service** in which the **virtual cluster** is created. Click on the **pencil** icon on the virtual cluster to be configured.



- Click JOBS API URL to copy the URL to your clipboard.



- Paste the URL into a text editor to identify the endpoint host. For example, if the URL is similar to the following:

```
http://dfdj6kgx.cde-2cdxw5x5.ecs-demo.example.com/dex/api/v1
```

Then the endpoint will then be as shown below.

```
dfdj6kgx.cde-2cdxw5x5.ecs-demo.example.com
```

- Once you get the endpoint of the virtual cluster, **login to the ECS master** and navigate to **/tmp/cde-latest** directory where the **cdp-cde-utils.sh** script is present.

```
[root@pvcecs-master ~]# cd /tmp/cde-latest
```

- Generate a self-signed certificate with the below command. Replace the endpoint_host with the endpoint of your virtual cluster that you got from the previous step.

```
[root@pvcecs-master ~]# ./cdp-cde-utils.sh init-virtual-cluster -h <endpoint_host> -a
```

For the example host we used above, this command will be as below.

```
[root@pvcecs-master ~]# ./cdp-cde-utils.sh init-virtual-cluster -h dfdj6kgx.cde-2cdxw5x5.ecs-demo.example.com -a
```

- These steps must be performed for each virtual cluster you create.

Configuring LDAP Users on CDE

This step is required to submit the jobs to CDE from the LDAP users.

- Log in to the ECS master host and navigate to the directory `/tmp/cde-latest`.

```
[root@pvcecs-master ~]# cd /tmp/cde-latest
```

- Install `krb5-workstation` package using `dnf`.

```
[root@pvcecs-master ~]# dnf install krb5-workstation krb5-libs -y
```

- Create a file named `<username>.principal` containing the user principal. As an example, we will consider `admin` as the username. Here `CLDRSETUP.LOCAL` is the realm provided during IPA setup. You need to replace it with the realm you configured.

```
[root@pvcecs-master ~]# cat>> admin.principal
cdpuser@CLDRSETUP.LOCAL
```

- Generate a keytab named `<username>.keytab` for the user using `ktutil`:

```
[root@pvcecs-master ~]# cat>> admin.keytab
[root@pvcecs-master ~]# sudo ktutil
ktutil: addent -password -p admin@CLDRSETUP.LOCAL -k 1 -e aes256-cts
Password for admin@CLDRSETUP.LOCAL:
ktutil: addent -password -p admin@CLDRSETUP.LOCAL -k 2 -e aes128-cts
Password for admin@CLDRSETUP.LOCAL:
ktutil: wkt admin.keytab
ktutil: q
```

- Validate the keytab using `klist`. This command should use the principals created with two encryptions provided above, namely `aes256-cts` and `aes128-cts`.

```
[root@pvcecs-master ~]# klist -ekt admin.keytab
```

- Validate the `keytab` using `kinit`. This command should get executed successfully.

```
[root@pvcecs-master ~]# kinit -kt admin.keytab admin@CLDRSETUP.LOCAL
```

- Make sure that the `keytab` is valid before continuing. If the `kinit` command fails, the user will not be able to run jobs in the *virtual cluster*. After verifying that the `kinit` command succeeds, you can *destroy* the Kerberos ticket by running `kdestroy`.

- Use the `cdp-cde-utils.sh` script to copy the user `keytab` to the virtual cluster hosts.

```
[root@pvcecs-master ~]# ./cdp-cde-utils.sh init-user-in-virtual-cluster -h <endpoint_host> -u <user> -p <principal_file> -k <keytab_file>
```

For the above example, the command would be below.

```
[root@pvcecs-master ~]# ./cdp-cde-utils.sh init-user-in-virtual-cluster -h dfdj6kgx.cde-2cdxw5x5.ecs-demo.example.com -u cdpuser -p
cdpuser.principal -k cdpuser.keytab
```

- Repeat these steps for all users that need to submit jobs to the virtual cluster.

Appendix

This appendix contains the following:

Appendix A – References Used in Guide

Cloudera on premises Base Getting Started Guide:

<https://docs.cloudera.com/cdp-private-cloud/latest/index.html>

Cloudera on premises Data Services Getting Started Guide:

<https://docs.cloudera.com/cdp-private-cloud-data-services/latest/index.html>

Cloudera on premises Machine Learning Overview:

<https://docs.cloudera.com/machine-learning/1.5.5/index.html>

Cloudera on premises Data Engineering Overview:

<https://docs.cloudera.com/data-engineering/1.5.5/index.html>

Cloudera on premises Data Warehouse Overview:

<https://docs.cloudera.com/data-warehouse/1.5.5/index.html>

Appendix B – Glossary of Terms

This glossary addresses some terms used in this document, for the purposes of aiding understanding. This is not a complete list of all multi cloud terminology.

Ansible	An infrastructure automation tool, used to implement processes for instantiating and configuring IT service components, such as VMs on an IaaS platform. Supports the consistent execution of processes defined in YAML “playbooks” at scale, across multiple targets. Because the Ansible artifacts (playbooks) are text-based, they can be stored in a Source Code Management (SCM) system, such as GitHub. This allows for software development like processes to be applied to infrastructure automation, such as, Infrastructure-as-code (see IaC below). https://www.ansible.com
AWS (Amazon Web Services)	Provider of IaaS and PaaS. https://aws.amazon.com
Azure	Microsoft IaaS and PaaS. https://azure.microsoft.com/en-gb/
Containers (Docker)	A (Docker) container is a means to create a package of code for an application and its dependencies, such that the application can run on different platforms which support the Docker environment. In the context of aaS, microservices are typically packaged within Linux containers orchestrated by Kubernetes (K8s). https://www.docker.com https://www.cisco.com/c/en/us/products/cloud-systems-management/containerplatform/index.html
DevOps	The underlying principle of DevOps is that the application development and operations teams should work closely together, ideally within the context of a toolchain that automates the stages of development, test, deployment, monitoring, and issue handling. DevOps is closely aligned with IaC, continuous integration and deployment (CI/CD), and Agile software development practices. https://en.wikipedia.org/wiki/DevOps https://en.wikipedia.org/wiki/CI/CD

IaaS (Infrastructure as-a-Service)	Infrastructure components provided aaS, located in data centers operated by a provider, typically accessed over the public Internet. IaaS provides a base platform for the deployment of workloads, typically with containers and Kubernetes (K8s).
IaC (Infrastructure as-Code)	Given the ability to automate aaS via APIs, the implementation of the automation is typically via Python code, Ansible playbooks, and similar. These automation artifacts are programming code that define how the services are consumed. As such, they can be subject to the same code management and software development regimes as any other body of code. This means that infrastructure automation can be subject to all of the quality and consistency benefits, CI/CD, traceability, automated testing, compliance checking, and so on, that could be applied to any coding project. https://en.wikipedia.org/wiki/Infrastructure_as_code
IAM (Identity and Access Management)	IAM is the means to control access to IT resources so that only those explicitly authorized to access given resources can do so. IAM is an essential foundation to a secure multi cloud environment. https://en.wikipedia.org/wiki/Identity_management
GCP (Google Cloud Platform)	Google IaaS and PaaS. https://cloud.google.com/gcp
Kubernetes (K8s)	Kubernetes is an open-source system for automating deployment, scaling, and management of containerized applications. https://kubernetes.io
Microservices	A microservices architecture is characterized by processes implementing fine-grained services, typically exposed via REST APIs and which can be composed into systems. The processes are often container-based, and the instantiation of the services is often managed with Kubernetes. Microservices managed in this way are intrinsically well suited for deployment into IaaS environments, and as such, are the basis of a cloud native architecture. https://en.wikipedia.org/wiki/Microservices
PaaS (Platform-as-a-Service)	PaaS is a layer of value-add services, typically for application development, deployment, monitoring, and general lifecycle management. The use of IaC with IaaS and PaaS is very closely associated with DevOps practices.
Private on-premises data center	A data center infrastructure housed within an environment owned by a given enterprise is distinguished from other forms of data center, with the implication that the private data center is more secure, given that access is restricted to those authorized by the enterprise. Thus, circumstances can arise where very sensitive IT assets are only deployed in a private data center, in contrast to using public IaaS. For many intents and purposes, the underlying technology can be identical, allowing for hybrid deployments where some IT assets are privately deployed but also accessible to other assets in public IaaS. IAM, VPNs, firewalls, and similar are key technologies needed to underpin the security of such an arrangement.
REST API	Representational State Transfer (REST) APIs is a generic term for APIs accessed over HTTP(S), typically transporting data encoded in JSON or XML. REST APIs have the advantage that they support distributed systems, communicating over HTTP, which is a well-understood protocol from a security management perspective. REST APIs are another element of a cloud-native applications architecture, alongside microservices. https://en.wikipedia.org/wiki/Representational_state_transfer
SaaS (Software-as-a-Service)	End-user applications provided “aaS” over the public Internet, with the underlying software systems and infrastructure owned and managed by the provider.
SAML (Security Assertion Markup Language)	Used in the context of Single-Sign-On (SSO) for exchanging authentication and authorization data between an identity provider, typically an IAM system, and a service provider (some form of SaaS). The SAML protocol exchanges XML documents that contain security assertions used by the aaS for access control decisions. https://en.wikipedia.org/wiki/Security_Assertion_Markup_Language

Appendix C – Glossary of Acronyms

ACL—Access-Control List

AD—Microsoft Active Directory

API—Application Programming Interface

CDP – Cloudera Data Platform

Cloudera on premises – Cloudera Data Platform Private Cloud

Cloudera on premises DS – Cloudera Data Platform Private Cloud Data Services

CDW – Cloudera Data Warehouse

CAI – Cloudera AI a.k.a. Cloudera Machine Learning

CDE – Cloudera Data Engineering

CPU—Central Processing Unit

DC—Data Center

DHCP—Dynamic Host Configuration Protocol

DNS—Domain Name System

HA—High-Availability

ICMP— Internet Control Message Protocol

LAN—Local Area Network

MAC—Media Access Control Address (OSI Layer 2 Address)

MTU—Maximum Transmission Unit

NAT—Network Address Translation

OSI—Open Systems Interconnection model

RHEL – Red Hat Enterprise Linux

Syslog—System Logging Protocol

TCP—Transmission Control Protocol (OSI Layer 4)

UDP—User Datagram Protocol (OSI Layer 4)

URL—Uniform Resource Locator

VM—Virtual Machine

VPN—Virtual Private Network

Cloudera Data Platform Cloudera on premises latest release note, go to:

<https://docs.cloudera.com/cdp-private-cloud-upgrade/latest/release-guide/topics/cdpdc-release-notes-links.html>

Cloudera Data Platform Cloudera on premises Base Requirements and Supported Versions, go to:

<https://docs.cloudera.com/cdp-private-cloud-upgrade/latest/release-guide/topics/cdpdc-requirements-supported-versions.html>

Cloudera Data Platform Cloudera on premises Data Services installation on Embedded Container Service requirements and supported versions, go to: <https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/index.html>

FreeIPA Reference

<https://www.devopszones.com/2020/03/how-to-add-freeipa-user-in-cli-and-web.html>

Add users on FreeIPA

- Log in to the IPA server and run kinit with admin and enter the password: **kinit admin**
- Run the below command to create a user. Enter the password in the password prompt.

ipa user-add --password <User Name>

```
[root@pvcbasemaster ~]# ipa
ipa: ERROR: Ticket expired
[root@pvcbasemaster ~]# kinit admin && klist -e
Password for admin@CLDRSETUP.LOCAL:
[root@pvcbasemaster ~]# ipa user-add --password kdsahu
First name: Kuldeep
Last name: Sahu
User login [kdsahu]: kdsahu
Password:
Enter Password again to verify:
-----
Added user "kdsahu"
-----
User login: kdsahu
First name: Kuldeep
Last name: Sahu
Full name: Kuldeep Sahu
Display name: Kuldeep Sahu
Initials: KS
Home directory: /home/kdsahu
GECOS: Kuldeep Sahu
Login shell: /bin/sh
Principal name: kdsahu@CLDRSETUP.LOCAL
Principal alias: kdsahu@CLDRSETUP.LOCAL
User password expiration: 20240321113054Z
Email address: kdsahu@cldrsetup.local
UID: 971200008
GID: 971200008
Password: True
Member of groups: ipausers
Kerberos keys available: True
[root@pvcbasemaster ~]# su - kdsahu
Last login: Thu Mar 21 01:00:40 PDT 2024 on pts/0

# 1 Add the user 'cdptest'
ipa user-add cdptest \
  --first=cdp \
  --last=test \
  --password
# (You'll be prompted to set and confirm a password for cdptest.)

# 2 Authenticate as an IPA admin
kinit admin

# 3 Generate a keytab for cdptest@CLDRSETUP.LOCAL
ipa-getkeytab \
  --principal=cdptest@CLDRSETUP.LOCAL \
  --keytab=/tmp/cdptest.keytab
```

```
#4 Verify the keytab contents
klist -k /tmp/cdptest.keytab
```

If instead you need a service principal named

cdptest (for example, to run as an SPN on a host), you would:

```
bash
```

```
# Create a service principal
```

```
ipa service-add cdptest/your-host.cldrsetup.local
```

```
# Pull its keytab
```

```
ipa-getkeytab \
```

```
--principal=cdptest/your-host.cldrsetup.local@CLDRSETUP.LOCAL \
```

```
--keytab=/tmp/cdptest.keytab
```

```
# Verify
```

```
klist -k /tmp/cdptest.keytab
```

command to get the principal file and key tab file with help of that

Here are the commands to list the principals in a keytab and extract individual keytab entries:

```
bash
```

```
# 1. List all principals in the keytab
```

```
klist -k /tmp/cdptest.keytab
```

```
# Example output:
```

```
# Keytab name: FILE:/tmp/cdptest.keytab
```

```
# KVNO Principal
```

```
# -----
```

```
# 1 cdptest@CLDRSETUP.LOCAL
```

```
# 2. Extract a specific principal's keys into a new keytab
```

```
# (useful if you want to split out one principal from a multi-principal keytab)
```

```
ktutil <<EOF
```

```
rkt /tmp/cdptest.keytab
```

```
|
```

```
wkt /tmp/cdptest-single.keytab Principal:cdptest@CLDRSETUP.LOCAL
```

```
quit
```

```
EOF
```

```
# 3. Verify the newly created single-principal keytab
```

```
klist -k /tmp/cdptest-single.keytab
```

```
klist -k
```

shows you all principals stored in a keytab file.

```
ktutil
```

```
can read (
```

```
rkt
```

```
) an existing keytab, list (
```

```
|
```

```
) its entries, and write (
```

```
wkt
```

```
) only the entries matching a given principal into a new keytab.
```

This way you can both inspect and slice your keytab as needed.

Free-IPA Command Reference:

```
kinit admin
```

```
ipa dnsrecord-add 16.172.in-addr.arpa. 231.31 --ptr-rec console-cdp.apps.pvcecsmaster.cldrsetup.local.
```

```
ipa dnsrecord-del 16.172.in-addr.arpa. 231.31 --ptr-rec console-cdp.apps.pvcecsmaster.cldrsetup.local
```

```
ipa dnsrecord-find 16.172.in-addr.arpa.
```

```
ipa dnsrecord-add cldrsetup.local *.apps
```

```
ipa dnszone-list
```

```
ipa user-del cadmin
```

```
ipa user-show
ipa status | start | stop
ipactl status | stop | start | restart
```

```
[kuldeep@pvcbasemaster ~]$ kinit kdsahu
Password for kdsahu@CLDRSETUP.LOCAL:
Password expired. You must change it now.
Enter new password:
Enter it again:
[kuldeep@pvcbasemaster ~]$
```

```
[root@ipaserver ~]# ldapsearch -H ldap://ipaserver.cldrsetup.local:389 -D "uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local" -w 'cloudera123' -b "cn=users,cn=accounts,dc=cldrsetup,dc=local" '(&(uid=admin))' | grep -v "#"
```

```
dn: uid=admin,cn=users,cn=accounts,dc=cldrsetup,dc=local
objectClass: top
objectClass: person
objectClass: posixaccount
objectClass: krbprincipalaux
objectClass: krbticketpolicyaux
objectClass: inetuser
objectClass: ipaobject
objectClass: ipasshuser
objectClass: ipaSshGroupOfPubKeys
uid: admin
krbPrincipalName: admin@CLDRSETUP.LOCAL
cn: Administrator
sn: Administrator
uidNumber: 971200000
gidNumber: 971200000
homeDirectory: /home/admin
loginShell: /bin/bash
gecos: Administrator
ipaUniqueID: e42d6b54-e094-11ee-9c71-0050568db389
memberOf: cn=admins,cn=groups,cn=accounts,dc=cldrsetup,dc=local
memberOf: cn=Replication Administrators,cn=privileges,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=Add Replication Agreements,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=Read Replication Agreements,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=Modify DNA Range,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=Read LDBM Database Configuration,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=Host Enrollment,cn=privileges,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=System: Add krbPrincipalName to a Host,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=System: Enroll a Host,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=System: Manage Host Enrollment Password,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=System: Manage Host Keytab,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=System: Manage Host Principals,cn=permissions,cn=pbac,dc=cldrsetup,dc=local
memberOf: cn=trust admins,cn=groups,cn=accounts,dc=cldrsetup,dc=local
krbLastPwdChange: 20240312172405Z
krbPasswordExpiration: 20240610172405Z
krbExtraData:: AAK1j/Blcm9vdC9hZG1pbkBDkRFBQVkJNEUy5DT00A
krbLoginFailedCount: 0
krbLastFailedAuth: 20240325064805Z

search: 2
result: 0 Success
[root@ipaserver ~]#
```

```
[root@pvcbsmaster ~]# ipa dnsrecord-find 16.172.in-addr.arpa.
Record name: @
NS record: ipaserver.cldrsetup.local.
Record name: 226.31
```


PTR record: ipaserver.cldrsetup.local.
Record name: 227.31
PTR record: pvcbase-master.cldrsetup.local.
Record name: 228.31
PTR record: pvcbase-worker1.cldrsetup.local.
Record name: 231.31
PTR record: pvcecs-master.cldrsetup.local.
Record name: 232.31
PTR record: pvcecs-worker1.cldrsetup.local.

Number of entries returned 12

Perform the PvC Base Cluster Validation:

<https://training-team.gitbook.io/setting-up-cloudera-data-platform-cdp/hive-validation>

<https://www.quora.com/How-do-you-load-data-into-a-Hive-external-table>

<https://stackoverflow.com/questions/17425492/hive-insert-query-like-sql>

https://github.com/mionisation/BI_BigData_2_HiveDatasetAnalysis/blob/master/createMovieLensTables.hql

<https://grouplens.org/datasets/movielens/20m/>

Validation:

```
[root@pvcbase-master ~]# dnf install -y wget unzip
[root@pvcbase-master ~]# wget https://files.grouplens.org/datasets/movielens/ml-20m.zip
[root@pvcbase-master ~]# unzip ml-20m.zip
[root@pvcbase-master ~]# cd ml-20m
[root@pvcbase-master ml-20m]# sed -i 1d *
[root@pvcbase-master ml-20m]#
```

```
[root@pvcbase-master ml-20m]# hdfs dfs -ls /
24/03/21 04:32:28 WARN ipc.Client: Exception encountered while connecting to the server : org.apache.hadoop.security.AccessControlException: Client
cannot authenticate via:[TOKEN, KERBEROS]
ls: DestHost:destPort pvcbasemaster.cldrsetup.local:8020 , LocalHost:localPort pvcbasemaster.cldrsetup.local/172.16.31.227:0. Failed on local exception:
java.io.IOException: org.apache.hadoop.security.AccessControlException: Client cannot authenticate via:[TOKEN, KERBEROS]
[root@pvcbase-master ml-20m]#
```

Locate the HDFS keytab

```
[root@pvcbase-master ml-20m]# find / -name hive.keytab
/run/cloudera-scm-agent/process/1546340988-hive_on_tez-HIVESERVER2/hive.keytab
/run/cloudera-scm-agent/process/1546340973-hive-HIVEMETASTORE/hive.keytab
```

List its contents

```
[root@pvcbase-master ml-20m]# klist -kt /run/cloudera-scm-agent/process/1546340973-hive-HIVEMETASTORE/hive.keytab
Keytab name: FILE:/run/cloudera-scm-agent/process/1546340973-hive-HIVEMETASTORE/hive.keytab
KVNO Timestamp Principal
```

```
-----
1 06/10/25 11:29:34 hive/pvcbase-master.redhat.local@REDHAT.LOCAL
```

Obtain a Kerberos ticket for the HDFS principal

```
[root@pvcbase-master ml-20m]# kinit -kt /run/cloudera-scm-agent/process/1546340973-hive-HIVEMETASTORE/hive.keytab
hive/pvcbase-master.redhat.local@REDHAT.LOCAL
```

Verify your ticket cache

```
[root@pvcbase-master ml-20m]# klist
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: hive/pvcbase-master.redhat.local@REDHAT.LOCAL
```

```
Valid starting Expires Service principal
07/01/25 06:17:08 07/02/25 05:17:24 krbtgt/REDHAT.LOCAL@REDHAT.LOCAL
renew until 07/08/25 06:17:08
```

```
[root@pvcbase-master ml-20m]# hdfs dfs -mkdir /tmp/movielens
[root@pvcbase-master ml-20m]# hdfs dfs -put * /tmp/movielens/
[root@pvcbase-master ml-20m]# hdfs dfs -chown -R hdfs:supergroup /tmp/movielens
[root@pvcbase-master ml-20m]# hdfs dfs -ls /tmp/movielens/
[root@pvcbase-master ml-20m]# hive
```

```
CREATE DATABASE movielens;
use movielens;
CREATE TABLE IF NOT EXISTS ratings ( userId int, movieId int, rating double, ts bigint)
COMMENT "Movie Ratings"
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\054'
```

```

LINES TERMINATED BY '\n'
STORED AS TEXTFILE;

```

```

LOAD DATA INPATH '/tmp/movielens/movies.csv' overwrite INTO TABLE movies;
LOAD DATA INPATH '/tmp/movielens/tags.csv' overwrite INTO TABLE tags;
LOAD DATA INPATH '/tmp/movielens/ratings.csv' overwrite INTO TABLE ratings;
LOAD DATA INPATH '/tmp/movielens/genome-tags.csv' overwrite INTO TABLE genome_tags;
LOAD DATA INPATH '/tmp/movielens/genome-scores.csv' overwrite INTO TABLE genome_scores;

```

Run the queries from HUE for create db, create table.

Upload data from Hive cli.

Run select query to fetch operations from HUE.

OZONE Validation:

```

[root@pvcbase-master ~]# ozone sh bucket list ozone11

```

```

24/05/26 07:32:42 WARN ipc.Client: Exception encountered while connecting to the server : org.apache.hadoop.security.AccessControlException: Client
cannot authenticate via:[TOKEN, KERBEROS]
24/05/26 07:32:42 WARN ipc.Client: Exception encountered while connecting to the server : org.apache.hadoop.security.AccessControlException: Client
cannot authenticate via:[TOKEN, KERBEROS]
24/05/26 07:32:42 WARN ipc.Client: Exception encountered while connecting to the server : org.apache.hadoop.security.AccessControlException: Client
cannot authenticate via:[TOKEN, KERBEROS]
24/05/26 07:32:42 ERROR client.OzoneClientFactory: Couldn't create RpcClient protocol exception:
... 42 more
org.apache.hadoop.security.AccessControlException: Client cannot authenticate via:[TOKEN, KERBEROS]

```

```

[root@pvcbase-master ~]# klist -e

```

```

Ticket cache: FILE:/tmp/krb5cc_0
Default principal: admin@CLDRSETUP.LOCAL

```

```

Valid starting Expires Service principal

```

```

05/15/2024 21:24:12 05/16/2024 20:35:44 krbtgt/CLDRSETUP.LOCAL@CLDRSETUP.LOCAL
renew until 05/22/2024 21:24:09, Etype (skey, tkt): aes256-cts-hmac-sha1-96, aes256-cts-hmac-sha384-192

```

```

[root@pvcbase-master ~]# find / -name ozone.keytab

```

```

/run/cloudera-scm-agent/process/1546347521-ozone-S3_GATEWAY/ozone.keytab
/run/cloudera-scm-agent/process/1546347511-ozone-OZONE_RECON/ozone.keytab
/run/cloudera-scm-agent/process/1546347517-ozone-STORAGE_CONTAINER_MANAGER/ozone.keytab
/run/cloudera-scm-agent/process/1546347273-ozone-STORAGE_CONTAINER_MANAGER/ozone.keytab
/run/cloudera-scm-agent/process/1546347267-ozone-OZONE_RECON/ozone.keytab
/run/cloudera-scm-agent/process/1546347277-ozone-S3_GATEWAY/ozone.keytab
/run/cloudera-scm-agent/process/1546344328-ozone-OZONE_RECON/ozone.keytab
/run/cloudera-scm-agent/process/1546344334-ozone-STORAGE_CONTAINER_MANAGER/ozone.keytab
/run/cloudera-scm-agent/process/1546344338-ozone-S3_GATEWAY/ozone.keytab
/run/cloudera-scm-agent/process/1546344034-ozone-STORAGE_CONTAINER_MANAGER/ozone.keytab
/run/cloudera-scm-agent/process/1546344028-ozone-OZONE_RECON/ozone.keytab
/run/cloudera-scm-agent/process/1546344038-ozone-S3_GATEWAY/ozone.keytab

```

```

[root@pvcbase-master ~]# klist -kt /run/cloudera-scm-agent/process/1546347521-ozone-S3_GATEWAY/ozone.keytab

```

```

Keytab name: FILE:/run/cloudera-scm-agent/process/1546347521-ozone-S3_GATEWAY/ozone.keytab
KVNO Timestamp Principal

```

```

-----
2 05/19/2024 22:11:49 HTTP/pvcbase-master.cldrsetup.local@CLDRSETUP.LOCAL
2 05/19/2024 22:11:49 s3g/pvcbase-master.cldrsetup.local@CLDRSETUP.LOCAL

```

```

[root@pvcbase-master ~]# kinit -kt /run/cloudera-scm-agent/process/1546347521-ozone-S3_GATEWAY/ozone.keytab
s3g/pvcbase-master.cldrsetup.local@CLDRSETUP.LOCAL

```

```

[root@pvcbase-master ~]# klist -kt /run/cloudera-scm-agent/process/1546347521-ozone-S3_GATEWAY/ozone.keytab

```

Keytab name: FILE:/run/cloudera-scm-agent/process/1546347521-ozone-S3_GATEWAY/ozone.keytab
KVNO Timestamp Principal

2 05/19/2024 22:11:49 HTTP/pvcbase-master.cldrsetup.local@CLDRSETUP.LOCAL
2 05/19/2024 22:11:49 s3g/pvcbase-master.cldrsetup.local@CLDRSETUP.LOCAL

[root@pvcbase-master ~]# klist

Ticket cache: FILE:/tmp/krb5cc_0

Default principal: s3g/pvcbase-master.cldrsetup.local@CLDRSETUP.LOCAL

Valid starting Expires Service principal

05/26/2024 07:36:13 05/27/2024 07:11:26 krbtgt/CLDRSETUP.LOCAL@CLDRSETUP.LOCAL
renew until 06/02/2024 07:36:13

[root@pvcbase-master ~]# ozone sh volume list

[]

[root@pvcbase-master ~]# ozone sh volume create ozone11

24/05/26 07:46:20 INFO rpc.RpcClient: Creating Volume: ozone11, with s3g as owner and space quota set to -1 bytes, counts quota set to -1

[root@pvcbase-master ~]# ozone sh volume list

```
[ {  
  "metadata" : { },  
  "name" : "ozone11",  
  "admin" : "s3g",  
  "owner" : "s3g",  
  "quotaInBytes" : -1,  
  "quotaInNamespace" : -1,  
  "usedNamespace" : 0,  
  "creationTime" : "2024-05-26T14:46:20.912Z",  
  "modificationTime" : "2024-05-26T14:46:20.912Z",  
  "acls" : [ {  
    "type" : "USER",  
    "name" : "s3g",  
    "aclScope" : "ACCESS",  
    "aclList" : [ "ALL" ]  
  } ],  
  "refCount" : 0  
}]
```

[root@pvcbase-master ~]# ozone sh bucket create ozone11/testkdbkt1

24/05/26 07:47:19 INFO rpc.RpcClient: Creating Bucket: ozone11/testkdbkt1, with server-side default bucket layout, s3g as owner, Versioning false, Storage Type set to DISK and Encryption set to false, Replication Type set to server-side default replication type, Namespace Quota set to -1, Space Quota set to -1

[root@pvcbase-master ~]# ozone sh bucket list ozone11

```
[ {  
  "metadata" : { },  
  "volumeName" : "ozone11",  
  "name" : "testkdbkt1",  
  "storageType" : "DISK",  
  "versioning" : false,  
  "usedBytes" : 0,  
  "usedNamespace" : 0,  
  "creationTime" : "2024-05-26T14:47:19.692Z",  
  "modificationTime" : "2024-05-26T14:47:19.692Z",  
  "sourcePathExist" : true,  
  "quotaInBytes" : -1,  
  "quotaInNamespace" : -1,  
  "bucketLayout" : "FILE_SYSTEM_OPTIMIZED",  
  "owner" : "s3g",  
  "link" : false  
}]
```



```
[root@pvcbase-master ~]#
```

```
*****
```

Cleanup Cloudera on premises Base Cluster:

<https://docs.cloudera.com/cdp-private-cloud-base/7.3.2/cdp-private-cloud-base-installation/topics/cdpdc-uninstallation.html>

UnInstall and Cleanup Steps (If Installation fails and not-able to resolve the issues)

Stop all Services

Delete the Cluster

On the Home page, Click the drop-down list next to the cluster you want to delete and select Delete.

Uninstall the Cloudera Manager Server

```
#### Cleanup DB
systemctl status cloudera-scm-server
#cd /etc/yum.repos.d
#rm -rvf cloudera-manager.repo
date
systemctl stop postgresql-17
#dnf remove -y postgresql-contrib postgresql-17-contrib postgresql-server postgresql-17-server
#userdel postgres

systemctl stop cloudera-scm-server cloudera-scm-agent cloudera-scm-server-db cloudera-manager-server-db
dnf remove -y cloudera-manager-daemons cloudera-manager-agent cloudera-manager-server cloudera-manager-server-db

systemctl daemon-reload
#rm -rvf /var/lib/pgsql/17/data/
mv -v /var/lib/pgsql/17/data/ /var/lib/pgsql/17/data_bkp_$(date +%Y%m%d)
```

Cleanup CDP-CM, Base Master and Worker nodes

```
#/opt/cloudera/installer/uninstall-cloudera-manager.sh
systemctl stop cloudera-scm-server cloudera-scm-agent cloudera-scm-server-db supervisor;
dnf remove -y cloudera-manager-server cloudera-manager-server-db-2 cloudera-management-agent cloudera-management-daemon
cloudera-manager-*; dnf clean all; systemctl daemon-reload;
for u in cloudera-scm* flume hadoop hdfs hbase hive httpfs hue impala llama mapred oozie solr spark sqoop sqoop2 yarn zookeeper; do sudo kill $(ps -u $u -o pid=); done
sudo umount cm_processes
sudo rm -rvf /usr/share/cmf /var/lib/cloudera* /var/cache/yum/cloudera* /var/log/cloudera* /var/run/cloudera* /etc/cloudera-scm-server
/opt/cloudera /etc/cloudera-scm-agent /var/lib/cloudera-scm-agent/cm_guid*
```

Cleanup CDP-CM, Base Master and Worker nodes

```
#do not run #sudo rm -rvf /var/lib/flume-ng /var/lib/hadoop* /var/lib/hue /var/lib/hbase /var/lib/navigator /var/lib/oozie /var/lib/solr /var/lib/sqoop* /var/lib/zookeeper /hadoop-ozone /impala /var/lib/{*ranger*,*oozie*,*ozone*,*tez*,*spark*,*nifi*,*hadoop*,*hdfs*}

sudo rm -rvf /tmp/kafka-logs /tmp/.scm_prepare_node.lock /var/local/kafka/data/meta.properties /hdfs/* /dfs/* /hdfs/mapred/* /hdfs/yarn/*
/var/lib/had*ozon* /yarn* /hadoop*
/etc/{*atlas*,*hadoop*,*ranger*,hue,impala,knox,hbase,*hive*,hbase-solr,hadoop-kms,*ozone*,oozie,*kafka*,*zeppelin*,*spark*,sqoop*,schemaregis
try,*solr*,hive-hcatalog,hive-webhcat,hue,*hbase*,*kudu*,*knox*,zookeeper,*tez*,streams*,*nifi*} /tmp/kafka-logs/*
/var/lib/cloudera-scm-agent/cm_guid

##### Only If you are doing end-to-end cleanup, including cloudera-manager and postgres DB, run on all
for user in hdfs httpfs sqoop kafka yarn hbase streamsrepmgr streamsmgmr livy kms atlas schemaregistry hue zookeeper accumulo phoenix mapred
druid ranger zeppelin oozie kudu Knox superset solr hive cruisecontrol impala rangerraz ozone tez dpprofiler flume nifi nifiregistry nifitoolkit spark flink
rangerrms omd hadoop kraft; do userdel -r "$user" 2>/dev/null; done

for group in hdfs hue httpfs sqoop kafka yarn hbase streamsrepmgr streamsmgmr livy kms atlas schemaregistry hue zookeeper accumulo phoenix
mapred druid ranger zeppelin oozie kudu Knox superset solr hive cruisecontrol impala rangerraz ozone tez dpprofiler flume nifi nifiregistry nifitoolkit
spark flink rangerrms omd hadoop kraft; do groupdel "$group" 2>/dev/null; done
```



```
java -version
python3 -V
```

Cleanup Cloudera on premises Data Services-ECS Cluster:

<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/installation-ecs/topics/cdppvc-installation-ecs-uninstall-pvc.html>

CLEANUP ECS

On each host in the cluster:

```
/opt/cloudera/parcels/ECS/docker/docker container stop registry
/opt/cloudera/parcels/ECS/docker/docker container rm -v registry
/opt/cloudera/parcels/ECS/docker/docker image rm registry:2
```

Stop the ECS cluster in Cloudera Manager

On each host:

```
cd /opt/cloudera/parcels/ECS/bin
./rke2-killall.sh # usually 2 times is sufficient
```

Use umount to unmount all NFS disks.

```
# umount /docker /lhdata /cdwdata #not needed in our case
./rke2-uninstall.sh
```

```
systemctl daemon-reload
```

```
rm -rvf /ecs/* # assumes the default defaultDataPath and IsoDataPath
```

```
rm -rvf /var/lib/docker_server/* # deletes the auth and certs
```

```
rm -rvf /etc/docker/certs.d /etc/docker/* # delete the ca.crt
```

```
rm -rvf /var/lib/docker/*
```

```
rm -rvf /etc/rancher /var/lib/rancher /var/log/rancher /var/lib/rancher/k3s/server/node-token
```

```
rm -rvf /run/k3s /opt/containerd /opt/cni
```

```
rm -rvf /docker/* /lhdata/* /cdwdata/* /ecs/*
```

```
rm -rvf ~/.kube ~/.cache
```

```
systemctl daemon-reload
```

Delete the ECS cluster in Cloudera Manager. To delete, follow the steps as below:

In Cloudera Manager, navigate to Cloudera on premises Data Services and click . Click Delete.

The Delete Cluster wizard appears. Click Delete.

#Clean IPTables on each host:

```
echo "Reset iptables to ACCEPT all, then flush and delete all other chains";
declare -A chains=( [filter]=INPUT:FORWARD:OUTPUT [raw]=PREROUTING:OUTPUT [mangle]=PREROUTING:INPUT:FORWARD:OUTPUT:POSTROUTING
[security]=INPUT:FORWARD:OUTPUT [nat]=PREROUTING:INPUT:OUTPUT:POSTROUTING );
```

```
for table in "${!chains[@]}";
```

```
do
```

```
echo "${chains[$table]}" | tr : "\n" | while IFS= read -r;
```

```
do
```

```
sudo iptables -t "$table" -P "$REPLY" ACCEPT
```

```
done
```

```
sudo iptables -t "$table" -F
```

```
sudo iptables -t "$table" -X
```

```
done
```

```
iptables -F; iptables -L
```

```
rm -rvf /etc/rancher /var/lib/rancher /var/log/rancher
```

remove agents from all hosts in CM UI (Optional)

```
systemctl stop cloudera-scm-agent
```

```
dnf remove -y cloudera-manager-agent cloudera-manager-daemons
```

```
rm -rvf /opt/cloudera/cm-agent/
```

```
rm -rf /opt/cloudera/ /var/lib/cloudera-scm-agent
```

Remove the hosts from CM-UI

Alternatively, an experimental script is available. This script combines steps three through five. The script is available here:

<https://github.com/cloudera-labs/snippets/blob/main/private-cloud/kill-2-rke.sh>

Reboot the host(s).

Before you install ECS again, ensure that the IP tables list is empty by executing the following command:

```
iptables -L
```

Cloudera on premises Base Cluster Error Handling

```
alternatives --set python /usr/bin/python2
```

```
openssl s_client -connect cldr-mngr.cldrsetup.local:8443 < /dev/null | sed -ne '/-BEGIN CERTIFICATE-/,/-END CERTIFICATE-/p' > knoxssoAmbari.crt
```

If you are using PostgreSQL, turn off Readline support by using the `-n` option. (`history-passwd`)
Start the Cloudera Management Service when the Reports Manager role is ready. See [Starting the Cloudera Management Service](#).

Ensure that the Ranger Solr and Ranger HDFS plugins are enabled.

Important

Ensure you complete the following tasks before you start performing the steps to configure TLS 1.2 on the Reports Manager for communicating with the database:

On the Cloudera Manager UI, navigate to Clusters > Cloudera Management Service.

Select the Configuration tab and search for `reportsmanager_db_safety_valve`.

Based on your database type you must override `headlamp.db.properties` file with JDBC URL properties. Enter the appropriate values in the following format to override the connection to use TLS 1.2.

PostgreSQL

```
com.cloudera.headlamp.orm.hibernate.connection.url=jdbc:postgresql://<DB-HOST>:<DB-PORT>/<DB_NAME>?useSSL=true&trustCertificateKeyStoreUrl=<PATH_TO_TRUSTSTORE_FILE>&trustCertificateKeyStoreType=<TRUSTSTORE_TYPE>&trustCertificateKeyStorePassword=<TRUSTSTORE_PASSWORD>
com.cloudera.headlamp.db.type=postgresql
com.cloudera.headlamp.db.host=<DB-HOST>:<DB-PORT>
com.cloudera.headlamp.db.name=<DB_NAME>
```

```
[15/Mar/2024 04:59:00 -0700] 10184 MainThread agent ERROR Heartbeating to localhost:7182 failed.
Traceback (most recent call last):
  File "/opt/cloudera/cm-agent/lib/python3.8/site-packages/cm/agent.py", line 1588, in _send_heartbeat
    transceiver = cmf.https.HTTPSTransceiver(
  File "/opt/cloudera/cm-agent/lib/python3.8/site-packages/cm/https.py", line 245, in __init__
    self.conn.connect()
  File "/opt/cloudera/cm-agent/lib/python3.8/site-packages/M2Crypto/httpslib.py", line 74, in connect
    sock.connect((self.host, self.port))
  File "/opt/cloudera/cm-agent/lib/python3.8/site-packages/M2Crypto/SSL/Connection.py", line 337, in connect
    if not check(self.get_peer_cert()),
  File "/opt/cloudera/cm-agent/lib/python3.8/site-packages/M2Crypto/SSL/Checker.py", line 122, in __call__
    raise WrongHost(expectedHost=self.host,
M2Crypto.SSL.Checker.WrongHost: Peer certificate subjectAltName does not match host, expected localhost, got DNS:pvcbasemaster.cldrsetup.local
pvcecs[1-5].cldrsetup.local; pvcecsmaster.cldrsetup.local: IOErrorException thrown while collecting data from host: Received fatal alert: internal_error
```

Solution:

```
openssl s_client -connect pvcbasemaster.cldrsetup.local:7183
```

```
[root@pvcbasemaster cloudera-scm-agent]# cat /etc/cloudera-scm-agent/config.ini | grep server
change hostname to dnsname in place of localhost and restart all agents (heartbeat issue resolved)

/opt/cloudera/cm-agent/bin/supervisorctl -c /var/run/cloudera-scm-agent/supervisor/supervisord.conf restart status_server

grep -v -e '^[[:space:]]*$' -e '^#' /etc/cloudera-scm-agent/config.ini
```

```
grep -v -e '^[[:space:]]*$' -e '^#' /etc/cloudera-scm-agent/config.ini2024-03-16 03:04:54,710 ERROR
pool-7-thread-1:com.cloudera.server.cmf.components.CmServerStateSynchronizer: Failed during cleanup : null
```

Solution:

Set java_home by searching java in configuration on the CM console.

Install Postgres and CDP base same day all together otherwise may cause ssl issue (observation)

Stale service status require restart of cluster

Ozone client config issue while deploy krb - known issue

It appears that you might have a proxy setup for the Administration Console. Specify the proxy url as the Frontend url or disable the HTTP Referer Check option.

Ranger, Atlas not running

Due to kafka issue and SOLR issue

SOLR error:

Initialize SOLR and create HDFS home dir from actions and start service will fix issue

Kafka error:

Kafka and SOLR depends on Ozone, (SOLR depends on Kafka as well) install this first

Tez error

tez -> action -> upload tez file to hdfs

CM > Hive > Action > Create hive dir

YARN queue manager error:

```
[root@pvcbase-master ~]# sudo mkdir /var/lib/hadoop-yarn/
[root@pvcbase-master ~]# sudo chmod +077 /var/lib/hadoop-yarn/
```

```
[root@pvcbase-master ~]# sudo chown yarn:hadoop /var/lib/hadoop-yarn/
```

Kafka error:

Solution: delete /var/local/kafka/data/meta.properties

Enable thrift server for hbase-hue

```
set wal property codec-hbase
```

HBASE master bad health:

The problem lies in Cloudera Management Monitor Service, not in Hbase itself. What I did is to restart Cloudera Management Monitor Service, and then restart HBase. After that everything seems to be fine.

Ozone error - Could not find or is not a file

Make sure that hdfs_service is enabled in the Ozone configuration. By having this enabled, the CM agent will put the core-site.xml into the process directory and that error will be gone.

Cleanup ozone directories before redeployment.

HDDS error Ozone

```
[root@pvcbase-master ~]# systemctl restart cloudera-scm-supervisord
```

Ozone ERROR datanode fail to start:

Solution: Perform Proper Cleanup on namenode and datanodes.

```
[root@pvcbase-master ~]# rm -rvf /hdfs/*had*oz* /var/lib/had*oz* /etc/had*oz*
```

```
[root@pvcbase-master ~]# sudo -u postgres psql -U postgres -p 5432 -h $(hostname)
```

Password for user postgres:

psql (17.11)

SSL connection (protocol: TLSv1.2, cipher: ECDHE-RSA-AES256-GCM-SHA384, bits: 256, compression: off)

Type "help" for help.

postgres=# \q

```
[root@pvcbasemaster data]# echo -n 'Sahu@123{admin}' | md5sum
```

c94251c29cd07ed2daf0b6edcf843362 -

```
[root@pvcbasemaster data]# sudo -u postgres psql -U postgres -p 5432 -h $(hostname)
```

Password for user postgres:

psql (17.11)

SSL connection (protocol: TLSv1.2, cipher: ECDHE-RSA-AES256-GCM-SHA384, bits: 256, compression: off)

Type "help" for help.

```
postgres=# \c ranger
SSL connection (protocol: TLSv1.2, cipher: ECDHE-RSA-AES256-GCM-SHA384, bits: 256, compression: off)
You are now connected to database "ranger" as user "postgres".
ranger=# update x_portal_user set password = '9746e519adb14ec3ffbf4aff051f104d' where login_id = 'admin';
UPDATE 1
ranger=#
ranger=# select * from x_portal_user where login_id = 'admin';
 id |   create_time   |   update_time   | added_by_id | upd_by_id | first_name | last_name | pub_scr_name | login_id |
password | email | status | user_src | notes | other_attributes | sync_source | old_passwords | password_updated_time
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----
  1 | 2024-03-17 12:15:27.272988 | 2024-03-17 19:15:43.854 |           |           | Admin     | Admin     | admin     | admin    |
c94251c29cd07ed2daf0b6edcf843362 |      | 1 | 0 |           |           |           |           |
(1 row)
ranger=# \q
```

```
[root@pvcbasemaster data]# sudo -u postgres psql -U rangeradmin -p 5432 -d ranger -h $(hostname)
Password for user rangeradmin:
psql (17.11)
SSL connection (protocol: TLSv1.2, cipher: ECDHE-RSA-AES256-GCM-SHA384, bits: 256, compression: off)
Type "help" for help.
ranger=> \q
```

```
[root@pvcbase-master ~]# tail -f /var/log/ranger/admin/access_log-pvcbasemaster.cldrsetup.local-2024-03-21.log
```

```
2024-03-19 23:59:28.861 PDT [7434] LOG: could not accept SSL connection: EOF detected
2024-03-19 23:59:28.861 PDT [7427] LOG: could not accept SSL connection: EOF detected
2024-03-19 23:59:28.861 PDT [7539] LOG: could not accept SSL connection: EOF detected
```

Caused by: java.io.FileNotFoundException: /var/lib/cloudera-scm-server/.postgresql/root.crt (Permission denied)

Solution:

Ranger UI error SSL issue : issue was with permission on cloudera-scm-server directory where root.crt was stored.

Postgres Connection Limit exceeded:

```
Operation error. response=VXResponse={org.apache.ranger.view.VXResponse@2ca9483cstatusCode={1} msgDesc={RangerKRBAuthenticationFilter
Failed : Exception [EclipseLink-4002] (Eclipse Persistence Services - 2.7.7.v20200504-69f2c2b80d): org.eclipse.persistence.exceptions.DatabaseException
Internal Exception: java.sql.SQLException: Connections could not be acquired from the underlying database!
Error Code: 0} messageList={null} }
javax.ws.rs.WebApplicationException
    at org.apache.ranger.common.RESTErrorUtil.createRESTException(RESTErrorUtil.java:56)
```

```
Request failed. loginId=null, logMessage=RangerKRBAuthenticationFilter Failed : Exception [EclipseLink-4002] (Eclipse Persistence Services -
2.7.7.v20200504-69f2c2b80d): org.eclipse.persistence.exceptions.DatabaseException
Internal Exception: java.sql.SQLException: Connections could not be acquired from the underlying database!
Error Code: 0
```

```
javax.ws.rs.WebApplicationException
    at org.apache.ranger.common.RESTErrorUtil.createRESTException(RESTErrorUtil.java:56)
```

```
2024-03-19 00:55:23,767 WARN
C3P0PooledConnectionPoolManager[identityToken->1bqot7nb2ns2s4q1pyen82|14b0e127]-HelperThread-#0:com.mchange.v2.resourcepool.BasicResourcePool: com.mchange.v2.resourcepool.BasicResourcePool$ScatteredAcquireTask@1ab83b08 -- Acquisition Attempt Failed!!! Clearing pending acquires. While trying to acquire a needed new resource, we failed to succeed more than the maximum number of allowed acquisition attempts (5). Last acquisition attempt exception:
org.postgresql.util.PSQLException: FATAL: sorry, too many clients already
    at org.postgresql.core.v3.ConnectionFactoryImpl.doAuthentication(ConnectionFactoryImpl.java:698)
```

standard_conforming_strings=off

```
2024-03-19 23:16:26,279 PDT [26314] WARNING: nonstandard use of escape in a string literal at character 261
2024-03-19 23:16:26,279 PDT [26314] HINT: Use the escape string syntax for escapes, e.g., E'\r\n'.
2024-03-19 23:16:33,719 PDT [24593] FATAL: sorry, too many clients already
```

Solution:

Increase max_connections to 1000 on postgresql.conf file

```
2024-03-26 03:35:19,203 ERROR - [main:] ~ GraphBackedSearchIndexer.initialize() failed (GraphBackedSearchIndexer:386)
org.apache.solr.client.solrj.impl.HttpSolrClient$RemoteSolrException: Error from server at https://pvcbasemaster.cldrsetup.local:8995/solr: Can not find the specified config set: vertex_index
```

<https://community.cloudera.com/t5/Support-Questions/atlas-webui-is-not-accessible/td-p/324743>

stop atlas> initialize atlas> start atlas

```
[root@pvcbase-master ~]# klist -kt /run/cloudera-scm-agent/process/1546342867-SolrServerGracefulShutDown/solr.keytab
[root@pvcbase-master ~]# kinit -kt /run/cloudera-scm-agent/process/1546342867-SolrServerGracefulShutDown/solr.keytab
solr/pvcbasemaster.cldrsetup.local@CLDRSETUP.LOCAL
[root@pvcbase-master ~]# /opt/cloudera/parcels/CDH/bin/zookeeper-client
```

Chart not showing--> install mgmt service.

```
logfile=/var/log/cloudera-scm-agent/supervisord.log
```

```
*****
*****
```

NullPointerException while starting zookeeper

```
Failed due to com.cloudera.cmf.command.CmdExecException: java.lang.NullPointerException
```

Solution: Zookeeper instances should be 3.

First Run Command

Investigate the failure step and once the cause is fixed, click Resume to continue

Status **Failed** Context [OnpremBaseCluster](#) Apr 16, 6:02:07 AM 9:37s [Resume](#)

Sending diagnostic data for this command helps Cloudera improve the product. [Send Diagnostic Data to Cloudera](#)

`java.lang.NullPointerException`

Completed 1 of 1 step(s)

☐ Show All Steps ☒ Show Only Failed Steps ☐ Show Only Running Steps

Run a set of services for the first time. Failed due to java.lang.NullPointerException:	Apr 16, 6:02:11 AM	5.02s
Execute 12 steps in sequence Failed due to java.lang.NullPointerException:	Apr 16, 6:02:16 AM	5ms
Execute 4 steps in parallel Failed due to java.lang.NullPointerException:	Apr 16, 6:02:16 AM	5ms
Start ZooKeeper Failed due to java.lang.NullPointerException:	Apr 16, 6:02:16 AM	5ms

Solr Error:

```
java.nio.file.NoSuchFileException:
/opt/cloudera/parcels/CDH-7.3.2-1.cdh7.3.2.p1032.62597146/lib/solr/server/solr-webapp/webapp/WEB-INF/lib/ozone-file-system-hadoop3-1.4.0.7.3.2.1032-3.jar
```

Solution: Find the correct gbn specific to cdh version installed.

URL for jar file:

[https://cloudera-build-us-west-1.vpc.cloudera.com/s3/build/52717809/cdh/7.x/maven-repository/org/\[...\]/6/ozone-file-system-hadoop3-1.4.0.7.3.2.1-246.jar](https://cloudera-build-us-west-1.vpc.cloudera.com/s3/build/52717809/cdh/7.x/maven-repository/org/[...]/6/ozone-file-system-hadoop3-1.4.0.7.3.2.1-246.jar)

Yarn Error:

```
Failed to execute command Install YARN MapReduce Framework JARs on service YARN
```

Solution: Find the correct gbn specific to cdh version installed.

Ranger admin UI not opening:

Solution: The issue is with the time synchronization, run "chronyc -a makestep" to sync the time.

Atlas & Knox UI not able to login:

Solution: need to update permission in order to access atlas and knox ui with default pam authentication enabled, use command.

```
chmod 444 /etc/shadow
```

Imp Links:

<https://community.cloudera.com/t5/Support-Questions/atlas-webui-is-not-accessible/td-p/324743>

<https://community.cloudera.com/t5/Support-Questions/how-to-change-default-Atlas-UI-admin-password/td-p/177312>

<https://community.cloudera.com/t5/Support-Questions/Knox-authentication-with-PAM/m-p/339556>

Cloudera on premises Data Services ECS Cluster Error Handling:

<https://spacelift.io/blog/kubectrl-port-forward>

<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/day-two-operations/cdppvc-data-services-day-two-operations.pdf>

<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/managing-ecs/topics/cm-manage-experiences-compute-service.html>

https://docs.cloudera.com/documentation/other/reference-architecture/PDF/cloudera_ref_arch_cdp_dc.pdf

<https://docs.cloudera.com/cdp-private-cloud-data-services/1.5.5/upgrade-ocp/topics/cdppvc-installation-update-pvc-ds.html>

Could not get current server API group list: Get "http://localhost:8080/api?timeout=32s": dial tcp 127.0.0.1:8080: connect: connection refused

Solution: kubeconfig error

```
[root@pvcecs-master ~]# journalctl -xeu kubelet
[root@pvcecs-master ~]# mkdir -p ~/.kube && cp /etc/rancher/rke2/rke2.yaml ~/.kube/config
[root@pvcecs-master ~]# mkdir -p ~/.kube && cp /etc/rancher/rke2/rke2.yaml ~/.kube/config
[root@pvcecs-master ~]# kubectl get pods
-bash: kubectl: command not found
[root@pvcecs-master ~]# export PATH=/var/lib/rancher/rke2/data/v1.26.10-rke2r1-e58e49f33617/bin/:$PATH
[root@pvcecs-master ~]# kubectl get pods
No resources found in the default namespace.
[root@pvcecs-master ~]# kubectl get all
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes  ClusterIP   10.43.0.1    <none>        443/TCP   13h
[root@pvcecs-master ~]#
```

Error in commands

```
kubectl -n kubernetes-dashboard get sa/admin-user -o 'jsonpath={.secrets[0].name}'
```

Solution:

```
kubectl -n kubernetes-dashboard get secret/admin-user-secret -o 'jsonpath={.metadata.name}' (working)
```

```
+ /var/lib/rancher/rke2/bin/kubectl --kubeconfig /etc/rancher/rke2/rke2.yaml -n kubernetes-dashboard get secret -o 'go-template={{.data.token | base64decode}}'
error: error executing template "{{.data.token | base64decode}}": template: output:1:16: executing "output" at <base64decode>: invalid value; expected string
```

Solution:

```
kubectl get secret -n kubernetes-dashboard admin-user-secret -o 'go-template={{.data.token | base64decode}}' (working command)
```

Error: INSTALLATION FAILED: cannot re-use a name that is still in use

```
++ K8S_DASHBOARD_CHART_FILE_NAME=kubernetes-dashboard-5.10.0.tgz
++ /opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/./installer/install/bin/linux/helm install
kubernetes-dashboard /opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/kubernetes-dashboard-5.10.0.tgz -n
kubernetes-dashboard --create-namespace -f
/opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/kubernetes-dashboard-overrides.yaml
```

Solution:

```
kubectl delete ns kubernetes-dashboard
```

Install python3.8 for ecs

Always mount correct volumes in config, created previously

Bootstrap error kubectl- remove /var/lib/rancher directory

```
/var/lib/rancher/rke2/agent/logs/kubelet.log
```

```
[root@pvcecsmaster ~]# kubectl get pods
```

Unable to connect to the server: tls: failed to verify certificate: x509: certificate signed by unknown authority

Solution:

After proper cleanup and reinstall issue will solved.

```
[root@pvcecs-master ~]# mkdir -p ~/.kube
```

```
[root@pvcecsmaster opt]# cp /etc/rancher/rke2/rke2.yaml ~/.kube/config
```

```
cp: overwrite '/root/.kube/config'? Yes
```

```
[root@pvcecsmaster opt]# /var/lib/rancher/rke2/data/v1.26.10-rke2r1-e58e49f33617/bin/kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
pvcecsmaster.cldrsetup.local	Ready	control-plane,etcd,master	61s	v1.26.10+rke2r1

```
[root@pvcecsmaster opt]# cp /tmp/ecs_util.sh /opt/cloudera/cm-agent/service/ecs/ecs_util.sh
```

Solution: cleanup was not done properly,after proper cleanup by script, it started running

Error: INSTALLATION FAILED: cannot re-use a name that is still in use

```
++ /opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/../installer/install/bin/linux/helm install
kubernetes-dashboard /opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/kubernetes-dashboard-5.10.0.tgz -n
kubernetes-dashboard --create-namespace -f
/opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/kubernetes-dashboard-overrides.yaml
```

Solution:

```
[root@pvcecs-master ~]# k get pods -A | grep dashboard
```

kubernetes-dashboard	kubernetes-dashboard-68876b7cb8-fl58c	1/1	Running	0	4m13s
----------------------	---------------------------------------	-----	---------	---	-------

```
[root@pvcecsmaster ~]# k delete pod kubernetes-dashboard
```

Error from server (NotFound): pods "kubernetes-dashboard" not found

```
[root@pvcecsmaster ~]# k delete pod kubernetes-dashboard-68876b7cb8-fl58c -n kubernetes-dashboard
```

pod "kubernetes-dashboard-68876b7cb8-fl58c" deleted

```
[root@pvcecsmaster ~]#
```

```
[root@pvcecsmaster ~]# k delete ns kubernetes-dashboard
```

```
[root@pvcecs-master ~]# k logs cli-v2m7w -n cdp
```

HTTPSPool(host='console-cdp.apps.cldrsetup.local', port=443): Max retries exceeded with url:

/api/v1/environments2/createPrivateEnvironment (Caused by NameResolutionError("<cdpcli.cdprequest.CdpHTTPSConnection object at 0x7fbb901047f0>: Failed to resolve 'console-cdp.apps.cldrsetup.local' ([Errno -2] Name or service not known)"))

Solution: Make sure to setup wildcard DNS in advance before proceeding with ECS Setup steps

Possible Cause and Solution:

Wildcard dns was not set up properly. Before proceeding with ECS installation, set up the wildcard DNS.

Map docker dependency if localhost:6443 apiserver error come in ECS

Bootstrap error

```
[time="2024-03-19T11:10:29-07:00" level=fatal msg="Failed to reconcile with temporary etcd: bootstrap data already found and encrypted with different token"]
```

Solution:

Cleanup not done properly.

Error in Initializing embedded Vault phase while installing data services cluster

```
failed to create secret namespaces "istio-ingress" not found
Error from server (NotFound): pods "vault-0" not found

Warning: deleting cluster-scoped resources, not scoped to the provided namespace
++ kubectl delete secret ingress-default-cert -n kube-system
++ kubectl delete secret ingress-default-cert -n istio-ingress
Error from server (NotFound): secrets "ingress-default-cert" not found
++ exit_code=1
++ kubectl delete namespace vault-system
++ kubectl create namespace vault-system
++ kubectl apply -f /opt/cloudera/parcels/ECS-1.5.5-b82-ecs-1.5.5-b82.p0.65068258/vault/vault.yaml
Warning: Detected changes to resource vault which is currently being deleted.
++ TMPDIR=logs
++ openssl genrsa -out logs/vault.key 2048
++ cat /opt/cloudera/parcels/ECS-1.5.5-b82-ecs-1.5.5-b82.p0.65068258/vault/csr.conf
++ envsubst
++ openssl req -new -key logs/vault.key -subj '/CN=system:node:vault.vault-system.svc;/O=system:nodes' -out logs/server.csr -config logs/csr.conf
++ cat
+++ cat logs/server.csr
+++ base64
+++ tr -d '\n'
++ kubectl create -f logs/csr.yaml
++ kubectl certificate approve vault-csr
++ '[' -z ''
++ sleep 1
+++ kubectl get csr vault-csr -o 'jsonpath={.status.certificate}'
```

Solution: To resolve the issue, manually create the istio-ingress namespace with the below command, once it becomes active, delete the vault-0 pod in the terminating state, then simply resume the installation process again from the UI, and it should proceed correctly from that point.

```
create ns istio-ingress
```

Issue while activating the CDW environment:

```
failed to create secret namespaces "istio-ingress" not found
Error from server (NotFound): pods "vault-0" not found
```

```
unable to create cluster initial state: unable to create the CDW resource pool: non-retriable error: failed to update the resource pool
'root.cdp-env-1.cdw', because: 'Request type: 'UpdateResourcePool', state: 'FAILED', last updated: '2025-04-21T07:35:46.134Z', finished: 'true', error:
'policy violation: child quota larger than parent quota', request ID: 'rp-req-e235fed3-757b-4cb9-91ad-f4570d1e2a25'
```

Solution: There was some issue with the cdw environment due to which it was not getting activated, we tried a workaround to create another environment to activate cdw service and it worked.

Commands:

```
kubectl get po -A | grep access
kubectl delete pod -n cdp cdp-release-cluster-access-manager-67c7b5d8cc-5gcrh
kubectl get po -A | grep pool
kubectl delete pod -n cdp cdp-release-resource-pool-manager-7dd8c84569-42d6c
```

Error: Resource Pool Manager Fails to Start After Upgrade

```
{"level":"ERROR","timestamp":"2025-04-29T10:09:01.448Z","caller":"cam/driver.go:265","message":"Error updating queueConfig","grpc error":"rpc
error: code = InvalidArgument desc = queue config update failed"}
...
"pool manager startup failure","error":"cam operation failed, returned state 'UNSET', info: ", grpc error: rpc error: code = InvalidArgument desc = queue
config update failed"
```

Solution: This issue was observed in the monitoring namespace under CDW.

This can be resolved by restarting the Cluster Access Manager followed by the Resource Pool Manager services in sequence.

Error: Vault Pod Not Running – Upgrade Process Stuck

```
Vault pod not in running state.
Upgrade checks hanging on "Vault unseal".
```

Solution: The Vault pod was sealed due to node issues during the earlier stages of the upgrade. To resolve this, manually unseal it using the command below, then verify the pod status before continuing with the upgrade.

```
vault operator unseal
```

Error: Upgrade Hook Job Stuck – Pod Deleted But Not Recreated

```
kubectl get job cdp-release-pre-upgrade-hook-job1 -n cdp
NAME                                STATUS  COMPLETIONS  DURATION  AGE
cdp-release-pre-upgrade-hook-job1  Running  0/1          12s       12s
```

Solution: If the pod was deleted, manually reapply the job YAML. Monitor the job's progress or troubleshoot it using below command. If necessary, delete and recreate the job.

```
kubectl logs job/cdp-release-pre-upgrade-hook-job1 -n cdp
```

Commands to delete and recreate the job.

```
kubectl get job -A | grep cdp-release-pre-upgrade-hook
kubectl get job -o yaml -n cdp cdp-release-pre-upgrade-hook-job1 > upgrade.yml
cat upgrade.yml
```

```
kubectl get job -A | grep cdp-release-pre-upgrade-hook
kubectl delete job cdp-release-pre-upgrade-hook-job1 -n cdp
kubectl get job -A | grep cdp-release-pre-upgrade-hook
kubectl apply -f upgrade.yml -n cdp
```

CDW Data Catalog Creation Issue

```
Initializing the schema to: 3.1.3000.2025.0.19.1-49
Metastore connection URL:
jdbc:postgresql://postgres-service-default-warehouse:5432/hive?createDatabaseIfNotExist=true&sslmode=verify-ca&sslrootcert=/mnt/certs/hms-root.crt
Metastore connection Driver: org.postgresql.Driver
Metastore connection User: hive
Failed to get schema version.
Underlying cause: org.postgresql.util.PSQLException : SSL error: PKIX path building failed:
sun.security.provider.certpath.SunCertPathBuilderException: unable to find valid certification path to requested target
```

```
| warehouse-cldrsetup metastore-0          • 0/1 Init:0/4      0 0 0 0 n/a 0 0 n/a pvcecs-worker10.cldrsetup.local 20s |
```

Solution: The CDW Data Catalog creation failed due to a certificate validation error between the Hive metastore and the PostgreSQL backend. This was caused by an invalid or missing truststore certificate (hms-root.crt). To fix this, extract the correct certificate, validate it, and reapply the secret before restarting the metastore pod.

Steps:

```
# Backup current file if needed
kubectl get secret pg-hms-db-certs -n warehouse-cldrsetup -o yaml > pg-hms-db-certs-backup.yaml

# Recreate the secret with the expected key
kubectl delete secret pg-hms-db-certs -n warehouse-cldrsetup

kubectl create secret generic pg-hms-db-certs \
  --from-file=TRUSTSTORE_PEM=root.crt \
  -n warehouse-cldrsetup
```

Websocket Connection failed:

```
livelog.ml-94961167-848.apps.ntnlab.local
```

Solution: Right-click on the page and select Inspect. Go to the Network tab, look for the livelog request, and add the corresponding livelog URL as an entry in the /etc/hosts file.

Kubernetes Command Reference:

```
export PATH=$PATH:/var/lib/rancher/rke2/bin
echo "alias helm='/opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/installer/install/bin/linux/helm'" >> ~/.bashrc
echo "alias docker='/opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/docker/docker'" >> ~/.bashrc
echo "alias helm='/opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/installer/install/bin/linux/helm'" >> ~/.bashrc
echo "alias kubectl='sudo -E /var/lib/rancher/rke2/data/v1.26.10-rke2r1-e58e49f33617/bin/kubectl --kubeconfig /etc/rancher/rke2/rke2.yaml'" >>
~/.bashrc
alias k=kubectl
. ~/.bashrc

kubectl get|describe|delete|create
all|pods|nodes|ns|namespaces|svc|service|pv|pvc|rb|rolebinding|sa|roles|csr|secret|hpa|netpol|statefulset|replicaset|crd      -n
vault-system | -A
kubectl get namespace vault-system -o json|yaml > tmp.json
helm list -n vault-system
kubectl api-resources --namespaced=true -o name | xargs -n 1 kubectl get -n vault-system
k delete ns vault-system --force
k get pods -A -o wide |grep dash
k get sa kubernetes-dashboard -n kubernetes-dashboard -o yaml
kubectl delete pods -n kube-system -l k8s-app=kube-dns
kubectl port-forward deployment.apps/kubernetes-dashboard 8443:https -n kubernetes-dashboard
kubectl port-forward deployment.apps/kubernetes-dashboard 8443:443 -n kubernetes-dashboard
k logs -n cdp cdp-release-dssapp-6b5b68bcfd-b9rdd
k apply -f secret.yaml
k create token default
kubectl api-resources --verbs=list --namespaced -o name | xargs -n 1 kubectl get -n vault-system --kubeconfig /etc/rancher/rke2/rke2.yaml
helm list -n kubernetes-dashboard
k get|delete|describe|edit      helmcharts.helm.cattle.io      -n vault-system

helm list -n kubernetes-dashboard
helm list -A
helm upgrade --install kubernetes-dashboard
/opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/kubernetes-dashboard-5.10.0.tgz -n kubernetes-dashboard
-f /opt/cloudera/parcels/ECS-1.5.5-b238-ecs-1.5.5-b238.p0.50802651/kubernetes-dashboard/kubernetes-dashboard-overrides.yaml --dry-run | grep
admin
find / -name kubernetes-dashboard*.tgz
netstat -lntup |grep 6443
openssl s_client -connect pvcbasemaster.cldrsetup.local:5432 -debug -msg
cp /etc/rancher/rke2/rke2.yaml .kube/config
envsubst
```



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